

Phil 110 Skepticism 101

Numbered Transcript

PHIL 110 • Dr. Michael Shermer

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Lecture 1: What Can We Know?

At a glance — Shermer asks what we can know.

- 1.1 All righty. Good morning, everybody. Nice to see you all. Welcome to my class, Skepticism 101: How to Think Like a Scientist. I've been teaching this class for many years at Chapman University, and I'm now on board with the idea of taking this out into the universe through the Peterson Academy. I appreciate that. I do think this is the future of higher education—that is, making content available to millions of people, rather than to the usual 25 students I have sitting there in a classroom. So that's kind of nice.
- 1.2 And in addition to teaching college for the last 30 years, I've also been publishing Skeptic magazine. We started in 1992, and our content is basically promoting science and critical thinking, investigating all kinds of claims of the paranormal, pseudoscience, fringe groups, cults, and claims of all kinds between good science, junk science, bad science, voodoo science, pathological science, non-science, and plain old nonsense. That's what we do. This is the the new design of the magazine. Um, here's some of the older issues we've had. So, we've covered a lot of hot-button topics in culture in, in culture wars, and so on.
- 1.3 My books largely, uh, search for the truth on various topics like conspiracy, conspiracy theories, the afterlife, morality, God, religion, evolution, creation, just pretty much you name it. I'm interested in, uh, to what extent we can say something meaningful or get out the truth about it. This lecture is called—this is the actual title of my next book that I'm writing now. So we'll be trying out some new ideas here. It's called Truth: What It Is, How to Find It, and Why It Matters.
- 1.4 So, I'd like to begin with some questions for you. Do you believe that truths are absolute or are they all relative? You don't have to answer. This is just, you know, in your head, you know. Do you believe in the germ theory of disease? That's been a popular topic since Covid 19 started. Do you believe that the 2020 election was rigged? Some nontrivial percentage of the American public thinks it was. Do you believe in a flat tax? Or maybe you believe in a progressive tax. Or maybe you believe in a regressive tax. Do you believe in democracy and capitalism?
- 1.5 These are rather different questions, if you see where I'm going with this. What do you mean by believe? OK. Do you believe there's absolute right and wrong? Do you believe that Jesus died and was resurrected for your sins? Do you believe there's objective meaning and purpose in life, or is it completely subjective and up to you entirely? And there's no, uh, purchase to find validity on some universal truths about meaning and purpose? Do you believe everything can be known, or are there unknowables? That is to say, not that we don't know them now, but that we can never know them?

- 1.6 All right, so these are just some of the—it's not like I have the answers to all these questions, because basically a lot of them, nobody has the answers or there's conflicting answers. So, what do you mean by truth? So I'm going to start with two types of truths to just kind of distinguish it a little bit. There's subjective internal truths. That is my truths. You've heard this expression. You know, that's my truth, right? Or objective external truths, truths that are not just true for me, but it's true for you. It's true for you. And we can we can get some kind of objective handle on that.
- 1.7 So let's look at some of these. Let's start with subjective internal truths. Now here's one for you. Dark chocolate is better than milk chocolate. Now, I happen to think this is an objective, absolute truth. But I understand from talking to people that some people prefer milk chocolate to dark chocolate. So that's probably one of these internal subjective truths. Now I'm dating myself here. I do happen to think Stairway to Heaven is the greatest rock song of all time. I know, I know what you're thinking. Free Bird. I love Free Bird. It's great, too. Or if you want to move up several generations, there's probably plenty to take the place.
- 1.8 There's some research that shows whatever songs you listen to a lot between the ages of 15 and 25 are, like, imprinted on your brain. It's those are the songs that touch all those deep memories from your teenage years and so on. So for me, it's classic rock. I just hit the classic rock station on my, on my phone and just listen to those. OK. But those are, those are subjective internal truths, obviously.
- 1.9 Then there are these subjective truths that could become objective truths. So my example for this is, meditation makes me feel better. And I'm using this as somebody else's example. I've tried it. It's not my thing. I sit there with my little chanting thing, like, omm, my God, I'm so bored. You know what I'm talking about. But a lot of people say, you know, it makes them feel better. It really improves their lives. So, but that's a subjective internal state. You're just telling me what works for you.
- 1.10 What people that are into meditation, like my friend Deepak Chopra, he wants to know it, does it actually work? That meditation really works not just for me. It's not my internal truth. Like, I like dark chocolate, you like milk chocolate and whatever, but that it really works. Right, so, to answer that question, and here's what Deepak did, is run experiments, like, at his center in Carlsbad, California. He brought in, like, 100 people. He had professional meditators that do this for a living. Then he had casual meditators that were instructed, and they tried it a little bit. And then people that didn't meditate at all.
- 1.11 Then they took objective measures—blood pressure, stress hormone levels, you know, heart, heart rate and all these kind of cardiovascular and medical measurements, chronic pain and so forth. And then to look, to see, are there measurable differences between the professional meditators, the amateur meditators, and the no-meditators?

- 1.12** Now that's a nice example. In the second lecture we're going to look at how science actually works of trying to tease apart the causal variables there. Does it really work, such that you could say something like, OK, meditation really works. You know, you do it five days a week, 20 minutes a day, and you use these techniques, and it has these measurable effects. Your blood pressure will go down this amount. Your chronic pain will decrease by two points on the scale or whatever it is. But you see what I'm going for there, is that you want to move it from inside me—it's just what I like—to, it's actually really true.
- 1.13** OK, so that's what we're trying to do in science. That's what I try to do, and in my work, is figure out what really works. Can we really get out of truth? Then there is objective external truths, like, there's X people here today. Well, I could just count them up and say, let's say it's, I don't know, 19. So we got two, four, six, eight, 10, 12, 14, 15, OK. It's 15. All you have to do is count, and we can get to the truth. So there really is an absolute objective truth to the question, how many people are here today?
- 1.14** You just look. Or the Dow Jones Industrial Average closed at 36,117.38 on Wednesday, December 7th, 2023, when I wrote this lecture. Well, that's true by observation because I just looked it up. The day we're recording this, I think it's over 37,000 now. Whatever, it's a moving target, but that's the idea, that you just observe. Just use your eyes there. It's either true or false.
- 1.15** Or something like, the dinosaurs went extinct around 65 million years ago. Now that's true by radiometric dating techniques for volcanic eruptions above and below dinosaur fossils. So you have a volcanic eruption, then dinosaurs are running around on top of it for years or decades or centuries or whatever. Then they die. The bones are lying there. Another volcanic eruption covers it. So you date, using these radiometric dating techniques that take the atoms that decay from one kind of material to another kind, like potassium to argon decays at this atomic rate, at some measurable rate.
- 1.16** So you can say, that level is 67 million years. And this level is 64 million years. There's dinosaurs in between. That's where they lived. After that, there's no more dinosaurs. So we could say around 65 million years ago, plus or minus. There's always these error bars. You know, your lab gets 64.3 million years ago, and your lab gets 64.2 million years, you know, so within the range. But nobody is getting like 64 million years ago and 10,000 years, like the creationists think that the age of the Earth is. So you get consistency. And every one of our labs is pretty close within the error bars.
- 1.17** That gives us some confidence that that's probably true. Not 100%. Could be wrong. Maybe in years to come, the labs will refine it, but it's not like they're going to suddenly discover the earth is only 10,000 years old, right? Or the Big Bang. Did it really happen? Well, this is true by a convergence of evidence. So we have the cosmic microwave background radiation. We have the abundance of light elements like helium and hydrogen. We have the distribution of redshifted galaxies. Turns out, pretty much every galaxy you look at, you take a picture of it and you put the light through the

spectroscope, and the light is shifted toward the red end of the spectrum, which means it's moving away from us.

1.18 I'm not a cosmologist or astronomer. I just tell you what I read from the professionals. They say that pretty much every galaxy is expanding away from us as if it was a big explosion. And then finally, it turns out that space itself is expanding, like it was an explosion. So, now again, it could be all these cosmologists are wrong, and that the Big Bang is not the correct theory for the origins of the universe. But I learned about this in college in the 70s, and it's been confirmed over and over and over and over again ever since. Not likely it's wrong at this point.

1.19 So we can kind of offer our provisional assent. That it's true, in the sense that the evidence is so substantial—here's my definition—that it's rational to offer your provisional assent. As I like to say, it's not impossible the dinosaurs died recently, just after the creation of the universe 10,000 years ago, as young Earth creationists think. But it's so unlikely we don't have to waste our time considering it. So this is my definition of truth, is a claim for which the evidence is so substantial, it would be rational to offer one's provisional assent.

1.20 OK. The key there is provisional. OK. In science, there's no absolute 100% truth. Just for now, this is what we think is true. Might be wrong. Maybe you're going to debunk it. Maybe you'll debunk it. Who knows? But for now, this is what the community thinks is probably true. So it's true with a small t. Let's just use it that way. And it could change.

1.21 And then, in science, we also want to make a distinction here between what we mean by proof. You'll often hear this word proof. Can you prove it's true? Well, what do you mean by prove? Mathematics, they talk about proofs like in Euclidean geometry, where the sum of the internal angles of a triangle is always equal to 180 degrees. That's just true by definition. You don't have to go out and measure triangles in the real world. You can just do it on paper.

1.22 Now I say Euclidean geometry because there's this other kind of geometry out in space where things, space time, is warped, and therefore you get more than 180 degrees in a triangle. But that's, we don't have to worry about that. And then another cut we could make on talking about truths is the difference between analytic truths and synthetic truths. So analytic truths are true by meaning of the terms in the claim or the equation. These are called a priori truths. And then there's synthetic truths, that is true by observation a posteriori.

1.23 So a priori, prior—I know it's true because it's right there in the definition of the words or the symbols. A posteriori—I don't know, I have to look and see if the claim is true or not. So one is just true by definition, the other is true, I got to go out and check to see if it's true. Right? So an analytic truth claim is something like the famous syllogism. "All men are mortal. Socrates is a man. Socrates is mortal." To know if it's true or not, you just have to know what the words mean. You don't have to go out and interview people

named Socrates and see what their gender is or anything like that. That would probably be a controversial thing to do these days, but let's save that for another, another day.

- 1.24** Or definitional truths. Pediatricians are doctors. All you have to do is know what the word means, or there can't be a married bachelor, because that's not what the words mean. Synthetic truths require evidence, like pediatricians are rich. Well, maybe. Maybe some of them are. Maybe some of them aren't. I got to go out and collect data and look at surveys and check tax returns or whatever to know if that's true. And the more evidence for that synthetic claim, the more you should believe it. And vice versa. The less evidence, the less you should believe it.
- 1.25** Or again, in mathematics, two plus two equals four. All you have to know is what the symbols mean. And then it has to be true, as opposed to, there's cheese on the plate. Well, I don't know. Let me look. If there is, I'll accept it. If not, then not. So you just look to see if it's true. So objective facts to determine provisional truths about the world are determined by methods developed over the centuries since the scientific revolution and the Enlightenment.
- 1.26** Just think about that for a minute. Before, about 400 years ago, nobody had any methods to determine what's true. It was mostly just my opinion, your opinion, our tribe versus your tribe, our nation versus your nation, my religion versus your religion, about what's true. Since the scientific revolution and the Enlightenment, we're now trying to get to knowledge as defined as justified true belief. That is, we want to know what we should believe is true. Do you have justification for it? Not just what we want to believe is true, but what's actually true.
- 1.27** Here's another way I like to say it. I don't want to believe in things that have to be believed in to be true, right? Now, I think that applies to most things. It's possible I'll come back to this later. There might be some political truths. You know, we all have to believe in democracy for it to work. You have to believe that the money you have will really be backed by the US government, because it's not backed by gold. So you have to kind of believe it to work. But I'm talking about other kinds of empirical truths. I just want to know what's actually true, not what I'm supposed to believe.
- 1.28** OK, so here's the problem. None of us are omniscient. I don't know if there's an omniscient God or not. But I do know this. It's not me. And I also know it's not you. OK, you're not God. I'm not God. Or if you prefer the secular equivalent, there is objective reality, only I don't know what it is, and neither do you. So we have to start with the assumption that we could all be wrong, and then work together to try to figure out what we should do, to determine truth.
- 1.29** The problem is, is that, as Charles Mackay said in his, really, this is the first modern work of skepticism, *Extraordinary Popular Delusions and the Madness of Crowds*. The book is about how wrong people get things when they get caught up in a, like a social contagion. There's a lot of these bubbles, like the tulip-o-mania he wrote about. And as

he said, "When men wish to construct or support a theory, how they torture the facts to their service."

- 1.30 OK, I have a whole lecture on how thinking goes wrong and how this motivated reasoning, confirmation bias, hindsight bias, and my-side bias clouds our rationality. But this is the problem that you and I all have to work together to solve. We want to know what's true. There's all these things interfering with our ability to know what's true. So we got to work together and be aware of these problems.
- 1.31 OK, so let me just lay down a few foundational suppositions to the entire course. And really, most of my work of what we start off assuming is true. That is universal realism. I mean, these are really broad. There is a reality, and we really can know something about it. Not just me, but you and you and your tribe, my tribe, our nation. Anybody can know something about reality.
- 1.32 Moral realism. I'm assuming there is a right and wrong, and that we can figure out what it is. Now, of course there's gonna be disagreements about this, but it's not completely relative. We can determine good and evil and right and wrong through arguments and evidence. That'll be my claim. And therefore, there's moral progress. That is, things really are getting better because we have measurable metrics that we can look at to see what leads to human flourishing. This, democracies lead to more human flourishing. Autocracies and theocracies lead to less human flourishing. One of them works, one of them doesn't work on average. And there we can measure moral progress or regress.
- 1.33 And then scientific naturalism, which will be a big part of the next couple of lectures. That is to say, we're looking for natural explanations for natural phenomena. It's not that there can't be a supernatural, but I don't know what that means. If it's outside of nature, then how are you going to measure it? I'll come back to that. And then finally, a foundational supposition is Enlightenment humanism. That is to say, these universal principles of reality that you and I can all discover and apply to making life better through these natural principles that can lead to something like a universal ethic that applies to everyone.
- 1.34 That's a test. That's a generalization, a test of any ethical principle. Does it generalize to everybody? It's not just good for me and my family, my tribe, my religion, my political party, but it's good for yours. Even though you disagree with me, it's still good for you. Then we can come to some agreement on that. So the general principle underlying all these is to have confidence in the truth value of any claim. There must be a source of external validation so that it can be treated as an objective truth. Otherwise it's just personal taste or opinion or feeling.
- 1.35 And again, if it's based on feelings and opinions, then you have all these cognitive biases that kick in, the my-side bias. This is what I want to be true. This is what my tribe believes. It's what my religion believes, what my political party believes. Then we're

diverging away from figuring out what's true. So without some form of external validation, knowledge becomes unjustified belief.

- 1.36** OK, let me tell you my favorite, one of my favorite stories. I call it Sagan's Dragon. This comes from Carl's book, *The Demon-Haunted World*, in which he set up this sort of thought experiment. I have a dragon in my garage. Would you like to see it? I mean, it's pretty cool. Dragons are just legendary in folklore. And what if they're really true, there really are dragons? And you say, well, yeah. OK, cool. Let me see it. I got to see it to believe it. So I take you to my garage. I open the garage. You look inside and there's, you know, kind of some empty paint cans, a few bicycles, a ladder, but there's no dragon.
- 1.37** And you say, well, where's the dragon? I said, well, OK. See this? This is an invisible dragon. Oh, you say. An invisible dragon? Yeah, yeah. Um. Well, OK. I propose, um, you propose, let's say we sprinkle some powder on the ground, and then when the invisible dragon is walking around, he'll leave footprints, and then we can say, well, there's the evidence. We can infer its existence from the footprints in the flour.
- 1.38** Well, I say, you see, this dragon levitates above the ground about a foot, and you're going, OK, let me get this straight. This dragon is invisible, and it levitates. Yeah, yeah, yeah. That's right. OK. You know those little thermometer things they used during Covid? You know, you'd point at people's forehead, get their—OK, I have one of those. How about we kind of point it around the garage and we can pick up the heat from the dragon because, you know, it's a warm-bodied animal. We get the heat. Well, no, this is a cold-blooded dragon. It gives off no heat at all. What about the fire? These fire-breathing dragons. Come on, we can measure the fire. No, it uses cold fire.
- 1.39** OK, at some point, you're going to go, Shermer, Sagan, you know. What's the difference between an invisible, levitating, cold-bodied, cold-fire-breathing dragon and no dragon at all? OK, so Carl's point of this is that, if there's no way to get at it, if you constantly do what's called hand-waving and you never provide any way to know if it's true or not, then what are we even talking about? OK. And you can apply that to any principle that you want. At some point we have to have evidence.
- 1.40** And as Carl said in his famous quip in *Cosmos*, in the episode on aliens, "Extraordinary claims require extraordinary evidence." We call this the ECREE principle, sometimes called the Sagan Principle. The idea is that, if you have an ordinary claim, it only requires ordinary evidence. If you have an extraordinary claim, it requires extraordinary evidence. So your extraordinary claim is not true until you prove otherwise. OK.
- 1.41** So, here we like to say that the burden of proof is on the claimant, not the skeptic or scientist to disprove the claim. So I just told you this crazy story. And if you said, well, do you have any evidence for this? I mean, did you pick up, like, one of the little widgets on the dashboard of the spaceship to show us? And if I say, can you prove I didn't go to the Pleiades? OK, that's an error in logic. The burden of proof is not on you

to prove that I didn't get abducted by aliens. It's on me to prove that I did. So we call that the burden of proof argument.

- 1.42 OK, what if somebody comes to you with a claim and they have no evidence at all, like the story I just told? Here I invoke what I call Hitchens' dictum. You recognize the great, late great Christopher Hitchens. "What can be asserted without evidence can also be dismissed without evidence." I like that. You know, I think Bigfoot is real, you say. I say, OK, that's interesting. Can I see the body? If you go, no, I see, here's the thing. I was in my tent. It was three in the morning. I heard this thing and I got out there and my phone didn't work, and but I took this blurry photograph. And if you kind of use your imagination, you can sort of see in the shadows this thing that looks like—OK, stop. I don't have to listen to this anymore, right? You got to show me real evidence.
- 1.43 Now, there's an interesting twist on this. In science, we have something called the null hypothesis. Sometimes we talk about these as negative truths. That is, the nonexistence of something is the truth. So, since we just went through this pandemic, let me use an example. Let's say you have, you claim you have a cure for Covid-19. You heard this from many people for the two years we went through that, you know, hydroxychloroquine or this or that. OK. Is it true?
- 1.44 Well, before the FDA will approve your drug for sale, you have to provide substantial evidence to reject the null hypothesis that you do have such a drug. So, the null hypothesis is, is that your hypothesis, that you have a cure for Covid-19, is not true. And we're not going to grant you permission to sell your drug to the public until you overturn the null hypothesis, which you can do through epidemiological studies, randomized controlled trials. You try it on 1,000 people that have Covid, that don't have Covid, a placebo group, and so on and so forth. That, you've heard these terms. That's how it how it works.
- 1.45 So what, I like to apply this principle to everything. The null hypothesis. The more extraordinary your claim, the more I'm going to say, I'm just going to assume the null hypothesis is true. Your claim is probably not true, until you show me evidence, and then I overturn the null hypothesis and accept it. So again, when people say to me that Bigfoot is real, I say, OK, I'll believe it if you show me. Otherwise, it's reasonable to be skeptical. In this case, Bigfoot not real is the null hypothesis.
- 1.46 And as I tell people, these cryptozoologists that show me these, like, footprints and grainy videos and fuzzy photographs, show me the actual body. I mean, just, come on, just take me, here it is. Not a guy in an ape suit at a distance through the camera that you know where it's blurry—and by the way, you know, the dispersal of over a billion smartphones with high resolution cameras, you would think we'd have tons of high resolution video and photographs of Bigfoot, Loch Ness, aliens, UFOs. In fact, we have nothing like that. That is one of these negative evidences, right? Probably the null hypothesis is true.

- 1.47 And then finally, on this category, I find it very telling that, in the millions of documents leaked through WikiLeaks and other whistleblowers, you would expect that if these conspiracy theories were true, there'd be some evidence that we weren't supposed to see it. But here it is, the whistleblower—there's no mention of a JFK assassination conspiracy or a UFO cover-up at Area 51 or wherever, or a fake moon landing. You know, if you fake the moon landing, and they did the whole thing in a studio like this, the people running the cameras and so on, they have to be paid. There'd be some paperwork. The whistleblower shows, oh, look, here it is. There's nothing like that.
- 1.48 Or that 9/11 was an inside job. Again, just think about all the people that would have to be involved to pull off 9/11 as an inside job. Not one of them wrote a memo? Not one of them told a story to somebody? Not one of them made a phone call or got a paycheck or something that's in the leaked documents? Nothing? So here are the evidence of absence or, sorry, the absence of evidence is evidence of absence. It's probably not true.
- 1.49 Now, I have two possible exceptions to this that we'll talk about later in the course as well. SETI, the search for extraterrestrial intelligence. I know these guys, these astronomers that work at this institute. They're great, I love it, I support them. Keep looking. They haven't found anything. No signals from the aliens or anything. No technosignatures and so forth. But the problem is, they just started looking. The universe is vast. You know, there's 100, 100 billion stars in our galaxy, and we've barely even looked at a few thousand, let's say, in our local neighborhood of our galaxy. So it's possible. It's reasonable to say the absence of evidence is not evidence of absence. Keep looking.
- 1.50 And my final example that I'll address in the last lecture of the course—God. Is there evidence for God? Is there an absence of evidence for God, enough that we can say there is no God? I don't know. I'm just going to put that out as a question, and we'll get to it in the last lecture. In the Q&A, you can tell me what you think, but it's one of those things where we have to be careful about what we conclude is not true, even if there's no evidence, because maybe we don't know enough yet. Right?
- 1.51 So this is what's called intellectual humility. I always want to be humble before the facts. If I lived 500 years ago, I probably would have believed in witches like everybody else. Now I don't, because pretty much nobody does. If I came, if I was chronically frozen, brought back 500 years from now, would I go, oh my God, look at the crazy stuff we believed in the 21st century, I can't believe it. Now we know, whatever it is, but if I knew that, I wouldn't believe, wouldn't believe what I believe now. So, intellectual humility.
- 1.52 OK. The key to reliable knowledge is external validation through experiment and corroboration by others. OK, again, I can run the randomized controlled trial, but I might be biased. Maybe I recorded the data incorrectly. So you got to do it, and you got

to do it, in your lab and your lab and your lab and so on. Then if we all get the same results, we're pretty close. Our confidence goes up.

- 1.53** But what about mystical truths? Mind-altering drugs like MDMA, ecstasy, ayahuasca, DMT. OK, let me just riff on this for a second, because people ask me about this. I know people that have done some of these things. Most of them swear that in that state they've experienced a different reality. There really is this spirit world. There are these animals and these creatures and these light beings and so on. And when I say, like Sagan's dragon, really? Show me, they go, well, take the ayahuasca, and you'll see for yourself. It's like, yeah, but that's still in my brain.
- 1.54** OK, so you know, I'm on Rogan and he's telling me, Shermer, you got to do, you got to try this because you won't believe what you'll see. It's like, I know, Joe, but I don't know. How am I going to know it's not just me hallucinating? And you were hallucinating and you took it, you hallucinated. These I find challenging because we are restricted in what we can know about reality to our senses. You know, species-specific perceptions about the world. Bats, they see things differently than you and I do. Dolphins see things differently than bats and so on.
- 1.55** But we assume there is a reality out there and we can know something about it. But what if there's something else? Now, I haven't done this yet, but I have, I have this invitation to go to this resort in Costa Rica and spend a whole week doing ayahuasca. And I read what people go through on this, like, oh, I don't know. This doesn't sound like that much fun to me. But the problem would be, how would—so I come back and I update to—you guys are invited back. OK, I changed my mind. You're not going to believe it. I took the ayahuasca. It was holy spirit beans and the colors and the love and oh. And you go, wow. OK, show us the evidence. Show us on the slides. Well, I can't. You got to take the ayahuasca and you'll see for your—OK, then we're in that loop.
- 1.56** OK. So when I was in college, I took a philosophy class from a professor named Richard Hardison who got me into all this business. And I wrote down what he said about mystical truths. By their nature, must be solely personal, and they can have no possible external validation. Each has equal claim to truth. Tea leaf-reading and astrology and Buddhism, each is equally sound or unsound if we judge by the absence of related evidence. This is not intended to disparage any one of the faiths, merely to note the impossibility of verifying their correctness.
- 1.57** Now, to me, that was meaningful because at the time I was a born-again Christian. I'm not anymore. But at that time I thought, huh, I wonder if my religious beliefs are more like internal subjective truths and not external empirical truths? This kind of got me thinking about that. And again, these mystical experiences may be one of these unknowables. That is to say, we may never know for sure because it's always going to be in your head when you do the ayahuasca or whatever. OK, so maybe I'll try it someday. Maybe not. I don't know.

- 1.58** OK. Additional distinction about beliefs and truths that I need to say. So I don't believe in the theory of evolution or the theory of the Big Bang. I mean, people say, do you believe in the theory of evolution? Well, no, I just accept them as true. So here, belief is something different than, let's say, I believe in a flat tax. You believe in a progressive tax, and she believes in a regressive tax. See, again, I want to make a distinction as we go along through the lectures. These empirical, objective truths in science—but religious truths, political truths, these may be in a different realm.
- 1.59** The truth is like, well, I don't know, whatever your democracy has voted to decide, this is what we want our society to do. We want to have a tighter social safety net and provide more social security for people and so on. So we have to raise taxes. So a progressive tax works better than a regressive tax or a flat tax for that purpose. So that's the truth. But it's not true, like really true, in some empirical way. It's more political.
- 1.60** And then as an example of religious truths that I'll touch on in a later lecture, the claim that Jesus lived and was crucified. Now, this may be an objective historical truth. I, I have many friends who study the Bible. They study the history of religion. Bart Ehrman, for example, has written many books on this. He thinks the evidence is fairly substantial that there was somebody named Jesus and he really was crucified. The Romans crucified everybody for everything, right? So that's not a particularly extraordinary claim.
- 1.61** But that he was resurrected from the dead, that may be a different kind of truth. It may be objective. But if so, it would be pretty extraordinary. I mean, no one has ever come back from the dead, as far as we know. So what's the evidence for that? But maybe that's not the right way to think about that. So I'm going to touch on that in a later lecture. Or how about the claim that Jesus died for your sins? How would you test that? I mean, it's just, that's just a religious kind of truth, not an empirical truth or measurable objective truth, that sort of thing.
- 1.62** I'll tell you a funny story. So I was at a conference on science and religion, and I'm on stage with Richard Dawkins, the famous evolutionary biologist and atheist, and Ken Miller, who is also a world-class evolutionary biologist, who was the primary debunker of intelligent design creationism. So he wrote a whole book about why the intelligent design creationists are wrong, and here's specifically why they're wrong. Here's their arguments, here's why they're wrong, and so on.
- 1.63** The last chapter of his book, he says, oh, by the way, I'm a Catholic, and I think Jesus not only lived, but he died, came back from the dead, and atoned for my sins. I accept him as my Savior. Wow. OK, so, on stage, Richard is like, OK, Ken, let's say we found a piece of the true cross. You know, in the Middle Ages, everybody had a little piece of the true cross, right? Piece of wood. And on the wood there's a little bit of flesh. It's Jesus' flesh. Now, we know that Jesus' birth was not normal. So if we extracted the DNA

from the flesh on the piece of the true cross, would it be different from a normal human DNA?

- 1.64** OK, so Ken could see where Richard was going with this, and he says, Richard, Richard, I'm not claiming this is true in some scientific sense. I'm a Catholic. This is what we believe. That Jesus died, was resurrected for our sins. That's it. And in a way, saying, I'm not claiming this is an empirical, objective truth. This is something else. I just believe it because that's what we believe as Catholics or Christians, right? So I find that interesting. Again, we can think about that and talk about it in the Q&A, where you can go with that.
- 1.65** OK. So let's just shift gears a little bit. Current debate you've heard since, say, 2016 or so. Are we living in a post-truth era? Here's the cover story of Skeptic on this issue. Steven Pinker at Harvard wrote exquisitely about this. Notice that we didn't frame it as the form of a question, why we are not living in a post-truth era. My original title was for this cover story was going to be, Are We Living in a Post-Truth Era? But Steve reminded me of Betteridge's law of headlines, which states, any headline that ends in a question mark could be answered by the word no. Headline writers are just trying to get you to flip open the magazine to the story. It almost always means no.
- 1.66** OK. The problem is, it's self-contradictory. It is, is the statement, "We're living in a post-truth era," true? If so, it can't be true. The evaluation of the proposition as true or false means we're still evaluating propositions as true or false, which means we're not living in a post-truth era. It's like the statement, humans are irrational. Is that a rational statement? If so, it can't be. Because if humans are really irrational, then who specified the benchmark of rationality against which humans don't measure up? And how did they conduct the comparison, right?
- 1.67** So this line of reasoning about reasoning was made most explicitly by the philosopher Thomas Nagel, in which he said, "Truth, objectivity, and reason are not negotiable." That is, as soon as you're making a case for them or against them, you're making a case, which means you're implicitly committed to reason. In other words, the minute you open your mouth, you've lost the argument, because you're trying to make an argument.
- 1.68** Nagel calls this a Cartesian argument. Recall that René Descartes' sort of extreme skepticism. You think I'm a skeptic, he was super skeptic. He sat down and decided to doubt everything he couldn't for sure know, and he got all the way down to, somebody's asking the question, is there a reality? And that has to be a brain doing that. So I have to exist. "Cogito ergo sum." I think, therefore I am. That's what Nagel means by this. Just as one might question one's own existence shows that you have to exist. That one is examining the question of rationality shows that one is committed to rationality. So in other words, we don't have faith in reason. We don't believe in reason. We just use it. It's just a tool. It's built into the system.

- 1.69** OK. Pinker also makes a nice distinction between what he calls the reality mindset and the mythology mindset. Mindset is that this kind of physical objects around us, people in our lives, our memories of interactions with others, people are fairly rational and reasonable in this reality mindset. The mythology mindset, Pinker says, it's harder to get at, and so it's more in a realm of, I just believe it because I don't know why. Because I don't really know. The remote past, the unknowable future, faraway peoples and places, remote corridors of power, the microscopic, the cosmic, the counterfactual, the physical, and so on—these are, these are things the average person, including us, really can't know much about.
- 1.70** You know, when my cosmologist friends tell me about the accelerating expansion of the spacetime universe, and that it's more than 13.8 billion years old, it's actually 46 billion years old because of the accelerating expansion of space, I have no idea if any of this is true. I just kind of, well, these guys are pretty smart and they compete with each other. They work in this community. They test each other's ideas. All right. I can't really get at it. So.
- 1.71** But what Pinker means is something slightly different, that we kind of have it in this mythology, it's part of my worldview, my politics, my religion, my just overall worldview, and I can't really know much about whether it's true or not. I'll give you an example. So this guy comes to my house, belong to this club together, have our kids play together, and he sees I have a chess set in my house. He goes, oh, you play chess? I go, a little bit. You know, maybe we could play sometime. Yeah, yeah. And he goes, yeah, I used to play grandmaster-level chess. Oh, well, you don't want to play me. I can't even beat the first level of my Mac.
- 1.72** And then he was talking to me. We play tennis at this club. He goes, oh yeah, I used to hit with Jimmy Connors, you know, all the time. We used, Jimmy and I, I'm like, OK, you don't want to play me because I'm nowhere near that level. And then we start talking about cycling because I'm a cyclist, and doping in sports, and the whole Lance Armstrong thing and all that. And he goes, oh, yeah, I used to be on the juice. I'm, you know, just totally just ratted out with steroids, when I was in the Special Forces. And he's going on about this.
- 1.73** And you have to picture it. This guy sitting across from me looks like Pee-Wee Herman. OK. It's like, OK, you know, three things like that. I'm like, I can't. I don't know how to fact. I'm not going to bother to fact-check this guy and look him up and call people. I just think, you know, three times, that's enough. I'm skeptical. Right. And this is, this is the point. Most of us can't check most things, so we just make a lot of assumptions about what's true.
- 1.74** But for the most part, people are more reasonable than we think. So, for example, a quantitative analysis of the role of fake news in the 2016 election found that fake news took up far less than 1%. So you heard this meme after Trump went, oh, well, it was,

you know, the Russian meddling in the election. And they changed, you know, the outcome. No, the research shows that they really didn't.

- 1.75** I mean, for example, on the Pizzagate thing, you know, that Hillary is running this child sex ring out of a basement of a pizzeria in Washington, D.C., the Comet Ping Pong pizzeria. Do people really believe that? One guy did. Edgar Welch. He went there with his gun to break up the crime and was surprised. There's no basement at the Comet Ping Pong pizzeria, much less anything untoward going on there. But people who already hated Hillary were not going to then change their vote. Hearing that, oh, there is no Pizzagate thing, right? Or vice versa.
- 1.76** So people mostly vote tribally, not empirically that way. It's often said that you can't reason people out of beliefs that they didn't reason themselves to in the first place, but that's not true. I get letters every day from people who changed their mind after reading something I wrote. It's possible to do that. Evidence can change people's minds, even on highly politicized issues. So the research on this, Nyhan and Reifler study, like, if you're talking to a climate denier about climate change, you can change their mind, first if you take the politics out of it. I mean, we often think that, the bundling of climate science through Al Gore's film *An Inconvenient Truth* made it a left-wing, liberal political issue rather than just a scientific issue.
- 1.77** And therefore, when conservatives hear climate change, their brain autocorrects to anti-capitalism, anti-America, anti-business, controlling the economy and so on, instead of CO₂ gases or melting glaciers or whatever. So you have to take that out of it and then just present the evidence as compelling and clear and simple, particularly visually, through graphs, to get people to change their minds. But they really do.
- 1.78** In this sense, then, hinting at that, beliefs also signal group loyalty. So things like climate change, evolution, capitalism, trade agreements. Most people don't know that much about them, so they take a position we call virtue signaling. So there's studies on this. Like, I'll give you some simple examples. Do you know how a zipper works? Everybody goes, oh yeah. Explain it. Well, you take the little hook thing and the doohickey and you pull it up. OK. That's not how zippers—or, you know, what's your position on NAFTA? Oh, I'm against it or I'm for it. What is it? Well, it's that trade thing with some of these countries. I forget who.
- 1.79** In other words, they're just echoing their political party position or whatever, as it, rather than actually really understanding it. And that's true for most things, I think. So it turns out science denial, as we call it, is not correlated with scientific illiteracy. What does predict it is political orientation. So, for example, a friend of mine studies beliefs in the theory of evolution or beliefs in creationism. Most college students say, yeah, I accept the theory of evolution as true. Explain it. Well, you know, the giraffe stretches its neck and the baby giraffe has longer—no, that isn't how it works.

- 1.80** That's a Lamarckian. A Darwinian model is natural selection, genetic variation, selection for the taller necks, and so on. But by saying they accept it, what they're really signaling is, I accept that the scientists usually get it right. That's good enough for me, because how can you check everything? Who can know everything, right? So unfortunately, a lot of these things are politicized. GMOs and vaccines, at least before Covid, were more of a left-leaning liberal. Skepticism on the right, anthropogenic global warming as a cause. Again, these are conflated with these political tribal issues that makes it difficult to know what's true.
- 1.81** All right. Final point on this. The problem is not post-truth, on this post-truth issue, but postmodernism or cultural relativism. I like this quote from Hannah Arendt. "The ideal subject of totalitarian rule is not to convince Nazi or to convince Communist, but people for whom the distinction between fact and fiction, in other words, the reality of experience, and the distinction between true and false, the standards of thought, no longer exist." So that's the concern about this postmodernism. It's not that we can't know anything. Of course we can, but that knowing what's true or not is less important than it used to be. That's our concern.
- 1.82** So a couple bullet points here that we like to, to say on this point. "Everyone is entitled to his own opinion, but not his own facts." The great Senator Daniel Patrick Moynihan. Right. Free speech. Have all the opinions you want, but the facts are facts, hopefully, that we can all know. And I like this from Philip K. Dick. "Reality is that which, when you stop believing in it, doesn't go away." Again, I don't want to believe in things that have to be believed in to be true. I just want to know what's actually true.
- 1.83** One of my favorite essays on this, from the great science writer and science fiction writer, Isaac Asimov. "When people thought the Earth was flat, they were wrong. When people thought the Earth was spherical, they were wrong. But if you think that thinking the Earth"—OK, it's not a perfect sphere. It's an oblate spheroid. It's fatter on the equator, on the equator, and so forth—"But if you think thinking the Earth is spherical is just as wrong as thinking the Earth is flat, then your view is wronger than both of them put together."
- 1.84** There really are ways to get at what's true and improvement in science. Or, as Richard Dawkins likes to say, "When two opposite points of view are expressed with equal intensity, the truth does not necessarily lie exactly halfway between them. It's possible for one side to be simply wrong." So back to our geology example with the dinosaurs. You know, geologists tell us the Earth is about 4.6 billion years old, plus or minus, based on all these radiometric dating techniques and so forth. Creationists say the Earth is 10,000 years old because this is what it says in the Bible. They think, OK, the truth is not, you add 4.6 billion to 10,000, divide by two and get the number. OK, one side is just wrong in that sense.
- 1.85** Another fun example comes from the theoretical physicist Simon Singh, who shows how science is self-correcting. OK, this is kind of a, the entertainment portion of this

lecture. Simon used this example from a singer in the UK named Katie Melua, who wrote a song, just a way of kind of metaphorically expressing the love for her partner. And it goes like this. There are 9 million—the name of the song is 9 Million Bicycles in Beijing. "That's a fact. It's the thing we can't deny. Like the fact that I will love you till I die." Then she continues, "We are 12 billion light years from the edge. That's a guess. No one can ever say it's true. But I know that I will always be with you."

- 1.86** OK. That's nice. So Simon then writes an op-ed piece about this in The Guardian, in which he kind of takes her to task in a fun way, a little tongue-in-cheek, you know. It's, you know, it's not 12 billion light years. We know pretty precisely, it's 13.7 billion light years away, you know, and there's no edge. The universe is just expanding. Light, space time itself is expanding. You're never going to reach the edge. And it's not a guess. Scientists have been working day and especially night to figure out what this is, through all these complicated techniques, and with well-defined error bars of our estimated speed of the galaxies and how old they are because of the number of light years they are away and so on.
- 1.87** So much to his surprise, after Simon published this little op-ed in The Guardian, he heard from Katie Melua, who wrote it, and said, I'm so embarrassed because I was in the astronomy club in high school and I really wish I could have got it right. So they went to a studio and they recorded a corrected version. I mean, it's often said, you know, what can science say about love? I mean, love in evidence? So here I'll quote my friend Tim Minchin, the singer-songwriter from Australia, who said, "You know what you call love without evidence? Stalking." OK. That's good.
- 1.88** All right, now let's talk about what we can know and what we can't know. You're probably familiar with our former secretary of defense, Donald Rumsfeld, in which he gave a very famous press conference in which he talked about these intelligence reports. And he said, "There are known knowns; there are things we know we know." But there are also, "We also know that there are known unknowns." That is to say, "We know there are some things we don't know." Then he adds this. "But there are also unknown unknowns—the ones we don't even know we don't know."
- 1.89** This became so famous that there were, like, t-shirts and hats, and there's a Wikipedia page dedicated to it. His memoir was called something like Known Unknowns. There's a documentary film by Errol Morris on his life called The Known Unknowns and so on. So here I'd like to add a fourth category: known unknowables. And I'm going to include in these: consciousness, free will, God, and why there is something rather than nothing. I'll come back to this in the last lecture of the course. But here's the idea.
- 1.90** Let's just take the hard problem of consciousness. That is, what it feels like to experience something. This may have to be postulated as a brute fact, about reality where explanation ends and epistemological wall, an ontological assumption. Let me tell you what I mean by this. This phrase, the hard problem of consciousness, was coined by a philosopher named David Chalmers. He contrasted with the so-called easy

problem of consciousness. Easy, as in, at least we can solve it. It's a hard, it's hard neuroscience. But like, figuring out where in the brain facial recognition happens, we know from experiments that it's in the temporal lobe, this little area here above the ear called the fusiform gyrus.

1.91 We know because, when there's damage to that from a stroke or tumor or a diseased brain, people can't recognize faces. You know, this, this face blindness. Sometimes they can't even recognize themselves in the mirror. They're looking at the mirror. They're not seeing themselves. Because this neural tract here is just damaged. So we know where that happens and where red is recorded in the visual cortex back here and so on and so forth. Those so-called easy problems were hard to solve, but at least they're solvable.

1.92 What Chalmers means by the hard problem is, what's it actually like to experience the red or the experience of seeing a face? And I don't know the answer to that, and I'm not sure it's knowable—in a weird way, because it's like asking, what's it like to be me? Or what's it like to be you? How would I know? I'm not in your head. Right? I'm going to come back to this in a later lecture on dualism and this problem of figuring out what's true for other people in the sense that it suggests there's a mind that is separate from the brain, and it floats out of my skull into your skull and tiptoes in there and looks at the screen in your mind to see what the red looks like to you. This is not possible.

1.93 You know, there's a famous paper, again, Thomas Nagel, who I mentioned. The one-thought-too-many guy that, what's it like to be a bat? Well, we're in a room where, you know, if I kind of face this way and I talk loud, I can kind of hear the sound bouncing off these walls. Maybe that's kind of what it's like to be a bat. If I was in a small closet and I yelled, then I'd really kind of get that effect, you know, but I'm still just a human, trying to figure out what it's like to be a bat or a dolphin or whatever.

1.94 You know, you could strap on one of those big monofins, and paddle along, and maybe have a snorkel, and a breathing apparatus and so on, so I can kind of slip through. But I would just be a human, kind of, I kind of see what it's like to be a dolphin, but I would really not have any idea unless I just became a dolphin, but then I would be a dolphin or a bat, not wondering what it's like. I wouldn't be a human in there, going, oh, this is what it's like to be a dolphin, no. I would just be a dolphin. And the dolphin wouldn't be like, oh, I wonder what it's like to be a human. They don't even ask questions like that.

1.95 So a lot of these questions we come up with in science, I think, might be in that known unknowable, that maybe we can never get at it because it's conceptually mistaken the way it's presented. And I think free will and determinism, again, I'm going to come back to these, this is kind of an overview lecture here, it may also be one of these. It depends on what you mean by determined freedom, volition, choice. You know, we live in a determined universe. That's what the scientists tell us. So yeah. OK. But I sure feel free. Like, I feel like I chose to come here, and you chose to come listen to me, OK? You made choices. You went left instead of right. You did this instead of that.

- 1.96** Now the determinists tell us, oh, that's all, was programmed in through your upbringing and your genes and your—and so on. OK. That may be, but the universe isn't predetermined, so I can have some effect on the future of the universe by changing things now, because there's a present me and a future me, and I go, OK, future me is going to want this. So current me is going to want, have to do something different to affect the future and so on. And I think that may be one of these problems.
- 1.97** And again, I come at this because I have all these people write books on these subjects on my podcast, and we talk about it for hours, and it never goes anywhere. I had, you know, the determinist guy and the free will guy, and they talked for three hours, and at the end, it's like, I don't really know which one's right. And I think it might, you know, this debate is 2500 years old, back to Aristotle and Plato and Socrates started to lay down the—and now, you know, the best minds in the world, a lot smarter than me, are writing books about this, and they completely disagree with each other.
- 1.98** Like, here's Robert Sapolsky in his new book, *Determined*. And it's like, oh man, this is so right. And then I get Dan Dennett with his book on *Freedom Evolves*, like, oh man, it's so right. Well, they can't both be right. Or more to the point, maybe the whole thing is conceptually mistaken. And so we end up at this one of unknowables. And then, there may be God. OK, again, I'm going to come to this many times in this course because it's one of the the great questions ever. You know, can we ever know if there is a God or not? I'll go through all the arguments in the last lecture of the course.
- 1.99** But what if we encountered—OK, back to the aliens. What if we encountered an extraterrestrial intelligence that's so advanced that it would be indistinguishable from being omniscient and omnipotent? What would you call it? Now, this doesn't seem very conceivable, but just take how far we've come in the last, I don't know, say, 100 years from the Wright brothers to landing on the moon and colonizing Mars or whatever we're going to do if Elon has his way.
- 1.100** This is just a start. Let's say you extrapolate that out 10,000 years of this kind of progress, you know, computer chips getting smaller and AI coming online and so on. Maybe in 100,000 years, we'll be effectively omniscient and omnipotent, able to manipulate the universe and so on. What would you call us in that case? Right. So again, this may be one of these things.
- 1.101** What do you mean by the word God? OK. This is one of those really hard ones. We're here with the Peterson Academy. Jordan is a friend. I know him, and he and I have had discussions about this, you know. Do you believe in God? He said, well, it would take me several hours to answer this question. So we get into it. You ask me, do you believe? No, I don't. OK. And it depends what you mean by these words. Right? So conceptually, I think this may be one of those known and unknowables.
- 1.102** And finally my fourth category here. Why is there something rather than nothing? OK, go ahead and try it. OK, so here we are. Picture nothing. OK. How are we going to do

that? First of all, we have to get rid of all this equipment and get rid of the walls, the building, the floor. Wait a minute. You and I are still in the room, so you got to get rid of the chairs and then get rid of us. And get rid of all matter and energy in the universe. And then what? I'm just floating there in some empty space? Just a brain, wondering why is there something rather than nothing?

1.103 And at some point you're going to have not just no matter, energy, space, or time, but no mental or Platonic concepts, no logic or mathematics and no sentient beings asking why there is something rather than nothing, because there would be no consciousness, no non-physical thing of any kind. Not even any Platonic ideas like God or triangles or math. Not even emptiness forever. And not even the existence of emptiness. And there's no forever because there's no time.

1.104 At some point, I don't even know what I'm talking about. Because these words have certain meanings. And if you just, well, they they just stop existing. I don't even know what that means. So, you know, we may be at the wall there, just like, I don't know, here I'm just going to speculate. There's this thing called the *Mysterion* mysteries. The *Mysterions* are these philosophers that think there are these unknowables. And I think they may be right about that.

1.105 You know, maybe if we had brains the size of, I don't know, this room, say, you know, some species, maybe we would know the answer to these questions. But given our limitations of our neural architecture, given our constraints by language and concepts and how we think about the universe, it could be we just will never know, you know. And so, and of course, the theists go, well, that's where the causal chain stops. And, you know, then you postulate, God started all that. You know, but why can't we take the causal chain back, just one more step?

1.106 Well, where did the God being come from? And the answer is, usually God is that which does not need to have a cause or a start. OK. Those are just words. Why can't the universe just be that which does not need a cause, or a start, or a prime mover or something? OK, I'm going to go over all this in the last lecture in more detail. But what I'm trying to say here is, again, I think if we apply intellectual humility and recognize we can't know everything, we're not omniscient, there may be things that we don't know now, but we will. Surely that's the case. But there may be things we'll never know. And so I think it's always good to be able to say, I don't know.

1.107 The correct answer to a lot of questions is, I don't know. Maybe we'll know. Maybe we'll never know. To which I say to my students, maybe you'll figure it out. That's what graduate students are for. You ought to figure this problem out. Beyond me, who knows? Right? So again, if we came back five, if we, if we came alive 500 years from now and we look back, what would we think about what people in the 21st century believe? Just think about all the hot button culture war issues—you know, race and gender and income inequality and immigration, abortion and so on. You know, are they still going to be fighting about that?

- 1.108** I mean, 500 years ago, it was like, the biggest debate was, you know, how can we determine which of these women are witches? OK, we got to look for the mark of the witch, and we got to look for these weird behaviors. And we don't have those debates anymore. And the reason is because we now know there is no such thing as witches, and women cannot cavort with demons in the middle of the night while flying on brooms. None of this has any connection to reality, so we reject it.
- 1.109** So my problem is, I don't want to be that guy that believes in witches equivalent in the 21st century and then centuries from now is like, oh my God, I can't believe that Shermer guy. Like, he gave this lecture and he said X, and look how stupid that sounds now. I don't want to be that guy, but if I knew which ones I'm wrong about, I would just not believe them. And that's the problem. So I'll end this lecture with a quote from one of my favorite philosophers, Douglas Adams. The Hitchhiker's Guide to the Galaxy. "Isn't it enough to see that a garden is beautiful without having to believe that there are fairies at the bottom of it, too?" Thank you.

Lecture 2: The Methods of Science

At a glance — The methods of science.

- 2.1** OK. Welcome back, everybody. This is lecture two of an eight-lecture series on Skepticism 101: How to Think Like a Scientist. What is the relationship of science and skepticism? Well, first of all, skeptics, scientists are naturally skeptical. I mentioned last time that the null hypothesis is what we start with in science. That is, your hypothesis about whatever is not true until you prove otherwise. The burden of proof is on the claimant, not on the scientist to disprove it. We assume it's not true until proven otherwise. And so scientists then are skeptical by nature.
- 2.2** But what is a skeptic? Since I publish a magazine of that title, I get asked this a lot. I mean, you can look it up in the dictionary, like here's from the Oxford English Dictionary. Look up the word skeptic. One who, like Pyrrho and his followers in Greek antiquity, doubts the possibility of real knowledge of any kind. One who holds that there are no adequate grounds for certainty as to the truth of any proposition whatsoever. Well, that's essentially nihilism or solipsism, that we can't know anything. And as you've already heard, I don't think that's the pathway to gaining knowledge, that is, justified true belief.
- 2.3** There are beliefs we can hold and justify, but what are the grounds for those? What's the evidence? What is it skeptics are skeptical about? Well, polls find various percentages of beliefs in different things. So for example, miracles, 76%. Ghosts, 42%. UFOs, 32%. Astrology, 26 %. Angels, 72%. The devil, 60%. The afterlife, 71%, and ESP, 41%. Interestingly, more people believe in heaven than hell. So that's optimism bias there.
- 2.4** So a lot of these things are called pseudoscience by skeptics or by scientists. But what do we mean by that? Well, there's different pseudonyms for pseudoscience—bad science, junk science, voodoo science, crackpot science, pathological science, non-science, and plain old nonsense. There's a lot of it out there. Nonsense, that is. Some people call us debunkers. But let's face it, there's a lot of bunk that needs debunking, and that's part of our job.
- 2.5** But in fact, most ideas that people come up with are not true, including those by scientists. So scientists are often just spitballing ideas, just throwing things out there. Could be this, could be that, try this hypothesis, try that hypothesis. Most of them are not true. So again, scientists are naturally skeptical because most ideas turn out not to be true. Most of us are wrong about most of the things we believe, so it's good, again, intellectual humility and skepticism about your own belief.
- 2.6** The problem is that no one in the history of the world has ever self-identified as a pseudoscientist. There is no person who wakes up in the morning and thinks to himself, I'll just head to my pseudo-laboratory and perform some pseudo-experiments to try to

confirm some pseudo-theories with pseudo-facts. This is a nice quote from Michael, my friend Michael Gordon, in his Pseudoscience Wars. In a way, it's—I like to say, no one in the history of the world has ever joined a cult. They join a group that they think is going to be really good. It's going to help improve my life, maybe save humanity, help the poor. We're going to do this and that, save the world. It's going to be great. And then, you know, down the line, it turns out it's a cult, but no one joins a cult, right?

- 2.7** So the problem is, is that people think differently. Most people believe what they say they believe, and we should take them at their word to that extent. Therefore, quoting my friend, the late great Carl Sagan, from his Demon-Haunted World, "You can get into a habit of thought in which you enjoy making fun of all those other people who don't see things as clearly as you do. We have to guard carefully against it." Carl said this back in the early 90s when we started the magazine, which is why I made Baroque Spinoza's quote our motto, which we included in that little video short. "I've made a ceaseless effort not to ridicule, not to bewail, not to scorn human actions, but to understand them."
- 2.8** This is hard to do, because the moment you have a belief, and somebody else disagrees with you, it's natural to think, well, they're crazy to believe this other thing. And then from there, you know, you can easily find yourself criticizing them, rather than thinking, well, they're just believing something different from me. So one of the things I try to do is, including debunking, is also, why is it people believe what it is they believe? It's one of the most important questions we can ask, including of ourselves.
- 2.9** OK, in science, we're looking for natural explanations for natural phenomena. Here's a crop circle made by, I don't know, let's think. What could it be? Could it be that the aliens traversed the vast distances of interstellar space and landed in Farmer Bob's field in Puckerbrush, Kansas, to make a crop circle that says skeptic.com to promote my web page? Or somebody with Photoshop made this for me. What's more likely? This is what we call methodological naturalism. This is what philosophers call methodological naturalism. That is to say, we assume that there are natural explanations for everything.
- 2.10** Another way to say it is, before you say something is out of this world, first make sure that it's not in this world. And again, what's more likely—the aliens traversed the vast distances of interstellar space to help George W. Bush in his campaign for the presidency, or that a tabloid newspaper just made it up? Well, we have no evidence of aliens landing anywhere on Earth. We have lots of evidence of tabloid newspapers just making stuff up. The what's-more-likely question gets us at an answer.
- 2.11** All right. I have to define science. This is the, kind of the wordy part of the lecture here. So here's how I define it. A set of methods designed to describe and interpret observed or inferred phenomenon, past or present, aimed at building a testable body of knowledge open to rejection or confirmation. All right, that's a mouthful. Let's kind of deconstruct that here. It's a set of methods to describe and interpret. A lot of things,

you can just look. Again, one of these kinds of truths that's empirical. I could just look and see if it's true or not. But a lot of times you can't see it. It's just interpreted. We think it's there.

- 2.12** The discovery of, of some of the external planets was made by the interpretation of their influence on other planets that we knew were there, therefore; or the existence of black holes was inferred and interpreted based on the motion of stars around this thing you can't see, so you infer that it's there. That's what I mean by observed or inferred. Sometimes you can see it. Sometimes you have to infer its existence. What's the point? Well, past or present, OK, a lot of science is done in the lab right here, right now. Let's see what the effects are. But a lot of it isn't. The examples I've given on the age of the Earth, the Big Bang universe, theory of evolution, the archeological digs, historians, and so on. These are studying the past, what already happened. You can't see it. It's gone. But we can infer that it's there from artifacts, from effects that happen and so on, that are measurable.
- 2.13** And finally, the point. It has to be testable again. Like Sagan's dragon, you have to be able to get at it. You know, you can't just hand-wave and go, oh, well, it's invisible. Oh, well, it's levitating. Oh well. And all this hand-waving, it doesn't help us. To know if it's true or not, there has to be some way to get at it, to reject it. One of my favorite questions to ask people, when they tell me their theory about whatever, is, what would it take to change your mind? You can't believe how dumbfounding this question is for most people. Well, nothing. Really? Nothing would change your mind, nothing? You know, that tells us they haven't really thought through in a kind of a probabilistic way. You know, this is what I believe now. But this evidence over here would, I would change my mind.
- 2.14** Right. That's what we're trying to do in science. It doesn't come naturally. But you can train people to do this. How to run experiments and how to reject the null hypothesis, how to change your mind, and so on. So much of this course is going to be about that. So in other words, to put that mouthful of definitions into a single short sentence, a method to explain the world that's testable and open to change. The testing and open to change is important. One of my favorite quotes comes from Sir Arthur Stanley Eddington, the astronomer who tested Einstein's theory of relativity. And when he said, "For the truth of the conclusions of science, observation is the Supreme Court of Appeal," I really like that. The Supreme Court is, that's the last court, that. They make the final decision. What is that in science? Observation.
- 2.15** OK. Just a little bit about this because it's a fun story. Einstein published his, his theory of relativity in two parts in 1905 and then in 1915. And he was not world-famous yet. He was still just a theoretical physicist and mathematician, and he predicted certain things would happen if his theory is true, but there was no way to test it at the time. And so, Eddington devised a method to run this test. One of Einstein's predictions is that, that space and time are one thing, spacetime, and a massive object in spacetime

will distort the spacetime around it. And that an object that comes near it will be bent around this thing, not because of gravity pulling it, but because the spacetime is being warped, and that's the natural path it has to take.

2.16 The analogy used is, let's say you have a huge sheet, rubber sheet here. You take a big, heavy bowling ball and throw it in there and it [sound] like that. And you roll the marble. The marble is going to go like this around it, not because the ball is pulling it, but because the ball is distorting the spacetime around it, right. So we have a big, huge, massive object in the sky called the sun. And Einstein had predicted that starlight from a distant star, passing close to the sun that comes to us on Earth, will be bent around the spacetime.

2.17 Well, the problem with testing this is that, if you're out during the day looking at the sun, you can't see the stars behind it because it's too bright. But there is a time, every once in a while, when in the middle of the day, the sun goes completely dark and you can see the stars behind it. And that's called a total solar eclipse. So Eddington devised this experiment, which he sent off two teams to different places where this total solar eclipse was going to pass, and then photographed the stars behind the sun during the total solar eclipse, and then compared the location of where those stars are to photographs taken earlier in the year, when it was night, when you could see it, when we're on the other side of the orbit.

2.18 And sure enough, the sunlight was bent, pretty close to what Einstein predicted it would be. And that's what put Einstein on the map. The New York Times ran this big cover story that the universe is all askew. The light is all askew in the universe. And that's what made Einstein world-famous, not his theory of relativity. It was the prediction that turned out to be true that made him famous, and Eddington got Time's Person of the Year for that.

2.19 So again, observation. Here's my favorite example from the great Richard Feynman, who explains what science is and how it works. He says if it disagrees with experiment, it's wrong. In that simple statement is the key to science. It doesn't make any difference how beautiful your guess is, how smart you are, who made the guess, or what his name is. If it disagrees with experiment, it's wrong. And that's all there is to it. And just for fun, here's the great Richard Feynman explaining that and saying exactly that in 1964. Cornell University. It's kind of fun to see how students dressed back then when they pan on the audience. Here he is.

2.20 Now I'm going to discuss how we would look for a new law. In general, we look for a new law by the following process. First we guess it, then we—don't laugh. That's really true. Then we compute the consequences of the guess to see what, if this is right, if this law that we guessed is right, we see what it would imply. And then we compare those computation results to nature. Or we say compare to experiment or experience. Compare it directly with observation to see if it works. If it disagrees with experiment, it's wrong. In that simple statement is the key to science. It doesn't make any difference

how beautiful your guess is. It doesn't make any difference how smart you are, who made the guess, or what his name is. If it disagrees with the experiment, it's wrong. That's all there is to it.

- 2.21** That's right. I mean, it doesn't make any difference, which, of course, helps if your name is Einstein or Feynman, but that just gets you a hearing initially. If you don't have good arguments, you don't have facts, your fellow scientists will not accept it. Let me give you a fun example. In the 1990s, when we were covering some alternative medicine issues in Skeptic magazine, there was one that came across our radar called therapeutic touch.
- 2.22** This was observed by a friend of mine named Linda Rosa, who was a nurse working in Boulder, and she was tracking this movement in which, in operating rooms, in the O.R., right there in hospitals, there were these, in addition to the surgeon, the assistant surgeon, the anesthesiologist, assistant anesthesiologist, and the nurses and so on, there was this, these therapeutic touch practitioners who were massaging the human energy field of the patient.
- 2.23** Now, when I say massaging, they're not touching the person. This person is actually being touched—in therapeutic touch, you could pass your hands a couple inches above the human body, allegedly massaging the human energy field. Now, this is based on some ancient Chinese medicine that has to do with chakras. You've heard of chakras. You know, your crown chakra and your throat chakra and your heart chakra and your solar plexus chakra and so on. Energy flows through in a pattern. Sometimes it gets blocked in a disease. You got to unblock the energy through acupuncture, acupressure, chiropractic, so on.
- 2.24** And so the therapeutic touch was this idea that you can kind of direct the energy and move it around and detect that here's the dark energy, here's the bright energy and so on. And people were claiming they could actually detect the human energy field. OK. We call this, by the way, the touchy-feely without the touchy. I mean, massage is kind of nice, right? It's very relaxing, but you don't even get touched. But the human energy field is what is being claimed here.
- 2.25** OK. My friend, Linda Rosa, had her daughter at the time. Emily was in fourth grade when she was watching her mother watch these videos of these therapeutic touch practitioners doing their whole thing. And Emily concocted an idea to test therapeutic touch. Now, you know, fourth-grade science project. You had to do some hands-on thing, the volcano or whatever, right? So Emily concocts an idea of how to test this. She's going to invite these therapeutic touch practitioners to come over to her house and be tested. And here's how she did it.
- 2.26** So she set up an apparatus. Mind you, this is not high-tech. This is a cardboard. This is a card table with a piece of cardboard here, in which the therapeutic touch practitioner on the left puts the hands through these two little slots in the cardboard, and then

drapes a towel over so you can't see anything that's going on there. On the other side is Emily, and she's just going to put her hand above the left hand or the right hand of the therapeutic touch practitioner. All she wants to know is, if you can't see where I am, can you detect my human energy field if I just hold it above there?

- 2.27** Now, before the experiment, they all said, oh, yeah, absolutely, of course. And then before she ran the experiment, she stood there and said, OK, can you detect? Oh yeah, yeah. It's a prepubescent pink and it's radiating. It's all great. And she goes, OK, now put him behind the cardboard thing. Put the two hands in there. And she used a high-tech random number generator to do this called a coin flip. You know, heads will go left, tails will go right. And then we'll just do it over and over and over and over. So she did. And here were the results.
- 2.28** So she tested 21 of these therapeutic touch practitioners. And OK, first of all, if you run 10 trials—left or right, left or right, left or right—how many should you get just by guessing? And of course, the answer is half. It's a coin-flip guess, right? And that's pretty much how it came out with these experiments, that most of them clustered around getting 5, 4, or 6 or 7 just by random chance, right? Nobody got 10. Nobody got zero. One person got 1, nobody got 9, and so on. And this would be the case if you ran hundreds of tests, which she later did.
- 2.29** Here are the results. She ran 280 trials, the initial experiment. They got 123 hits and 157 misses. In other words, they did worse than random guessing, OK? If they just flipped a coin, they would have done better. OK. Astonishingly, Emily then gathered more data, tested more of them, and made it a real study. and they submitted it to the Journal of the American Medical Association, where it was published, and 1996, and Emily was 11 at the time. She is still, as far as I know, the youngest person ever published in a major scientific or medical journal at age 11.
- 2.30** We published the results in Skeptic to no great fanfare because our people were pretty skeptical of this anyway. But this turned into a big media event. It was called the Emily event. Why? OK, first of all, none of the therapeutic touch practitioners said, oh my god, I was wrong. Obviously I can't detect the human energy field. No, they said, oh, well, this is the presence of the skeptic. Or it was the problem with the test or whatever. And a lot of them, though, were kind of innumerate, as we say. They thought getting half meant passing the test. It's like, no, you have to do better than random guessing because that's the real effect. The null hypothesis is that you'll do no better than guessing. And the test results showed you did no better than guessing. The null hypothesis stands.
- 2.31** Anyway, this turned into a big thing. She was featured on national television shows and so on. It was, it was quite amusing. But what did she really test? Now, a lot of people think, well, she debunked therapeutic touch as a modality of healing. No, she didn't. To do that, you'd actually have to have some study that went for years and you'd have epidemiological data and you had randomized controlled trials, and you see some

objective measure of what it is—the criteria. Like, what? Your heart disease went away or your cancer went into remission or? She wasn't doing any of that.

- 2.32** All Emily wanted to know was, can they detect the human energy field if they don't know I'm there or not, if I'm there or not with my hand? And the answer is no, they can't. So that's a, I find, a really instructive demonstration of how science works. It's just observation. Again, back to those kinds of empirical truths. Let me look at it and see if it actually works. OK. That's one of my favorite stories here.
- 2.33** A few more foundational suppositions to science and skepticism. Again, back to where we began in lecture one. We assume that there is a reality and we can know something about it, that there is moral realism and moral progress. Therefore, scientific naturalism is the right worldview to look at. And Enlightenment humanism is the ultimate derivative of that scientific naturalism. But these require a commitment to the constitution of knowledge. Let me tell you about one of my favorite books here, by my friend Jonathan Rauch, one of the great defenders of free speech and open inquiry. His book is called *The Constitution of Knowledge*.
- 2.34** So it's more than just observation. There's a lot more to it than that, that has to do with the assumption of fallibilism. That is, the ethos that any of us might be wrong. Again, I've mentioned this several times already. The most, one of the most important things you could ever say is, I was wrong, or I might be wrong, or I don't know. Objectivity. Again, a commitment to the proposition that there is a reality and we can know it. Now, again, I'm willing to acknowledge there may be some truths that are not objective. I've given you some examples of political truths or religious truths, but for the most part, what we want to know is, what's actually true? Not what I just hope is to be true. And that requires objectivity.
- 2.35** Again, this one, disconfirmation. What would it take to change your mind? OK. In science we use peer review. All right, and it's actually blind peer review. And the reason for that is, I might be biased when I'm reading this person's paper, and if I know who it is and I don't like this person, maybe I'm going to be super critical of it. So it's good that I don't know who wrote the paper, and it's good that that person doesn't know who's reviewing it, because if he finds out it was me and I gave it a harsh review, then if he reads my paper, he's going to give me a harsh review. So, you know, blinded peer review.
- 2.36** And then replication. You know, I have a lab. I run the experiment. Oh my gosh, we found these incredible results. But are we biased? You run it and then you run it and you run it and then you run it and we'll all publish and we'll see what the results are. In journalism we have fact-checking and editing. You may have noticed there's a lot of online sites that are not particularly adherent to rigid fact-checking and editing these days, but that is a tried-and-true thing. Do you have a quote for that? Do you have a backup quote for the quote? The person you talked to, were they a direct observer of this event that you're writing about, or are they second-hand? And if they're second-

hand, can you get the first-hand person that actually saw the thing you're writing and so on?

- 2.37** And to do that, you have editors who are editing the reporters, and then editors editing the editors, and the fact-checker and so on, and that's how it normally works. You have to do that because everybody gets it wrong. In the legal system, we have adversarial lawyers. That is, the whole judicial criminal justice system is based on adversarial argument. I have my attorney, the state has their attorney, or you have your attorney, and they go at it. And here it's a little bit different, the difference between science and the law. In science we just want to know what's actually true, what really happened.
- 2.38** The law works slightly differently. You make your best case and you can distort the arguments as much as you want. But that lawyer is going to do the same thing on the other side. And then the jury will say, OK, I heard this, I heard that, here's what we think is the truth. That's how the legal system operates. And it works pretty well, you know, not perfectly, but it works pretty well. And I'll see, when we get to signal detection theory, we'll show exactly how that kind of cashes out in terms of whether somebody is really guilty or not. But it's a, it's a pretty good system.
- 2.39** You know, you see these shows, all these law shows that are on TV, and, you know, like, the one-hour show and it's about some mysterious murder. And like the first 15 minutes, 20 minutes, you're sure. That's the person. Oh my god, they so did it. They are so busted. They were so good. I just know. And then you look at the clock and you go, wait a minute. There's like 40 minutes left in this show. That can't be. You know, they're setting this up and then of course, they—but that's how it works in criminal justice. You hear the first lawyer go, oh, well, then you hear the other lawyer. Oh, well, OK. Could be different.
- 2.40** And then finally, red team review. This is used in a lot of corporations where the CEO or whoever has this idea, we're going to launch this new product—wait a minute. What if this is a crazy idea? Let's have a red team try to debunk it. What are all the reasons we should not introduce this product, why it could fail and cause us to go bankrupt or whatever? OK. Continuing accountability. We have to hold everybody accountable for their mistakes because everybody makes them. Pluralism. Viewpoint diversity. This is a big bugbear in academia these days. Diversity is a big buzzword in academia. We want diversity. Yeah, I like diversity. What about viewpoint diversity? Well, that has been kind of neglected and that's caused some problems in academia recently.
- 2.41** So the pluralism of viewpoint diversity in science means, have people on your team or in your lab or your colleagues or friends who don't agree with you, and then say, look, I got this idea about the universe, and I think here's my theory. And then you go to somebody who actually knows this. Have I gone off the deep end here? Have I lost my mind? Oh, Shermer, you forgot this. Oh, yeah. OK, so you have to have that viewpoint diversity.

- 2.42 Civility. Argue calmly. Arguing is good, but be respectful. Be rational about it. There's a thing about, you know, not, you know—you've heard of ad hominem. Don't attack the person, attack the arguments. And so there's another one called ad Hitlerum. You know, the moment you accuse somebody of being Hitler, the conversation is over, right? No one thinks that they're a Nazi or that they're Hitler, right? So we got to argue calmly. Stick to the points.
- 2.43 Professionalism. Credentials count. There's a reason people go to school, law school, medical school, graduate school, and so on, because nobody can invent the world anew completely. We all build on other people. And so the professionalism of earning credentials to learn what other people have already done, and then you build on that and create new things.
- 2.44 And my favorite one here, this is a technical term now, no bullshitting. Seriously, this is a whole area in psychology that's studying bullshitting. What bullshit is, is, OK, lying, the liar knows what the truth is and cares that that's true and doesn't want you to know what the truth is. So he lies. The bullshitter doesn't care what's true or false. The bullshitter is just saying whatever is true just for its own sake. And so you can't—none of that. We want to, we want to get rid of that.
- 2.45 OK, let me give you a fun example of this testing of ideas in this kind of weird realm. Let me introduce you to the gopher golf ball finder. All right, here it is. This is a backstory to this. Back in the late 90s, Dateline NBC sent a film crew to our offices to test this object. This thing is called the gopher golf ball finder. And this particular one was given to me by a dentist. And we were, office was in Pasadena, and he lived in the Valley. And he said, he uses this to find golf balls. And I said, well, where are you finding these golf balls? He said, well, the golf course where I play golf. Well, that's a shocker. Imagine that, finding golf balls.
- 2.46 Well, how does it work? So he goes, well, you walk along like this on the greens and the fairway, and it goes like that. And you go, oh, OK. And then you walk in that direction. And if you keep looking, eventually you find golf balls. Oh, really? OK. How many times did it point to something, and you didn't find golf balls? He goes, I don't know. All right. So you don't know what the misses are. You only remember the hits. OK, then I kind of —this was a dentist. I felt like saying, in medical school, did they ever give you, like, how science works? Basic methodology testing? Anyway, no. Apparently not.
- 2.47 And, but the reason this became a story is because this Quadro corporation that owned the gopher golf ball finder started—this was, I think, 50 bucks, something like that. Started selling these to high school administrators to detect marijuana in students' lockers. Now, the theory behind it, this is all bullshit, is that inside this chip here is a little piece of golf ball. And the golf ball vibes go out like sympathetic magic, and then they come back, and this kind of directs it in this direction or that direction or whatever.

- 2.48** And so, then you swap out the chip that has a little bit of THC, the active ingredient in marijuana, and then you sell it to these high school administrators, which they were selling for 900 bucks. It's just a piece of plastic with an antenna. So they would walk down the hallway with banks of lockers, you know, and it would just kind of do this sort of thing like [sound] and or [sound]. Wait a minute. No. And oh, and if you open enough lockers, guess what you eventually find. OK, so they said, well, we got hits. I mean, we actually found marijuana in students' lockers. OK, how many lockers did they open, and there was nothing? I don't know.
- 2.49** OK, so we put this to the test right there. So here we go. I'll show you how it works. All right, you hold this golf ball for a second here. I'll show you how it works. It looks something like this. So I'm just randomly walking along here and, aha, OK, but maybe it's a right-biased leaning here, so maybe I'll go like this. Try it this way. Aha. Now, is that a test? Is that a good test? Of course, we know what's actually going on here. If you've ever played the Ouija board, this is called the ideomotor response. Your muscles are constantly moving slightly. You don't even realize it. Just thinking I want it to go right, it just goes to the right. Or thinking I want it to go left, it just goes to the left. It's pretty easy on a sensitive test like that.
- 2.50** So what we want to do, what we need to do in science, is do it blind. Right? Where I don't know which object or where it's located. So here's what I do in my classes. Usually at class, I run a whole—students have to come up. We run an experiment, goes on for quite a while. So I have two opaque cups and I just put them on a table or on the floor where it's beneath one of them or the other one. So let's say we just do it this way here. I'll do it at an angle, I think. And let's say I do the other one over here.
- 2.51** Now I know it's already over there. So when I do something like this, you know it's going to go [sound], like that. But if I don't know, the question is, can I do better than 50-50? We're back to our coin flip model, right? So if it goes this way or it goes this way—now remember the little bell curve, right? If I ran 10 of these, for each of you, maybe some of you get 5, some of you get 7, some of you get 3. That's not a test. We need to run hundreds and hundreds and hundreds and then look, statistically, is it above randomness?
- 2.52** And that's what we did for this TV show. And of course, astonishingly, he got no better than 50-50, guessing. Sometimes he got 6 out of 10, sometimes. And he thought, well, I got 6 out of 10. That's random. You know, you need to do like 7. 5 out of 10 over the course of 1,000 trials. And then, you know, we could say something significant. OK. So that's where that story ended for a while. It's just nonsense. Again, it's the ideomotor response. It's not doing anything, like the Ouija board. You're pushing it and so on.
- 2.53** To my utter astonishment, years later, this Quadro Corporation sold the company and the golf ball finder to a guy named James McCormick, who converted it into a bomb detector, in which he was then selling it to security agencies—this is post-9/11—security agencies for hundreds of thousands of dollars. Not 50 bucks, not \$900, but

tens or hundreds of thousands of dollars, to detect bombs. And so there, they told the security agencies, there's a little chip in here that has some explosive devices or whatever it is they wanted to sell. And then they spiffed it up a little bit and called it the ADE-651. It sounds all sciencey, right?

- 2.54** And astonishingly, there were security agencies for years that were having these guys, like, here's your car, and this kind of, OK, is there something in there? And they look and they look and they find nothing. They find nothing. They find nothing, and they find a package. Aha! It works. Again, you're just remembering the hits and forgetting the misses. The question is, how many misses did you get? And is the hits statistically, significantly different from what you would expect from randomness? And the answer is no.
- 2.55** That guy James McCormick is in prison in the Old Bailey in London for fraud. They estimated he made about \$80 million, basically selling a golf ball novelty device to security agencies. And you can watch films of this online. It's just quite amusing of these people trying to analyze it. Like, we got a quantum physicist to analyze whether—you don't need any of that. You could do the Emily experiment. Can it do any better than 50-50? That's all you need to know. You don't need to know physics or anything like that.
- 2.56** OK, so what I've been talking about there is a pretty basic, simple—it's called a t-test, a statistic to see if there's a difference between these two groups or whatever. A slightly more sophisticated tool is what's called signal detection theory. It's very much related to statistical detection theory, like I just hinted at, in which we want to know, is there a signal in the noise or is it just noise? That's the general principle. And let me come at this from a rather amusing way that, a few years ago, a film crew came to my office, wanted to get my opinion about cursed horror films. Cursed horror films.
- 2.57** In other words, horror films in which bad things happen on the set, to the actors, to the producers, before the filming, during the filming, after the filming or whatever. And they had a lot of examples of this. And they said, so what is your opinion? So I said, well, how many horror films are not cursed? And they're like, what? I said, and how many non-horror films had bad things happen on the set? And they're like, no idea. I said, right. This is signal detection theory. Here's what it looks like. I kind of outlined them for this way.
- 2.58** So we have a two-by-two cell, in which you have four two-by-two matrix, four cells. The one they're interested in is the upper left, horror films that are cursed. Poltergeist. Poltergeist Two. The Exorcist. The Omen. The Crow. Twilight Zone, the movie. Bad things happened in all these movies. Twilight Zone, the helicopter crashed, cut off, decapitated some of the actors. The Crow, Brandon Lee was killed by one of these guns with the little fake bullet. The Omen, I think one of the actors died in a plane crash afterwards. Something like The Exorcist, there were all kinds of problems on the set in which the actors were getting injured, and so on and so forth.

- 2.59** All right. So I said, OK, good, you're on to something. Maybe. But what about the horror films that are not cursed? I mean, It? Have you seen It? I mean, this is one of the scariest movies. Or The Ring? You know, when I watched The Ring—I'm a total skeptic. I don't believe any of this stuff. And still I'm sitting there, my heart rate is like, my blood pressure is up. I'm like, I'm sweating. Oh my god, this—and I don't even believe it. These films are so well done. Or The Ring, you know, when she pops up out of the screen, or the Sixth Sense.
- 2.60** The Shining. I mean, come on. That scene where the little boy is pedaling his tricycle down the hallway, and there's the two girls. The two girls had been murdered, and the blood comes gushing out of the walls, and, but nothing happened to any of those sets or actors, producers, before, during, or after. And then you have non-horror films. Superman. You know, Christopher Reeves after that fell off his horse, broke his neck, paralyzed for life. You know, The Wizard of Oz, Rebel Without a Cause. James Dean is killed in a car crash shortly after that. Apocalypse Now. Martin Sheen has a heart attack on the set, and he's a young man. It's like, how? But those aren't cursed, right? These are non-horror films. I mean, those are cursed, but they're not horror films.
- 2.61** And then, of course, in our fourth category, you have non-horror films that are not cursed. The Godfather, Star Wars, Casablanca, Citizen Kane. Pick any movie. Most of them are not horror films. Most of them are not cursed. You know, when you do that, it becomes clear, well, there's really nothing to explain. Right? There's no signal in the noise. It's just all noise. The selection process is by the person interested in that particular quadrant, that one cell and the two-by-two matrix.
- 2.62** OK, there's a lot of things in the world that are nicely analyzed by signal detection theory. Let's look at something slightly more important than cursed horror films. That is nuclear war. Here I would like to introduce you to Stanislav Petrov, who is the man who saved the world. This is a documentary film about his life that shows on September 26th, 1983. He saw on his radar screen blips of incoming missiles, missiles incoming from the United States. This is what is called a preemptive strike. In kind of game theory, Cold-War thought, the idea is that we should strike them before they should strike us. And if we do it, we have to take out their missiles.
- 2.63** So the other guy thinks, well, I'm going to have so many missiles hidden in so many places that even a first strike can't wipe them out. And therefore we could counterstrike, and therefore they won't launch a first strike because we can counter them. And this is what ended up in this Cold War arms race. This is the logic behind it. And so there he is, right after that Korean Airlines was shot down. It veered over Russian territory and was shot down by Russian fighter jets. So tensions were pretty high between the United States and Russia.
- 2.64** And there was a number of other events. Again, 1986, this is right in the middle of the Reagan administration. The Cold War was ramping up. Production of nuclear weapons was quite high. We had the nuclear triad, the submarines, the jets, and the land-based

missiles. They had theirs. The height, I think, was about 87,000 nuclear weapons between the United States and Soviet Union. It's down now, it's less than—it's about 10% of that. It's about 8,700, now, so that's good.

2.65 But what would you do if you're sitting there, you're the officer. You're looking at the radar screen, and there's blips. What is it? Again, back to our premise of the course. I'm not God, neither are you. He doesn't know. All we have are blips on a radar screen. Are they missiles or not? This is a signal detection problem. So let me give you the language of signal detection theory. There's a signal. Blips. What is it? Are they missiles or are they birds or something else? And then you have to make a decision. Yes, I think they are, or no, I think they're not. Right, so there's our two-by-two grid. You know, missiles are birds. Yes or no? All right.

2.66 So if the missiles are real and you say, yeah, I think they're real, and you hit the, you know, you contact the boss and he says, OK, we're going to launch our counterstrike before they hit, that's a hit. If the missiles are not real, turns out they're birds or whatever, and you say, no, I think they're not real, and you don't hit the button, that's a correct rejection. OK. Those are the two good things. You want to get it right in both directions. But if the missiles are not real, and you think that they are, and you go hit, let's launch a counterstrike against the United States, that's a false alarm. A false positive. That's called a type one error. And then finally, the missiles are real, and you say that they're not real, that's a miss. That's a type two error. False negative.

2.67 All right, so here is what it looks like in our two-by-two grid. So up in the upper left cell, you have real missiles. And you say they are real. OK. That's a hit. And the missiles are real, but you fail to recognize them. And that's a miss. That's bad because your cities might be destroyed. Or the missiles are not real, and you think that they are. That's a false alarm. And maybe you launch a counterstrike and that leads to cities being destroyed. Also not good. Or the missiles are not real, and you correctly reject them as such. So we want, in the cells, we want the two quarter ones. That's what we aim for. And we want to avoid the two types of errors, type one and type two errors, and so on.

2.68 Or you could just think about a medical issue. Cancer. There's—you get an X-ray and there's a little blip on the X-ray. What is that? Do you have cancer or not? I remember back in the, I think it was the late 90s when people were getting full-body scans because they came online at relatively cheap. It's like, I want to see, I want to get preemptive. If there's any cancer, I'm going to do something about it now. Catching it early is a good thing for cancer and so on. It turns out, people were finding all kinds of things in there that weren't cancer. It turns out, all of us have little blobs of stuff that are just in there, and they're not cancer. So people were panicking and getting surgeries and radiation. Turns out, they weren't cancer, right.

2.69 So this is another one of these signal detection problems, right. So upper left cell, you really have cancer. And the doctor says you do, and you agree, and you get the treatment. You save your life. That's good. The problem is, if you really do have cancer

and you go, no, I don't think it is, I'm not going to do anything, well, then, that's untreated cancer. Maybe you die from that. Or you don't have cancer and you get the treatment anyway, just in case. That seems like a good thing, except treatments for cancer are not without their side effects and risks. So that's a tough cell to be in. And finally, of course, the other bottom one, you don't have cancer and you agree with your doctor after a second or third opinion, yeah, probably don't. So you don't get the treatment. That's a good, that's a correct rejection.

2.70 Or criminal justice. This is a signal detection problem. There's the, there's the guy sitting in the courthouse. Is he guilty or not? He's been charged with a crime. Did he do it or didn't he? That's a signal detection problem. So let's say he really did do it, and you convict him. That's good. That's a hit. Justice has been served. But let's say the person really did do it. And you go, well, there's not enough evidence. We're going to acquit the person. Well, that's a miss. You've just sent back a criminal into the world. Maybe they'll re-offend. Or it's not the person. He really is innocent. He didn't do it, and you convict him. That's a false alarm. And that is very costly, because that's a true injustice, an innocent person in prison. And then, of course, what we're aiming for is that the person is innocent and you acquit them. That's good. That's another form of justice.

2.71 Well, there's some interesting research on this, on people's intuitions about where to draw the line of the signal in the noise. And it goes back to William Blackstone's famous quote, "It's better that ten guilty persons escape than that one innocent suffer." So, roughly speaking, we want to avoid that kind of error, that false positive, that type one error with a 10:1 ratio. We're going to set the bar really high, right? It's not just a preponderance of evidence in a criminal case, which is often said to be 50% plus a feather, just, just barely. Right?

2.72 In the criminal justice system, where the state is trying you, it's beyond-a-reasonable-doubt evidence. It's got to be way more than 50%. The reason for this is because most of us think it is really bad to put innocent people in prison. You can see the morality of terrorism in signal detection. You know, after 9/11, the United States moved its signal detection criteria for possible terrorists far to the left. That is, they wanted to catch more real terrorists, but also more innocent people then get nabbed in the net.

2.73 I'll tell you a funny story. You know at the airport, where they always play that little message, "Do not leave your bags unattended"? OK. So, yesterday I'm going through the Phoenix airport, and I take off my shoes and all that stuff because the shoe bomber—again, this is a signal detection problem, OK? The one guy made that shoe bomb and it didn't even work. It burned his, whatever. There was another guy, the underwear bomber, but thank god they don't make us take off our underwear.

2.74 But, you know, the government decided, OK, we have zero tolerance. We can have no more shoe bombers. Zero. We're going to catch every single one of them. How are we going to do that? We're going to make every single person take the shoes off every—urg, all right. So you get the TSA pre-approved. Maybe you don't have to take your

shoes off or whatever. But what we're talking about there is, how much are we willing to tolerate one kind of error versus another kind of error? And I think what's happened with, since post-9/11, is that we've, we've moved the signal detection criteria to the left. What do I mean by this?

- 2.75** All right. Now I'm going to shift from my two-by-two cell to overlapping bell curves. So this is a little more jargon on the signal detection theory. So the left curve is the noise, and the right bell curve is the signal. Bell curves are nice because, as you saw in the little Emily experiment, there's just natural variation statistically in how people behave. So again, some people got 7 or 8 right. Some people got 2 or 3 right. Most got 4 or 5 or 6 correct. So you'd expect statistically. This is true also in our measurements—cancer detection, missile detection, and so on. You're going to get random variations. Some are better at it than others. Some equipment works more accurately than others, and so on.
- 2.76** So, what we want to do is then set our response criteria. Remember, that's our yes or no. I think it is a true thing. I think it's not a true thing. And then, so we end up with our two-by-two grid, but instead we have four overlapping, the two overlapping bell curves with four options under the shade. The correct rejections which is good, and the hits which are good. But we want to avoid the misses and false alarms. OK, so response criteria, whether you say yes or no, is based on the risks and benefits of each of the four. What is it you're trying to do as a society, or whatever. You want to combat terrorism, you want to catch more cancer. You want to make sure that our nation is safe from incoming missiles, whatever it is.
- 2.77** OK, so if you set the response criteria to the left, as you can see here in the upper part of the screen, you get more hits, but you get more, more false alarms. Or if, on the bottom, if you move the response criteria, the response criteria is measuring just basically how many of these we're going to get hits and false alarms or correct rejections and misses. So if you shift it way to the left—that is, you make it quite lax—you're going to try to get everybody, you're going to get a lot of hits, but you're also going to get a lot of false alarms.
- 2.78** We saw this, for example, not just in the terrorism. Everyone take their shoes off. But like, say, stop and frisk in New York City when Rudy Giuliani introduced that. We're going to catch more of these criminals by just stopping and frisking people and seeing if we catch more of them. Well, they did, but they also got a lot of innocent people that were not doing anything wrong. So you get more false alarms. More radar blips as missiles, more scan blobs as cancer. So each yes carries costs and benefits.
- 2.79** Or you push it way to the right, a tighter yes or no response. You get fewer hits, but fewer—fewer false alarms is good. Fewer hits is not so good. So you assess more radar blips as false alarms, or you diagnose few scan blobs as cancer. Fewer false alarms, but fewer hits, right? So how do we decide where to place the response criteria line of yes

or no? It depends. You have to conduct a cost benefit analysis of the hits, misses, false alarms, and correct rejections.

- 2.80** Let's look at a recent event we've all just been through, the social pandemic that the whole pandemic and social isolation, right. So OK, let's say it's, I don't know, February 2020. Everybody knows there's something bad going on. This Covid-19 virus is spreading at a geometric ratio. I mean, it's just doubling, doubling, doubling, like every week. What should we do? This is a signal detection problem. Is Covid-19 going to be incredibly deadly, like Ebola or AIDs? Remember, HIV at one point was 100% fatal. 100%. If you got it, you're dead until the drug cocktails came, right?
- 2.81** So, imagine you're the president, the governor of the state, the mayor of the town, the head of the school district, you know, whatever. And you're at the press conference, and they stick the microphone in your face and go, what do we do? Do we close the schools tomorrow? Do we close the restaurants? What should we do? OK. It's a signal detection problem. Here we go. Let's say Covid-19 in the upper left is a really deadly disease. Tens of millions of Americans will die if you don't do something. And you do something, and it turns out it was deadly. That's good. You saved a lot of lives by shutting everything down. And that's what, of course, every politician wants to get it right. Or public health official.
- 2.82** But what if Covid-19 is really deadly and you go, no, you know, I'm just, I'm just not feeling this. I just think it's going to be, you know, I'm just going to go away, and we're just going to carry on, business as usual. Schools open, restaurants open, no social isolation, no social distancing. Don't worry about masks and so on. And then, I don't know, 50 million Americans died. That's on you. You just made that decision. OK. Or it turns out Covid-19 is not that bad, and you think it is. And you shut everything down.
- 2.83** Well, we know that that's pretty much what it was. It was, you know, not just the flu. It was worse than that. But it wasn't super deadly, like Ebola or HIV. And so the economy was hurt. And now we have inflation. We have kids that are two years behind in school. And so, so each of these cells has a cost. It's not costless to close all the schools, shut down the economy, and just socially isolate for months on end. Right. So each of these, I think, has important implications for—you can apply this to pretty much everything in your life, including, just so you know I'm not all boringly technical about world-changing ideas.
- 2.84** You can even apply signal detection theory to your dating life and marriage. OK, I just kind of have a little fun with this, particularly with my 18- to 20-year-old students who are trying to figure out what they should do about this. Who should you marry? OK, it's a signal detection problem. It seems kind of silly, but OK, so in the upper left corner there, we have, this is the person that's perfect for me. And I asked them to marry me, or they asked me to marry them, and I say yes, and we lived happily ever after. Of course, that's what everybody wants, right? Upper left. OK.

- 2.85** But on the upper right, this is the perfect person for you, and you turn them down. Oh. You missed. OK. But you can try again. I mean, it's not like there's just one person out of the 8 billion that would be a good match for you. I guess it would be 4 billion. It's half and half on the—anyway. Or the person on the left corner there is not a good match for you, and you marry them anyway. OK, unfortunately I have a lot of expertise in this area, so lots of people do. Lots of us make mistakes, but eventually—or in the bottom right, this is not the right person for you. You turn them down, you try again. Correct rejection.
- 2.86** Right, so we want the hits and the correct rejections. That's what dating is for, is to try to figure out—what you're doing when you're doing that is collecting more data. Which cell is this person in? Right. And that sometimes takes a while and maybe takes some mistakes. So you don't—again, you do a cost benefit analysis. It's, oh, just get married. What the hell. What's the—what could it cost? Well, yeah, it's a big decision, right?
- 2.87** Anyway, OK, I think I'll end with one of my favorite quotes that I read in college, from a book on logic written by Alfred Mander, what we're after here. "Thinking is skilled work. It's not true that we're naturally endowed with the ability to think clearly and logically without learning how or without practicing. People with untrained minds should no more expect to think clearly and logically than people who have never learned and never practiced can expect to find themselves good carpenters, golfers, bridge players, or pianists."
- 2.88** And that's what we're after here, in this course, in all of science and critical thinking, is teaching people how to think. Most of us don't know how to do it. These basic principles of remember the misses, not just your hits. You know, single data points are not a trend. You know, just try to keep track of the four different cells. What are all the options? I heard about cursed horror film. Yeah, but what about these? You know, what about these other things? What would it take to change your mind? What kind of evidence would you consider to possibly change your mind? Could you admit that you could be wrong? Could you say, I don't know, right? These are all thinking skills. In the next several lectures, I will have another couple dozen of little tools you can apply to thinking critically. All right. Thank you.
- 2.89** Question regarding the signal detection and Covid. How do you incorporate new information? You know, at the beginning of Covid, for example, you know, we were going blind, and we're trying to decide what to do, and then we're getting more information later. What is the process to, you know, add new data and reconfigure that two-by-two square?
- 2.90** Yeah. The process should be what they didn't do, which is to say, adopt the new information and change your mind. So I'm going to talk about Bayesian reasoning in the next lecture, which has to do with assessing the probabilities of something based on your priors, that is, your prior knowledge or information; and then changing the credence or your confidence in the belief based on new information that comes in.

- 2.91** So one of my criticisms of the CDC and Anthony Fauci and others—you know, I think, I do think they're good people trying to do the right thing. I think they made some big mistakes in not presenting this as, like, this is a huge unknown for us. This is what we think we should do today, this week, this month. We're going to try this, and we may change our mind, but they didn't do that. I would have preferred they said something like, well, we think the mask thing are a bad idea because here's the problem. We may not have enough masks for the medical community. Instead, he didn't say that. He basically lied. So now, particularly the right is just hammering these medical institutions for that.
- 2.92** And then, you know, should we, how about close the schools, restaurants, for a month and see what how it goes, and then open them and see what happens? And there were some that's called the comparison method. Comparative method where you compare different countries, what they did, different states, what they do. You do this with gun control or whatever you're interested in studying, and then you see what happens. Well, so there were some countries that shut down more than us, others that shut down less than us. And those were different effects.
- 2.93** I would have liked to see some more honest reporting on that. Instead, you get, you know, these right-wing media sources that say one thing and the left-wing media sources say it's completely different, and I don't know what to—what am I supposed to believe? And that, I think, would have been a huge boost to that, just, you know, acknowledging we don't know a lot. Just like, just a two-by-two cell, you know. It could be deadly. I don't want this on me. What if 50 million Americans die and, ugh, that's not good. So we think we should do this. And somebody goes, well, no, wait. It's not a cost-free decision. Right? And that kind of open debate, we don't really have, to that extent.
- 2.94** So you mentioned for Covid, you think that there are good people trying to do the right thing. I think there's a growing sentiment online that some of the people in positions of power don't actually have good motivations. And so the information that they're disseminating isn't necessarily true. And so how are skeptics or people who are trying to use critical thinking supposed to approach the information that we're receiving? If there's this growing sentiment that the information we're getting is not actually true enough to base the decision on anyway. Like, what are you supposed to do?
- 2.95** Yeah, that's a huge problem. That's a great question. I don't know. It depends on who the, what the source is, I guess. I mean, the current state of our politics is so polarized. It's really disturbing. I mean, you can watch, you can flip, toggle back and forth between Fox and CNN, and they're talking about the same event. And it's like two different, completely different events I'm listening to. What am I supposed to believe? And I think my sense of Fauci and Francis Collins, CDC people, they're just trying to do the best they can.
- 2.96** They're not—I mean, there's this whole conspiracy theory—I have a lecture on conspiracy theory—this whole conspiracy theory, that all of this was a test to see how

much they could get us to obey the orders of the command on top. You're going to wear a mask. Now, we know that you don't need the mask, but we just want to see if you'll do it. Obey our orders, you know. And so I don't believe that. I think people thought, well, you know, the masks, the 1918 flu. Look what happened. A helpful thing, so let's try that again. I think they're just trying to do the right thing, often feebly, and operating under conditions of great uncertainty leads to the problem.

- 2.97** The, yeah, the distrust of others. This has changed. We know, since the 90s, the polarization by all measures has gotten worse. And that even worse is that the people on the other side of the aisle, politically, they're not just different from me. Oh, you think this tax rate should be that, and I think it should be this; or I think the border should be more open, and you think the border should be more tight. But, you know, we're going to play golf on the weekends and go to our kids' softball games and have fun and have dinner together. We just have these differences between liberals and conservatives. It's not like that anymore.
- 2.98** It's not only that you have a different opinion, you're wrong. And not only are you wrong, you're immoral, you're evil. You want to destroy America. It's like, I mean, what is it? Is it Hannity's book or Tucker's latest book on, you know. No, no, it's, what's his name? Mark Levin. You know, Democrats. What, Democrats hate America. Democrats want to destroy America. I mean, this is not helpful, you know, because this is not what Democrats are trying to do. Well, I mean, one of things we're trying to do is just teach people to think critically. More, more exposure in the media of just thinking like this, even just sort of really basic stuff, you know, just be willing to change your mind. What would it take to change your mind? What kind of evidence would you evaluate? You know, and so on. And most people, they don't really think along those lines.
- 2.99** Yeah. Or even asking, you know, can you, like the zipper or NAFTA, you know, explain it in your own words? It's really hard to do. Later I'll be talking about steel manning other people's positions, rather, straw-manning somebody's position, where I summarize what I think you believe, but in a way that I can easily debunk it. I try to articulate it in a way that you would go, yes, that is what I believe. In other words, give the best argument you can for this other person's case, to the point where they would go, yeah, yeah, that is what I believe.
- 2.100** And that's hard to do for a couple of reasons. One, we don't really listen to what other people are saying. We're just waiting for them so I can get my next point in. And also, it's hard to know. It's hard to put yourself—the curse of knowledge. I think I know what you know, but I don't actually know. And the last point on that, you may not even know why you believe these things. You just believe them. And when I articulate why I think, why I think you believe this, then you listen to that and go, no, that isn't why I believe that. Or yeah, that's a good argument, when you didn't actually have that argument. So yeah.

- 2.101** I'm just kind of wondering, most of these tools don't seem extremely complicated. Surely in practice they're a little more difficult. But given that it's not impossible, I think, for the average person to grasp these tools, why do you think they're not more publicly discussed and used? Like you said, Fauci, for example, could have said, well, this is how we do cost benefit. Here's how we came to this conclusion. Essentially, show their work and adjust over time. Why not?
- 2.102** I like that. Show your work. That would be nice. Oh, I think there's a couple of things. First, I think there's an underestimation of the public's intelligence and ability to reason by most people in the media. I mean, I encountered, I've been, I've done a million television show interviews and stuff and pitched my own Skeptic TV show. Most producers, they think the public doesn't want to hear this, you know, UFOs or whatever. You know, here's why I'm skeptical. Oh, no, no, we don't want to say that. They don't want to hear this. And I disagree.
- 2.103** You know, there's this whole meme about, you know, well, people can't reason using statistics and probabilities and numbers and math. This is not true. Look at the sports page. I mean, it's just full of statistics and probabilities and trends and graphs and charts. The weather page. The business section. It's nothing but numbers and data, statistics, probabilities. People do understand and can reason about stuff they actually care about. You know, so I think that's a mistake.
- 2.104** You know, and I think a lot of it's just not taught. If I, if I was the education czar of America, which we don't have, you know, I would maybe even get rid of, I don't know, early middle school geometry, algebra, pre-algebra. I took all that. I never used it. How about a class in, you know, critical thinking? Just like what I just did, just a basic signal detection. Here's how to—the golf ball finder. You don't need to do high math. Just, oh, yeah, of course. Just obvious. You know, just the basic economic, how does economics, what is money? How does it work? You know, the idea of compound interest. People have no idea how to kind of project.
- 2.105** You know, that, you know, this is one of those cognitive task things, you know, if you have a field of weeds that doubles every day for 30 days, which day is it half full? Day 29. It's doubling every day, right? People, oh, yeah. Or if you say, OK, let's say you invest 100 bucks a month, 100 bucks a week, every week for 40 years. How much will you have? And most people, they kind of do the numbers in their head and they come up with, well, it's like a 400,000 or something. And it turns out it's like 2.1 million because you're accumulating. You're building it each week. The whole principle is larger than it was before. And that 10%, I forgot to add the 10% growth. 10 % growth each month is on a larger or larger number. And people underestimate how rapidly that accumulates over a long period of time.
- 2.106** It's like the doubling rate of the Covid spread. It was—everybody was shocked. How did it spread so quickly? I mean, by February, we were here and in March and April, boom, everybody's got it. That's a doubling effect. You know, this exponential growth curves

that go up, like population growth curves. People just intuitively don't grasp that very well. And so I have a whole bunch of examples of this in the how-thinking-goes-wrong lecture coming up. And so again, it's thinking is skilled work. But it's not that hard. Again, as you said, nothing I've said is exactly rocket science, right. Anybody can understand this. My kid could understand the golf ball finder.

2.107 OK. So yeah, it's more. Well, that's what we're doing here. Education. We like to think it matters. Hopefully it matters, to an enlightened democracy, that you understand the issues. You know, there's this, there's this theme about the rational voter. It's a myth. There's no such thing as a rational. I don't think so. I think people can reason their way around these issues, if you treat them with respect. I think people do want to know what the truth is.

Lecture 3: How We Make Decisions

At a glance — How we make decisions.

- 3.1** All righty, everybody. Welcome back to Skepticism 101: How to Think Like a Scientist and a Skeptic. This is lecture three of our eight lectures. This one's on rationality and irrationality. You've often heard it said, we are irrational. Remember in the first lecture, that humans are rational. Is that rational? What do we mean by these words? Well, let's just start with that word, rationality. Popular definitions tend to be rather tautological. That is, for example, rationality is using reason. Oh, yeah, that makes sense. But what's reason? Well, reason is using rationality. Right. OK, well, what exactly is that?
- 3.2** All right. So, what is rationality? Popular definitions are not particularly helpful. Rationality is using reason, or what is reason? Well, that's using rationality. How could we break out of the loop? I think a better definition, which comes from Steven Pinker's latest book called Rationality, in which it's the ability to use knowledge to attain goals. Use knowledge to attain goals. Now, remember from lecture one, what is knowledge? Justified true belief. Right. So, what is justified? Well, is there evidence for it? Are there reasons for your belief?
- 3.3** Beliefs then are held in the service of a goal. That is, we want something. Rational agents have goals. By this definition of rationality, your choice becomes, do you have goals or don't you? Or do you want things or don't you? That's it. If you do, rationality is what allows you to get them. Now, it's reasonable to ask, well, why should we want things? Why should we use rationality to get things? This is yet another example of one of Thomas Nagel's one thought too many. This is like asking, why do computers compute? That's what they do. That's just a full stop. Stop asking any other questions. That is the explanation.
- 3.4** In that sense, reasoning with logical rules and empirical evidence is executed by the mechanism that's hardwired in the machine, or, in our case, the brain, and runs because that's how the circuitry works. In that sense, rationality needs no other justification that this is what brains do to attain goals for organisms. And evolution wired the circuitry to work this way. Let me just make it all the way back. Just the simplest of organisms to survive have to capture energy from the environment. The second law of thermodynamics, entropy, I'm going to have a whole discussion of this in another lecture, means everything is running down.
- 3.5** And so, the first law of life is to push back against entropy. That is, capture energy from the environment in order to survive and reproduce and get your genes into the next generation. That's what evolution designs us to do. In that sense, your genes are immortal. The body that houses the genes is mortal. This comes from Richard Dawkins, The Selfish Gene book, which is still a classic, in which he said, "The target of evolution

is not the body, it's the genes." The body is just a survival machine for the genes to perpetrate themselves indefinitely into the future. So genes wire up bodies to do certain things, like capture energy, reproduce, and so forth. Those are the goals.

- 3.6** From the simplest single-cell organism that's just following a chemical gradient in a pathway, in the water, wherever, that's a goal, and it's, what's the best way to do that? Now, of course, single cells are not thinking anything. They're just following the gradient. But the more complex the organism and the more degrees of freedom, that I'll get in a minute, the more different pathways you could take to get to the goal. We don't think of single cell organisms or simple structures using rationality to get those goals. But as we build on that, you could see where something like human cognition and rationality comes into play.
- 3.7** Here's how William James, the distinguished psychologist, really one of the founders of modern psychology, distinguished a rational agent from a non-rational force. "Romeo wants Juliet, as the filings want the magnet. If no obstacles intervene, he moves toward her by as straight a line as they. But Romeo and Juliet, if a wall built between them, do not remain idiotically pressing their faces against its opposite sides like the magnet and the filings with the card." Just picture that, the card between the magnet on one side, the filings on the other, and the little filings are going, they want to get through, but they don't know how.
- 3.8** "By contrast, Romeo finds a circuitous way by scaling the wall or otherwise, of touching Juliet's lips directly. With the filings, the path is fixed. Whether it reaches the end depends on accidents. With the lover, it is the end which is fixed, the path may be modified indefinitely." OK, so at this point, let me make a little side path down to talk more about free will and determinism. The iron filings have very few degrees of freedom of movement. They could just go [sound]. They just move straight to the magnet. Humans, like Romeo and Juliet, have many degrees of freedom of movement.
- 3.9** Just think of a machine that's programmed to do just one little thing, make donuts or whatever, and it just keeps cranking them out, cranking them out. It doesn't have very many degrees of freedom. It doesn't have much choice about what it can do. But the more complex machines can do many more things. And a general intelligence machine, this is the idea behind general AI, will be able to do an infinite number of things, and that's more like us than the iron filings or simple machines.
- 3.10** This idea was nicely developed by the Tufts University philosopher Daniel Dennett, a long-time good friend and one of the smartest interesting people I've met, who makes some really good arguments for what's called compatibilism. I mentioned determinism, like Robert Sapolsky; he's a determinist. He has a 500-page book explaining all the varied ways that you and I are really not as free as we think we are. Then you have the free will people over here, which say we're completely free. This is sometimes called libertarian free will, but not libertarian in the political sense, just that we are truly free. There's a little homunculus in there that's free to do whatever it wants.

- 3.11** Most professional philosophers are not libertarian free will supporters. Most are compatibilists, along the lines of what Dennett argues in several of his books, but this one in particular, *Freedom Evolves*. That is, the evolution of organisms, from the simple cell just grinding along through chemical gradients, very few degrees of freedom, all the way up to us, just is an increase in the degrees of freedom, the amount of options you could take at any one point along the pathway or the gradient you're following to get to your goals.
- 3.12** And Dan has a nice metaphor he calls the Cartesian theater. Remember I mentioned last time, Rene Descartes' dualism. That is the belief that there is corporeal and incorporeal stuff. There's mind and brain; there's body and soul. It was Descartes who really put this idea on the map through his work in which he has these diagrams like this, where the sensory input from the senses into the brain ends up in this little thing here, like the pineal gland, where that's where the magic happens. That's where the soul is or mind or something that's spooky. The ghost in the machine, as it's called. Or the fire, A, causes the signal, B, in the foot to go up. Up here is this little thing called F, and that's where the mind processes the pain from the heat of the fire and so forth.
- 3.13** I guess that wasn't too irrational and unreasonable in his time, 1700s, because we knew pretty much nothing about the body and the brain. And so it was as good a guess as any. But the other problem is that dualism is quite intuitive. Paul Bloom, the Yale psychologist, thinks that we are natural-born dualists, that it comes online very early in childhood. That is, it feels like I have thoughts floating around up there. I don't feel my brain operating. It just feels like there's thoughts. It seems like that's normal. It seems like, I must have some kind of soul because I can't imagine not existing.
- 3.14** The first lecture, I said imagine nothing. There's nothing here, no room, no spacetime. At some point, you're just like, I can't even conceive of nothing. But in a similar way, if I said, conceive of not being alive anymore, picture yourself dead, this is an actual experiment people have done. People go, OK, I see myself there, and I see the casket, and there's my body, and all my loved ones are hopefully sad, and everybody's there, and it's a nice—well, of course, none of this would happen because, to perceive anything, you have to be alive. You have to be sentient. But at some point, it's not possible to conceive of not being alive. All we know is that we are a self that's sentient and conscious and continuing from one moment to the next.
- 3.15** There's another little sidebar in my book, *Heavens on Earth*. I talked about this mind uploading. We're going to live forever by scanning the connectome of your neural network, of all the synaptic connections between your neurons. That'll be a snapshot of everything you are, your memories, yourself, everything, and upload it as a digital file into the cloud and turn it on, and you'll be up there. Johnny Depp played a scientist who did this in this movie called *Transcendence*. He wakes up and he's in the computer, looking out the little camera hole or whatever. None of this is possible.

- 3.16** First of all, it's not possible, technically, even remotely. It's possible that the digital file would be so big, all of Google's computers couldn't hold it. But even that, it's not really you. Because you are you from a point of view. It's called the POV self, the point of view self. From one moment to the next, looking out of your eyes, that's you. If I take a snapshot of you now and upload it to the cloud, well, that's just a copy of you. Let's say we do this while you're still alive. Slide you into the fMRI brain scanner, turn it on, scan it, upload it to the cloud, pull you out and go, well, you're up there now. You're like, no, I'm not. I'm standing right here. What are you talking about? That's just a copy of me. So it wouldn't be you, anyway. It'd just be a copy of you.
- 3.17** None of this can be done—it's science fiction. But it's, again, this idea that, somehow, we're something other than the stuff. We're not just electric meat, as the '60s philosophers called it. It feels like there's something more to it. And yet we have no evidence of this other stuff. Now, again, I want to be Bayesian about this. I don't want to say, I know for sure that after I close my eyes for the last time, I won't wake up in some other realm. If I do, yay. I hope that happens, I guess. I think I hope that happens. And hopefully, my friends are there, my parents will be there. And so on and so forth. But there's no evidence that this happens, but the sense that it does.
- 3.18** So like, Paul Bloom's famous experiments with these little children. I mean, they're really young, they're three. And they give them these little puppet shows where the alligator munches the little mousy puppet. Where is the mouse now? Oh, the mouse is in this other place, and this is his mommy, and he's hungry, and he's scared. From a very early age, children have a sense of dualism. There's something else that continues.
- 3.19** Even in pop culture, we're all natural-born dualists. If you watch a movie like *Freaky Friday*, where Jamie Lee Curtis and Lindsay Lohan switch bodies, and now the middle-aged mom is back in high school with all the turmoil about boys and so on. Now the teenage girl is in the corporate boardroom and hilarity ensues. We get it, it's funny, but it's not possible, because it's not like your body and your mind. It's just your body and brain. That's it. There's nothing moving anywhere.
- 3.20** Dennett is fond of pointing out that, if you had an organ transplant, the brain is the one transplant where you want to go to the body. You don't want somebody else's brain because you're in the brain. There's no self. But the intuition about that, or the Steve Martin, Lily Tomlin movie. What was it called? *Something Me*. I forget the name of the movie now, where Lily [Tomlin] and Steve Martin switch bodies. Now this guy knows what it's like to be a woman, and she knows what it's like to be a guy. Oh my god, you guys think about sex this much? Hilarity ensues.
- 3.21** But this is not possible. There's nothing switching because there's no soul. This is that Cartesian dualism that we just hold. We give an essence to things like that, like in that Tom Hanks movie where he's stranded on the island, and he gets attached to Wilson, the volleyball. Wilson's like his best friend, and Wilson has an essence to him. Then he

cries when Wilson floats off. We all get that because, yeah, it feels—like, you get mad at your computer like it's a person. We imbue things with essences like that.

- 3.22** Research on organ transplants. When people get a new heart, particularly, they feel like the personality, characteristics of the person, the donor, is coming with the heart. It's just that essentialism. Another psychologist friend of mine did one where he asked subjects, Would you wear Hitler's jacket? You got it right here in a box. Would you put it on? Almost everybody goes, oh, no. God. It's like the evil essence would come with the jacket. Then he also asked, Would you put on Mr. Rogers' jacket? Oh, Mr. Rogers, our neighborhood friend. Yeah, I would wear his jacket because the essence of goodness comes with the jacket.
- 3.23** He also did a funny one where he put Brad Pitt's shirt on sale on eBay, washed or unwashed. Well, guess which one got the higher bids? He's like, oh, I want the unwashed essence of pure Brad-ness. But that essentialism, that dualism, that there's something beyond just the physical being that we all believe, and yet science says, well, we don't see it. That's why, when Dennett says, when you discard Cartesian dualism, you really must discard the show that would have gone on in the Cartesian theater, as well as the audience.
- 3.24** I like that because that is how we think of memory in the lecture on cognitive biases and how thinking goes wrong. One of our cognitive flaws is that we think of memory as a high fidelity recording of something that happened in the past, and you play it back in the theater of your mind and you watch to see what happened. You remember accurately what happened. This isn't even remotely like how memory works. It's a constantly edited process in which there is no original. It just is constantly updated, updated, updated, and so forth as you go through your life. But that's what it feels like. People get things wrong about how memory works, and it's all part of that.
- 3.25** In any case, this idea that there is somebody up there would not give you free will in any case, because that would just mean the little homunculus is making the decisions, not you. As I like to say, is there a mini me in the brain? Well, if there's a little mini me up there, then who's making the decisions for him? There's got to be a mini mini me inside mini me's brain, and then an itty-bitty mini me inside mini me's inside, and so forth. That's the problem with that.
- 3.26** Where does the choice come in? Well, I've got to go through one more thing. Let's revisit that idea of the hard problem of consciousness. What's it like to be something? I used an example of what it's like to be a dolphin. I've swum with dolphins before at this resort in Mexico. I live in Santa Barbara. They're out there all day. It's big fun. I'd love to know what that's like. But I can't, for the problem I already mentioned, that this idea of what's it like to be something. This is the short phrase for the hard problem of consciousness. What's it like to be something?

- 3.27** In a way, it also touches on this, what's called a philosophical zombie problem. Zombies, not like in the movies, but the philosophical zombies are— you have to understand, philosophers just do these crazy thought experiments. Here's the thought experiment. Everybody in this room is a zombie. The lights are out. You're not sentient, conscious at all. You're just a machine, giving responses to me that makes me think you're sentient and the lights are on, but they're not. There's no way for me to know for sure, other than, well, you just look like you are. The same could be for me. I could just be some robot and some super AI and so on.
- 3.28** What's the solution to that? Well, my solution is that I apply the Copernican principle. The Copernican principle is that we're not special in the universe. To myself, I'm not special. I'm pretty much like you. We all have similar brains. We know how neurons work and swap chemicals at the synaptic gaps and so on. We know how neurons talk to each other and how brains work. More or less the same for all of us. If I see you expressing emotions, there's a pretty good chance that you're feeling something like what I feel when I express those emotions.
- 3.29** There's this other weird problem. Can I really ever know that your red looks like my red? Again, there's no mini me to transport out of my skull into your brain, and on the Cartesian theater of your mind, I see, oh, his red looks just like mine. None of this is possible or even realistic. It's, I think, conceptually flawed. We can never know what it's like to be something else. That's OK because that's just the nature of the machine and how it works.
- 3.30** Now, how do we get free will out of this, freedom and choice? Let me tie it back into rationality and irrationality. Let me introduce you to Benjamin Libet, who changed the conversation of free will and determinism. He ran these experiments in which initially he had subjects with EEG-reading cap, the cap on their skull, where they could measure the motor cortex activity. Then he would have them make a decision, like I'm going to raise my finger or raise my hand or push a button, and I will tell you when I push the button or when I first made the decision, and then however long it takes to press the button.
- 3.31** Well, Libet could tell milliseconds before the subject said, I'm now making my decision to raise my hand or push the button, before the subject had even thought of it, or at least he thought he had thought of it. Now, again, they're measuring the motor cortex. The decision to press the button or move your hand is coming from below that. It takes a little time for the pathway to go up to the motor cortex that says, OK, he's now going to move his hand. That decision is made.
- 3.32** Since then, instead of using these EEG caps, they use fMRI brain scanners. Like John-Dylan Haynes, for example, extended the time between brain activation and conscious awareness of a choice to several seconds, and sometimes 7 seconds. Again, just picture this. These scientists are outside in the other room where the computers are that run the MRI machine because they can't be in the room because it's a big magnet. They

could tell seconds before the person said, OK, I've decided to press this button instead of that button. They could tell which one he was going to press before he did.

- 3.33** That's pretty weird. It's like, what? That sounds pretty spooky. We're not really in control. All right, how do we work around this? Well, I have three paths to free will. First of all, our brain is not one unit, it's a modular unit. Instead of thinking of it, one model is, analogy is the Swiss Army knife. There's a bunch of different tools. Another one is the smartphone with all the different apps. It's one device, but it has a whole bunch of different things it could do.
- 3.34** If a subcortical region of my brain sends a signal to the cortical region of my brain to inform it of a preference, it's still my brain making the choice. It's me making the choices, regardless which part of me is actually making the decision. I'm responsible for me and my choices. In other words, there's a lot of mini mes in there, a bunch of them.
- 3.35** Then there's something that's called free won't. There was a famous paper on this called Free Won't. So, free will is defined as the power to do otherwise. I decided to come here to be with you today, but I could have not. I could have just said, no, I'm not going to do it. I'm not interested. That's free will, the power to do otherwise. Could I have done that? Free won't is the capacity to reject a particular action arising from the unconscious neural network, such that any decision to act one way instead of another way is an authentic choice.
- 3.36** These two scientists, Marcel Brass and Patrick Haggard, did a similar experiment to the other guy with an fMRI brain scanner, in which they're lying in there with the little keypad. You have to understand how these things work. It's a huge magnet that's just tracking the hemoglobin, the iron in your blood, and the blood is moving around to different parts of your brain that need it for the oxygen when it's tasked to doing something.
- 3.37** These fMRI brain scans always give subjects things to do. You got to do this, and then you got to do that. You got to look at this, you got to look at that. Then that make different parts of the brain light up, light up. It's just the blood is moving more there than some other place, and the scanner can control that. Then it's all digitized on a file, they put artificial coloring in there, and so on. It's a little deceptive of what's going on there. They're not measuring neural activity. They're measuring blood flow, with the theory that the neurons need oxygen, so the blood is called to there.
- 3.38** Anyway, these guys, Brass and Haggard, gave subjects the choice. OK, so you make your decision. I'm going to push this button or that button or whatever. Subjects are wearing goggles, and the goggles have little TV monitors in them that correspond to the computer outside the room. Then they give them choices, this or that, and they press this or that of what they're looking at, left or right or whatever. But they were then tasked with, after you've made your decision, you can change your mind if you want.

- 3.39** When subjects chose to veto their initial decision, a specific area of the brain lit up. Particularly, it was the left dorsal frontal medial cortex, the DFMC, as it's called, normally active during decision-making, especially in the intentional inhibition of a choice. It's right up here. Just think of your prefrontal cortex, and this whole lobe of the brain up is where rationality, decisions, control over the urges bubbling up from down below. Just think of, to a certain extent, a lot of what we do is driven by these lower impulses coming from the amygdala and the lower areas of the brain, the more emotional, let's say, intuitive, instinctive parts of the brain that you don't even know where are acting.
- 3.40** But those signals can then be modulated by these higher regions of the brain, particularly the DFMC. That's it on that little red blob there, the artificial coloring to show you where the blood was going when they were tasked with. You could change your mind and it goes, yeah, I changed my mind. It's still the right one, I'm going to push the left one. Here's what they've included. "Our results provide the first clear neuroscientific basis for the widely held view that people can refrain from doing something, even if they genuinely wish to do it. We speculate that the DFMC may be involved in those aspects of behavior and personality that reflect self-control."
- 3.41** Look at that phrase that they tell us, "even if they genuinely wish to do it." Just think about your own life. I'd really, really like to do that, eat the chocolate chip cookies or whatever. But I really don't want to do it either, because it's going to have some consequences down the line. We're always faced with these constant bubbling-up urges. I'd like to do this, but I don't want the consequence. I'd really like to do that, but these other things are also appealing. I want to see the movie, but I want to read the book. I want to have this dinner, but I want to have that dessert. Life is full of these kinds of choices.
- 3.42** And even Benjamin Libet, in his memoir published at the end of his life, really came down more as a compatibilist than a determinist, despite his research being used by determinists to say, see, you're not free, you're determined. He says, "The role of conscious free will would be then not to initiate a voluntary act, which seems to just bubble up from where we know not, but rather to control whether the act takes place. We may view the unconscious initiatives or voluntary actions as bubbling up in the brain. The conscious will then selects which of these initiatives may go forward to an action or which ones to veto and abort."
- 3.43** There's interesting research by neuroscientists on, well, this is going to sound weird, but on serial killers and rapists that are on death row in prison. The scientists take these portable fMRI brain scanners and they scan the brains of these guys. They're all guys. To a man, they have pretty much no prefrontal cortex. I mean, it's just quiet. It's just shut down. They're just urges bubbling up constantly. So they're always in fights. They have no self-control. You can train some of it a little bit, but basically, that area that's lighting up is just gone.

- 3.44 Why? Well, there's different theories about this. Dropped on their head as children or just debt-deprived environments, drug abuse, whatever. All kinds of genetic and environmental factors seem to contribute to this. But this idea of the urges are bubbling up, it's an interesting idea here, and that something puts the brake on. That's where your self-control or free will or whatever you want to call it, volition comes in. Yes, I could do that, but I'm not going to do it. I'm not going to do it today. I'm not going to do it tomorrow. I'm going to build up a habit of resisting my temptation. That's the idea.
- 3.45 Now, of course, there's variation on that. Some people, as I just mentioned, just have no self-control at all, and others have way more. But then again, a lot of the urges that these guys in prison have that end up in violence and fights and so on, I just don't have those. I just don't feel it. Most people don't walk around with the urge to punch the next person that walks by. Most of us don't think like that. The question that I have, which no one has the answer to, why is that? Why do some of these guys just, urge to constantly punch somebody, and most of us don't? I don't know the answer, but maybe you'll figure it. We'll send it to the graduate students.
- 3.46 That's one of those hard problems of neuroscience. To what extent can we predict people's future behavior, like a minority report? We can predict what the criminal is going to do before he does it and then intervene. OK, this is not possible because of all these different influencing factors. Then my third point is that we are self-causing agents. That is, we're self-determined. We're part of the causal net of the universe. Yes, the universe is determined. Sure, fine. But we're also part of that causal net, and we're also one of the causes in the causal net, and we're aware of the causes. We're aware that we're aware of the causes and so on, and you can choose among them.
- 3.47 This is what's called causal agency. That is, the universe is determined, but it's not predetermined, so we have a role in the unfolding of the universe. This is how I justify whatever position you want to call it, compatibilism, a form of volition or free will, self-determination. The universe is determined, but we're self-determined. It's not predetermined, so I can alter it. But I know that I'm going to be tempted by this tomorrow, so I'm going to do something different today, so I resist the temptation tomorrow. I'm aware that I'm doing this, and I'm aware that I'm aware, and so on and so forth, and most of us can do this.
- 3.48 OK, I'll come back to that a little bit in a moment where we're talking about using rationality to control those sorts of things. But first, I want to return to our friend David Hume, one of the heroes of this class and the great Enlightenment philosopher, one of the great skeptics. This is one of his most famous quotes that I think people misunderstand. "Reason is, and ought only to be, the slave of the passions and can never pretend to any other office than to serve and obey them."
- 3.49 OK, this guy, Hume, is like one of the champions of rationality. What does he mean here? I mean, is he saying, just go out, party like it's 1999? No, that's not what he

means. Something a little deeper. He's saying that reason is the means to an end, but it cannot tell you what the end should be or even that you must pursue it. Rationality and reason can help Romeo find Juliet and vice versa, if they want, but it doesn't tell you that you should want Romeo or Juliet. That comes from somewhere else. Who am I attracted, who do I want to be in love with? That's not a rational decision.

- 3.50** This is what Hume meant by the passions. He's referring to the likes, the wants, the drives, the emotions and feelings wired into us, without which reason would have no goals to figure out how to attain. Again, it may just be one of these deeper, ultimate causes that we pursue for some evolutionary reason. We pursue food for caloric intake, we pursue sex to reproduce, keep the genome going in the species, and so on and so forth.
- 3.51** The problem is that we have conflicting goals. That is to say, people want more than one thing. So evolutionary theorists talk about proximate goals and ultimate goals. So proximate goals are immediate, nearby, and ultimate goals are more long-term and far future. I just think about food. So, approximate. Why do certain foods taste sweet? Fruit, cake, cookies, whatever. The approximate reason is because of the molecules, the structure, and where they land on the tongue, and so on and so forth. The neural pathway that registers this is a pleasurable experience.
- 3.52** The ultimate reason has to do with caloric capture in an energy-stingy environment. Remember the second law of thermodynamics and entropy. The universe doesn't care about us. It's not providing us with food because we want that. We have to get it ourselves. In our natural environment, foods that taste sweet or fatty or really juicy or whatever are very rare and hard to get and unpredictable. We have essentially a mismatch between our evolutionary goal-directed behavior to consume as much of those foods as you can, because who knows what tomorrow may bring.
- 3.53** We don't have a satiation point in our brain that says, that's it. No more of that Ben and Jerry's ice cream, Shermer. And I'll just sit there with the whole bucket and the spoon in front of the sports. It's hard to stop. And so evolution designed our bodies to want those sorts of things. You have the proximate and ultimate. Or lust for an attractive sexual partner. Obvious reasons why that feels good, having to do with our neural networks in our brain, related to organs in our body and so on.
- 3.54** But the ultimate reason, the evolutionary reason, is to get your genes into the next generation. Now, no one thinks about that. You don't think, well, this is what I'm trying to do. What we're talking about here is that we're using reason and rationality to choose between competing goals. In this case, between your present self, this is what I'd really like right now, and your future self, which reasons, huh, maybe I'd better not do that because next week, next month, next year, this will be the consequences of that. Dessert now, slim, healthy body later. Sexual encounter now versus risks later.

- 3.55** OK, in my podcast, I'm always referring to these chocolate chip cookies and my temptations of my future self, because there's this health store that makes these big, huge, walnut chocolate chip cookies. Anyway, then, lo and behold, I get in the mail last week this box of chocolate chip cookies from a fan that has a bakery, and I'm like, oh. I mean, I'm glad. It's like, oh, boy. And then I'm like, oh, no. How am I going to resist this? It's sitting right there, and one of them says skeptic right on the cookie. How can I not eat these cookies? That was before I came. They're mostly gone now. That's the little bumper sticker. Life is short, eat dessert first. Yeah, maybe, but what if life is not short? What if it's long, and I'd like to live longer, and therefore, maybe I shouldn't eat the cookies.
- 3.56** Remember this company, Ashley Madison? Life is short, have an affair. They went bankrupt when their database was exposed, and it turned out that something 99% of their customers were all men, and that they were having these sexual encounters online with bots, not with women. Anyway, so that was the end of that. But the idea that this is equally distributed between men and women clearly has no basis in reality. There's a reason that has to do with that. Of course, why you shouldn't do that, there's very good reasons why you shouldn't do that if you're married. The consequences are not trivial.
- 3.57** All right. so what we're facing here is decisions about self-control, free won't, delay of gratification. All this has to do with something called time preference. Again, degrees of freedom. We're complex organisms. One of the things that makes humans different from all other species is we can really project far into the future. My dog, he's dumber than a post, but I love him. His name is Hitch, named after Christopher Hitchens. He's a chocolate lab. He's a good guy. But his future involves the next five minutes. That's it. Ten minutes maybe, and then it's nap time, and then it's ball time, then it's eat time, go to bed, wake up, do it all again. He just doesn't have that.
- 3.58** Chimps, gorillas, a little more far future, but not much. There's something unusual about the human species, having to do with the size of our cortex, probably, in which we can project ourselves far enough into the future to make decisions now about what we want the outcome to be later. But we're not always rational about that, and it's hard to know what is the right thing to do. Is a bird in the hand worth two in the bush? Take what you can now. Or is a penny saved, a penny earned? You should invest for the future. What should you do? Pension fund at the company sounds like a good idea, but what if the company goes bankrupt, and you lose all your investments, your money?
- 3.59** Or what if everything they said was bad for you turns out to be good for you? Remember that? Eggs, shrimp, nuts, coffee. Bad, bad, bad, bad. No, good, good, good, good. I'm glad I didn't sting on those because I really like those, right? Or you're only young once and you can't take it with you. Yeah. But what if you save up for the best 90th birthday party ever and you drop dead at 89? You don't want to do that either.

We're in this constant tension between my current self, short-term self, long-term self, and far future self, and making these decisions.

- 3.60** These are called the marshmallow test. Do you know the marshmallow test? OK, this is Walter Mischel's famous experiment when he was a professor at Stanford University. He had the children of the professors there, who had a little day school there, in this experiment, which he gave them a choice. Would you like one marshmallow now? But if you wait 15 minutes, I'll come back and I'm going to bring a second one for you, and you can have both of them.
- 3.61** OK, if you just Google this on YouTube, the funniest videos. OK, I'll be right back, and the door closes and the kids are like, oh, and they touch it and they smell it and they nibble. Oh, no, I can't. Then they cover their eyes and they look away and they're like, I really, really, really want it now, but I'd really like the second one. How long is 15 minutes? They're checking, looking at the door. It's just hilarious. Life is a series of marshmallow tests for all of us.
- 3.62** Now, what made Walter's experiment famous is that he did a follow-up later and followed these kids throughout their lives. It turns out that those who delayed gratification, as it was called, they waited for the second marshmallow, ended up having better grades in school, higher GPAs, higher SAT scores, better GPA in college. They earned more money in their life. They had better jobs, better marriages, better health outcomes, and so on. The principle was, don't eat the marshmallow.
- 3.63** But then there's been some critical studies of that. Well, it depends on the environment. But if you're in a home where you have a bunch of siblings, and if you don't eat the one marshmallow now, there'll be zero in 15 minutes. Better to take the one now. It's better one in the hand than two in the bush. It's very context-dependent. Life is a series of these marshmallow tests between sooner small reward now and a later large reward.
- 3.64** Here's a funny story about that. A man is sentenced to be hanged for offending the sultan and offers a deal to the court. If they give him a year, he'll teach the sultan's horse to sing, earning his freedom. When he returns to the dock, a fellow prisoner says, "Are you crazy? You're only postponing the inevitable. In a year, there'll be hell to pay." And he responds, "Well, I figure, over a year, a lot can happen. Maybe the sultan will die, and the new sultan will pardon me. Maybe I'll die. In that case, I wouldn't have lost a thing. Maybe the horse will die, and then I'll be off the hook. And who knows? Maybe I'll teach the horse to sing."
- 3.65** As Homer Simpson told Marge when she warned him that he would regret later his heavy drinking now, "That's a problem for future Homer." Man, I don't envy that guy. Or as Oscar Wilde said, "I can resist everything except temptation." Or as St. Augustine prayed to God "to give me chastity and continence, but not yet." What do we do about this?

- 3.66** Well, recall Odysseus' temptation. Remember Odysseus on the ship? They're sailing by the rocks where the siren songs would lure them in to these beautiful temptresses on the island. Well, he'd really like to hear the music. He's worried that his sailors, if they hear the music, they're not coming back to the ship. They'll be gone. He has his sailor's ears plugged with wax. Rational ignorance. In a way, he's using rationality to decide what we don't want to know. It's better they don't hear it at all. I'd really like to hear the siren songs. He has himself strapped to the mast of the ship, without the wax, so he can hear the songs and then he won't be tempted because he can't do anything about it. This is called Odyssean self-control. In other words, be your own Odysseus.
- 3.67** Here are some examples. I'll give you some real-world tips here from this research. Food-shop when you're satiated, so you skip the dessert part of it. When you go in a supermarket, you're supposed to stay on the outside where the vegetables and fruits and healthy stuff is, and stay away from the inside where the, I don't know, canned and junk food stuff is located, something like that. Well, if you eat first before you go shopping, then you're more likely, research shows, you're more likely to buy healthy food. Again, my present self, my future self. I know in half an hour, I'm going to buy the bad food, so I'm going to eat now, and then that's fullish. That's pretty well.
- 3.68** Set out your workout clothes the night before. This is one of my favorite examples. I ride with these guys that like to go out early on training rides, and they leave at 7 AM. So I'm basically going to get up at 5:30, and get my coffee, this and that. And I know I'm going to flake out. So I get everything ready. I get the water in the water bottle in the car, the bike is in the car, the tires are pumped up. The bike clothes are right there next to the bed. I get up, get my coffee. As few obstacles as possible, because I know. Current Shermer knows future Shermer is going to be very weak at 5:30 in the morning. So I'm modifying the environment.
- 3.69** Use friends and apps like Fitbit and Strava to monitor your activity. That's why these things are so successful. I thought it was the dumbest thing I'd ever heard until I started using them. I found that I get a lot more workouts in when I check Strava every day to go, oh, my buddy got out this afternoon. Now I got to go. It's, again, that constant feedback to alter your future self. Or count to 10 before speaking or acting impulsively. Or have your employer withhold from your paycheck money for retirement, health insurance, life insurance.
- 3.70** This is called libertarian paternalism. It's not an oxymoron. Sometimes it's called choice architecture. This is Cass Sunstein from Harvard and Richard Thaler from the University of Chicago. Cass Sunstein worked in the Obama administration when they launched this program about using choice architecture. That is, using rationality to design things a certain way to get the kind of outcomes you would like.
- 3.71** For example, I'm from California. On my driver's license, I had to punch the little tab that says, I will donate my organs if I am killed. In Oregon, you are donating your organs unless you punch the tab that says, I opt out to donate my organs. Oregon has much

higher rate of organ donations in California. This is their example of a rational choice architecture. We have to make a decision one way or the other. As a public, we want to solve this collective action problem. We need more organ donations. What can we do to get that? Here are some options.

- 3.72 Or restaurants with their menus. There's been a lot of research on designing a rational choice architecture for restaurant menus. What do you put up front? What do you put in the back? Where are the desserts? Or cafeteria layout. You don't put the pie first. You put the healthy stuff. The pie is way down at the end, whatever. People still could choose. I still have the bad stuff. Or the mandated calorie of every food item at the store, at the fast food store.
- 3.73 That really does—at first, I thought, oh, this is pathetic, these people. Then I thought, I started looking at them, going, holy, oh my god, look how many calories if I get the fries. I'll just get two burgers and skip the fries. Again, that rational feedback allows us to make better decisions. Their book, *Nudge*, is quite instructive in this way, of all the different ways societies can structure choice architecture to get better outcomes. That's an example of applying rationality to solving human problems.
- 3.74 Or have a future sacred goal so valuable that it trumps any immediate impulse. Big ones, like I want to be involved in some political cause or social cause or cultural cause, and always we should be realistic. Here's my fun example of this, to my students, especially, because they're at that age. Should you use birth control or should you just practice abstinence? OK, let's look at this. Why can't people just say no to sex? That is, just use abstinence as a form of birth control, or time their sex for when it's safe during the woman's monthly cycle. Well, they can, of course. But as the old joke goes, what do you call couples who use abstinence or the rhythm method of birth control? Parents.
- 3.75 My favorite paper when I was writing about this in one of my books was titled *Like a Virgin (Mother)*, analysis of data from a longitudinal US population representative sample survey, published in the *British Medical Journal*. They found that one out of 200 adolescent girls—there was, I think, 4,800 people, a total at the end of 4,800—one out of 200 adolescent girls reported they became pregnant without sex. Right. Who were these immaculately conceiving, parthenogenic Marys?
- 3.76 The adolescents who reported a virgin pregnancy were twice as likely as other pregnant women to have signed a chastity pledge, and they were significantly more likely to report that their parents had difficulties discussing sex or birth control with them. As I advise to my students, abstain if you like, but use birth control just in case. Again, current self, future self. You sign that abstinence pledge at church on Sunday morning. The next Saturday night, you're out with your friends, a few drinks, and then before you know it.
- 3.77 Rational ignorance is another category here. Odysseus had himself tied to the mast so he could still hear the siren songs. His earplugged sailors didn't even know what they

were missing. This kind of a form of ignorance is bliss. What you don't know can't hurt you. Spoiler alerts in films and sports. You ever done this, where you record the Super Bowl or the NBA final or something because you can't watch it, but you got to be super careful not to talk to anybody or look on social media or whatever? Because the moment you find out what the outcome is, now I don't even want to watch. Rational ignorance, better not to know.

- 3.78** Many parents choose not to know the sex of their baby or genetic testing in deadly diseases. Would you want to know? I did the 23andMe. I got the cheap swab and all that, and I ticked the box for, I wanted to know if I had the Alzheimer's genes. There's two. I didn't have them, so I thought, oh, boy, good. That's good. I won't get Alzheimer's. Well, no, that's not what that means at all. If you have the genes, you're slightly more likely to get it. If you don't have the genes, you're slightly less likely to get it. But it's not at all that hugely determined.
- 3.79** But there are some you can get tested for, like Huntington's disease. If you have the genes, you're getting it 100%, and you will be dead, dead, dead by this time. Pretty close to that year. Would you want to know? Or if we had some super sophisticated test that will tell you, you will be dead on February 12, 2036. Would you want to know? Most of you are nodding. No, I think I don't want to know. I think the ignorance is nice. I just want to go along. Now, maybe if I knew, OK, I do know, within 50 years, I for sure won't be here. But maybe I'd lead my life slightly differently if I knew it was coming sooner rather than later.
- 3.80** But on the other—this is one of those, there's no right answer. I mean, people just—how will it affect your life? We mentioned before, blinded experiments and blind peer review. Again, this is a form of rational ignorance. It's better that I don't know who wrote this paper in case I know this person. Now, to be sure, to be realistic, you can often tell who wrote a scientific paper. If you look at the references, and all of a sudden, there's a whole block of one name, it's like, that's the guy. He's citing himself. We all do that. It's really funny. Anyway. But it's better not to know, blind review, and so on.
- 3.81** Even in the experiments. It's better that the subjects don't know which condition they're in because that could affect the outcome. It's better if the experimenters, the immediate experimenters, don't know which group they're in because that could affect how you collect the data. They're coded with numbers and letters and so on as random code. Then you decode them later when you're running the data experiment.
- 3.82** Or juries are sequestered. Why are juries sequestered? This is a form of rational ignorance. We mentioned this adversarial judicial system. There's rules about what evidence can be shown or not, based on how good it is or how irrational or unreliable the evidence is. If the juries are really going to make a rational decision about the guilt or innocence of somebody, there has to be a controlled environment, which is why, if

it's a big case, they sequester them in a hotel. They're not allowed to have their phones or newspapers and so on.

- 3.83** There was that famous example when Charles Manson was on trial in LA, and he brought in a newspaper headline. It said, Nixon declares Manson guilty. It was the cover of the LA Times. He's there in LA, and the judge immediately said, stop, that's it. Jury, you're out. We have to get a whole new jury now. Because you can't have the President of the United States telling people who he thinks is guilty because that's going to contaminate their rationality. That's the idea behind that.
- 3.84** Or choose to be ignorant to prevent our rational faculties from being exploited by rational adversaries. For example, the sign on Brinks trucks that pick up money, "Driver does not know the combination of the safe." It's good that the criminal knows that you don't know, so you know that he knows that you don't know, so he won't bother you and so on. That's a form of rational ignorance. Then there's rational irrationalities. Some of these game theory examples are really quite funny, like the game of chicken. You know the game of chicken? You're in your car, and the other person is there, and you're going as fast as you can toward each other. You're going to have a head-on collision until one of you diverts. Who will it be?
- 3.85** This was played out in *Rebel Without a Cause*, James Dean's movie. Of course, the solution to that is, you get up to speed, and then you lock your steering wheel in and you climb in the back seat, and you make sure the other guy sees that you're now out of control. Then he has to veer because he knows you're not going to veer. Of course, then he puts himself in the back seat, and then this is the problem with most of these game theory outcomes, when you apply it to nuclear exchanges and so on.
- 3.86** So we end up in a lot of these insanely, seemingly irrational ideas of, well, we're going to launch our missiles. They'll launch them, and we'll only lose 20 million Americans. You should read these Cold War documents from the Department of Defense in the early '60s about how many Americans they were willing to just let die if we could just make this happen over there. And literally, it was like 20, 40 million, it'd be OK. That's why that movie, *Dr. Strangelove*, was so good because it was exactly what the calculations were, where the one general is saying, "Mr. President, I'm not saying we wouldn't get our hair mussed, but I'm talking maybe 20, 30 million tops, depending on the conditions."
- 3.87** Really? That really is the kind of calculations they were making. That's that—you end up with these rational irrationalities, bargaining dilemmas. You start much higher than you think you can get. They start much lower than they think that you'll give and so on. So you end up somewhere in the middle. Or even something like paper. What is it? Paper, rock, scissors. I played this with my kid. It's a big fun. It's best to not know yourself which one you're going to throw. Try to be as random as possible because they may pick up on some pattern that you have and then exploit that.

- 3.88** Or Nixon and the nuclear weapons. This was called the Madman Theory. He might just be crazy enough to do it. But that, of course, is a rational decision. Let them think that I'm so crazy. Maybe that's what Trump did with North Korea. I'm just going to act like a total madman. Maybe that works. The problem with doing this with terrorism, if there's a religious fervor behind it, is that mutual assured destruction only works if you presume the other person doesn't want to die. Again, there's this great scene in Dr. Strangelove where our side finds out that the Russians have a doomsday machine. It's like, why didn't you tell us you have a doomsday machine? It only works if we know you have it, so that we can't do what you don't want us to do.
- 3.89** Finally, let's apply rationality to the question of meaning and happiness in life. Believe it or not, there's people that do research on this subject in psychology. First, they distinguish between meaningfulness and happiness. These are not quite the same things. Having a purposeful life or a meaningful life is different than having a happy life or being satisfied. These are words that carry a lot of baggage about what we want. You have to break them down a little bit.
- 3.90** We think, well, I want to be happy. What do you mean? Are you thinking about just walking around with a beatific smile on your face, going [noise]? Most people actually don't want that. That's not all that satisfying. Maybe for a couple of hours or something, but not for a life. The research makes a distinction of five different topics that inform us on this. There is a difference between wants and needs. Satisfying your needs may lead to greater happiness. Look, I'm really hungry. When I eat, then I'll be satisfied. That'll make me really happy for a couple of hours.
- 3.91** But this seems to have nothing to do with meaningfulness and purposefulness. These are like Maslow's basic pyramid, lower levels. You got to eat, you got to have shelter, just the basics of life. But that's just a minimum. To have something deeper, you have to go beyond that. My example is flying first class. I don't usually do that, unless someone else is paying. If it's a long ways, otherwise, if I'm doing the calculation, well, let's see, it's \$10,000 to fly first-class to Europe. I could buy a brand new carbon-fiber, all-electric bicycle that the Tour de France guys use. I'd get two of them, if I flew first-class both ways. I'm sitting in coach.
- 3.92** Or money. Money is good, but not intrinsically. It opens up possibilities and doors and things that you can do that bring meaningfulness and purposefulness. But by itself, it doesn't seem to do that much. We know the research on this. Above a certain level, people's happiness flattens out. But something else happens if you use the money to do things, philanthropy or start a company or whatever that you want to do, that that makes you feel purposeful and meaningful for some long-term goal that you have.
- 3.93** Good health makes people happy. I'm in good health. But it's a minimum. It's not like that is what gives my life meaning. I don't have any problems with my body. No, that's not enough. That doesn't lead to meaningful. That's a minimum baseline. Then there's time frames and actions. Happiness is a present state of mind. Meaning transcends the

here and now, reaching back into to the past and forward into the future. Thinking a lot about past and future does not make people more or less happy, but it does lead them to feel more meaning, as if what they did or will do makes a difference.

- 3.94** Here, it's just, is it long-term or short-term? What am I doing this afternoon? Some of what I'll do this afternoon is just for fun. I have lunch, we do a workout. All that's short-term. I feel happy, better that I did these little things. But tomorrow, it doesn't make any difference at all. A month from now, I won't even remember what I had for lunch today. But I will remember, I started work on this chapter of this next book this afternoon. Five years from now, when that book comes out, I go, I remember I was writing that chapter when I was at the hotel at Miami, and so on, and so forth. Short-term, long-term, time frames make a difference.
- 3.95** Thinking only about the here and now can make people more or less happy, but seems to have no effect on meaningfulness. Although happiness is partially inheritable—that is, about half as accounted for by the genes—it appears that meaning is almost entirely environmental, what you do more than who you are. Let me unpack that for a second. I'm going to talk about in the next lecture how we determine causality in science. Why do you act the way you do? Why did your life turn out the way it did? There's never a single variable.
- 3.96** Genetics is one of it, is part of it, of course, not simply, but in a complicated way. But roughly speaking, when we run these identical twins, separated birth experiments, and compare those to identical twins raised in the same environment, regular genetic siblings raised in the same environment, genetic siblings raised in separate environments, adopt these in the same home, and then you compare them across hundreds of different measures, it turns out on the happiness scales, about 50% of the variance between people on happiness is genetic.
- 3.97** You know people that are just always bubbly and happy and doesn't really matter what's going on in their lives, they seem to have a nice countenance or a temperament. Other people, they're just constantly grumpy. You know the type. No matter how good things seem to be going for them, they find every little thing right there. I'm going to be pissed off the rest of the day. It just seems to get to the—a lot of that's just genetic, but that's just having to do with happiness. In terms of, like, what am I going to do for my life that's going to bring meaning to it? It's more of what I just environmentally choose to do. I'm just going to make the decision to do this thing. Maybe it doesn't feel good now, but future me will appreciate that.
- 3.98** All right, family and friends. Yes, this is a huge part. There's these longitudinal studies on longevity. How long do people live? Well, what should I do? If you don't smoke, moderate drinking, exercise every day, the basic stuff. Get your vaccines and have a decent environment, eat healthy foods, and on and on. It turns out that all of those effects are washed over completely by having close friends and family, people that love you, people that you love, friends that you hang out with pretty much regularly, every

week or so, having a little list of maybe three or four people, maybe six at most, that you could call on any given day, and they'll take your call and they know what you're calling about.

3.99 They know something about you. That seems to have a huge effect on longevity. People that have that seem to live a lot longer and healthier lives than—it seems to count even more than not smoking in the opposite direction, which is pretty weird, but that's important. My other example of this is caring for children or aging parents. I was, had four parents. I had two bio parents, two step-parents, and I ended up caretaking for a couple of them, and they're all gone now.

3.100 But I was taking my dad around to all these doctors and clinics and hospitals and all this stuff. It was not fun. I wasn't like, oh, this makes me so happy. No, it was tiring and exhausting and frustrating and finding a place to park in the hospital parking lot, the doctor is late. But I feel like a much better person for it. This is a meaningful, purposeful thing to do. You don't do it because it makes you happy. You do it because it's the right thing to do. It's a principled thing to do. We want to be virtuous, principled people, which you get by living that, by actually doing it.

3.101 This is Aristotle's virtue ethics. You just live it, you just do it, and then you feel it, and then it makes you a better person, and then you're more virtuous, and that makes you feel more virtuous, and so on and so on. So much of life is not— in terms of rationally calculating what you should do, short-term, long-term, and so on. Balance it, mix it up. Some stuff you do because it makes you feel better in the next hour, but a lot of it, really, it's, how am I going to feel 10 years from now?

3.102 That brings up number four, struggles and problems. Having good things happen to you leads to both happiness and meaningfulness. But having bad things happen to you reduces happiness but may lead to increased meaning, depending on the outcome. In other words, having challenges to overcome, it's not fun at the time. Almost everybody goes, oh god, that was the worst experience. But man, I feel so much better now about what I learned and how strong I could be. I had no idea I had that kind of fortitude to do that, and on and on.

3.103 You'll hear these in the greatest generation that went through World War II, my father's. This was the most miserable thing they ever did and the best thing they could ever remember having done. It's like, how can that both be? Again, short-term at the time, miserable. Long-term, that was a good thing to do. Purely happy people may lead a stress-free life, but they may not have a meaningful existence. Whereas those who seem unhappy or stressed out may be experiencing purposefulness and fulfillment.

3.104 Retirement, for example, typically leads to an increase in happiness, but a decrease in meaningfulness until the person finds something goal-directed, something meaningful to do. Again, what's our purpose in life? Accomplish goals. Go for goals. That's wired in. That's how the machinery works. That's what evolution designed us to do. Seek goals.

Because we have big brains, we can have thousands of different goals, not just the basic ones.

3.105 Finally, self and personal identity. Actions that are expressions of your identity and sense of self are most directly related to meaning, but least related to happiness. That has to do with who you are as a person, how you identify your core being. I've talked about the self and what that means for those long-term goals. There's no answer for me to give you. That's just one of these things. You have to do the calculations in your head. I'm going to use rationality and reason to figure out what it is I want to do. But no one else can tell you that. Again, it's the goal versus the means to the goal. We're using rationality as a tool to get to the goal. What the goal is, that's entirely up to you.

Lecture 4: Cause and Effect

At a glance — Cause and effect.

- 4.1 All righty. Good morning, everyone. Thanks for coming back. This is lecture four of an eight-lecture series. Just to remind you, this is Skepticism 101: How to Think Like a Scientist. We spent the first three lectures talking about truth, the different kinds of truths there are. Are we living in a post-truth era? And rationality and irrationality. Today, I'm going to talk about science and how it works. Recall, one of our truths are empirical truths, objective truths, external truths, truths that you can see, I can see, we can all agree, that's probably true just by looking at it.
- 4.2 That is how science works. Basically, since the scientific revolution, we've developed a number of techniques to try to figure out, are we getting it right? Are we getting it wrong? In the next lecture, I'll have a whole series of ways that thinking goes wrong. But in this lecture, I want to show you how thinking goes right in terms of how science works. I'm going to start with a series of questions for you.
- 4.3 Does smoking cause lung cancer? Now, I know you know the answer to this question, but for decades, it was not clear. It turns out, it was not that easy to prove that smoking causes lung cancer, especially since the tobacco company flooded the scientific community with studies showing it doesn't. That may sound like a familiar refrain, but that's how it often goes. This is why, in scientific papers, they now require you to put where your funding comes from, just in case it's from somebody who may not like the findings if they go a certain way.
- 4.4 Does drinking coffee cause heart disease? I hope not because I have coffee every morning. And does burning fossil fuels cause global warming? OK. These are questions about causality. That is to say, what we want to know is, does X cause Y? We have an X, smoking. Does it cause lung cancer, Y? Does drinking coffee, X, cause heart disease, Y? Does burning fossil fuels, X, cause global warming, Y? We have an X and a Y that are in constant conjunction.
- 4.5 This is how the great Scottish Enlightenment philosopher David Hume defined causality. That is, there's a constant conjunction. X happens, then Y happens. X happens, then Y happens, or A and B, or however you want to phrase it. When X is constantly in conjunction with Y, we naturally infer a causal connection. For example, where there is fire, X, there is heat, Y. So we infer that the fire caused the heat.
- 4.6 Now, part of the problem is, back to our signal detection problem, is that, is there really a signal in the noise? That is, does X really cause Y, or does it just look like X causes Y, but it actually doesn't, and you've made an error in assuming that constant conjunction equals causality? My thought experiment on this is, imagine you're a hominid in the plains of Africa three and a half million years ago. Your name is Lucy. You're a little

Australopithecine, walking bipedal for the first time. You hear a rustle in the grass. Is it a dangerous predator or is it just the wind?

- 4.7 Well, if you think that the rustle in the grass is a dangerous predator, and it turns out it's just the wind, that's a type one error, a false positive in our two-by-two matrix for signal detection. But that's a relatively harmless error to make. You just become more skittish, you run off, you're cautious and so on, but no harm. On the other hand, if you think the rustle in the grass is just the wind and it turns out it's a dangerous predator, you're lunch. You've just been given a Darwin Award for taking yourself out of the gene pool early.
- 4.8 My argument is that we evolved the propensity to make more type one errors than type two errors. That is to say, it's safer to assume the worst, just in case. And if it's not, it's not that costly of an error compared to the other way. So natural selection selected us to make causal connections through constant conjunction, of which we make errors. One type of error is worse than another type of error.
- 4.9 Now, why can't you get it right? Why can't the animal just sit there and collect more data until it knows for sure whether it's just the wind or it's a predator? The answer is because predators don't wait around for prey animals to collect more data about them. That's why they stalk them. That's why they sneak up on them. That's why they stay down upwind instead of downwind. That's why they're camouflaged and so on. In the split second life or death situations in which we have to determine whether the signal is really a signal or noise, and the errors we make, we're more likely to assume one error than another, just in case. That's the basis of, generally, learning, and we make errors.
- 4.10 We assume A is connected to B. Oftentimes, it's not, but that's a decent error to make. How can we get around this problem? The answer is science, really. It's redefining causality in another way, in which Hume added that the constant conjunction doesn't always suffice for causality. Now, his amusing example of this comes from Geoffrey Chaucer's *The Canterbury Tales*, which included a rooster named Chanticleer, who came to believe that every time he crowed in the morning, the sun rose. So he crows, the sun rises, he crows, the sun rises, and day after day, he goes, oh, boy, I have the power to make the sun rise.
- 4.11 And Hume said, well, not so fast. So he offered this counterfactual causation definition of causality. That is to say, in his 18th-century style of writing, "We may define a cause to be an object followed by another, or in other words, where, if the first object had not been, the second had never existed." This is counterfactual causality. That is, in his case, silence Chanticleer, the rooster, so he can't crow in the morning, and then see if the sun rises. You silence Chanticleer, the sun rises; you go, OK, sorry, Chanticleer. It wasn't you after all. There's some other cause to the sun rising.
- 4.12 Back to our examples. Does drinking coffee, X, cause or be related to heart disease? Well, some scientists thought that was the case until other scientists pointed out that,

in fact, coffee drinkers also smoke more, and they exercise less, and they don't eat as healthily. So, coffee was only a proxy for other factors that caused heart disease. It wasn't really the direct cause. Or maybe non-coffee drinkers who get heart disease, maybe because they smoke, then we can infer that it wasn't the coffee causing the heart disease.

- 4.13** In other words, you have your two variables, X and Y. Take X out and see if Y happens or not, or maybe change X to something else and see if Y happens or not. That's what experiments are all about, is controlling for these what are confounding variables. Because causal chains are often more than just an X and a Y. For example, if you burned yourself with a flame on your stove, it isn't the fire that caused the burn, it's the heat from the fire that caused the burn. Here we would infer that fire, X, caused heat, Y, and that caused the burn, Z. All of a sudden, we have three factors in our equation instead of two, an X, Y, and Z instead of just an X and Y.
- 4.14** To give you a fun example of this, I'll tell you a story about the time that I walked across 1,000-degree hot coals for a TV show on fire walking. You've seen the fire walkers? They look like this. There's hundreds of images of this. People do it all over the world. You have a bed of wood that's been burning for hours, and they rake them out. Then you put the thermometer up there, it's 1,000 degrees, and the flames are on the edge. You just take your shoes off and you walk across it.
- 4.15** Now, this sounds completely insane. When I agreed to do it for this TV show, it was months ahead of time. As I got closer, I thought, what am I doing? This is insane. I went around barefoot all summer, so I got some good calluses going. Then it came time to do it, and I took my shoes off, and there it is, 1,000 degrees. I almost backed out, but other people had already done it, so I did it. I just walked across hot coals, and it didn't burn my feet. I didn't even get blisters. Most people don't.
- 4.16** The question is, how can that be? The answer is because dead wood is a poor conductor of heat. It's not the heat that's burning the skin. It's the material, the conductivity, the transfer of the heat from one substance to another. In this case, from dead wood that's been burning to the bottom of your feet. The analogy we use is, just to explain this particular, apparently, paranormal phenomenon, which it's not— let's say you bake a cake in the oven, you turn it up to 350 degrees, put the cake in there.
- 4.17** You let it cook for a while, you open the oven. If you put your hand in there, just where the air is, you don't get burned, but that's 350 degrees. You can put your hand on the cake. The cake is also 350 degrees, and you don't get burned. But touch the cake pan, also 350 degrees, you get burned instantly. It's not the temperature, it's the conductivity of the material. Metal is a good conductor of heat. Cake is a poor conductor of heat, as is air, because it's soft and fluffy.
- 4.18** I did it. Actually, I've done two of these now, and it is always anxiety-producing, but it works for the most part. Very few people get burned. It helps if they have, which we

did, had a mat of wet grass on one side and a mat of wet grass on the other side, so you get your feet nice and cool and wet, and then you traipse across. As long as you don't stop and take selfies, when I did it, there were no smartphones yet. Now people have gotten burned, I read, because they stop and take a selfie, and then the temperature builds up if you stand there. If you walk fairly quickly, you don't get burned.

- 4.19** In this case, the causal chain of fire, X, to heat, Y, to burn, Z, is interrupted by many confounding variables. The material being burned, in this case wood. By the way, no one ever does a fire walk with metal plates because no one could do it. You'd be burned. It's wood. It's always wood, and you burn it for hours. The calluses on your feet, that also helps. Sometimes people get a little chunk between their toes or whatever, then they get burned because it sits there. The speed with which you walk, again, walk briskly. Don't stop and dawdle and take selfies. And the cool wet grass, that helps, too.
- 4.20** That's an interesting tale. First of all, we have a natural explanation for apparently supernatural phenomena, walking on hot coals. It's nothing to do with positive thinking or anything like that. In fact, when I did it, I wasn't thinking positive at all. I was thinking, oh my god, what have I got myself into? Have you ever gone on a diet with your friends, and some lost weight and others did not? OK, of course we have. We've all done it. The most we could say is that X is associated with why, that is, dieting is associated with weight loss and likely depends on many other factors.
- 4.21** For example, perhaps people who diet also exercise more. It's the exercise, not the diet, that contributed to the weight loss. Or losing weight causes people to exercise more because their overall motivation is higher. Or maybe it's not the diet per se, or maybe it's the particular type of food. You've read these. They used to do the all-protein, the Paleo diet, because our Paleolithic ancestors, that's all they ate, was meat. Turns out that's not the case at all. Our ancestors were—they ate "see food." If they saw it, they ate it. I see it; I eat it. Basically, they ate everything they could.
- 4.22** But maybe all protein, or maybe it's all carbs, or maybe it's all veggies, whatever the particular— maybe it's that and not the diet per se. Or maybe it's going on the diet, not the diet, the going-on part that helps people lose weight. That is, you're doing something. You're measuring your weight every day, you're watching what you eat, and that motivates you to exercise and lead a healthier lifestyle since you're doing all this other stuff already. It's those factors, not the diet or the particular food. It's the going-on part. I'm doing something my weight, something like that.
- 4.23** Here's another causality problem that I like to bring up to my college students. Is a college degree worth the money? Well, let's see. Pepperdine University, where I went to school, is \$90,000 a year now. Unbelievable. Chapman, where I've been a professor for 10 years, I think it's \$75,000 a year now for everything. These are private colleges. But even the state universities are pretty expensive. Is it worth it?

- 4.24 All right, this is a causality question. Should I go to college, and will it pay off? Well, what's the measure you're using? Well, let's just say how much income you'll make over a lifetime. Well, that's a causality problem, and we can look at that. In fact, the studies show that a bachelor's degree is worth nearly \$1 million more in lifetime earnings compared to those with only a high school diploma. If you go to graduate school, you'll earn about \$1.5 million more than those with just a high school diploma.
- 4.25 Yeah, OK. OK, I guess, if I pay, I don't know, a quarter million dollars for my bachelor's degree and I earn \$1 million more in a lifetime, OK, then it was worth it. Of course, there are other confounding variables that complicate the relationship between these two, going to college and lifetime earnings. For example, smarter parents are more likely to send their kids to college. We know there's a positive correlation between intelligence and income. So maybe it was the inheritance of a high IQ that led to greater lifetime earnings and not the education.
- 4.26 OK, so just let me comment on that. IQ is highly heritable. Maybe at least 50, but maybe 70 to 80% of the variance between people and IQ is accounted for by genetics. Maybe it's not the college education; maybe it's that smarter parents send their kids to school, college more likely, and they pass on their intelligence through their genes to their kids. Or even if you don't accept that and that it's mostly home environment that leads to higher IQ, well, then those parents that have more enriched home environments and send their kids to school, maybe it's the enriched environment.
- 4.27 You see where I'm going? There's a lot of different factors here. In fact, the number one predictor of how much money you'll make in your life is how much money your parents made in your life. That is, parental income is a predictor of offspring income. Maybe it isn't IQ or education or some other factor that leads children to having higher incomes, but it's other things like guidance on investing or parental loans or role modeling or inspiration or inheritance.
- 4.28 I remember when my father explained to me how the stock market works, and some other advisors told me, here's what you do. As soon as you can have a couple of hundred extra bucks, buy some stuff. Just let it sit there for the next 40 years. Here's what will happen, 8% to 10% every year for 40 years. They showed me that cumulative, and it's like, wow, OK. I didn't know that. They don't teach that in school, at least when I was there. Maybe it's that, just having guidance like that.
- 4.29 Or maybe healthier people earn more money in a lifetime. Maybe some combination of genes and environment makes some people healthier and also makes them wealthier. In fact, in all likelihood, it's a combination of these factors and probably many more besides, including luck. I've been doing research on how lives turn out. It turns out luck, if you study successful people, luck is a huge factor. You went left instead of right. You took this class instead of that class. You met this person instead of that person. You took that job instead of that. The timing of the job and so on. It's just astonishing how much that is a role.

- 4.30 So if you got lifetime income, is it education? Well, it's maybe partly education, but also IQ, health, parental income, parental upbringing, the neighborhood that you grow up in, the culture at large, your lifestyle, and of course, luck. Just think about some of these success stories. Just let me riff for a second on how hard this is to study scientifically. Why are some people successful and not others? The luck.
- 4.31 It's said that, had Bill Gates been born 10 years earlier or 10 years later, he would not be the richest person, most successful person. Because he came online with his particular skill at that moment to write software for these computers at that time, it really came down to a month that he was right there at the right time. Same thing with the Google guys, Larry and Sergey. Had they been born 10 years earlier, 10 years later, they'd probably be successful doing something. They were smart guys. They are smart guys. But that particular thing, search engine, Google, now one of the wealthiest companies ever? Probably not. Timing and luck.
- 4.32 People do these studies. What makes somebody successful? Here's the characteristics. I call this the biography bias. I wrote about this when Walter Isaacson's biography of Steve Jobs came out. Let's go through Steve Jobs' life and see what's the magic formula. OK, so you go to a liberal arts college, an elite liberal arts college. You went to read in Oregon. Then drop out early, move back to your parents' house, and do a startup in the garage with your buddy. Before you know it, you have Apple computers, the richest company in the history of the world. You got to treat your employees really badly because that, apparently, is really good. And so on.
- 4.33 These are all just hindsight bias, just back-engineering after the fact. Here's what he did, and it became a correlation between here's the things Steve Jobs did and here's how successful Apple computers are. But this is irrelevant. How many people, again, in our two-by-two matrix, how many people dropped out of college, moved back to their parents' home in the '70s, and did nothing? The answer is, we have no idea because no one writes biographies of those people. There's a selection bias, a survival. It's called a survivorship bias. We only look at, notice the ones that made it, not the ones that didn't make it.
- 4.34 OK. All right, let's look at correlation and causation. In science, we use this thing. You may vaguely recall this from your class on statistics, something called the correlation coefficient. What we want to know is how two variables that we are interested in studying, like, again, education and lifetime earnings, let's say, how they co-relate. We call it a correlation. So somewhere between zero and one—zero is, there's no relationship at all. One is a perfect relationship between the two, and there's no other factors influencing it. Of course, there's never zero or one. It's always 0.01 to 0.99. So correlations fall in between there.
- 4.35 They use the little word *r*, the little letter, italicized *r*, for the correlation coefficient. Let's take something like height and weight. Of course, the taller you are, the more you get away, it's just more mass. But the correlation is not perfect. If you had the curve

like this and you had dots on a perfect line, then weight would depend entirely, 100%, on height. But it doesn't. The correlation is 0.7. That's pretty high as correlational coefficients go. But still, it's 0.3 below a perfect 1.0. Obviously, there's other factors at work, such as diet, nutrition, exercise, lifestyle, upbringing, the quality of the country you live in, and so on and so forth.

- 4.36** Now, there's one other move here that social scientists use with the correlation, is you square it. That is, you take the little r and you square it. This is called the coefficient of determination, or the proportion of the variation in the dependent variable that is predictable from the independent variable. In this case, height is the independent variable. This is the thing we're interested in, in measuring as the causal agent. And weight depends on height. We say weight is the dependent variable, in as much as it depends on height.
- 4.37** If you square 0.7, our square, that's 0.49. 7 times 7 is 49. So, 0.49, or roughly 50% or close to 50% of the variation between people of the same weight is accounted for by their height. That tells us there's something else going on here, a lot more. That is, 50% of the variation of people is not based on height. It's based on something else. Again, it's probably a whole nest of other causal factors, diet and nutrition and the quality of the air and the quality of the neighborhood and your country you live in and so on and so forth.
- 4.38** Let me give you another example that's kind of interesting. There's a lot of—I'm interested in religion, so I've studied this for decades. There are interesting correlational studies between identical twins, raised together, raised apart, siblings raised together, raised apart, and so forth, on a whole host of things—IQ and height and whatever, personality, temperament, and whatnot, but also how religious you are.
- 4.39** Now, it's not like there's a religious gene, exactly, but that genes code for different preferences and choices in life, and some people choose to be more religious, some people choose to be less religious. Anyway, the bottom line here is that, when you run these studies, it's called behavior genetics, with these identical twins raised in the same home, identical twins raised apart and so forth. You end up with a correlation of about 0.49.
- 4.40** That is, about, if you square it, about a quarter of the variation between people on their religious preferences and interests is accountable by their genes, which means that roughly 75% of the variation is determined by the environment. Things like the religiosity of your parents, grandparents, your peers, teachers, mentors, the culture at large, where you happen to have been born, what age you were born in, in history and so forth, accounts for religiosity. I'm just giving you the examples to give you a sense of, it's really hard to say X causes Y . It's always probably a rat's nest of half a dozen to a dozen different factors, and it's really hard to know which is the right one to conclude as the causal agent.

- 4.41 OK, so correlation, co-relate, how two dependent random variables co-relate, very much depends on the different factors that you're looking at. These are best visualized in a scatterplot, which two variables have here co-related. Let me just give you an example here. This is just one of hundreds of beautiful slides at Max Roser's Our World in Data. Max is an economist, and he's interested in different correlational factors that go into important things, like, how long will you live? That's an interesting question. What should I do to live longer? Everybody wants to live longer.
- 4.42 Well, so, he looked at this one variable here, a correlation between life expectancy in years at birth, by liberal democracy index. That is, is it better to live in a democracy or an autocracy or a theocracy or some other? Well, even within democracies, some democracies are better, some democracies are worse. Shown is each country's life expectancy at birth, plotted against the liberal democracy index, zero to one, higher or better. There's organizations that track every democracy in the world, and they rank them. How good is your democracy? How fair are your elections? How corrupt are the elections? And so forth and so on. Can everybody vote? Does everybody vote? And so on.
- 4.43 You see, if you squint, you can see a best fit line that goes like this. The better your democracy, the longer that people live. But it's not a perfect line. Not every country falls on that line, like here's China. The size of the blob is the size of the population of the country. Obviously, it's possible to have a crappy democracy, which China has not really even a real democracy. Still, the people are now living a lot longer because they've made other economic changes that led to longevity and better health care and so on in the last two decades.
- 4.44 But that's just to give you an idea of how messy social science data can be. All right, now, how do we go from co-relation—there's just association— to causation. That is, it could be that A is correlated with B because it causes it. That is, A caused B, but maybe B causes A, and that there's a reverse causation there. Or maybe C, some confounding variable causes both A and B. Back to my example. Exercise is correlated with weight loss. Maybe exercise causes weight loss, but maybe you start losing weight, you think, well, now I'm really getting motivated here. I think I'll start exercising.
- 4.45 So it wasn't the exercise, it was the reverse, or maybe there's something else. Now I'm eating better, and that leads to both weight loss and exercise because I'm motivated, something like that. In fact, in science, most effects have multiple causes, so it's always risky to say, "Now I know what caused Y. It was X." It's almost never like that. How do we do that? Well, in science, the gold standard, to go from correlation to causation, is the randomized controlled trial.
- 4.46 OK, what is that? Well, that's where you randomly assign subjects in different conditions. If I had you guys here, I would assign you each a number, and I would just use a random number generator to say, you're going to be in this group, you're going to be in that group, this group, the neutral group, the control group, the experimental

group, and the placebo group, and so on, because I don't want to select you or have you self-select because that could bias the conditions.

4.47 Let's say we randomly assigned you to, you're only going to eat meat, you're only going to eat vegetables, and then we're going to measure something over the next six months. Maybe it's your weight or your cholesterol level or whatever measure we're interested in. Then people randomly assigned to the exercise group or the non-exercise group, OK, you guys are going to exercise 30 minutes a day, every day, seven days a week. You guys are going to just sit on the couch and do nothing, or something like that. Then we see if there's some outcomes that are different between the groups.

4.48 You can already see the problem with this. I can't make you do stuff. Maybe you do it, maybe you don't. Over the next six months, will you really never go to McDonald's? Now, maybe you tell me you do, but you didn't, but you did. A lot of this is self-report data. Fill out this form. How many times did you eat junk food? Oh, never. Then you find out from the spouse or whatever, they actually did eat. OK, so people exaggerate, they misremember what they actually did. These are really hard to do.

4.49 Some of the diet and nutrition experiments are done on prisoners because they're not going anywhere. We know what they're going to feed them. But even there, new prisoners come in. Prisoners are released. If they're all in the same group, hard to know, really, if you've controlled for all the other confounding variables. The next best thing to do in the randomized controlled trials are natural experiments. That is, things that happen anyway, and then after the fact, you study them. I can't make you, this group, smoke, and you guys don't smoke. But there are people that smoke, and some people that don't smoke. You just divide those in and you just, after the fact, look at their health outcomes. Of course, as you know, there are huge differences in health and life outcomes from smoking or not.

4.50 Or you look at the blue zones. Blue zones are areas around the world where people live a lot longer. I mean, decades longer, like into the nineties or hundreds, versus our average of 72 and 74 for men and women in the Western world. What are they doing? So scientists go to these areas. There's one in Greece, there's another one in Asia. There's one in Loma Linda, southern California, where the Adventists are, who never drink or smoke or anything like that. All right, what are they doing? They don't drink, they don't smoke, or they drink in moderation. They exercise, they walk a lot, they have a lot of friends, they do Sudoku puzzles. Whatever it is, you're just looking for what is it that those guys do.

4.51 Or if you look at the 50 different states in the United States, well, they vary. Let's say you're interested in gun control. Do gun control measures lead to a decrease or an increase in, let's say, homicide and suicide rates? Well, this has been done by a lot of researchers. The problem is because the 50 different states have different gun control laws. You just look to see what they did and what the outcome was. But as you could guess, the problem is, there's so many other confounding variables. How many guns

per capita does each state have? That varies a lot. What is the historical culture of violence or not?

- 4.52 It turns out, for example, Southern states have higher rates of homicide and gun violence, in part because they have a higher culture of honor. That is, you settle disputes yourself rather than going to the authorities who largely cannot be trusted. There's studies on this showing the difference between New England states, where democracy came early and the government largely defanged the population and had a monopoly on the legitimate use of force, the definition of a state, versus Southern states that were populated by a different group of Europeans, and the long arm of the law was slow to get there, so they had to take justice into their own hands. Therefore, violence goes up, just because people want justice, and they settled scores themselves and so forth.
- 4.53 There's a lot of variables there. For example, this self-help justice idea. If something is illegal, let's say, you want a trade in drug exchanges. You have drugs, you want to sell them and make money. Just good old free market capitalism. But it's illegal. If I'm doing it illegally, and then you come in, and you come into my territory where I'm selling drugs, I can't go to the courts and sue you for violating our agreement that this is my territory and you stay over there. I have to go to your house at midnight with my gun and go, you better stop. And you see what happens in that escalation.
- 4.54 OK, those are the kind of variables that are really difficult to control. We have identical twins reared apart, as I mentioned, or you compare those to identical twins reared together or bio-siblings reared together, adoptees reared together. There's a pretty high correlation, let's say, on almost anything you want to measure. I have examples of this. I'll just show you some of the data on IQ correlations. This is one reason we think IQ is highly heritable. But even so, you can see with these, it's not just genetics.
- 4.55 The top line is the same person, the correlation of the same person. That sounds like a weird thing. But remember, there's current you and future you. You take the IQ test now, you take it a month from now, a year from now, 10 years from now. The correlation is about 0.85. It's pretty high. You square that, and 72% of the variation in your scores and IQs is counted for by the fact that it's you. You are a perfect identical genetic copy of yourself, obviously. Your identical twin is, too, almost 100%. It's not 100%, but it's pretty close to 100%. There the correlation and the r^2 is largely the same.
- 4.56 But then you have identical twins reared apart, and all of a sudden, the correlation drops to 0.76, and the variation that's accounted for between identical twins reared apart drops to about 58%, accounted for by the genes. Obviously, there's other factors. All of a sudden, you start to see, oh, OK, so the environment does make a big difference in IQ outcomes. Or fraternal twins. These are twins that are not identical twins, just two different eggs fertilized at the same time.

- 4.57 Then it's just exactly the same as regular siblings, except closer in time. Instead of birthed a few years apart, they're birthed a few minutes apart. They're going to be more identical. You could see the correlations are still reasonably high, but not as high as identical twins. Then you just have regular old siblings raised in the same home versus reared apart. Don't worry about the details of this particular example. The point is, you can see how a social scientist thinks, like, how can I figure out if it's genes or environment and what percentage is which?
- 4.58 On the natural experiments case, I can't make one country become a democracy and another country be an autocracy and a dictatorship, and let's see what happens. But that happens naturally. Take North and South Korea. You can see where they divided here and adopted different economic and political systems in the late '70s and early '80s, and how rapidly the blue line, South Korea, developed in terms of, and this is per capita income, that is how much money does each of your citizens make on average every year over the course of decades. It's quite obvious which one has an effect.
- 4.59 What is specifically different about democracies versus autocracies? It's difficult to say because democracies are better in every which way you can conceive of. Better education, better roads, better air, more electricity, better banking and financial resources, and on and on and on. You can even see the difference from space. You can see which one is North Korea, the dark one, which is South Korea, which is lit up and alive. You can see the difference in height. North Koreans are 4-6 centimeters shorter on average than South Koreans. That's just a diet. They just have crappy diets and crappy environment and so on compared to North Korea. That's what money and democracy buys you.
- 4.60 Another tool we have, and this is one of the most important ones, and I'll take us to the end of this lecture, is called Bayesian reasoning. I find this really a useful tool for just thinking. Now, technically, there's these massive equations, and I'm not going to go through all that. I just want you to get the idea of Bayesian reasoning that, this is developed by the Reverend Thomas Bayes in the 18th century, of trying to put a probability of something on a claim being true.
- 4.61 So recall where we began. There is objective reality that we can know something about it, but not 100%. We're going to get it wrong. Then we have to put a probability, a likelihood on it, of not just yes or no, but 80% probability, 20% probability or so on. This is what Bayes' calculations, Bayes' formula, or Bayes' theorem tries to do. That is, you describe the probability of an event based on the prior knowledge of the conditions that might be related to the event.
- 4.62 The priors, your priors, is your initial degree of belief. I've noticed over the years that the language is getting adopted more and more. I'll hear newscasters or opinion editorialists on TV say, well, my priors were this, and now I've adjusted my priors to change my credence. It was like, whoa, they're talking like Bayesians. This is incredible. Even if they don't know how to do the math, which is fairly complicated, you don't have

to. So these probabilities that we have in our heads, the likelihood or not of something happening, determines the credence of belief, that is the credibility or strength of the belief as a percentage.

- 4.63 With Bayesian reasoning, you have your priors, that is what I previously believed based on what I knew at the time. That determines my credence or confidence that the thing I'm studying is true. Then, of course, I want to adjust my credence based on changing evidence or changing priors. As new information comes in that changes the probability, you can update your priors and thus your credence in the claim. The rule, the theorem, Bayes' theorem or Bayes' rule, guides us on how much to revise our probabilities, or to put it differently, change your mind, when we learn a new fact or observe new evidence.
- 4.64 You'll hear this quip attributed to different people, but "When the facts change, I change my mind. What do you do?" In reality, most people don't change their mind, which is part of the point of this course, is we got to be willing to change your mind if the evidence changes. Most of us don't do that. That's the point of teaching Bayesian reasoning. In science, you have to change your mind. If there's new evidence, you have to say, you know what? I think it was wrong. Look at this. I didn't know about that. That's OK.
- 4.65 In fact, that's a better mindset. It was reasonable for me to do—just think of the example I used yesterday of COVID in the early stages, where we just didn't know. Wear masks or not? Close the schools or not? Shut down the economy or not? What should we do? In a Bayesian way, they should have said, well, we're not really sure what's going to happen. For now, our credence is that it could be deadly, so we're going to do the following, but we're going to update our priors. If new information comes in, turns out masks don't work or they do, or that children are not affected, but older people with preconditions are, and so on. But they really didn't do that. They don't even have to do the math. Just talk like a Bayesian. Here's our probability.
- 4.66 All right, I'm going to give you a technical medical problem that's given to med students and the population at large, on how to calculate these Bayesian probabilities. All right? Let's go back to our signal detection problem. There's a blob on a scan. Is it breast cancer or is it something else? Remember our two-by-two grid. On the upper left, you really do have cancer and you agree with your doctor, I do have cancer, we're going to treat it. That turned out to be the right thing to do, that's a hit.
- 4.67 Recall our bottom right that you actually don't have cancer, the blob was something else, and you agree with the doctor, it's not, we're not going to treat it, that's a correct rejection. Those are the two things that are good to do. On the upper right, you actually have cancer and you decide you don't, so you don't treat it, and that could be deadly, so that's a bad miss. Or you don't have cancer, but you do treat it, and then you have to experience the side effects of radiation, chemo, and surgery. That's also not risk-free.

That's just to recall our signal detection theory and how it ties in with Bayesian reasoning.

4.68 Here's the Bayesian medical diagnosis that is given to med students and so forth. Breast cancer prevalence is 1%. The test sensitivity for breast cancer is 90%. That is, if you actually have breast cancer, the test will detect it 90% of the time. It's not perfect. No information, no measurement is ever perfect. The false positive test rate, recall in our signal detection problem, that's the cell in which you actually don't have cancer, but the test says you do. The false positive rate of these tests is 9%.

4.69 A woman tests positive. What's the percent probability she has breast cancer? Most people, including physicians, say the answer is 80 to 90%. The actual answer is 9%. Just imagine that. You go to your doctor, you have this test, and you're told you have a 90% probability you have breast cancer. The real answer is, it's 9%. What has gone wrong here? Why does almost everybody get this problem wrong?

4.70 The answer is base rate neglect. The rate is only 1%. It's very low. Probabilities are counterintuitive. They apply to populations of people, not to one person. Here, let me give you an example. Let's say I have a silver dollar here and I flip the coin and I do it like this. What are the probabilities it's heads? It's 50%, right? But hang on, I've already flipped the coin. It's 100% one or the other. It's just we don't know what it is. I don't know, you don't know. Then I look.

4.71 This is the problem with one example versus populations. When I say, when we go 50%, it means I'm going to flip it a thousand times, and 50% of the time is going to come up heads. But if I just flip it once, it's not 50% heads, it's 100% one or the other. What we're talking about is my ignorance of what it is, not what it actually is. OK, so here's the problem. Let's work it out together.

4.72 Not a woman, but 1,000 women in our population, 10 of whom have breast cancer. That's the base rate, 1%. Of these 10 women, 9 will test positive. That's the positivity rate, the 90% sensitivity rate of the test. Of the 990 women without breast cancer, 89 will test positive. That's the 9% false positive rate. You just multiply or divide 89 by 990, you get 0.09 or 9%. A woman tests positive. Does she have breast cancer or not? That's the problem. That's the wrong question.

4.73 Out of 1,000 women tested for cancer, 98% of them test positive in all. That's 89 from the positive test, 9 from the false positive or vice versa. Nine of them have cancer, you divide 9 by 98, and you get 0.091, or roughly 9%. When you frame the problem this way, that is to say, a population of people rather than one person, most people get the answer right.

4.74 Another way to think about it, here's another one of these medical examples that people use. You have a bulls-eye rash on your arm. Do you have Lyme disease? Here's a Bayesian way of thinking about it. Well, Lyme disease is on the increase, and it's not uncommon, so we could say it has a high prior, based on what we know. The bulls-eye

rash doesn't occur 100% of the time, but it is a common symptom, so it too has a high prior. Bulls-eye rashes are uncommon otherwise. In other words, other diseases don't have a bulls-eye rash. Our credence that you have Lyme disease, given that symptom, is very high.

- 4.75 Again, you can do these calculations, these mathematical equations that just roll over your brain without doing them a lot. But I'm just going for the principle here. Here's another principle that is Bayesian. If you hear hoof steps outside, think horse, not zebra. Now, as it's been pointed out to me, if you live next to a zoo or you're in Africa with a lot of zebras, well, maybe the hoof print is—but that's the question, what's the base rate? Here in Miami, it's much more likely to be a horse than a zebra, unless there's something I don't know about—there's a zoo nearby with a bunch of zebras. But the idea is the base rate. What's the base rate comparison?
- 4.76 The problem is, again, we're not natural Bayesians because base rate neglect causes people to forget about the prior probability. Again, I threw you off by saying a woman has this. What are the probabilities? A woman. Statistics and probabilities are done on big groups, not a person. We're not naturally Bayesian in our reasoning, and instead, we use cognitive heuristics or shortcuts. The pioneering work in this area is done by my friend and colleague, Daniel Kahneman. His classic—I'm off by one. No, that's good. There we go.
- 4.77 His classic book on this is *Thinking, Fast and Slow*. The idea is that there is two systems at work in our brains, system one, system two. System one is rapid cognition, intuitive, instinctive. This is the sense I have of what's true or false or right or wrong or whatever. Then there's system two, which is the slow, deliberative, rational way of thinking and solving a problem. It was Kahneman and his colleague, Amos Tversky, who died, and therefore only Kahneman got the Nobel Prize. He's the only psychologist to ever win a Nobel Prize.
- 4.78 It's because it's not in psychology. There is none in psychology. It's in economics, because pioneered behavioral economics. Largely, the study of why people get things wrong so often, because of things like base rate neglect in this particular case. That medical example comes from Danny's book, *Thinking, Fast and Slow*. His idea is that, as we'll see in detail in the next lecture, is that we have these shortcuts, what he calls heuristics, rules of thumb. If this, then that, I'm going to make a snap decision.
- 4.79 No one has time to do deliberative Bayesian-like calculations for everything you do. You'd never leave the house. If you go shopping and you're going down the aisle, here's how I do it. I just go as fast as I can. But there are other people who are very cautious. They look at every label and to see this and what the price is and what the ounces are and so on and so forth. I'm using more system one rapid cognition. I like the blue toothpaste or whatever. Other people are doing the calculations and so on.

- 4.80 There's research by neuroscientists showing that there's some of these subjects that have brain damage in the amygdala, emotion parts of the brain, in which rapid decisions based on instinct and intuition, these rules of thumb, these heuristics, is damaged. These people have the hardest time making decisions. They'll sit there in the aisle in front of the toothpaste rack going, oh, gosh, there's Crest, and this one, and this one has this many ounces, and this one costs that much, and this one has fluoride, more fluoride, less—they just can't even get out of the store.
- 4.81 So Kahneman's point here, Tversky and Kahneman's point, is that almost all of life is just rapid cognition, just to get things done, just to move along. So we use our intuitions. The intuitions are not always right. These cognitive biases I'm going to talk about in the next lecture are a result of these rules of thought, these rapid cognitions. When you present problems like the Bayesian medical diagnosis, we get it wrong because we're being tripped up by these stories. A woman, but that's not how the calculations are done. It's like, on 1,000 woman. Oh, I see. OK. The coin flip example and so on.
- 4.82 All right, let me apply Bayesian reasoning to a more fun topic than medical diagnosis. UFOs. You've seen in the news of late a lot of UAPs. These are unidentified aerial phenomenon, or now it's called unidentified anomalous phenomenon. This is called the euphemism treadmill. Steven Pinker calls this the euphemism treadmill. The label changes constantly as it becomes pejorative. So, UFOs have always been associated with these fringe, loony, tinfoil hat-wearing goofballs in their basements, thinking the aliens are talking to them. People who are into this stuff, they don't like that affiliation, obviously. They've changed it to UAP, unidentified aerial phenomenon.
- 4.83 What are these things? What is the phenomenon? This is the movie poster for a documentary film that a colleague and friend of mine named James Fox made. He's a hardcore UFO believer. I'm a skeptic. I heard about the movie. I said, hey, can you get me a trailer, so I can watch it and review it for you before it comes out? He writes to his colleagues there. "Hello, gentlemen. Can we get Michael Shermer the screener for The Phenomenon? I understand he's ready to join our club of cult believers."
- 4.84 I did a little Bayesian calculations for him. This is what I actually wrote to him and his UFO followers. "My priors, that all such evidence is terrestrial in nature, is very high, based on decades of studying the phenomenon and claims made for ETI visitation. So my credence in the hypothesis that we've been visited is correspondingly low. But like a good Bayesian, I'm willing to adjust my priors if there's new evidence to consider." I mean, it's really that simple.
- 4.85 I've been studying this, we started Skeptic magazine over 30 years ago. I've been hearing about disclosure of the aliens coming any day now for 30 years, and it never happens. When I hear a new report, again, what's the base rate? Well, it never happens. I'm just going to be skeptical. Now, OK, bring on the space. Maybe the aliens walk right here in the studio and go, Shermer, we're here from planet Vega, or

whatever. Oh, OK. I change my mind. Of course, I might first check to see if the guy's wearing a suit, some dude in an alien suit or something like that. We have to do that thing and so forth.

- 4.86 In fact, I even tweeted about this exchange we had, in which I said to James Fox, hubcaps, because in his movie, he has these pictures here, classic UFO pictures, that are presented in his documentary. I said, "These again, hubcaps and toys tossed in the air as evidence of UFOs." I debunked these back in the '90s with our own such sightings. Here, these are from a TV show in which we faked UFO photos. This little thing right here is a lid of a little candy box. I just took this clear plastic and I just took some thread and dangled it or just tossed it like a hubcap. We tossed hubcaps and so on.
- 4.87 My point was that the photos presented in the documentary as real UFOs are indistinguishable from the hubcaps and candy lids that we fake for UFO photos. How can we tell the difference? Again, Sagan's dragon. What's the difference between the invisible dragon and no dragon at all? What's more likely? Here's back to our Hume's question of the, what's more likely? UFOs represent spacecraft from other worlds or anomalies from this world. Again, remember our little affirmism. Before you say something is out of this world, first make sure it's not in this world.
- 4.88 That was taken from my house when I lived in Altadena, southern California. Here's another one of these UFOs coming in from the top of the screen. It's a little light in here, but you can see there's downtown LA here from my house in Pasadena. Here's another one of these UFO spaceships, and here's one with the engine is blasting off from the UFO. All right, and here's one hovering over the San Gabriel Mountains in southern California, just near the Rose Bowl there in Pasadena.
- 4.89 In fact, here's what we were actually doing. We were tossing hubcaps off the side of the cliff where I lived and just photographing them. Again, I'm not saying all UFO photographs are fake, but if you can't tell the difference between the real ones and the fake ones, then what are we even talking about? All right, for a different show, this is my adult daughter on the left, cameraman on the right. I'm holding a sheet of glass with some coins, double stick tape to the glass. You just photograph. This is, again, how to fake your photo. You aim your camera through the glass and focused on the distance in the cityscape and so on, and you get these little coins.
- 4.90 Now, for some reason, historically, UFOs always travel in threes in a triangular shape, or it's a single triangular shape. For whatever reason, that's popular. So I took a quarter and two nickels and taped them on there. If you just photograph it through there, it looks like UFOs hovering over Los Angeles. The point is to make fun of UFO photographs, that how can you tell the difference? Is there some way we could tell? It's really hard to tell when photos are faked. It's getting even worse with the deepfake stuff. Be careful of what you see online.

- 4.91 Finally, I think it's good to remember what the U in UFO stands for, or UAP. It doesn't mean extraterrestrial. It just means, I don't know. I saw something. What was it? I don't know. "I don't know" doesn't mean your pet theory about what you think you saw is right. Just means it's unknown, and that's it. Again, it's always OK to say, I don't know. But most people have a hard time with that.
- 4.92 Another thing I tweeted about these UAPs because these videos, shot by Navy pilots off the coast of San Diego, in particular, have got people very excited. Here's one of this little blob here, in the middle of the camera, all of a sudden shoots off to the side. People have gotten very excited about this, that these are traveling at speeds that our aircraft cannot possibly do. In fact, they make these left turns and they go so fast that the G-forces that the pilots would be pulling would just squash them and just turn them into nothing and kill them.
- 4.93 How can that be? There must be somebody on Earth has this super advanced technology. Maybe it's the Russians or the Chinese. Or maybe the aliens are here and they have this super fast technology of traveling. But as somebody pointed out here, a guy named McWest, when you look at the two frames where the UFO shoots off, there's a little number here, 1.0. That's hard to read for you here on the screen, but it's zoom. This is the camera zooming. It goes from 1.0 to 2.0. The thing that it's videotaping isn't moving at all. The camera is just zooming in on it, which makes the object, makes the whole frame, double in size so the object appears to move and double in size, as you could see.
- 4.94 When you go through all these UAP videos, they all have explanations like this. I find it astonishing that in the halls of Congress, senators and representatives are listening to these people, showing these videos, and nodding in agreement. Oh my god, this is unbelievable. We are being visited by aliens, the US government. These are senators and congressmen who are the US government, going, the US government is hiding stuff. OK, really?
- 4.95 And again, as I always point out, there's over a billion smartphones in the world. I think it's actually more, two or three billion now, that all have high-resolution cameras that can take high-resolution photographs and videos, and yet all we have are these grainy, blurry photographs and videos. That's not good evidence in science. So here's how I think about, in a Bayesian way, what we're talking about with these UAPs. What are they? They are ordinary terrestrial phenomenon, camera lens effects, visual illusions, balloons, geese, whatever.
- 4.96 Or they're extraordinary terrestrial. They're spy planes or drones capable of unheard feats of physics and aerodynamics. Or three, they're the thing that everybody wishes they were, including me, they're alien intelligence. The aliens have visited Earth. That would be pretty cool if that were true. But I don't want to believe in things that have to be believed in to be true. Is it really true? If they're really here, OK, show me. The likeliest hypothesis is number one, that they're ordinary terrestrial phenomenon.

- 4.97 The reason has to do with Bayesian reasoning and the base rate effect. Let's come back to that. Let me introduce you to Leslie Kean. She is a pro-UFOlogist, a journalist who's been studying this for decades. She's most famous now for having published a New York Times bestselling book called UFOs, Generals, Pilots, and Government Officials. Go on the record. She's the one that wrote that 2017 New York Times cover story just below the fold saying, "The Pentagon has a UFO program that it has funded for decades." People were like, what? The government spending money on UFOs? Oh, my god.
- 4.98 And it was an eye-opener. I read her book and I found this interesting passage about what she thinks most of these sightings are. There are literally tens of thousands of sightings since the 1950s, late '40s, really. She says 90% to 95% of UFO sightings can be explained as weather balloons, flares, sky lanterns, planes flying in formation, secret military aircraft, birds reflecting the sun, planes reflecting the sun, blimps, helicopters, the planets Venus or Mars, meteors or meteorites, space junk, satellites, swamp gas, [inaudible], sun dogs, bolt lightning, ice crystals, reflected light off the clouds, lights on the ground, lights reflected on a cockpit window, and more.
- 4.99 That's our base rate. 95% of all UAPs can be explained by number one, ordinary terrestrial phenomenon. A UAP appears again. What's more likely? One is our explanation. It's because of the base rate. Again, you hear hoof prints, hoof steps, you think horse, not zebra. You see another one of these grainy videos, you think probably one of the 95%, probably not part of the other two, which are extraordinary terrestrial.
- 4.100 The Chinese or the Russians have super advanced technology. OK, let me riff on why that can't be. OK, first of all, there's the counterfactual. What else would be true if, let's say, the Russians had incredibly advanced UAP jets and drones and so on, that DARPA and the US government has no idea how to do? One, we would see them in Ukraine because they're at war and they would use them, and we don't.
- 4.101 Two, that's not how governments work. That's not how intelligence agencies work. Nobody gets very far ahead of anybody else. The reason is because we all steal each other's secrets. We all back-engineer each other's technology. There's this whole claim that, oh my god, the government has this crash retrieval program in which they take stuff and they take it to a warehouse and they back-engineer what's actually going on. That's absolutely true. This has been true since the First World War. The government has captured, let's say, German biplanes and going, OK, what are they making? Because this is better than ours. We're going to just steal their ideas. We've been doing that, not just we. The Russians do it, the Chinese do it. All governments do this.
- 4.102 Think about the Manhattan Project, the most secret scientific technological program in the history of humanity, where we made a nuclear atomic bomb. The Russians had one in three years. How did they get that? They stole it. They had a spy in Los Alamos. A guy named Klaus Fuchs, who passed on the secrets that our physicists had developed. As we'll see in the conspiracy lecture, a lot of our physicists came from Europe. These are

defected German scientists that we stole. Then after the war, we took even more of them. That's a separate, that's a kind of conspiracy.

- 4.103** It's not possible that these UAPs represent anything unbelievably more advanced than what our government is capable of doing. In any case, we know DARPA, the CIA, and all our security agencies are vastly more funded than the Chinese or Russians are, or at least equivalent. If we don't have it, it's not likely they would have it. Even if, back to the Manhattan Project, we know from historians that physicists everywhere in the world knew there was something up about atomic research, because physics journals, all of a sudden after the war started, stopped publishing articles on nuclear fusion. They went, OK, somebody has something, and they've silenced these journals. We would know.
- 4.104** Finally, number three, why can't it be aliens? It could be. Not likely. The reason is because the base rate is so high for the other one, and why would the base rate be low for this? Well, first of all, we have no base rate. We have no evidence ever for sure that aliens have visited us. Let's just do a little geography lesson here. The universe is vast. It's mostly empty space. If you had, let's say, our Earth shrunk to the size of an orange, and you put it in Los Angeles, the closest star would be in Chicago, another orange. In between, there's just nothing, empty space.
- 4.105** So the chances of anybody finding us here on Earth, much less, why do they always go to the United States desert? And they crash. Somehow they have super advanced technology. They can traverse the vast distances of interstellar space, and they can't seem to land. They're just constantly crashing. As we'll see in a later lecture, the chance of them getting here and their being pretty much like our technology, just slightly more advanced.
- 4.106** This whole conspiracy theory about how, at Roswell in 1947, the aliens crash-landed, and our government back-engineered the technology. From that, we got modern computers. We made the transition from vacuum tubes to silicon chips, and that was it. We got it from the aliens. But you're telling me that the aliens somehow had advanced technology, traversed the vastnesses of interstellar space and all this super advanced stuff? They just have technology, like, five years ahead of ours? That's it? OK, anyway. So it's very likely not two or three, very likely number one.
- 4.107** OK, I will close with a fun example of another email I got. This is from an email. I get these pretty much every week from people who write me. He writes, Dear Michael, I believe you may find interesting to take a glimpse of a few free excerpts from my rare spiritual prophecy book, titled My Dreams Come True. I related my prophetic, paranormal dreams predicting real events in the near future, offering proof of a higher power which can only know that future. Then he sends me the link to his Amazon. He wants me to buy his book. It's a self-published book here.

- 4.108** Basically, I wrote him. An email back in a Bayesian way. Dear loan, which of your prophecies did not come true? What major events happened that you did not prophesy? You recognize our two-by-two grid, right? I asked, square number, upper left, you made prophecies and things happened. That's a hit. I'll give you that. OK, but how many misses did you have? That is, you made predictions and nothing happened. Or how many big things happened? Invasion of Ukraine or the Hamas invasion of Israel, whatever. No one seemed to predict it. It's always after the fact.
- 4.109** And astonishingly, he wrote me back. Thank you for your prompt reply to me. It's hard for me to remember any major events which happened during a length of time since I began having these dreams, which began, and so it just rambles up. Basically, he says, I have no idea. I never kept track of my hits and misses. OK, we're back to the golf ball finder. How many misses did you have? Of course, we all remember the hits. I had this dream, and then this incredible thing happened. OK. How many dreams did you have, and nothing incredible happened? How many times did the plane crash or the war started or the earthquake or whatever, and you had no dreams about it and so on? OK.
- 4.110** So finally, remember, before you say something is out of this world, first make sure that it's not in this world. Extraordinary claims require extraordinary evidence. Keep an open mind, just in case. The final point on Bayesian reasoning is something that I enjoy, called Cromwell's Rule of Bayesian reasoning. Oliver Cromwell was a great English politician and jurist who famously said, "I beseech you, in the bowels of Christ, you might be mistaken." It's a famous quote from English history. But in Bayesian reasoning, this is called Cromwell's Rule. Never assign a zero or a one to anything, just in case you might be mistaken.
- 4.111** Now, I would give the alien thing like a 0.01, 0.02, very, very low, but not zero, because it could be. They could walk in today, and you and I would go, oh, my god, we got it on camera. The aliens are here. OK. But I've been talking about this for 30 years that they've never come in when I'm—so I don't give it a zero, but pretty low based on my priors. Therefore, my credence is, they're probably out there somewhere, given the fact that there's hundreds of billions of galaxies, each of which have hundreds of billions of stars, each of which have half a dozen different planets. We're talking then trillions and trillions of planets in the cosmos. The chances that we're the only ones, pretty low.
- 4.112** But the chances of them coming here, also very low. I think it's reasonable to say, they're probably out there somewhere, they probably haven't come here. That's a nice Bayesian way of thinking about that, but I could be wrong. Thank you.

Lecture 5: Mistakes Our Brains Make

At a glance — Mistakes our brains make.

- 5.1 Welcome back, everyone, to Skepticism 101: How to Think Like a Scientist. We're halfway through the course. This is lecture five on how thinking goes wrong. We've seen how thinking goes right, how we can solve problems, but there are obstacles. Again, back to where we began. No one's omniscient. We make mistakes. Here are some of the ones we make, and I'm going to start off by showing how you, too, can make mistakes in your thinking by audience participation time. You have your index cards there, you have your pens.
- 5.2 In our first experiment, I'm going to divide you into two groups. No one's going to be talking here. You are Group 1, you are Group 2. Easy to remember. Group 1, you watch the screen. Group 2, close your eyes or cover your eyes or whatever. No cheating. No cheating. Remain silent. Group 1, here's our first question. You're going to write down your answer on your index card. OK, Group 1, continue. We need your second question. Everybody write down a number? Yes?
- 5.3 OK, now, Group 1, please close or cover your eyes. Group 2, you can open your eyes and look at the screen. All right, so you guys, Group 1, close your eyes, avert your eyes. Group 2, you are now looking at the screen. Here we go. Your first question, on your little index card. OK. Now, Group 2, here's your second question. All right, everybody got one? You wrote down two different things on your index card. Everybody do that? OK, now we're going to stop. I'm going to do the calculations.
- 5.4 You all give me your cards. I'm going to collect them and do some little higher math. These are anonymous. Believe me, it doesn't matter who you are, what you wrote. You all wrote a number. Here we go. First, let me do this calculation here. All right. Welcome to an example of a powerful cognitive bias that causes our thinking to go wrong. This is called the anchoring effect, the phenomenon which having a number in mind, even an implausible one or one you consciously try to disregard, affects your thinking.
- 5.5 You were divided into two groups. Each group read silently and answered two questions. The second question was, how many countries are there in Africa? Currently, there are 54 countries in Africa. The first question was different for the two groups. Group 1, I asked you, are there more than 100 countries in Africa? And group 2, I asked you, are there more than 10 countries in Africa? What was the difference between Group 1 and Group 2? Are you ready? Here we go.
- 5.6 Group 1's average for the number of countries in Africa was 48.1. Group 2, the average was 30. How can that be? I just gave you a random number. The answer is, it's this anchoring effect. I primed your brain with a number, completely arbitrary, 10, 100, whatever. I pulled up your estimate or pulled down your estimate based on the

number 10 or 100. That's bizarre. Here's just this, I've been doing this for years. I did it for this lawyers' group. The difference, again, between, you're primed with 100, your group is primed with 10. In this lawyer group, I got 45 versus 23.

- 5.7** At Claremont Graduate University, where I taught PhD students for a few years, here's the effect on one of our experiments, 56 versus 34. At Chapman University, where I've been teaching for the past decade, this particular class I've done it a bunch of times, 54 versus 29, 69 versus 26, 59 versus 27, 62 versus 28, 71 versus 29, 58 versus 28, 58 versus 42. So your 48 and your 30 average is right down the barrel of what you would expect. This experiment has been done numerous times, replicated over and over. That is bizarre.
- 5.8** This is what I've been talking about, how most of us really don't know much about anything. How many countries are in Africa? How the hell should I know? I don't really know. It's not something I study. I'm just going to do some ballpark estimate off the top of my head. If I prime you with a number, is it more or less than 100? That's going to up my just intuition because I don't really know, or can lower it. That's the effect.
- 5.9** Back to our friend Danny Kahneman in his book, *Thinking, Fast and Slow*. This is called the anchoring effect. This example comes from his book; other people did this research as well. For example, this anchoring effect, a heuristic. Again, it's a rule of thumb. It's this kind of system one, rapid cognition. Just what do I feel is the right answer? What's my quick sense? That's where this anchoring effect, priming you with a high number or a low number, has its effect.
- 5.10** For example, if I asked you to write down the two digits of your social security number, last two digits of your social security number, and I asked you to bid on a price of wine—this has actually been done—people with a high two-digit, last two digits of their social security number between 80 and 99, they bid \$39 for a bottle of wine. Compared to people with a low two digits of their social security number bid only 12 bucks on average. I mean, that's ridiculous. That that should have no effect at all on determining what's the real value of it.
- 5.11** But I'm priming you to think a certain way. That tells us that our cognition is very much influenced by other factors. This is why, by the way, restaurants typically include on the menu a very expensive bottle of wine. They know you're not going to buy that one, but now you'll go up from the lower one that you were going to buy, up to the higher one because they primed you with this higher number. Really? But they do that? Yeah, of course they do that.
- 5.12** Or the seller strategy, where you start off with a high number. You don't think the buyer is going to pay that, but you want to bring them up from what they were thinking that they would pay. Or prosecutors asking for a higher sentence from the jury. They know the jury is not going to go all the way up to here, but we ask for that, then they'll come down, and we'll get what we really want.

- 5.13** OK, here's experiment two. You could just do this right here without having to write it down. What continent is Kenya in? Of course, you all know that's Africa, right? What are the two opposing colors in the game of chess? Black and white, right? Name any animal. OK, this is our old friend. If you hear the hoofprints, of course, most people say zebra, but only after they've been primed with the first two questions before answering the third.
- 5.14** By the way, thank you for doing my little experiment with the countries in Africa. Honestly, every semester, I get some student that puts, like, 1,000. I have to throw that out. This is a little trick of statistical research. Whenever you have a big sample size, there's some outlier that pulls the whole average up into some direction, which is not good. Again, we're interested in the average, not the extremes.
- 5.15** Another one of these is called the availability bias or the availability heuristic. This explains a lot about how we consume news content in our evaluation of whether we think the world is getting safer or less safe. Are things getting better or worse? Are we making moral progress or are we in moral regress? Part of the problem is that the availability heuristic or bias influences how we think about that. That is, we judge the probability of events by the ease with which instances come to mind.
- 5.16** If you think of your memory as like a Google search engine, if I say, what are you most terrified about? Most people just scan their Google search in their brain of, well, what was on the news last night? Well, let's see, terrorist attacks, shark attacks, killer bees. These are the stories that get covered in the news. Then that is what comes to mind. We fear the wrong things for that reason. Just some annual deaths worldwide, I pulled up off the Internet.
- 5.17** 1.25 million people die in automobile accidents worldwide every year. In airplanes, it's less than 100 per year, all flights. What do people fear? Flying. Do people fear driving? No. This is completely irrational. But the news doesn't cover car accidents unless there's some massive pileup on an interstate freeway where there was a big storm or something and hundreds of cars. Then that gets covered. But just the grindingly slow, day-to-day car crashes that kill people never gets covered. Plane crash, man, you get total coverage wall-to-wall for hours or days.
- 5.18** How many people have died in nuclear power accidents over the past 60 years? Total, maybe in the thousands, maybe, if you include Chernobyl. Nobody died at Three Mile Island. Unfortunately, the film China Syndrome with Jane Fonda and some other big stars came out the week after the Three Mile Island disaster, and everybody had in their mind this conflation. Again, that availability. What's in my mind when I think of nuclear power? I think of, oh my god, the China Syndrome. It's going to melt down and go all the way to China. Oh, it's—right.
- 5.19** Fukushima, the primary risk from that, were the walls being breached from the tsunami, not the nuclear power plant itself. Chernobyl is different, but far fewer people

died of that than we expected. You can go to Chernobyl now and tour there. It's relatively safe. There's no two-headed fish and stuff around that area. It's relatively safe. It was not the disaster it was. By contrast, coal kills over a million people a year from lung diseases and accidents and things like that. And yet, nuclear power is something almost everybody seems to be against. People are, well, we got to have coal. Clean coal, whatever.

- 5.20** OK, so we're not good at doing those calculations because of what comes to mind when it gets covered in the news. In terms of terrorism, everybody's afraid of terrorism. Well, you can see from the CDC and other sources that more people die than from terrorism in lightning strikes, deer collisions, peanut allergies, bee stings, and my favorite, ignition or melting of nightwear. How does that happen? People smoke in bed, big mistake, and their clothing gets ignited.
- 5.21** After 9/11, it has been calculated that there were 1,500 extra deaths by car because people decided to drive somewhere rather than fly. OK, why? Well, because we're not rational in how we calculate the odds of risks because of this availability heuristic. In general, we fear what we're familiar with, rather than the unfamiliar. Better the devil you know than the devil you don't know. Controlled versus imposed. When I'm driving, I feel I'm in control. When I'm flying, it's like, my life is in somebody else's hands. That makes me more anxious.
- 5.22** A natural versus man-made or human-made. This is, I think, what distorts people's thinking on GMOs. This is a debate we've followed in Skeptic for years. Why are people against GMOs? We've been modifying organisms for 10,000 years. This is how we have big, beautiful ears of corn and wheat and so forth from what naturally grows, but it feels differently, the words organic, as opposed to what? Everything comes from Earth, it's organic, but inorganic sounds worse, and so on. Is it natural? Is it human-made? Everything is human-modified, but people get confused about that.
- 5.23** We dread what's uncontrollable, what's potentially catastrophic, what's deadly, inequitable. That potentially catastrophic is interesting because, back to our hominid on the plains of Africa, you hear a rustle in the grass as a dangerous predator, or is it just the wind? Better to err on the side of caution, just in case. People tend to think, well, what could go wrong? I can think of all sorts of things that could go wrong. But if I asked you, how could your life get better this afternoon? You go, well, I don't know. Maybe just slightly better here, slightly better there. But what could go wrong? I could get hit by a car when I leave here. Our brains naturally autocorrect to riskier things, dangerous things, things we can't control, and so forth.
- 5.24** Another one of these biases, what's called the representativeness heuristic. Heuristic, again, rule of thumb, just instinctive, just quick intuition about something. You can also call it a bias. This is where people judge the probability that an item belongs to a category or an event, according to how representative the item or the event is of that

particular cognitive category. That is, how close it is to the prototype of the category. We call this reasoning by stereotype.

- 5.25** People are very critical, for good reason, of prejudice and bigotry and so on, and that's right. But reasoning by stereotype, like, these things are all like that, these things are all like this. That's perfectly normal. That's how our cognition works. Nobody can do the calculations and collect data on every single thing that there is to think about. We don't have the cognitive capacity to do that or the time. We have these heuristics, these rules of thumb, these shortcuts. I'm just going to think about this as a stereotype of a particular category, or the similarity of the data to the hypothesis.
- 5.26** Let me give you an example. This is an actual example given to subjects to do a Bayesian calculation of what the odds are of something happening. What's more likely? A nuclear weapon will be launched in 2024, or a nuclear weapon will be launched in 2024 by North Korea, following a breakdown of talks and a military attack against North Korean targets, which Kim Jong Un mistakes for an all-out invasion.
- 5.27** Amazingly, most people will say, well, the second one, obviously. But it can't be. The probability of some nuclear weapon being detonated somewhere, anywhere in the world, by any country, is going to be higher than that this one particular thing happened in these very specific circumstances. Just by odds. But the second one tricks us because it's a story in which we can say, oh, that sounds more plausible than this other one.
- 5.28** OK, I'll give you another example of this. This is the famous Linda problem that Amos Tversky and Danny Kahneman concocted to represent this representative heuristic that also touches another one called the conjunction fallacy. So here's the story, the little thought experiment subjects are told. Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.
- 5.29** What is more probable? Linda is a bank teller, or Linda is a bank teller and is active in the feminist movement? Well, you can already see where I'm going with this. But the thing is, most people chose option two. Why? Let's go back to it again for a second. It just seems like she has to be a feminist and is active. Not just a bank teller. Read the story. But if you just do the calculation, which is pretty easy to do, the probability of two events occurring together in conjunction is always less than or equal to the probability of either one of them occurring alone.
- 5.30** So, here's the calculation. Let's say we assign a low probability of Linda being a bank teller. Say, probability Linda being a bank teller equals 0.05, plus the high probability that she's a feminist. Let's even put that at P, probability Linda is a feminist, at 95%, 0.95, assuming they're independent things. So P, that Linda is a bank teller and that

Linda is a feminist, you multiply them, 0.05 by 0.95 , where you get 0.0475 , which is lower than the 0.05 of Linda as a bank teller by itself. OK, that's a bit of a mouthful.

- 5.31** In other words, the probabilities of this one simple thing being true is always going to be higher than when you have multiple elements added to it. That makes the probability lower, just by the calculation. And Kahneman and Tversky, this representative heuristic is that— but read the story. She can't just be a bank teller. Come on. She's demonstrating against nuclear power and so on and so forth. She just has to fit this other category. That's incorrect.
- 5.32** The deeper point of this is, when you hear a whole lengthy story about something and then you're tasked with thinking about it, you're going to be distorted in how you think about it. How representative is that of this larger category? Option two seems more representative of Linda, based on the description of her, even though it's clearly mathematically less likely.
- 5.33** All right, so just for fun, I did this in my next lecture on conspiracy theories. I'm going to talk about 9/11 truthers. But I realized when I first encountered the 9/11 truthers, that they were committing the conjunction fallacy. Therefore, I already know that they're wrong before I hear any of the details of what they believe. For example, 9/11 truthers claim that planes hit the World Trade Center buildings, and explosive devices were used to bring down those buildings.
- 5.34** Here, there's this picture of the architects and engineers for 9/11 Truth. This is the biggest so-called 9/11 truther group. They really are a group of intelligent, educated people that are professionals in architect and engineering. They really think two things happen. The planes hit the buildings, and some operative within the Bush administration worked to plant explosive devices in the buildings that also helped bring them down. That's what they believe, the two things happened. In the next lecture, I'll go in detail about why that can't be.
- 5.35** Here's the question, are we truly irrational? I mean, if you read Kahneman and Tversky's work, you think, God, we are hopeless. We get it wrong all the time. Well, not really. It depends on how you think about these things. I mean, type one thinking, this rapid cognition, works pretty good in natural environments. For the most part, it's not irrational to use your intuition and your instincts, your rapid cognition, your rules of thumb, your heuristics.
- 5.36** But these scientists put subjects in these artificial conditions that seem to trick them, like I tricked you. I gave you the 100 prime, I gave you the 10 prime, and I tricked you into getting it wrong. If you think about these cognitive biases as illusions, like optical illusions, these are byproducts of mechanisms that recover a 3D shape in the world from a 2D retinal image. You've seen, in these textbooks in Psych 101, all the optical illusions that are pretty cool, like this one. In which, if I said, well, what looks like the longer line? Of course, like most people say, well, the top one looks longer. Of course,

you know by now that I'm tricking you and that they're the exact same line. But we're being fooled by the converging lines of the railroad tracks.

- 5.37** I thought just for fun, I'd make an actual example of this. This is the train track near my house up in Santa Barbara. Cuts right through Montecito, which is where I'm standing right there. But if I take that lower board there and I just copy and paste it and put it up there, it really looks longer than the lower one. But all I did was just do a little copy and paste and stick it in the exact same image. But our perceptions are fooled by the context in which the image is falling on the retina because that never happens in the real world. We never evolved the capacity to work around these illusions because you don't have to. It never happens.
- 5.38** The question here then, in terms of this, are we naturally rational or are we just gullible beyond belief and we fall for all these things? Are cognitive illusions like perceptual illusions? That is, they're byproducts of adaptive cognitive mechanisms? Sometimes, but sometimes not. This is what's called bounded rationality, that is, information processing has limitations. We have motivated reasoning.
- 5.39** That is, so here's the question. Did we evolve rationality and reason to get the world right, to accurately perceive reality? It's called veridical perception. This is the way the world really is. Or are we just using reason and rationality to win arguments for our team? That's the debate in cognitive circles. Again, I think it's both. I think we do have a capacity to understand the world the way it really is. So that's my commitment to universal realism, and that science is the best way to understand that.
- 5.40** But I also recognize that we have expressive rationality to signal our tribal solidarity. Again, my example, I can say, well, I doubt climate change is real. I'm a global warming skeptic. Do you actually know anything about it? Most people don't. People that say that, they're just signaling their commitment to, "My tribe is doubtful, and I agree." Or the opposite. Most people who say they accept anthropogenic global warming, if you ask them, well, what's the evidence for it? They really don't know. They're just publicly signaling, "I don't know, my tribe tends to accept it. I think the scientists usually get it right, close enough."
- 5.41** I think rationality is a little bit of both. We're not that dumb or irrational. The problem is that everything I've been teaching you so far, like Bayesian reasoning, probability theory, statistics, these are all developed in the 18th and 19th centuries. That's not that long ago compared to our evolutionary past of hundreds of thousands of millions of years. Also, calculating using large data sets, you take 100 million medical patients and then you do the calculations of getting cancer or whatever. That requires computing, large statistical analysis, huge data sets. We never had that when we were evolving.
- 5.42** So heuristics are these adaptive rules of thumb using information available in a natural environment. In that sense, we get it right often enough. It's good enough. You don't have to perceive the world perfectly, 100%. Just get it close, you'll survive and flourish

and reproduce and so on. We evolved to notice events, not statistics. Again, no news media source sends a camera crew to a school where there's been no shooting. Here we are again. It's another Tuesday and nothing has happened here. They send them to where the event happened, and it's always bad news. Events are what trigger our thinking about the world, rather than statistics.

- 5.43** Solving social problems, not probabilities. I'll give an example of this. This is what the founders of the field of evolutionary psychology, John Tooby and Leda Cosmides, call the cognitive niche. It's to say, all these tricky problems that people like Tversky and Kahneman give subjects to do that trips them up, like the example I gave with you guys —this is a little bit like the optical illusion where I copy and pasted the board up here. That never happens in the real world. This is a way of tricking things.
- 5.44** Let me introduce you to the Wason logical puzzle. This has been studied for decades. Every card here on the screen has a letter on one side and a number on the other. I'm not going to ask you to write this down. It's just, follow in your head. Here's the rule. If a card has a D on one side, then it has a three on the other side. Here's the problem you're tasked with. Is that true or false? That is to say, which cards must be turned over to check? This experiment's been done thousands of times on hundreds of thousands of undergraduate subjects, which is how much of psychology is studied.
- 5.45** Most people say D, or D and three. But the rule doesn't, it doesn't—look at the rule again. The rule doesn't say, if a card has a three on one side, then it has a D on the other. It says if the card has a D on one side, then it has a three on the other. Most people say D or D and three. But in fact, the correct answer is D, of course, and 7. That is, you want to try to falsify it. If the 7 had the wrong letter on the backside, then that would falsify the rule. In other words, most of us are tasked to test the hypothesis. That is to say, we should be trying to falsify the hypothesis. Most people just try to confirm it by turning over the D.
- 5.46** OK. Now, that's a mouthful. When I read this, I go, OK, I got to read this again. What? What am I, what is it I'm looking at here? This is my problem I have, not just me, with the whole Tversky-Kahneman paradigm of, let's see how we can trip up subjects to show how irrational they could be and all their biases. So Tooby and Cosmides presented it in a different way. Let's say you're a bouncer at a bar. Here's the rule. If the bar patron is drinking beer, he must be over 21. Well, we know that's true. That's a real rule. You are the bouncer in a bar, and you have to enforce the rule. Who should you check? Someone drinking beer and how old they are? Someone drinking Coke and how old they are? Do you have to card them? Someone who's over 21 and just see what they're drinking, or someone who's under 21 and see what they're drinking?
- 5.47** When you present it this way, someone drinking beer, let's see your driver's license, see how old you are, or someone who is, you check your license and they're under 21, and OK, what are they drinking? That's the rule. When you present it this way, it's the same problem as the Wason logic cards. But almost everybody gets this one right.

Almost everybody gets the first one wrong. The reason is because people are illogical with abstract symbols and the rules and the—it's the what now? I turned this over and the, huh? The D and the three and the number and the letter, and I get all confused.

- 5.48** Oh, I'm a bouncer at a bar. OK, yeah, of course. Obviously, I'm going to card somebody here. I'm going to look what they're drinking over there. This is what Tooby and Cosmides mean by this ecological rationality. Well, that's the environment we evolved in. That's our history. We live in groups. We're with other people. Those are the social problems we're good at solving, not logic and math, which is all pretty new, historically.
- 5.49** OK, experiment three here for our audience participation time. Both groups can participate at the same time. I'm going to ask you to study a group of words for 30 seconds and memorize as many as you can. You don't have to write them down, just keep them in your head, however many you can. OK, you ready? Here we go. 30 seconds. Let me put my timer on here. We'll go 30 seconds and go.
- 5.50** OK. Think of all the words from the list that you can recall. I say write them down, don't worry about that. Just think of how many you can remember there. Now, which of these are you confident were on the list? Was the word taste on the list? Was the word point on the list? How about the word sweet? OK, you can see where this is going. This experiment involves memory. Unlike a computer or video camera, our memory of information is influenced by many mental effects.
- 5.51** Note the original words here. What's the theme there? Candy, sour, sugar, bitter, good, taste. Oh, taste. There's the word taste. It's on there. The word sweet is not on there. But in these experiments, in this experiment, almost everybody goes, well, yeah, the word sweet had to be on there. It's not on there. But conceptually, it falls into that category. Again, it's part of the representative heuristic. Yeah, that word represents that category, and we think in concepts like that. Again, it's reasoning by a stereotype. That's a stereotypical candy, sweet, dessert. Sweet should be on there, even though it's not. There's our words. Point was not on there. Taste was, and so on.
- 5.52** Let me just comment for a minute before I show you this little video here. Again, back to the way the mind works. Our brains, our memories are not like video cameras. We have these cameras here. They could rewind and play back exactly what happened right here on the stage. Our memories are not like that. You will remember this event differently from, not just Group 1 and 2, but each of you will have a different memory of this experience because it's a constantly changing, updated system of recall from what happened in the past.
- 5.53** Again, back to the Cartesian theater, there is no Cartesian theater. There's no little mini me up there watching the screen, and you play back the memory on rewind and little mini me goes, oh, no, no, that's not what happened. Here's what happened. There's nothing like that. We have all these effects that distort what we think we're looking at. OK, fun example here from my friend Richard Wiseman, who is a professor of

psychology, who also happens to be a professional magician. Now, magic is fun because like the optical illusions, they're very skilled at tricking us into falling for something that's not true because they know how the mind works. OK, let's see if this is will play here.

- 5.54** Hi, I'm Richard. This is Sarah. We're going to perform the amazing color-changing card trick with this blue-backed deck of cards. Now, the idea is very simple. I'm just going to spread the cards in front of Sarah and ask her to push any card towards the camera. All right, OK, let's see. I'm going to go for this card here. OK. Now, Sarah could have selected any card at all from the deck, but she selected the card which is now face-down on the table.
- 5.55** What I'm going to ask her to do is show us which card she selected. I'd like to say the card that I chose was, in fact, the three of diamonds. The three of diamonds, OK. Excellent choice. That card goes back into the deck. Now I'm just going to spread the cards face-up on the table. Do a little click of the fingers, and you'll see that Sarah's card here has now got a blue back. Not particularly surprising. What's slightly more surprising is all of the other cards have got red backs. And that is the amazing color-changing card.
- 5.56** Hi, I'm Richard. This is Sarah. We're going to perform the amazing colour-changing card trick with this blueback deck of cars. Now, the idea is very simple. I'm just going to spread the cards in front of Sarah and ask her to push any card towards the camera. All right, OK. Let's see. I'm going to go for this car here. OK. Now, Sarah could have selected any card at all from the deck, but she selected the card which is now face down on the table. What I'm going to ask her to do is show us which card she selected. I'd like to say The card that I chose was, in fact, the three of diamonds. The three of diamonds, OK. Excellent choice. That card goes back into the deck. Now, I'm just going to spread the cards face up on the table. Do a little click of the fingers and you'll see that Sarah's card here has now got a blue back. Not particularly surprising. What's slightly more surprising is all of the other cards have got red backs. And that is the amazing Color Change in Color.
- 5.57** OK, that's a lot of fun. Did you notice the gorilla sitting in the back? OK. This is a riff and Richard's cheeky nod to Christopher Chabris and Dan Simons' gorilla tape. This is the famous tape where, probably most people have seen this now, where two groups of students, one in white shirts, one in black shirts, are passing a basketball back and forth. I tasked you with counting the number of passes between the white-shirted students, ignoring the black shirts. It doesn't really matter what the task is. I just want you to focus on one thing.
- 5.58** Anyway, in the middle of the video, it's only one minute long. A guy in an ape suit walks across and pounds on his chest and waves and walks off. This has been replicated countless times. People don't see, half the people don't see the gorilla. Probably half of you didn't see all the changes Richard made there in the color-changing card trick, if

any of them. The reason is what's called selective attention. We did not evolve a perceptual system that sees every single thing that happens. You'd never get out of bed and leave the house if you did this. You have to just take in what's important and just move on.

- 5.59** If I tell you, follow my gaze like the magician does, they go like that. They look at it. You can't help but following my eyes and my hand. This is how they make coins disappear and cards disappear and so on, because that's what our sensory apparatus evolved to do. Pay attention to the important things. If I can get you to really focus on just one thing, you'll miss lots of things in our environment. Again, this is part of memory and perception and why we get things wrong.
- 5.60** Let me introduce you to my friend and colleague, Elizabeth Loftus. She's a professor of psychology, one of the world's leading experts on memory at UC Irvine. She did a number of famous experiments in the '90s. For example, in eyewitnesses to an automobile accident, the choice of adjectives to describe it. Group one, maybe I say, "How fast would you estimate the cars were traveling when they smashed into each other?" If I said to this group, "How fast would you estimate the cars were traveling when they collided?" That group will give me a higher speed than this group. How is that possible? Just the adjective alone.
- 5.61** Now, believe me, lawyers know these. When they're talking to juries or interviewing witnesses, they'll use different adjectives to get a different answer. Because, again, there's no recording in there, and you play it back and go, oh, I see the cars are traveling. No, it's nothing like that. Just the language you use to ask somebody—just asking somebody a question leads them to construct something that never happened.
- 5.62** Elizabeth's most famous experiment was called Lost in a Mall. Here she has adult subjects, who she's interviewed extensively, including their family. She knows for a fact that none of these people were ever lost in a mall when they were children. But for half the subjects, adult subjects she's interviewing, in a series of questions, she says, "Do you remember when you were—we heard from your family, your parents, that you had this traumatic event growing up. You were lost in the mall."
- 5.63** Over half of them came up with detailed memories about what a horrible experience, this, "Oh, yes, it was terrible. I was so scared, and I heard a voice coming over the speaker, and I wanted my mommy, and this man in a flannel shirt helped me." None of this happened. They're just making it up, right there in the interview room, because she asked them, "What was it like to be lost in the mall?"
- 5.64** Or she'll show the famously just short video clips. Here's a 15-second clip, just the scene in a park, somebody sitting on a bench, somebody runs up and grabs the woman's purse and runs off. Boom. Done. Then you ask a series of questions. What kind of hat was the burglar wearing? Oh, it was a baseball cap. Oh, it was a green cap. It

was a beret cap. It was this or that. The guy wasn't wearing a hat at all. Just asking about the hat plants the memory that, oh, well, of course, there must have been a hat.

- 5.65 Again, you're just looking at this scene, you're just taking in just two or three bits. OK, person on the bench, this guy comes in off the side, steals the purse. That's it, three data points. I don't know if he had a hat or not, what shirt he was wearing, what color the bench was, or whatever. This is important because, in the 1990s, well, in the 1980s and 1990s, there were a number of mass movements that were very disturbing that psychologists then began to study. There was the Satanic panic of the 1980s, in which there was a belief that there was a whole movement of Satanic cults in American cities. It's like, oh my god, what's going on here?
- 5.66 Then in the 1990s, there was something called the recovered memory movement. Are people really recovering memories from the past of traumatic events that happened? Then there was the McMartin preschool case. OK, let me just go over some of these, and why it's important. Because a lot of these involve crimes. The most famous example here of memory distortion and eyewitness testimony is this couple who I met. They're not a couple. They just know each other through a traumatic event that happened, Ronald Cotton and Jennifer Thompson.
- 5.67 When Jennifer Thompson was in college, she was raped, violently raped, and reported to the police, who gave a lineup, that that's the guy who did it, Ronald Cotton. He was tried and convicted and sent away to prison for 20 years. Ten years into his sentence or so, he meets somebody else in prison who says he did it. So he asked for a second trial. He gets another trial, and he's still convicted. By now, fast forward into the mid '90s when the OJ trial was going on, and everybody was introduced to the idea of DNA testing. In rape cases, you have samples, and then you can now test the DNA to see if this is the person that's standing there in the court.
- 5.68 Again, signal detection. DNA testing is a way to accept or reject whether somebody is guilty or not. So, Ronald Cotton managed to get, through the Innocence Project, his DNA tested, and they still had a sample from the rape, of a bodily fluid that they could test. It wasn't him. It was this other guy. You just have to watch this story. When they gave a talk at this conference we were together at, and every person in the room was crying. It was just unbelievable how they met, and she's just sobbing that she put this poor guy away, 11 years in prison. He didn't do it.
- 5.69 The problem is, is that, the moment she decided that was the guy—you get the possible suspects there, you have to pick one out of a lineup—we're not very good at that. You have in your image, this is what I think he looked like. It's not reliable. There's no Cartesian theater in there and you go, oh, yeah, there he is. I could see it in my—it doesn't work that way. Elizabeth and other cognitive psychologists, memory experts, began to realize that this is not how the memory works. We have to study this. This really goes wrong.

- 5.70** We did a whole Skeptic issue on recovered memory movement. Let me tell you a little bit more details about that. This was in the 1990s. There were a series of incidences where, these are mostly adults, they were all adults, who were going to psychotherapy for various psychological issues they had, anxiety or depression, sleep problems, weight problems, whatever, variety of—they're going to get help. A handful of therapists decided— it's kind of a Freudian model—that traumatic events that happen in your early childhood can be repressed. They're in there somewhere, again, on the screen somewhere. If you could just pull them up, you could see what happened.
- 5.71** It's hard to do. You have to do it in these therapy sessions where the therapist asks you leading questions. I'll just shorten it here, but it looks something like this. Could it be that the problem you're currently experiencing is a result of some childhood trauma? I don't think so. Well, a lot of people that show your symptoms report having been, say, molested as children. Well, no, I wasn't molested. I know, because you might be repressing it. You can't remember it, but it could have happened. It might be in there.
- 5.72** Here's a book about this recovered memory stuff. Go back and think about it, and then they come back in another session. Do you think this happened? What color was the hat? OK, here's the hat. Were you lost in a mall? Yes, I was lost in a mall. A number of these subjects came back after a few months and said, "Yes, you know what? I think I was molested by my father as a child." OK. If it stopped there, that would be one thing. But by the late '90s, there were people being tried and convicted and imprisoned for molesting their children based on nothing more than one of these recovered memories.
- 5.73** In other words, there was no corroborative statements or accounts from family members, from the mother, from siblings. It's, oh, yeah, I know this happened. Because, of course, this does happen. It's, again, one of these signal detection problems. The molestation of children is real. This person on the docket that's being tried, are they guilty or not? One of the things that Elizabeth Loftus and other psychologists began to study was, these are not reliable. We published this article by a woman named Laura Pasley, who went through this. She went all the way to convicting her father of molesting her. Then she realized it never happened, and that the whole thing was a false memory planted by the therapist.
- 5.74** OK, so by the early 2000s, this is how this particular mad social movement came to an end. A lot of these now broken-up families began to sue the therapists for malpractice, and that put an end to the whole thing. There was no more recovered memories. It's flawed from the beginning because that's not how memory works. It's not like, again, there's a screen up there and you can play it back and go, oh, yes, there it is. I can see my father molesting me. It's a terrible thing. It isn't like that at all. These people were just making this stuff up. I don't blame them. It's the therapists that are planting those ideas.

- 5.75** I mentioned the Satanic panics of the 1980s. This was the old blood libel from the Middle Ages. Here, these Jews are sacrificing this Christian child for the blood libel. That thing carried all the way through into the 20th century. I mean, the whole Nazi thing with anti-Semitism, much of it was part of the blood libel and the stab in the back, the Jews did it to us and so on. This got all the way up into this belief that there were these Satanic cults around America, until the FBI finally looked into it and said, we're reading these stories. It's on the news every night. There was a mutilated cat or a butchered dog or some weird thing that happened in this town. Oh, it's a Satanic cult.
- 5.76** The FBI said, OK, we're going to get to the bottom of this. There must be hundreds of thousands of these. There were zero. It was all a panic, a moral panic. This led to, before the OJ trial, the longest court trial in the history of California, the McMartin preschool case, that involved this family, the McMartin family, who ran this little preschool in Manhattan Beach. Now, you have to picture this. It's just a little tiny converted house that's now a little preschool, with a couple dozen kids running around. In Manhattan Beach, which is packed, it's just solid. There's people everywhere, roads, buildings, offices, restaurants, and so on.
- 5.77** Somehow for years, the family was supposedly running this Satanic cult out of the basement of the McMartin preschool, in which they were sacrificing children, drinking their blood, and sacrificing animals. Then off of the coast of California, there's Catalina Island, it's about 20 miles out. People go out there for vacation. Allegedly, they were taking the children out to Catalina during the day and had these secret tunnels underneath the school. This went on and on and on.
- 5.78** Where did they get this information? From the children. Not right away, but they used these bizarre interview techniques, where they would say, OK, these poor children, they've suppressed the memory of these horrible things. Here's a little anatomically correct doll. It looked like this. The children are in the room with the police, without their parents. The kids are scared to death. OK, here's the doll. Show me where he touched you. These poor little kids are like, I just want my mommy. OK, he touched me there. Can I go now? Aha. They found evidence that all this was going on.
- 5.79** Until finally somebody said, wait a minute. Secret tunnels underneath the school? OK, where are these? They're in there. They went out to Catalina, 20 miles out there and back in a day, and no one noticed? Then this went on and on and on. They finally, the whole thing just ended up as a trial bust because there was no evidence for this. This went on for years, based on this idea of fake memories.
- 5.80** OK, final two examples of how thinking goes wrong in this category. Also in the 1990s, there was a movement called facilitated communication. OK. This was with severe autistic children who cannot read or write or do math or anything. All of a sudden, there was a movement at a school of education, at a university back east, in which the person running this said, I've developed a technique in which these autistic children can communicate to the rest of us. It turns out, they're completely normal. They're just as

smart and rational as everybody else. They just can't communicate. They don't have the verbal skills.

- 5.81** But if you use this technique, what was the technique? Facilitated communication. Here's how it works. You have a keypad or a sheet with numbers or an alphabetic, like, a QWERTY keyboard or just an ABC keyboard on the right there. This is all from Skeptic magazine when we covered this. The autistic child is sitting here. The facilitator, who's helping this happen, holds the hand of the autistic child, forms it like here's a finger, holds it above the keyboard, and then asks the question. OK, tell me what three plus two is. Five. Aha.
- 5.82** Now the question is, is the child moving the hand? Well, I have to hold the hand because the autistic children, they do this self-stimulating when they move around a lot. So I got to hold it stable and then, boom. So this went on for, again, years. And all of a sudden, there were these parents of these autistic children who are, of course, heartbroken that their little child is developmentally disabled and so far behind their peers and their benchmarks in school and so on. What can we do?
- 5.83** Oh, it turns out, if I hire this person, get the therapist in there with the facilitator, my little Johnny, my little Susie is just as smart as the other kids. Look, they're doing math. They're reading Shakespeare. They're answering questions about novels and so on. This went on for—it was astonishing. Then finally somebody said, let's actually test it. Again, this is so simple. This is like the golf ball finder. This is like the Emily experiment. It's so simple. You just have the facilitator sitting there ready to go, and then you have a divider between the therapist and the child. And you show the same image to both of them, and you see what gets typed, dog or table or whatever.
- 5.84** Then you change it. So the facilitator sees one thing and the child sees something completely different. Guess what got typed? Every time, without exception, whatever it is the facilitator saw and never what the child saw. It was very tragic. Again, this ended up in lawsuits. The whole thing was a bust, that these parents got their hopes up. You just can't believe how far this movement went. There were these autistic children, this went on for so many years, they went to college. They got college degrees with the facilitator in the classroom doing all the note-taking and so on. The kid has no idea what they're even doing.
- 5.85** Then finally, it's an example of this. Back to where we were with the golf ball finder, have you ever played the Ouija board? It's a Satanic thing, supposedly. You just have the little planchette that you put on there. It's just an A through Z, and then one through nine and zero. Hello, goodbye, yes or no, and so on. You just put the planchette there. There's a famous Rockwell painting of this, of the Ouija board. Of course, the question is, who's doing the moving of the planchette? I'm not moving it. I'm not moving. You sit knee-to-knee, you put the board on there.

- 5.86** I remember I played this with my little sister when she was seven or eight. It was really moving. We were asking crazy questions like, who's moving the board? And it types out Tessa. Who's Tessa? Oh, the spirit guide from the Middle Ages. This went on and on. It was hilarious. She said, I'm not moving it. I said, I'm not moving it. But of course, it's the ideomotor response. It's like the golf ball finder. You are moving it. You're just pushing it ever so slightly that you're not even aware of it, and that's how that works.
- 5.87** So, again, the facilitators and facilitated communication could be completely honest people just trying to help. They just don't know that they're doing it. I'm moving it, not the child, and so on, because they don't perceive that. A little bit like the water dowsers. When these are tested, they claim, again, they remember the hits. Oh, I did it this time and I found water, and I did that time. And these farmers hire me and they pay me to tell them where to drill for the water well and so on, and it always works.
- 5.88** OK, now we're going to test it. Here's a field. Here's a bunch of opaque boxes like the golf ball thing. Some of them have jugs of water in them, some of them have nothing in them. Tell us which ones have water in them. Guess how well they do? No better than random guessing. They also, well, it normally works. I don't know what's going. It must be the vibes from the power lines or whatever. It's back to the old golf ball finder. It's the same principle.
- 5.89** OK. Let's look at some, back to probabilities and statistics and problems with figuring out what's likely to be true. This is one of the most famous examples of this statistical probability innumeracy called the Monty Hall problem. Back in the 1970s, '60s, '70s, '80s, I think the show has even been revised. A show called Let's Make a Deal. It looks like this. Monty Hall is the host of the show at the time, so it's called a Monty Hall problem. Monty has behind him on the stage three doors, with an audience like this.
- 5.90** And so I get a volunteer from the audience and I tell everybody, including my volunteer, "Behind one of the doors is a brand new car. Behind the other two doors are goats. Pick a door." Here's what it looks like. There's three doors, behind one of which is one car and behind two of which are goats. Now, Monty knows what's behind all three doors. Monty is not going to reveal what's behind the door you just picked. Let's say you pick door number one. He's going to then reveal one of the other doors you didn't pick, and he's only going to show you a goat. He's obviously not going to show you the car.
- 5.91** Let's say you chose door number one, Monty shows you what's behind door number two. It's a goat. Here was the question. Behind door number 1, you don't know. He's not going to show you that. Behind door number 2, it's a goat. Behind door number three, who knows? Do you want to keep your choice or switch? That's it. Now, this has been done experimentally. Almost everybody stays. The reason is, is they reason that they went from three doors and a one out of three chance to getting it right, to two doors, and now it's a 50/50 choice. That's our intuition, and it's wrong.

- 5.92** OK, this is a funny story because you remember Parade magazine? It's the highest publication subscription service magazine in the world. It was in every newspaper until recently, everything went digital. The Parade magazine is read by tens of millions of people. In there, they have a weekly column by a woman named Marilyn vos Savant. She is the world's smartest person. She scored over 200 on IQ tests. She wrote about the Monty Hall problem, and she said, "Our intuitions tell us it's now 50/50, so it doesn't matter if you stay or switch, but that intuition is wrong. Here's why."
- 5.93** She got a heap of hate mail. Over 10,000 letters were sent to Parade magazine, over a thousand of them from PhD mathematicians and scientists and so on. I'll read a couple of these to you. This is, "Dear Marilyn, you blew it, and you blew it big. Since you seem to have difficulty grasping the basic principle at work here, I'll explain. After the host reveals a goat, you now have a one in two chance of being correct. Whether you change your selection or not, the odds are the same. There's enough mathematical illiteracy in this country that we don't need the world's highest IQ propagating more. Shame!" writes Scott Smith, PhD from University of Florida.
- 5.94** Or how about this one, from W. Robert Smith, PhD from Georgia State. "I am sure you will receive many letters on this topic from high school and college students. Perhaps you should keep a few addresses for help for future columns." And then, "Maybe women look at math problems differently than men," says Don Edwards from Oregon. OK. I'm afraid the world's smartest person was right, and the mansplainers were all wrong, PhDs aside. Here's why.
- 5.95** OK, at the start, you had a one-third chance of picking the car and a two-thirds chance of picking the goat. Switching doors is bad only if you first chose the car, which happens only one-third of the time. Switching doors is good if you first chose a goat, which happens two-thirds of the time. Thus, the probability of winning by switching is two-thirds or double the odds of not switching.
- 5.96** If that still seems like a mouthful, let's change it to, now there's 10 doors, behind one of which is a car, behind none of which are goats. Again, Monty knows what's behind all 10 doors. You choose door number one. Monty reveals what's behind doors number 2, 3, 4, 5, 6, 7, 8, 9. They're all goats. OK, so there's our choice. You chose door number one. If Monty then shows you the rest as goats, would you switch? Of course you'd switch, right? Or you can make it 1,000 doors. And Monty shows you 999 of them. Of course, you'd switch.
- 5.97** Now, thankfully, because of the Internet, you can actually go on and play the Monty Hall problem and try. If you just go stay, stay, stay, stay, you'll get this many right. If you go switch, switch, switch, switch, you'll get far more right. Even with that, I wrote about one of my Scientific American columns on this, about why people—I said, "By now everybody knows you should switch because the Monty Hall problem has been discussed so often." And I explained why it is that people get—I wasn't interested in the Monty Hall problem. I was interested in writing about why we get it wrong, because we

didn't evolve to think probabilistically and so on. We misunderstand randomness, statistics, and whatnot.

- 5.98** Anyway, then I got a heap of mail to Scientific American. Shermer should be fired. He's so illiterate and innumerate and so on. I had to write a post explaining, go online to this site and just try it. Anyway, it's kind of funny. But this helps us explain a lot of other weird, paranormal experiences like death dreams. I get these letters and inquiries. How do you explain death dreams? I woke up at 3 in the morning and I had this horrible dream about my grandmother died and so on. Then I woke up feeling sad. Then my mom called and said, bad news, grandma died. Wow. Last night? Yeah, about the time I had this dream. How do you explain that?
- 5.99** This is our signal detection problem. OK, what about all the times you had dreams, bad things, and nothing happened? Or the time bad things happened, and you had no dreams about that? Again, back-of-the-envelope calculations, let's just say we have about five dreams a night. I'd say day but night. That's 1,825 dreams a year. Let's say you only remember about one out of 10 of your dreams. That's 182 and a half remembered dreams a year. When I did this calculation, there were 295 million Americans, so that results in 53.8 billion remembered dreams a year.
- 5.100** Sociologists and anthropologists tell us that each of us knows about 150 people relatively, fairly well. That's a network grid of 44.3 billion personal relationships. Then I took the annual US death rate, all causes, all ages, it's 0.008. That's 2.6 million people die every year in the United States. Here we go. It's inevitable that some of those 53.8 billion remembered dreams will be about some of these 2.6 million deaths among the 295 million Americans and their 44.3 billion relationships. In fact, it would be a miracle if some death premonition dreams did not come true.
- 5.101** Here's a show you will never see. Next on Ellen or Oprah. "We have somebody who's had these incredible dreams about horrible things, and so far, not one of them has come true. But stay tuned because one of them eventually—" You'll never see that. Again, the survivorship bias, the selection bias. We're only interested in focusing on the ones that happened, the hits. You can think about that with miracles, which we're going to discuss in a later lecture. But back-of-the-envelope calculations, the information is pouring in about one per second. There's 30,000 seconds in an eight-hour day. That's about a million events you witness or experience every month. Most of them are just totally insignificant. When you're driving here today, the car pulls left in front of us, whatever. This stuff happens, most of it is insignificant.
- 5.102** But we would expect, by these calculations, a million-to-one odds, about a miracle to happen on average once a month. You got a million things a month that happen. Well, one of them is going to stand out as something really weird. With the confirmation bias, we remember the unusual and forget the usual. Again, this problem of, we don't understand probabilities, even the simple coin flip model we've talked about.

- 5.103** Here's another experiment. You ask subjects to imagine flipping a coin in your head. Just go ahead, pretend it's random. Most people go heads, tails, heads, tails, heads, heads, tails, tails, or whatever. It looks something like this, where humans trying to emulate random sequences will almost never place more than four heads or tails in a row. In true randomness, here's what it looks like. There's lots of streaks— 6, 7, 8 heads in a row, 6, 7, 8 tails in a row, and so on. In fact, a string of at least five or more identical outcomes is 0.999. It's almost inevitable that will happen.
- 5.104** But people underestimate, especially gamblers, these streaks. So if you're at the roulette table, you think, well, I've gotten five heads in a row, five reds in a row, and then they make two errors. Well, black is due to come up, as if the machine is keeping track. I've had five of these in a row. I'd better switch over because this guy's going to— it doesn't work that way, or the opposite. Red is hot. I think I'll bet on that again. Neither of these is true because the roulette wheels don't have a memory. But our intuition's about this. It's easy to spot which is the human-generated one. It's the one that looks less random.
- 5.105** OK, a couple other examples that are really weird. My friend Martin Gardner, the late great Martin Gardner, one of the founders of the modern skeptical movement, he wrote the Scientific American column before I did for 25 years, wrote a lot of books about why we should be more literate, statistically and probability, but he tells an incredible story about this book. One day, Martin was perusing a used bookstore when he came across a familiar title. His father had given him a copy of that book when he was a child, but he lost it during many a move, many years later. As he considered purchasing his replacement copy, Gardner flipped through the pages and found his own handwriting in the margins. It was the very book that he had lost. That's pretty weird.
- 5.106** And even weirder is Anthony Hopkins' book story. This is, of course, the famous actor who was set to star in the film adaptation of George Feifer's novel, *The Girl from Petrovka*. Hopkins went to several booksellers to purchase a copy, but none had it in stock. Heading for home at the Leicester Square tube in London, he chanced upon a book on a bench. It was *The Girl From Petrovka*. That's not the weird part. It gets even weirder. Two years later, when Hopkins met Feifer and mentioned the coincidence to him, the author noted that he had lent his last copy to a friend with handwritten edits in it for the American edition, changing labour with the U to the American spelling of no U or L-A-B-O-R, for example. Guess what happened? When Hopkins flipped open his copy, he saw Feifer's hand annotations. It was the very same book.
- 5.107** OK, so, these are these statistically highly improbable events. How do we explain those? OK, to bring this to a close. There's several laws of probability at work here that distort our rational thinking about these. The law of inevitability, if you make a complete list of all possible outcomes, then one of them has to occur. Here's everything that could happen today; one of them is going to happen. Or the law of truly large

numbers. With a large enough number of opportunities, any outrageous thing is likely to happen. Again, we noticed the unusual things, forget the usual things.

- 5.108** Here's another back-of-the-envelope calculation I did. According to Google books, there are about 130 million book titles in the world. If the average print run was around 5,000 copies each, that total is 650 billion books. With a world literacy rate of 84.1%, that means around 5.9 billion people are reading some of those 650 billion books. It would be miraculous if somewhere, someone, sometime, did not find a surprising connection with some of those books. Then, of course, the most unusual ones pop out. The rest, there's nothing to it.
- 5.109** OK, final weird statistical thing is something called regression to the mean. OK, so we have a mean, a measure of whatever it is we're interested in studying. Let's say height, back to our example of height. Turns out that for two tall parents, of course, their children are going to be tall, like them. But for every four children they have, three will be shorter than their average for every one that's taller than their average. This would be true for two short parents. Of course, their children are going to be shorter. But for every four children they have, three of them are going to be taller than the average of the two parents, for every one that's shorter from the average of the two parents.
- 5.110** Same thing with IQ. Two really smart parents, of course, are going to have smart children. But for every four children they have, three of them will be under their IQ average than for everyone that's smarter than the parents. Hollywood films, producers and directors get fired after one bad film, even though their films under production were a success. We have this average, and then there's variations around the mean. Sometimes the average is above the mean, the spike, the data is above the mean, sometimes it's below the mean. It's hard to tell what's going on without keeping in mind this regression to the mean.
- 5.111** It's like the famous Sports Illustrated cover curse. That is, if you're on the cover of Sports Illustrated as an athlete, you're going to have a bad season or a bad week or bad whatever next time, and people track this. But what does it take to get on the cover of Sports Illustrated? You have to have had the best game, the best week, the best month, the best year you've ever had. Are you going to repeat that? No, there's going to be a regression back to your mean, which is going to be lower than your best ever.
- 5.112** This is like golfers. They remember the best game they ever played, and they almost never play that well. They always think, damn, I usually make that shot. Actually, no, you usually don't make that shot. You're remembering the one time you did make this shot. The person that really figured this out is a guy named Francis Galton, cousin of Charles Darwin. He developed what's called the Galton board, which is to drop a bunch of beads around these little pegs, and you'll see how they fall out into a bell curve, a statistical bell curve. You can see that, for every one of these pegs, the little bead has to go left or right, left or right, left or right.

- 5.113** It's inevitable you're going to get something like a bell-shaped curve. Because to get to either extremes, let's say, height, super short, 4.9, super tall, 7.2, each bead has to go—when it has a 50/50 probability, left or right, it has to go right, right, right, right, all the way down to 7.2 or left, left, left, left, all the way down to 4.9. The chances of that happening are very slim, so most of us are in the middle. You can use that for anything you want.
- 5.114** The best tennis players or the best baseball player, whatever sports' performance, there's a reason why it's called six sigma or six standard deviations out from the mean to describe someone like Novak Djokovic, greatest tennis player of all time. He had to go through every one of those choices all the way out here. How many people are going to be at that level? Three others, two others? According to him, none others. He's the GOAT. And so on. So much of life falls into this statistical probabilities of how things happen.
- 5.115** OK, all of this is clouded by the fact that we have confirmation bias, where we look for the hits and forget the misses, seek and find confirming evidence, and ignore disconfirming evidence. Again, this is part of what's called motivated reasoning. The problem is that we fail to try to disconfirm a hypothesis. Again, always ask yourself, what would it take to change my mind? Almost none of us do that.
- 5.116** All right, my last fun example here comes from when I was in ninth grade, 1969. There was a story that Paul McCartney was dead. Oh, my gosh, the Beatles are over. No, they found this replacement for him, and they're giving us clues that he's still alive. I mean, he's dead, but the replacement. In the cover of Sergeant Pepper's Lonely Hearts Club Band album, they have the original Beatles, and they have the new Beatles. There, above Paul McCartney's head, is this guy holding a palm, supposed to be a sign of death, telling us that he's really dead.
- 5.117** Everybody was going through the album covers. We all looked at the cover of Abbey Road, and it looks like a funereal procession. And poor Paul, he's barefoot, and he's got a cigarette. That's a sign. He's probably the corpse. He's really dead, and the rest are dressed like they're part of the funereal. If you look at this Volkswagen up close, first of all, there's a police car there. Oh, there must have been something. And the license plate of that little Volkswagen is 28IF, IF. Paul would have been 28 if he hadn't died. And then, and this goes on and on like this, the back of Abbey Road. If you look here, it looks like a three. There's three Beatles. They're giving us a clue. And in the shadow here, it looks like a skull. They're giving us a clue that Paul is dead.
- 5.118** Anyway, as a teen, I'm thinking, oh, this is unbelievable. We played the album, some of the records backwards, Revolution Number 9. Play it backwards, it says, "Turn me on, dead man. Turn me on, dead man." Now there's a book about all this stuff called Turn Me On, Dead Man. It's really funny. About all the little—of course, it's the confirmation bias. You're just looking for and finding all the weird anomalies. We call this anomaly hunting. Instead of all the stuff where nothing happens.

- 5.119** Final example, the Bible Code. There was a claim that, by this author Michael Drosnin, of this book called *The Bible Code*, that if you took the original Bible of Hebrew letters and you strung them all out into a huge long string in the computer, and then you did a search selection by some random number of what the block would be—I think this was like, how many letters across that you want to put the string of type into a single block—and then search for connections. The letters spell out certain things when they're close to each other.
- 5.120** In this case, he's claiming that it says, twin towers, airplane, terror, strikes, will collapse, will burn. He's finding these letters. Again, the claim is that this 4,000-year-old document predicted 9/11. OK, is that true? Well, anyway, I wrote one of my *Scientific American* columns showing why this is total claptrap. It's just seeking and finding connections and things totally randomly. Again, call that patternicity. You can spell out almost anything you want, President Kennedy to die in Dallas.
- 5.121** But all that depends totally on the random decision of how many letter strings you want to go across for the block of type. You change that by one letter, and all of a sudden, these are no longer connected. They all shift a little bit. So this is totally random. Anyway, the funny part is not that part, but this guy, Michael Drosnin, was not happy with my column. You libeled me or whatever. It's like, well, write us a letter and explain why I'm wrong. And he goes, oh, never mind.
- 5.122** But I did get a letter from a comic book illustrator, John Byrne, who wrote about his experience finding weird connections in his long years of writing and illustrating comic books. Here's what he wrote me. I'll end with this. "Reading Drosnin's response," he did send us a response, "to Shermer's column on the Bible "code" and its ability to accurately predict the future, I could not help but laugh. I've been a writer and illustrator of comic books for the past 30 years, and in that time, I've predicted the future so many times in my work, my colleagues have actually taken the referring to it as the Byrne Curse."
- 5.123** It began in the late '70s while working on a Spider-Man series titled *Marvel Team-Up*. I did a story about a blackout in New York. There was a blackout the month the issue went on sale, six months after I drew it. While working on *Uncanny X-Men*, I hit Japan with a major earthquake, and again, the real thing happened the month the issue hit the stands. Now, these things are fairly easy to predict, but consider these. When working on the relaunch of Superman, for DC Comics, I had the Man of Steel fly to the rescue when disaster beset the NASA space shuttle.
- 5.124** The Challenger tragedy happened almost immediately thereafter, with time, fortunately, for the issue in question to be redrawn, substituting a space plane for the shuttle. Most recent, and chilling, came when I was writing and drawing *Wonder Woman*, and I did a story in which the title character was killed as a prelude to her becoming a goddess. The cover for that issue was done as a newspaper front page with the headline *Princess Diana Dies*. Diana is *Wonder Woman*'s real name. That issue went

on sale on a Thursday. The following Saturday, I don't have to tell you, do I?" Of course, that was when Princess Diana herself was killed in a car accident.

5.125 And John Byrne writes in conclusion, "My ability as a prognosticator, like Drosnin's, would seem assured, provided, of course, we reference only the above and skip over the hundreds of other comic books I have produced which featured all manner of catastrophes, large and small, which did not come to pass." That's a good point to end it. Thank you.

Lecture 6: Why We Believe Conspiracies

At a glance — Why we believe conspiracies.

- 6.1** All righty, thank you, everybody, for coming back. This is lecture six of our eight-lecture series on Skepticism 101: How to Think Like a Scientist. I thought, in the last three lectures, I'd do specific topics of applying then our tools of reason and science to understanding what's true or not true in some probabilistic way. This is on conspiracies and conspiracy theories. You may have noticed that these have been in the news the last couple of years. Actually, they've always been in the news.
- 6.2** Just to take the example of when President Barack Obama was President, he was a busy man during his eight years in office. According to conspiracy theorists, he concocted a fake birth certificate to hide his true identification, where he was really born. He created death panels for his health care plan. Who would live and who would die? They would make the decisions for you. He masterminded Syrian gas attacks as a pretext to going to war against Syria, which for some reason we never did. He blew up the Deepwater Horizon offshore drilling rig, so he could implement his environmental policies.
- 6.3** He built concentration camps for Americans who resist his control over the American public, like FEMA camps in Texas, to put people that own guns and who would refuse, when the Second Amendment was banned, to give up their guns. They were going to put them in these FEMA camps. Turns out, when you look at the photographs of these so-called FEMA camps, they turn out to be cattle ranches, or what look like prison camps, but in North Korea, not Texas. He also attends a Christian church. Why? To hide his secret Islamic faith.
- 6.4** Well, anyway, I wrote a whole book about this problem, which I called The Conspiracy. That is, why the rational believe the irrational. If you look at the cover, by the way, not only does it spell out conspiracy, you can get CIA. CIA is hidden in there. And Soros, S-O-R-O-S, George Soros. This is an example of connecting the dots that are not really connected. What I really wanted to do in this lecture, based on what I did in the book, is explain, why do people believe conspiracy theories? I mean, you hear these and go, how can anybody believe that? It's just astonishing.
- 6.5** I gave you an example in a previous lecture of Pizzagate, that Hillary and these Democrats are running this child pedophile ring out of a pizzeria, Comet Ping-Pong pizzeria in Washington, DC. Does anyone really believe that? Well, I mentioned Edgar Welch did. He went there with his gun to break up the crime. Most people, when they say they believe it, do they really believe it or are they just doing something else? I'm going to argue it's a deeper kind of rationality behind it. They don't really believe that specific one. It's a representative of something else.

- 6.6** But before we get into that deep dive, I start off in the book defining what a conspiracy is and a conspiracy theory and a conspiracy theorist or conspiracist. The reason I do this is because, over the decades of watching commentary and reading books and articles about conspiracies, people are just hopelessly confused about these words. What do they mean? Oh, that's just a crazy conspiracy. Or that's one of those nutty conspiracy theories. What are you talking about?
- 6.7** There are conspiracies, if you define it carefully. There's nothing wrong with having a conspiracy theory because there really are conspiracies, so it's good to know if they're true or not. I'm trying to, by using the definitions I'm about to give you, destigmatize the concept of conspiracy and conspiracy theory, and just talk about it rationally. It's a signal detection problem. Here's a conspiracy theory. All we want to know is, is it true or not? What's the likelihood? Maybe it is, maybe it isn't. What's the evidence? And so on. Just treat it dispassionately as a problem to be solved.
- 6.8** Therefore, conspiracy theorists or conspiracists, they're not tinfoil hat-wearing weirdos in the basement of their parents' house, concocting these crazy ideas. Most conspiracy theorists, as we'll see, are totally rational people. You and I are all conspiracy theorists about some things as I'll show you. I'm trying to make this a rational problem to be solved, not some issue to moralize about.
- 6.9** What is a conspiracy? Well, it's two or more people plotting or acting in secret to gain an advantage or harm others, either immorally or illegally. That's about as simple as it gets. Most people that write about or research conspiracy theories, yeah, that's pretty much what we're talking about. It's not me by myself plotting to do something against you. When I'm involved with other people, that makes it a conspiracy.
- 6.10** A conspiracy theory then is just a structured belief about a conspiracy. I think this is what's going on. Whether it's real or not, that's a separate question. That's the signal detection problem. I think it's true. Is it true or is it false or can't tell? Whatever. Then a conspiracy theorist or conspiracist is somebody who holds a conspiracy theory. Again, whether it's about a possible conspiracy, again, whether it's true or not, that's a separate question. So again, nothing wrong with being a conspiracy theorist or a conspiracist.
- 6.11** I'm a conspiracist about a lot of things that I suspect are going on. You are, too, as I'll show you with tons of examples in this lecture. It's not irrational to believe conspiracy theories. Just think back on that single detection problem. You make one error or another error; different errors have different costs; and assuming there's a conspiracy afoot when there isn't, is a less costly mistake than assuming there isn't a conspiracy when there is. That's going to be my argument. OK.
- 6.12** All right, first of all, let's get the lay of the land. How many types of conspiracy theories are there? Well, there's so many that they have to be classified. I have a whole bookcase in my office of just conspiracy books that I've collected over the decades. It's

astonishing how much interest there is in this topic. There's aviation conspiracy theories, the black helicopters, and the chemtrails. You see these jets with the trails, vapor, behind them. What is that? Well, we know what it is, but a lot of people think it's chemicals being dispersed out of these planes by big pharma or big industry, either to cheat the system to save money or to control the American population and dumb us down, make Americans even dumber. It appears to be working. No, just kidding.

- 6.13** Black helicopters are these, this mysterious, it's probably just a military helicopter. But anti-Semitic conspiracy theories, like the famed protocols of the learned elders of Zion, this hoaxed document in the early 20th century out of Russia, that then got perpetrated by Hitler and the Nazis as the alleged secret documents of what the Jews were doing to take over the world. Unfortunately, even just since I wrote this lecture and my book, there's been this spike in anti-Semitism in America and the West. It's quite disturbing because of the Hamas attacks on Israel. And already, there's lots of conspiracy theories about what's really going on there. It's everywhere. It's in the news.
- 6.14** Alien conspiracy theories. Again, I mentioned last lecture, UFOs, UAPs, alien abductions, Area 51, Roswell, Biblical conspiracy theories, lots of them, but the most famous one, made famous by Dan Brown in his novels, that Jesus was actually not killed on the cross. He just went into a coma. His disciples took him off and put him in this tomb. He came back alive. They come back the next three days later, the tomb is empty. Well, they actually whisked him off to France, and he married Mary Magdalene, and they had children, and their descendants are still with us today.
- 6.15** If you want to go full Dan Brown, there's actually a non-fiction book about this. It was called Holy Blood, Holy Grail. That was a big bestseller in the 1970s. They ended up suing Dan Brown because they said he stole our ideas. It's like, yeah, but I'm a novelist. I'm just making stuff up. They lost their case. But that's an actual Biblical conspiracy theory. Then there's conspiracy theories about mysterious deaths, supposedly mysterious deaths. Who really killed JFK, RFK, MLK, Jimmy Hoffa, Princess Diana, Kurt Cobain? You've seen RFK Jr. on the campaign trail. He talks about constantly how he thinks the CIA murdered his father, RFK, and his uncle, JFK. OK, is he right or not? I think he's not, but we'll see why.
- 6.16** There are government conspiracies, of course, false flag operations, 9/11 was an inside job. These FEMA camps are not really federal emergency places to put people that need help; rather, they're camps to put Americans away. I mentioned that last time. Medicine-related conspiracy theories. Fluoridated water. In the 1950s, the American government decided, through the American Dental Association, that putting fluoride in water will reduce the number of cavities that children get. But all kinds of conspiracy theories erupted over that. Same thing with vaccinations, cancer quackery, there's a lot of conspiracy theories about this. The medical establishment wants to keep us ill so they can make money treating us and so forth.

- 6.17** Secret organizations are in this conceptual world of conspiracies. The Federal Reserve, the World Bank, the Bilderberg Group, the International Monetary Fund, the Rothschilds and the Rockefellers, the Trilateral Commission of the United Nations, and the European Union—what are they up to? You see people like Alex Jones and his camera crew outside the building, screaming at them, "You're not going to take over the world," and so on. There's a lot of people that are suspected, these groups like the Knights Templar, the Freemasons, the Illuminati, the New World Order, the Council on Foreign Relations, and especially the Zionist Occupation Government, ZOG. These are mostly far-right extremist groups who think that the Jews are secretly running America. They've thought this since the 1930s.
- 6.18** All right, so there's plenty of topics to go around there, that we have to classify them. I first make a cut at making a distinction between realistic conspiracy theories and paranoid conspiracy theories. Realistic conspiracy theories are like government agencies or corporate entities consolidating power, or personal profiteering for one's group, one's company, one's family, whatever. These are the sorts of things that go on all the time. Tax-cheating schemes. Does a week go by that we don't hear about some group doing something to save money, cut taxes, and so on?
- 6.19** Regulatory capture, one of the things that RFKJ talks about a lot, that he's not wrong, about is on the opioid crisis. How is it that the regulators didn't stop this from happening? The answer is, essentially, they just bought off the regulatory state. I'm talking about Purdue Pharmaceutical and the whole Sackler family that really popularized the use of these opioids as painkillers. How did this happen? The answer is, they captured the regulatory state and bought it off. That's a conspiracy. RFK Jr, he says a lot of things that are just way out there, but a lot of them are not way out there. It's like, yeah, that's right. They did do that. OK, so that's not irrational.
- 6.20** Insider trading. Again, does a week go by we don't read about these things? Corporate malfeasance. Intelligence agencies spying on US citizens. Not only US citizens, foreign citizens. Remember, in WikiLeaks, when Ed Snowden revealed through his whistleblowing activities that the NSA was spying on Angela Merkel, the Chancellor of Germany, basically monitoring her cell phone? She was not happy about this. You're not supposed to do this. Our government was doing that.
- 6.21** Of course, most recently, the Twitter files. After Elon bought Twitter, he opened up the files to Bari Weiss and Michael Schellenberger and a few other journalists. They went through there and found all sorts of things that the US government, the NSA, and Homeland Security, and FBI, were actively engaged in either shadow-banning or outright blocking tweets from groups or people that they didn't think they liked, for whatever they justified as homeland security or spying or whatever, including, and especially most famously, the Hunter Biden laptop story, which was starting to unfold just before the 2020 election.

- 6.22 Some people thought, well, it's Russian fake news. But other people thought, well, no, I think there's actually something to this. We're journalists. We should publish everything, both sides, so American people know about this story. And we know from the Twitter files, they actually have emails from Homeland Security, National Security Agency, so two Twitter employees saying, "We want this story blocked, or we want this guy that's promoting it, like the New York Post who first published the Hunter Biden laptop story, those tweets that promote the New York Post story should be shadow-banned or blocked, so they can't be spread around and go viral."
- 6.23 This really happened. Our government agencies, before an election, really did that. Again, so when people like Alex Jones and RFKJ go off on their crazy ideas about the lizard aliens secretly running the world or whatever, that's crazy. But then they also throw this in, and you go, "Yeah, but actually, they did do that. That's right." So again, we have to be careful of what we're dismissing as a crazy conspiracy theory, because some of them are true.
- 6.24 Then there is, again, if it's too paranoid, if it's too out there, evil demonic forces are running the world, or the shapeshifting lizard aliens' world domination schemes, Bill Gates wants to control the world, or Jeff Bezos, or Elon Musk or whoever the latest, richest guy is, has a lot of power. No one can run the world's economies. If you just take an Econ 101 course, you realize actually nobody understands why economies operate the way they do. You have the left-wing economists and the right-wing economists, and the economists that have no wings at all. The middle, how can they know? Because it's a complicated system no one really truly understands, and it's a difficult problem.
- 6.25 In science, the idea that some group or one person is actually running the government, not possible. Or as an example I've given before, that QAnon exposed the Satanic pedophile ring, allegedly being run by Democrats, or that George W. Bush orchestrated 9/11 as an inside job as a pretext to going to war in Iraq and Afghanistan. And of course, most recently, the Jews are doing something bad.
- 6.26 And what's been so frustrating for me this last two weeks, since the embarrassment in Congress of those three university presidents from Penn, Harvard, and MIT, who could not seem to muster up the moral fortitude to say, "It's wrong to call for the extermination of the Jews." He just could not do it. If you mis-pronoun someone, "Whoa, OK, we're going to expel you from—but, oh, well, the Jews." This is astonishing, given the history of actual conspiracies to exterminate the Jews. It's not paranoid for Jews to worry about this. Again, what's true and not.
- 6.27 Different classificatory systems of how to think about conspiracy theories. Here's one from Jesse Walker's Book, The United States of Paranoia, four types of conspiracy theories. Enemies outside— these are foreigners coming into our country to do something bad. I'm not just talking about America. All European countries experience waves of conspiracy theories about what outsiders are going to do when they come into our country. This is going on in Germany now. My wife is from Cologne, Germany.

When Angela Merkel opened up the borders for the Syrian refugees to come in because they needed help, there was all these conspiracy theories about what those people are going to do to our country and so on. Enemies outside.

- 6.28** Then there's the enemies within, our own citizens. Not everybody, of course, but the people on the far right, the people on the far left, the Jews or whoever. They're going to do something to us from within. Then there's the enemies above, usually government agencies, or big, large, corporate, powerful agencies are going to do something bad to us from the top-down, or the people below, the enemy below. These are those, the socialists, the communists, the far-right, whatever. These citizens are going to do something bad to us. Everybody seems to concoct enemies from somewhere. It has to do with fears of a threat from enemies.
- 6.29** Who believes in conspiracy theories? Well, about a third of Americans, for example, believe the birther conspiracy theory, that Obama is a foreigner. Now, these vary. This one's old enough now, you could see the wave of supporter skepticism for it. Of course, when Obama was still campaigning, it was pretty high, and it remained high, thanks to Trump keeping the story alive throughout Obama's presidency. Then it went down after that for a while, and it comes back every once in a while.
- 6.30** The same thing with 9/11 is an inside job. By the way, about the third of America, there's more on the right, believe the first one, the birther. On the left, it's more of people on the left. I'm sorry, on the bottom one, 9/11 as an inside job. That's more supported by liberals and people on the left. That shows you we have to distinguish who's believing what. It depends on the particular conspiracy theory and what deeper issues it touches on.
- 6.31** A 2016 study, for example, found that when you ask people, "The government is concealing what they know about—," so 54% said 9/11, what they really know about what really happened in 9/11. Almost half, about what the government is covering up about JFK. 42% about what the government really knows about aliens. 32%, about a third, for there's a one-world government secretly being plotted. Again, the birther conspiracy theory, about 30%. 24% on the moon landing. My point of this data dump is that these are not trivial numbers. This isn't like 1%, some weirdos off in the basement. This is people you know, people you work with, whatever. Lots of people believe these things.
- 6.32** In one study, this was a study done by my colleagues at Chapman University, in which they looked at the relationship between the number of conspiracy theories people believe. On the horizontal axis here, no conspiracy theories, 1-3, 4-7, or 8-10. Then some other variable here that they're interested in measuring. "Belief that the world will end in my lifetime." These are both right-wing, religious, end-of-the-world scenarios, or left-wing, secular, end-of-the-world scenarios, due to whatever, climate change or AI or whatever. The more conspiracy theories people believe, the more they think that the world will end.

- 6.33** Again, with all the caveats about causation, correlation or causation, the arrow could go the other way. It could be that people that think the world is going to end are more likely to believe conspiracy theories, rather than vice versa. All we know is that they correlate; there's an association there. The more conspiracy theories people believe on the horizontal axis there, on the vertical axis, the more they distrust others and fear others are doing something to them. Then finally, the more conspiracy theories people believe, this one bothered me the most, the more guns they buy. Not will they buy a gun or not, everybody has a gun, it's how many guns do you have? Again, it could be that gun owners are more likely to believe conspiracy theories than vice versa. So it depends on exactly which way the causal arrow goes.
- 6.34** Characteristics of conspiracists. First of all, according to researchers, they cut across gender, age, race, income, political affiliation, educational level, and occupational status. It depends on the particular conspiracy. Liberals suspect the media and political parties are pawns of rich capitalists, and sometimes they are. So it's not completely irrational. Conservatives tend to believe that academics and liberal elites try to control media and politics. And guess what? Sometimes they do. So there's some truth to that.
- 6.35** GMO conspiracy theories tend to be embraced mostly by those on the left. Climate change conspiracy theories are endorsed mostly by people on the right. African Americans are more likely to believe that the CIA planted crack cocaine in inner-city neighborhoods, created AIDS, and that the Jews control the media. White Americans are more likely to believe the government is conspiring to abolish the Second Amendment.
- 6.36** This one always astonishes me. During Obama's administration, there was this constant patter about, "He's conspiring to take away our guns." The result was the biggest spike in gun ownership in the history of America. It's like, well, whatever he's doing is not working. He's having the opposite effect. Of course, these gun sellers are like, "Get your guns now before he takes them away." I got to buy a—crazy, but that's the result of conspiracy theories. They really affect people.
- 6.37** Education has an effect. It's encouraging to me as an educator to see this difference between those without a high school diploma, 42%, versus those that score high in conspiracies, versus only 22% for those with postgraduate degrees. 42% with only a high school diploma are higher in conspiracism than those with graduate degrees. But that 22%, that's one in five Americans with a postgraduate degree, believes in conspiracies. That tells us something else is going on here. It isn't just education or race or political orientation or gender. There's something else going on.
- 6.38** Why do people believe conspiracy theories? There's a number of psychologists and sociologists who work on this problem, so I summarize their research in my book. One thing is called a monological belief system. This comes from my favorite paper I read in researching my book, called *Dead and Alive: Beliefs in Contradictory Conspiracy Theories*. What the authors think is going on is this monological, one large thing, that's

underlying all the specific conspiracy theories. For example, they write, once you believe one massive, sinister conspiracy could be successfully executed in near-perfect secrecy, it suggests that many such plots are possible.

- 6.39** What they're after here is, the surveys show, if you tick the box for one conspiracy theory, whatever it is, "I think the government's hiding what it knows about aliens or 9/11" or whatever, you're more likely to tick a whole bunch of boxes. "And they're hiding what they know about 9/11, the moon landing, and fluoridation," and so on. So people that are monologically paranoid, suspicious of the government, are going to go, "Anything that's powerful is probably doing something." So you end up picking a lot of those.
- 6.40** So for example, "a belief"—they write—"that a rogue cell of MI6 was responsible for Princess Diana's death is correlated with belief that the moon landing was a hoax, and that governments are covering up the existence of aliens." These have nothing to do with each other, but somehow there's something underlying it or overarching it that pushes people into that paranoid direction. My favorite finding from this particular study is that people who think Princess Diana was murdered also believe that she faked her death. OK, they can't both be true. She can't be murdered and dead somewhere and living with Dodi Fayed or Elvis or wherever she is. It's just crazy.
- 6.41** But it's not completely crazy because it's not the specific thing that's important. It's touching something underlying it, something deeper. Then there's also this conspiracy, proportionality bias. That is, there's a dissonance created in a mismatch between the size or importance of an event and its cause. OK, so think about the Holocaust. This is one of the worst genocides in the history of humanity. What was the cause of this? The Nazis, one of the worst political regimes in the history of humanity. There's a balance there. Yeah, that makes sense. Big effect, must be a big cause.
- 6.42** But JFK and RFK and MLK, the best evidence shows they were all assassinated by lone nuts, just mentally disturbed people. But that just doesn't feel right. It's like, but JFK, leader of the free world, most powerful person, famous and articulate and handsome and so on. And some crazy, just nobody communist gunned him down? That doesn't feel right. Or you'll hear this with Princess Diana. Princess Diana's cause of death, drunk-driving, speeding, no seat belt. But come on, that's how ordinary people die. But she's a princess. There must have been something else, the MI5, the MI6, the royal family, the Arabs, the Palestinians, the Jews, whatever. Somebody had to be in on it.
- 6.43** I often hear this with 9/11. Are you telling me that you think 19 guys with boxcutters pulled this off? Come on, look what they did. The two World Trade Center buildings, the Pentagon. My god, it was huge. It completely changed history. It just doesn't feel right. There had to be some other things going on to adjust to that. That cognitive proportionality bias distorts how we understand and think about conspiracies.

- 6.44 Also, research shows that anxiety and loss of control leads to conspiracism. Subjects that are induced into feeling anxious or lacking control, we're more likely to see meaningful patterns in random stimulation. You just have this noisy-looking image. There's a bunch of them, some of which have actual objects in them, embedded in them, some of which are just random dots. People that are put in a condition of feeling anxious or out of control then are more likely to see illusory patterns in these random patterns. The theory is that finding a conspiracy theory, or believing it, is an explanatory model for the apparent chaos of the world, and it makes you feel like you're more in control. That's the fear.
- 6.45 For example, conspiracism runs higher after natural disasters like earthquakes or when people fear that they may lose their job. There's something going on behind the scenes. Somebody's doing something. Somebody made that earthquake happen. It must be that government station in Alaska that's some sort of secret DARPA project to control the weather. They're the ones that made whatever happen. That's not unusual.
- 6.46 Two other cognitive processes that I've hinted at earlier in this course—patternicity, the tendency to find meaningful patterns in random noise, and agenticity, the tendency to infuse those patterns with meaning, intention, and agency. Again, we're pattern-seeking primates. We look at a random series of dots, we connect the dots. It's super easy to do. Our brains are just wired to do that. The world is not that random, our brains think. There must be reasons behind why things happen. Again, what's the rustle in the grass? It's just the wind. That's an inanimate force. But a predator is an intentional agent. So infusing into those patterns agency, there's somebody behind that wind. He means me harm, so I'm going to be super cautious, risk averse, and careful, and so on.
- 6.47 This explains a lot of conspiracism. That is to say, you just look at random dots. This poster is one of the most interesting. You can't see what the heck is going on here. But people are connecting the dots between various things. I'll just blow it up here, and you can all of a sudden see the pharaohs of Egypt and Solomon's Temple, and Aleister Crowley, the Rockefeller Foundation, and the Pentagon, Nazi Germany, the Brave New World, and Henry Ford. And then you blow it up even more. You've got the FBI connected to COINTELPRO, and Hiroshima and Nagasaki, and all of a sudden, the base in Antarctica, and the Council on Foreign Relations, and Johnny Rockefeller.
- 6.48 If you have enough stuff, anybody can find connections between anything. We call it anomaly hunting. Anytime anything happens that seems kind of big, people go searching for something weird that they think, and they connect it to something else, and then before you know it, you have a whole conspiracy theory.
- 6.49 Then we have the innumeracy bias, that is the inability to understand randomness. Again, in the last lecture, we talked about coin flipping models. Most people underestimate the streakiness of randomness because it just doesn't make sense to us. But if you think about constellations in the sky, the stars are completely randomly

distributed from our perspective on Earth when you look at the night sky. But they don't look random. They look like, well, constellations, clusters of animals and objects and things, big dippers, little and small, and horses, and so forth. But that's what randomness looks like, and it fools us.

- 6.50** It's like cancer clusters. Finding the signal in the noise is this cancer cluster of, why are there so many people in that neighborhood getting cancer? There must be something there. Maybe there was an industrial plant nearby and they contaminated the whatever. Sometimes that does happen, but maybe it's just randomness. If I took a big, huge pile of coins and just threw them up in the air and landed them on the floor here, they would not be perfectly evenly distributed. They would be clustered. Some would be closer, some would be further away. That's what randomness looks like. But the brain can't see that. We see connections. Oh, why are they clustered over there? There must be some magnetic or whatever.
- 6.51** My favorite story of this one, Steve Jobs and Apple introduced the iPod. They added a random shuffle feature that would play your music list randomly. People complained that it wasn't random because certain songs were played more than other songs. Apple had to explain, well, no, that is randomness. They will be clustered together. Some songs played more, others are played less. So they had to reprogram the system to not be random, to make sure every song got played at least once every cycle through the playlist. But you have to program it to make that happen. You have to program out the randomness so the human brains realize, oh, now it's random. I've heard every song once. That's not random.
- 6.52** Then there's the hindsight bias. It is the tendency to reconstruct the past to fit with the present knowledge. Sometimes it's called the curse of knowledge. I already know what happened. Now I'm going to go back and look at history to see why it happened that way. This is a very dangerous thing to do in conspiracism. If you take the Japanese attack on Pearl Harbor on December 7, 1941, which, by the way, technically was a conspiracy. They didn't tell us ahead of time. They plotted in secret to do this and so on. But instead, Roosevelt, President Roosevelt, was accused of knowing that it was coming. He let it happen on purpose as an excuse to get the United States into war.
- 6.53** Why? Because of this so-called bomb plot message intercepted in October 1941 by US intelligence. Specifically, a Japanese agent in Hawaii was instructed by his superiors in Japan to monitor Pearl warship movements. In 9/11 truther circles, by the way, they have this phrase, "Let it happen on purpose, or MIHOP, made it happen on purpose." So the debate was, did Roosevelt let it happen on purpose, or did he make it happen on purpose? OK, neither is going to be the case. It's just the hindsight bias at work.
- 6.54** In fact, between May and December of 1941, 58 messages were intercepted regarding Japanese ship movements targeting the Philippines. 21 messages involved attacks on Panama. Seven messages affiliated with attacks in Southeast Asia and the Netherlands' East Indies. Seven messages connected the United States West coast. So there were so

many intercepted messages that the army intelligence stopped sending memos to the White House out of concern that there might be a breach in security.

- 6.55** Only in hindsight, after the attack on Pearl Harbor, you go, oh, look, there's the memo. Why didn't they act on that one? But that's the curse of knowledge. If you didn't know and you had this room full of intel messages—they're going to attack here, here, here, here, here— you don't know which one is going to happen. Same thing happened to George W. Bush. After this happened, it was revealed that there was a message from Condoleezza Rice, Secretary of State, Condoleezza Rice, to Bush that Bin Laden determined to strike in the United States. This was dated August 6th, 2001, less than a month before.
- 6.56** Why didn't they act on that? I'll tell you why. They knew it was going to happen, and Bush wanted to get us into war. He made it happen because he wanted to evade Iraq and Afghanistan to help his buddies at Halliburton with Dick Cheney, so they're going to make money. They're going to—oh, all this stuff is going to happen. And reading the memo in hindsight is eerie. It includes references to hijacked planes, bombing the World Trade Center, attacks on Washington, DC, and LAX. The only question for a conspiracist was, did he let it happen on purpose or did he make it happen on purpose?
- 6.57** In fact, here's my theory. It was a COWHOP. He capitalized on what happened on purpose. Oh, OK. So now we have motivation to invade Afghanistan. You know why we're over there? We might as well go into Iraq because now we have the support of the American public. That's what politicians do. The real conspiracy is that governments capitalize on what happened on purpose to do some agenda that they want. Roosevelt wanted to take America into war against Germany because England needed it, and we are allies, but he could not do it. Congress would not approve it. You had to have the approval of Congress to go to war. Pearl Harbor, done deal. OK, that's how politics really works.
- 6.58** All right, let's go back to our old friend signal detection theory problem. So conspiracy theories that are really true, in our upper left corner, and you think that they are true, that's a hit. That's always good. Conspiracy theories that are true and you think they are true, you've missed. That's a type 2 error, false negative, not good. Conspiracy theories can be bad, so it's good to identify them. Conspiracy theories that are not true, but you think that they are, or sorry, that you agree that they are false, that's a correct rejection. Or that they're false and you think that they are true, that's a false positive.
- 6.59** Again, my argument on the two right cells there, it's better to make the one error than the other. Assume the worst. Assume conspiracy theories are true just in case, even if they aren't. This is why we tend to make errors about that. What can we do about it? Well, I've developed a little model called the conspiracy detection kit. Conspiracy theory detection kit. This is modeled after Carl Sagan's baloney detection kit that he presented in his Demon-Haunted World, baloney being the nice word for BS.

- 6.60 How do you know? When I hear about a conspiracy theory, first of all, I think, again, could be true. Cromwell's rule. Sorry, let's go back. I could be wrong. You never know. Don't assign a zero or one to anything. Let's see what the evidence is. For example, I asked, how many people would need to be involved in the conspiracy theory to pull it off? You have a conspiracy theory, how many people would have to be involved to pull it off? The more people you need, the less likely it is to be true. People are incompetent, they change their mind, they chicken out. Something happens, they go to the wrong place, they get the wrong thing, they forget. That's normal human behavior.
- 6.61 How many elements must come together to pull it all off? If it's a super complex one, and you've got to be standing over here and you're going to be behind the bushes and you're on the grassy knoll and you're behind the picket fence and you're up on the Dealey Plaza and you're on the sixth floor, and it all has to come together, and the guy is going to open his umbrella, and that'll signal to us we have to do this and this, you got 50 different things. These are conspiracy theories about JFK.
- 6.62 Like, 50 different things that have to happen just at the right place at the right time, and everybody is confident and pulls it off. Not likely, because we know how people actually behave in the real world. They screw things up, they forget, randomness happens, shit happens. It almost never goes the way people would think it is. Then, again, how perfect needs to be the timing of the people and the elements.
- 6.63 Finally, how big is the conspiracy theory? OK, so just take 9/11 as an inside job. Back to our conjunction fallacy, that two things have to happen at the same time. It's far more probable that the one happened than that both of them happened. We all saw the planes hit the buildings. It can't also be that they planted explosive devices, and the planes hit the buildings. So it had to be just the planes. By the way, I should note, parenthetically, there are conspiracy theories—they're called the no-planers— that there were no planes that hit those buildings. This was all CGI, fake videos. OK, even the 9/11 truthers I deal with, they think this is completely crazy. It's really fun to see them both go, oh, those guys, they're completely crazy. We're the real conspiracy theorists.
- 6.64 It reminds me of that book, *The Three Christs of Ypsilanti*. It was written by a psychiatrist in the late 1950s, who was at this mental hospital in Ypsilanti, Michigan, in which he had three different patients, each of whom thought he was Jesus Christ. So he got all three of them in the room together, and every one of them said, "Those other two guys are completely paranoid. They are crazy. I'm the real one." This is how the mind works. Somehow or another, in addition to the planes hitting, in the floors that they hit at, at an angle, the conspirators would have to know, ahead of time, what floors, at which angle, to plant the explosive devices, because you can see on the videos of the building's collapse. They start the collapse right at the floors where the planes hit. That's where the collapse begins in both buildings.

- 6.65 So, somehow the conspirators knew ahead of time? How did they do that? Well, these planes were flown by remote control devices, so they're very accurate. Well, where were the passengers? Oh, they took them off the plane because we didn't want to risk them charging the cockpit. What about all those phone calls they made from the cell phone? Oh, those are all faked by—the CIA has these fake phone message devices that can call people and sound like them. This just goes on. How did they get into those buildings? The terrorists tried to blow up the World Trade Center building in 1993. So ever since then, these are super secure buildings. I've been there in the new one. There's this police all over, security, bag checks, everything.
- 6.66 How do they do that? Well, they went in under the pretense of elevator repair. I'm serious. Elevator repair. "Oh, we're just fixing the elevators." On all those floors, in both buildings? Where 50,000 people work every day, no one noticed, "Hey, they're breaking into the drywall and they're carrying all these explosives. What is that?" Not to mention, not one of them wants to tell anybody? That "you're not going to believe what I was involved in." Not one of them wants to go on 60 Minutes and be a whistleblower and go, "OK, this is it. I know the guy that orchestrated the whole thing. It's so-and-so in the CIA or FBI or the Bush administration," whatever.
- 6.67 All right, let's look at what I think are the three primary underlying reasons why people believe conspiracy theories. In an earlier lecture, I made a distinction between proximate causes and ultimate causes. Proximate causes are more immediate reasons for why people believe conspiracy theories, which I've been giving you— lack of control, anxiety, political orientation, so on. These are deeper, I think more ultimate ones. That involves what I call proxy conspiracism, tribal conspiracism, and constructive conspiracism. Let's look at each of these individually.
- 6.68 Proxy conspiracism is where the conspiracy theory is a proxy for a deeper type of truth. Tribal conspiracism is where the conspiracy theory reinforces truths held by a particular tribe, party, religion, or ideology. Then finally, constructive conspiracism, where the conspiracy theory has some element of truth to it, so it pays to believe it just in case, better safe than sorry. Again, that's on my "better to make one type of error than the other" model, that we are naturally risk-averse and look for threats. So, constructive conspiracism is a, it's kind of rational on this level.
- 6.69 Examples of each of these. Many conspiracy theories are proxies for a different type of truth, a mythical, psychological, or lived experience truth. The details of the particular conspiracy theory are not that important. It's what they represent underlying it. Again, back to the QAnon and the Pizzagate and all that stuff. If you showed somebody who said they believe that, the Democrats are doing it, that Democrats are not actually doing that, it's not like they're going to go, "Oh, in that case, I'll vote for Hillary." They were never going to vote for Hillary.
- 6.70 The conspiracy theory is just a proxy for, a stand-in for something else. "I just don't like those Democrats. That's the kind of thing Democrats would do. If they didn't do that

one, they'd do something else because we don't trust them." That's how it works. It's a proxy for something else. I like, as a model, the OJ trial. In a way, OJ was acquitted on a conspiracy theory. That is, his legal defense was that the police planted the bloody glove and the other evidence at the scene of the crime to conspire to put him away. The fact is that that's not completely irrational. There's a reason African Americans don't trust the police when you look at the history of how those things have unfolded.

- 6.71** I have used the example of—there was this great ESPN documentary series called OJ: Made in America, that tracked the whole history of the African American community post-World War II moving into the Southern California, particularly Los Angeles area, and how bad it was. It's just really horrible the way police treated African American community in LA, all the way up, really, until the '80s, late '80s.
- 6.72** By the time the OJ trial happens—what was it, '94, '95?—the idea that the police might plant evidence against African American suspects is not irrational. In fact, that used to happen a lot. In a way, even though if you watch the trial, the evidence, well, obviously, he did it, but the acquittal, I think, had something to do with a deeper truth. You know what? They do that kind of thing. Whether they did it that time, that's the secondary issue. These police are, yeah, OK. There's a rationality to that.
- 6.73** Again, anti-vaxxers. I've spent a lot of time debunking anti-vaxxers. But there's a reason they don't trust big pharma, as I gave as the example, the regulatory capture by pharmaceutical companies in the opioid crisis, that happened. They really did do that. You could just go back in time to the tobacco industry. They lied about the effects of tobacco. Chemical companies have lied about the effects of chemicals on the environment, and so on and so forth. They really do these things.
- 6.74** There's this great book called Merchants of Doubt, Naomi Oreskes, a historian of science, who tracked the whole campaign by big tobacco to lie and misinform the public about the effects of tobacco on health, particularly lung cancer. Then they showed that, in the '80s, there was a similar movement for the chemical industry in which, well, a lot of house fires were caused by smoking in the house or falling asleep with a cigarette or whatever, couch catches on fire. The solution was not, get rid of the cigarettes. The solution was, let's coat everything in the house with these anti-inflammatory chemicals that we can make a lot of money and stop the fires and still make cigarettes.
- 6.75** What Naomi discovered was, in many cases, it was the same people. These companies hired the same people the tobacco companies hired to plant a seed of doubt, merchants of doubt. There was a famous memo she uncovered called, "Doubt is our product." Internal memo in the tobacco industry. Doubt is, "We're just going to tell people the jury is still out on this." What jury? Where? There's no jury. It's science. Then all the way up to climate change. Some, these oil companies hired the same people. It's astonishing. There's an underlying rationality why people don't trust big corporations, big pharma, big regulators, and so on. There's a logic to it.

- 6.76** JFK conspiracists don't trust big committees like the Warren Commission, or 9/11 truthers don't trust big government. QAnoners don't trust the deep state. Again, with each of these, I may be critical or skeptical of some of their specific claims, but I get why they think that. Because when I think about it—I was at this political event this last summer with, RFK Jr was there. There he was, campaigning. I was at this conference. I got this little thing on my phone. "You're invited to go with RFK Jr tonight at dinner." I'm like, god, me? Maybe just a few other friends. I'll get to ask him questions and maybe push him a little bit on the vaccine stuff. I've read all his stuff.
- 6.77** Turns out, there was a thousand of us close and personal friends at this. I was like, OK, he's way over there, surrounded by 100 people anyway. But again, he starts off talking about this stuff. Then before you know it, "The CIA killed my father and my uncle," and off he goes, most of which I don't think is true. But then he comes back and goes, "But you know the big pharma?" I'm like, yeah, actually, that's right. Again, this proxy thing.
- 6.78** OK, tribal conspiracism. Many conspiracy theories represent political, religious, social, tribal identity. So this is virtue signaling, it's called. I'm signaling my affiliation, or I'm saying, "I believe this because everybody I know believes it or everybody in my group believes it." There's a logic to that. That gives us that cross-pollination of conspiracy theories and the propensity for people who believe one to believe many. "Yeah, I ticked the box. Yeah, I don't trust the big government, so therefore they were involved in this, this, this, this." Whether they're true or not, I don't even know anything about them. How would I fact-check this? Where did they film the fake moon landing? I don't know. But it's the kind of thing the government would do, that kind of thing. Endorsement of conspiracy theories serves as a social signal or loyalty to the tribe.
- 6.79** I'll introduce you to Melissa Rein Lively. I worked a couple of years ago, two years ago, I guess now, on this Netflix series on mind control in the brain, how it works. One of the episodes was on conspiracy theory, why people believe them. We found this woman, Melissa Rein Lively, who was a very successful, ran a very successful PR agency in Dallas, and happily married, got kids, great house, great job, and everything is going great. Covid hits, shuts down her business. She's got nothing to do. She's home all day, searching online, and ends up going down the rabbit hole of QAnon, the deep state, the pedophile ring, Pizzagate, the whole thing.
- 6.80** She decides that the most important thing she could do with her life now is to stop this Satanic pedophile crime that's being committed and that Donald Trump is going to stop this. You remember when all that was happening. QAnon was being cued by Trump, and they were going to expose the deep state, and all this stuff, horrible things that were going on. She went down this whole rabbit hole, to the point where not only was her business gone, mainly due to Covid, but to the point, and she even picked up her phone and played the voicemail message, her husband left her. "If you don't stop this, I'm taking the kids and leaving."

- 6.81** She told him, "Go ahead, because this is the most important thing I will ever do. I am going to save the world. That's more important than raising my kids." The reason we have her is because she came to her senses and went, "Wait a minute. Wait a minute. I'm leaving, I'm giving up my family? This is crazy." And she started fact-checking and realized that this was all nonsense. But that's how captured somebody could be. This is what my tribe is doing. The QAnon deep-staters, "We're going to put a stop to this crime. This is far more interesting and important than anything I'll ever do."
- 6.82** That's another one of these appealing things about conspiracy theories. Most of us have lives that are routine, boring. I get up, go to work, take the kids to school every day. There's nothing particularly exciting about this. But now, oh, my God, I'm in on this. I found out about this thing that's going on. If I can be part of it, I can put a stop to this, it's going to change the world. Sort of like what I said about, no one joins a cult. They join this group; it's going to be incredible. Then they find out later, "Oh, my god, I went down the rabbit hole."
- 6.83** Then finally, constructive conspiracism, that is, enough conspiracy theories are true that it pays to err on the side of belief. The reason is because, when when we evolved our conspiratorial cognition, it was better to assume that a conspiracy theory is real when it's not. Again, that's that signal detection problem. One error is worse than the other. This manifests itself in another cognition called the negativity bias, or why bad is stronger than good.
- 6.84** You'll notice there's no conspiracy theories about some secret cabal is plotting in secret to make the world a better place. Who knows? No one ever says that. No one ever thinks that. It's always something bad. They are doing this bad thing to us. That's what all conspiracy theories are about, conspiracism is about. Why is that? Here I probe into this deeper problem of why bad is stronger than good.
- 6.85** There's papers and research on this thing called the negativity bias, why we notice negative things more than positive things. In behavioral economics, this thing called loss aversion, where losses hurt twice as much as gains feel good. Let me repeat that. Losses hurt twice as much as gains feel good. To get me to make an investment in something, the potential payoff has to be at least twice as how painful it would be for me to actually lose my investment. In a way, once you own something or have the asset, it goes up in value because I have it, and therefore, its loss would feel much worse. Rather than not having it, and now I gain it, that feels good, but not as good as it would feel bad in the opposite direction.
- 6.86** Negative stimuli have a stronger influence on neural activity than positive stimuli. Lots of experiments on this. In an analysis of over 17,000 research papers in psychology, 69% dealt with negative issues. Depression, anxiety, and sadness, and violence, and aggression, and homicide, and suicide, and rape, and all the bad things that people do. And only 31% were positive. The reason is because bad things have a greater impact on human thought and behavior than do good things, so it's easier to get published.

- 6.87 How do I get research money and published for my study? I'm going to study why people are happy. Oh, come on. How about a societal problem? How come we have people committing suicide? That, we will fund you for. It's like cancer. There's not much funding in medicine for wellness. Let's study why people are not sick. No, we got to cure cancer. Negative versus positive. Bad smells elicit more animated facial expressions than good odors do. Bad impressions form faster and are more resistant to change than good ones. Memory recall is better for bad behaviors, events, and information than it is for good ones. Losing money and friends has a greater impact on people than gaining money and friends. Criticism hurts more than praise feels good.
- 6.88 Again, I can go through my Twitter feed. I'll post something. It's like, oh, great post. I agree. Shermer, you've gone off the rails. You're completely wrong. Who is this guy? All of a sudden, this one bad one pops out. Reviews for my book, good, good, good. Yeah, yeah, yeah. What is this? This guy doesn't like my book. Who is this? And I'm going to—that just pops right out. I can't even remember what the good ones are, but I remember the bad ones.
- 6.89 Bad events have a greater impact than good events on everyday life. Having a good day does not necessarily lead to a good mood the next day. Start the day over, we'll see how it goes, but a bad day carries over its consequences into the next day. Why are we so focused on bad things? OK, let me go as deep as I can go for an ultimate causal explanation. And that is the second law of thermodynamics. Entropy. OK, it's a law of nature. It's not a force that's pushing things to happen a certain way. It's that, if you do nothing, things just fall apart.
- 6.90 Think of, you make a pot of coffee that sits there. It doesn't get warmer by itself, it gets cooler by itself. Because heat is produced by atoms, sorry, molecules just bouncing around off each other and generating heat from the impact. If it just sits there, there's less jiggling, and at some point, you just get a uniform temperature with the room. That's entropy. There's nothing making it happen. That's what happens if you do nothing. To do something, you've got to put it in the microwave and heat it up. That's actually active. So entropy, the purpose is to push back against entropy.
- 6.91 Just think of the sandcastle. There's only so many ways you can push together grains of sand and water to congeal enough to make a nice structure that resembles something like a castle. There's a very limited number of ways you could do that. There's an infinite number of ways that wind and rain and dogs and small children can turn your beautiful sandcastle into a blob of amorphous sand. This is a metaphor for life. There's just way more ways for things to go wrong than there are for things to go well. So your first duty in life is to push back against entropy.
- 6.92 Since this is the Peterson Academy, I will invoke Jordan's rules here about make your bed, clean your room, do the basic things in life. OK, well, first of all, I find it astonishing people don't know this, but apparently a lot of people are never told this when they're growing up, because that really is solid advice. If you do nothing, your room just decays

into the clutter. Your garden will just turn into weeds if you don't weed it. Your body will turn into flab if you don't work out every day, and so on and so forth. You have to consume calories to have energy to live.

- 6.93** Or poverty. There's tons of books about, why is there poverty? You don't need any books on why is there poverty. Poverty is what you get when you do nothing. That's what entropy leads to, just nothing. Wealth is what needs an explanation. What am I going to do to put energy in the system to reconfigure atoms and molecules into stuff that people can use and that produces wealth? The point of all this, with conspiracies, is that things can go south pretty quick. Your body can fall apart, you have a stroke, you get hit by car, whatever, heart attack. There's tons and tons of ways, but there's only so many ways for you to just keep going, and we don't notice that.
- 6.94** I don't notice that I'm standing here and my heart keep beating just like normal. It's only when it goes wrong that I think, uh-oh, I have a problem here. So pain stands out far more than, say, pleasure. It's not pleasure so much as, I'm just standing here and I'm feeling nothing. That's actually good. Nothingness is good news. There's no pain. That means things are just running along. The whole point of pain is to say, there's a problem. You should stop doing what you're doing and do something else to make the pain go away. There's actually theories about depression and anxiety, about these maybe evolved emotions to get you to stop doing what you're doing and do something else because this is not leading to good outcomes. Again, the goal, that's what we do. We have goals built in the system. Using rationality, I should do this instead of that.
- 6.95** There's an asymmetry between good and bad in our evolutionary history. I'm trying to give a logic of negativity. Progress is made mostly in incremental steps, small steps. Here I'll just refer a minute. There's tons of books on utopia and lots of commentary on utopians. Forget utopia. This is a terrible idea. You can't get there anyway. There's no there, there, what a perfect society would be. Just go for protopia. This is Kevin Kelly's idea of this small incremental steps. Make tomorrow just that much better than today. Go three steps forward and two back, and then you'll be one ahead tomorrow. That's achievable.
- 6.96** It isn't exciting. All these protests that we're witnessing on the streets and campuses today. No one wants to follow me when I go, "What do we want? Slow, gradual legal change. When do we want it? Eventually. Come on, everybody." Well, where is everybody? It's like, "No, we're going to tear down the system. We're going to march to the police. We're going to tear down the campus administration, whatever, defund the police." This is what people wanted. It's noticeable, it's big. But in fact, that isn't how the world really works. Again, regress can happen easily, in just one colossal calamity.
- 6.97** There's a cost of overreacting to a threat. It's less than the cost of underreacting. We are on the side of overreaction. Again, it's that signal detection problem, and this is called pessimism. The fact is, in our evolutionary history, the world was far more dangerous. It paid to be risk-averse and highly sensitive to threats. That, again, to wrap

it up on that subject is the second law of thermodynamics, also known as another way shit happens. OK.

- 6.98** Then finally, to bring this to a close. Again, a lot of conspiracy theories are true. That Lincoln was assassinated by a cabal. There was a whole group of these Southerners who were going to, right at the close of the Civil War, the South is going to rise again. We're going to decapitate the Lincoln administration. We're going to kill him. We're going to kill his vice president, kill the Secretary of War, kill the Secretary of State, and on. They had it all planned.
- 6.99** This is how it typically goes. One guy just changed his mind. Somebody else chickened out. Somebody else went to the wrong place to get his weapon. The guy who was supposed to kill the vice president or Secretary of State got upstairs and tried to stab him, but he had his neck brace on because he had hurt himself, fell out of a carriage. The knife hit the neck brace instead of killing him. Then that guy tried to escape, and he was captured. The only guy who succeeded was John Wilkes Booth, who shot Lincoln. Within hours, they knew who it was, and they tracked him down, and within a few days, he was captured, and they were put on trial.
- 6.100** That's how conspiracies normally go. People screw up, we find out about it, and so on. World War I was triggered by a conspiracy. This was the assassination of Franz Ferdinand in Sarajevo, in which a conspiracy of Serbian nationalists called the Black Hand, conspired to assassinate the Archduke. He's the heir apparent of the Austrian-Hungarian Empire that wanted to incorporate more lands that the Serbian nationalists didn't want to be part of, and so on. So they plotted to kill him.
- 6.101** Here's what happened. This is how it normally goes. So, there was seven of these guys, and they each had instructions: "Okay, each of you, you go to this address, you give this secret pass. They'll give you the hand grenade, they'll hand you the gun, and then you're going to go stand right there because we got the parade route"—they published it in the newspaper, like JFK: "Here it is. Here's where he's going." He's in this big, huge, beautiful convertible coupe sitting in the back seat, no lid—I mean, no top, no glass, and so on. They're driving along the parade route.
- 6.102** One of the guys pops out, and he pulls the pin in the hand grenade and throws it, and it bounces off the back of the trunk of Franz Ferdinand's car and goes underneath the car behind it. Boom, it goes off, and those guys are hurt. Then Franz Ferdinand is like, well, that wasn't very nice. They take off, and he goes to this place where he's supposed to give this speech, and he gives the speech. He's like, "Thanks a lot. I come here to give you guys support and you try to kill me." Then he finishes his speech, and the mayor's like, "Oh, I'm so sorry, we didn't know they were here."
- 6.103** Anyway, so then he gets back in the car and goes, "Well, let's go back to the hospital and see how our friends are doing who got hurt in the car behind us." They take the same route back. Now, mind you, the other conspirators, one of them, the gun failed to

go off, another one jumped in the river to drown himself, and that failed because the river was too shallow, and this and that. The one guy, Gavrilo Princip, was sitting there at the curb, just despondent, like, "Well, that was a total screw-up. We failed." He's sitting there on the parade route, and the car is coming toward him and then decides to make a turn, but it's the wrong street.

6.104 This coupe happened to have no reverse. The driver puts it neutral and just backs up at three miles an hour. There's the guy sitting right there, like from me to you. And he's like, holy crap, boom, dead. That's how that happened. Then because of all the treaties and so on, we have to take revenge against these, and that country is backed by these countries, and so we have to attack them. Before you know it, you have World War I and eight million people dead. That's how conspiracies normally go.

6.105 Or as I mentioned, the Japanese attack on Pearl Harbor, that was a conspiracy. Operation Northwoods under JFK was a conspiracy. This is one of the more interesting things to come to light since the 1970s started to reveal some of the documents associated with JFK's assassination. Due to the fact that the Zapruder film showing Kennedy's assassination was finally made public—we're talking November 1963. It wasn't until Geraldo Rivera on a TV show said, "We are going to show you the Zapruder film and be prepared." A trigger warning. It's pretty gruesome, and it is. See his head just blow off. It's like, oh my god. Back and to the left, back into—sure looks like they had somebody up front, and so on and so forth.

6.106 That led to investigations and the Church committee to really look into what the CIA and FBI were really doing over all those years. They found this, Operation Northwoods, where members of Kennedy's own cabinet came to him and said, "We have some ideas of some false flag operations we can pull off as an excuse to invade Cuba and kill Castro." Here were some of the things they suggested. We'll stage a fake attack on US military base at Guantanamo Bay, or we'll employ a fake Russian MiG to buzz a real US civilian airliner and blame it on the Russians or the Cubans. We'll fake an attack on a US ship and make it look like the Cubans did it, or we'll shoot down an enemy, empty US airliner with a tape claiming to be under attack by Cuban forces.

6.107 To his credit, JFK says, "We're not doing this. This is insane. I can't believe my government's telling me to do these things." But at the highest level, they're saying, basically, this sounds like crazy Alex Jones stuff. "It's a false flag operation." Yeah, but it is. Now, they didn't do it, but that they were suggesting it. The fact is, the CIA attempted to assassinate Castro at least a dozen times, by the craziest things. We'll put in a hand grenade, an explosive device inside this giant seashell, and we'll put it down on the bottom of the ocean where we know he goes scuba diving there. He picks it up. Or we'll put some poison in his wetsuit. He puts the wetsuit on, he'll die. Cigars. We'll put poison in his cigars.

6.108 Castro himself said, "Boy, I survived so many assassination attempts. If this was an Olympic sport, I'd be the gold medalist." Everybody knows. When Kennedy gets

assassinated, they go, "Oh, it probably was Castro and the Cubans, preemptive killing before—" Anyway, so again, there's a rationality behind believing, "Wait, our government was doing this?" Or take the COINTEL program. This was revealed when a group of activists broke into a FBI headquarters in Philadelphia under the pretence of doing something else.

- 6.109** They were looking for something else, and found all these documents about the FBI was spying on US citizens, infiltrating civil rights groups, women's rights groups, animal rights groups, Native American rights groups, and so on, not only just spying on them, but also to try to make them look bad. Let's do some really stupid things, and then the government can have a pretext to shut them down because they're causing trouble and civil unrest and so on, all the way up to, astonishingly, the attempt to blackmail Martin Luther King.
- 6.110** Again, this is our own government doing this. This is the FBI, signed off by not only J. Edgar Hoover, but RFK himself said, "Yeah, OK, we'll wiretap King's phone calls." There's a new biography of King. It's just fabulous. Jonathan Eig. I just had him on my podcast. It's great. He has detailed phone conversations that King had with his civil rights co-leaders about, "Here's what we're going to do. We're going to march, and we're going to protest this, and we're picking these targets. It's the place where we think we can get the most progress on civil rights."
- 6.111** We know how he did it. Where did all those phone calls come from? The FBI. They tapped his phone. Even worse, let's just say, King wasn't the most faithful husband in the world, right? So he's traveling, he's in hotel rooms. He has women in these hotels. The FBI recorded his sex and then threatened him in this letter, "We are going to expose and release your sex tapes if you don't—" And then they hint at the bottom, "If you don't do what you know we want you to do, that is, either kill yourself or just go away and stop being a civil rights activist." Let me say that again. This is our government, to a US citizen protected by the Constitution, and they're threatening him with sex tapes? This is illegal. And they did it anyway. It's astonishing.
- 6.112** There was this whole program called MKUltra. This is a CIA program having to do with mind control. In the 1950s, we're worried about the Russians or the Chinese or the North Koreans getting ahead of us on mind control, brainwashing. The CIA launched a whole program, MKUltra. Let's see if we could close the gap between the Russians and us on mind control. There's a number of books about this. Wormwood, The Poisoner-in-Chief, the story about Sidney Gottlieb and the CIA's search for mind control.
- 6.113** Wormwood is a documentary series about the actual LSD dosing of a US, not just a US citizen, a CIA employee. They're off on some trip with the other guys. They're at this resort and they're at the bar and they're just drinking, having fun, yucking it up. They give him some LSD in his drink and he loses his mind. Then a week later, he's in New York City and he dies mysteriously. He fell out of this New York high-rise, or did he? So Errol Morris' film, Wormwood, ends with the two different scenarios. He fell or he was

pushed. In any case, the CIA under President Ford paid off the family of this guy \$750,000, under the pretense, well, he never gave permission, uninformed consent. He never gave informed consent, so that was the pretense. But again, wait, our government was doing what? This is all super illegal.

- 6.114** Or Operation Paperclip. This is Annie Jacobsen on the right there. All of her books are great. I can't recommend them enough. But this is the US government's attempt to get Nazi physicists, chemists, and biologists to move to America and work for our government before they got swept up by the Soviet Union to work for their government, which they also were doing. The most famous, of course, is Wernher von Braun here, who was in charge of the Apollo rocket mission thing.
- 6.115** But these are guys who committed the same kinds of crimes that the people we put on trial in the docket at Nuremberg and hanged them for war crimes. Other ones, they got a paper clip on their file going, this one, let's see if we can get them to work for us. We'll give them a house in Ohio, and you'll have a family, and it'll be great, and a pay check. Everything is great. Let's just hope the American people never finds out about it. Now we know about it. Watergate was a conspiracy. Iran-Contra was a conspiracy. Of course, JFK, everybody thinks that he was assassinated by a conspiracy. I don't think so. I devote a whole chapter to why I do think Oswald was acting alone, but I'm in the minority position on that.
- 6.116** But I will note that there's so many JFK conspiracy theories, and they're so popular that there's even a joke about this. Here's the thing. A guy dies and goes to heaven. And God says, "Son, you were such a righteous, good man that I'll grant you one wish. I'll answer any question you'd like answered totally truthfully." He goes, "All right, who really killed JFK?" And he goes, "Lee Harvey Oswald, acting alone, using his M91 Carcano rifle." The guy goes, "This goes even higher than I thought."
- 6.117** I'll recommend the two best books on this. If you want to do the deep dive, you'll spend the next 10 years going through all these. Case Closed by Gerald Posner really unveils exactly what Lee Harvey Oswald was like and why he really did do it, and no evidence for anybody else. Vincent Bugliosi, the trial attorney who convicted Manson, also wrote this 1,600-page book. You can get the abridged version on Audible that's only 30 hours long, which I did, in which he addresses every one of the conspiracy theory claims, the grassy knoll and the CIA and the Mafia, the Cubans, the Russians, this, that, this memo, and so on and so forth, and there's no evidence for any of those.
- 6.118** I'll end this lecture by just showing one of the more famous pictures. This is the opposite side of where the Zapruder film was shot. Zapruder, you could see, this is from the middle of the grassy thing. This is Dealey Plaza. This is the picket fence. This is the grassy knoll. Here is Abraham Zapruder standing on that concrete pillar with his camera. You've surely all seen the Zapruder film. That's where it's filmed from. From this is a Polaroid picture taken by a woman named Mary Mormon. This has been gone

over in great detail by conspiracists that I think properly represents the problem we're talking about here.

- 6.119** They claim that, in that Mary Mormon picture, they could see where the assassins are all lined up. And all I see is patternicity and agenticity. There's just nothing there. If I don't show you where they're supposed to be, can you see where the assassins are? And the answer is no. I mean, this is like the UFO photos. This is like the Bigfoot photos. OK, if you use your imagination and squint, you can sort of, kind of see what I'm telling you I want you to see. Yeah, no. I'm not seeing any of that. And I think that's the problem. Thank you.
- 6.120** You mentioned memory recall. Do we have any data on people who claim to have a photographic memory and whether or not they're recording a little bit better than the rest of us?
- 6.121** Yes, OK, snapshot memories or these light-bulb memories. There's some evidence that that can happen, but there's also distortion. A lot of times people say, "Oh, I remember exactly where I was," and so on and so forth, and it turns out, that wasn't the case at all when they fact-checked with family members or so on. There are, of course, these weird cases of these idiot savants, so-called idiot savants, who do have exact memory recall every day for the past 20 years. You can ask them, "What was the weather in Philadelphia on Tuesday, February 28, 1968?" They go, "Oh, it was good."
- 6.122** I mean, that is really bizarre that any human brain could do that. The one guy, after he died, who was really good at this, he had one of these photographic memories. He was one of these lightning calculators. He could square huge six-digit numbers faster than he could punch it into the calculator. How is this possible? He had no corpus callosum. He had just this one giant lobe. Maybe the information, he speculated, was in there and able to be communicated faster, something like that.
- 6.123** The problem is, again, memory research is hard to do. Even I have some examples from my book, *Conspiracy*, of the journalists there interviewing people that were right there. "I was standing there in the grass, I saw them." Then they're interviewing them later, and "we saw this and we saw that." The example I use is the guy's wife is like, "We didn't see that. We were in the parking lot when that happened." Oh, yeah, that's right. Even then, just at the moment, within an hour of the event. Again, it's back to Loftus' research. If you just ask the question a certain way, people just—
- 6.124** Here's the problem. Now, that's the new memory now, the one you just said. There's no, like, rewinding back to the other one and erasing what you—so poor Jennifer with Ronald Cotton, it took her a long time to look at Ronald Cotton and go, "You're not my rapist. It was that other guy." Because she had for 11 years, "That's the guy, 100%." And then the problem is never 100%.
- 6.125** You mentioned a couple of lectures ago the word luck. Is luck something that you create, either good or bad? Things just happen or things happen just.

- 6.126** Yeah, that's a great question, and the answer is yes. OK, so first, there's no such thing as cosmic luck. There's somebody up there or some mysterious force that brings things together just for you at just the right time. That doesn't happen. What does happen is a self-fulfilling prophecy. We have research on this, that people who self-identify as lucky, "Yeah, I'm a lucky guy. Good things seem to happen to me," also score higher in openness to experience. They like trying new things. They're more sociable, gregarious. They score higher in interacting with other people and having more social contacts with others.
- 6.127** You can see how that could play out. If I'm high in those, I'm more likely to go to the party, meet new people, make connections. "I got the job." Then later, I'm like, "God, I was so lucky. I just happened to be at this party. I just happened to meet this person that just happened to have a job opening, and I got the job. It changed my life. I met my spouse, I got my career," whatever. But in fact, you really made that happen. Other people that are shy and not as social, they may just not go to the party. Then like, "Oh, I'm so unlucky. How come I never meet anybody?" Well, so yeah, I think it's mostly self-fulfilling. Yeah, for sure.
- 6.128** Of course, there is, again, the other kind of luck, by which I mean randomness. Again, Bill Gates just happened to be born and come of age right at the moment when computers needed software, and that's what he was doing, and so on, and whatever. Most of our lives are like that. We were lucky we were born, let's say, in America and not Afghanistan or Somalia, something like that, war-torn country, or Syria. I mean, that's just bad luck. You can't really take credit for that.
- 6.129** If you have good parents that sent you to good schools and they're smart and they raised you in an enriched environment, you didn't do anything to deserve that. So there's this, kind of this just-world theory that conservatives tend to hold, that people that are wealthy and successful, they earned that. Well, they did a lot of things, yes, but on the other hand, you can't take credit for your intelligence because that's mostly born. You mostly inherit that, both genetically and culturally. We can't take credit for that you happened to go to good schools because that's where your parents sent you, and you're raised in a good environment, and the neighborhood was clean and crime-free and so on. You don't get any credit for that. But that's luck.
- 6.130** When Obama gave that famous speech that made conservatives' heads explode, "You didn't build that," what he was saying is, look, don't forget the role of government. We build the roads that you drive to to get to your company. We build the Internet that you have all this prosperity from. The government did all that. You didn't do that. He doesn't mean people don't work hard and so on. Of course, that's also a role. But that luck, that we just happen to be here, OK. I don't get any credit for that. What's the right word? Gratitude. Yeah, gratitude. Just lucky me.
- 6.131** I'm just curious, how do you tend to, or maybe I'm not sure how often you have done this, but I presume it's happened—when somebody suggests that rationality is a tool to

be used to form a narrative, rather than the way we determine the correct or maybe most likely narrative, how do you tend to defend that?

- 6.132** Yeah, because that is true. Rationality is often used in the service of some narrative or story that we want to be true. Absolutely. That is a form of the confirmation bias. I'm just going to gather and present the facts that support my worldview. OK, everybody does that. That's why you need the adversarial system. You need the red team. You need the fact-checking. You need the peer review. You need some system that says, OK, maybe. Your facts presented in the support of this particular narrative might be true, but let's look at the other side.
- 6.133** I'll give you an example. This is pretty controversial, but there's a film that just came out called *The Fall of Minneapolis*. OK, it's about the George Floyd case. This is going around the internet. OK, I better watch this. See what this is all about. I watched it. First of all, these are the kind of questions I asked. Who made this film? Oh, it's by this journalist who was the star reporter at this Minneapolis TV station. OK, that's good. She's probably a real journalist. She's not there anymore. Oh, I wonder what happened. Now she's at this right-wing conservative media source. That's not a definitive problem, but it could be. And then, OK, and her husband also is producer of the film. Who's he? Oh, he's head of the police union in Minneapolis. OK, all right. I'm going to get a slant for the pro-Derek Chauvin police. All right. But maybe they have a point. Let's just set that aside.
- 6.134** OK, so now they start. You've all heard about the autopsy report that George Floyd died by asphyxiation, the knee on the neck. They go, OK, here's this other autopsy report that says he died of a fentanyl overdose, cardiovascular issues. He had high blood pressure. He took fentanyl, and it was the stress of being, tangling with the police caused this heart attack. He didn't die of asphyxiation. Derek Chauvin should not be called guilty. OK, but there's this other autopsy report that says he did die of that. Which is the one I'm supposed to—what am I looking at here? Who's the right one?
- 6.135** Well, these attorneys presented this one, and these attorneys presented that one. Hang on. It should be a scientific question, just a factual question. What is the cause of his death? The fact that you can have two attorneys hire two different autopsy docs and get two different reports, I'm like, come on. There's something wrong there. This goes on and on like this. That the, Chauvin's defense was that this maximal restraining technique, MRT, is part of the Minneapolis police training. Now, in the trial, the head of the police said, "No, we never did this. We didn't train anybody. We don't do that. This knee on the neck thing, we don't endorse that."
- 6.136** OK, and then in the film, they go, "Here's the manual." They got Derek Chauvin's mother who's got his manual. Here's the book he was trained on. She turns the page. There it is, knee on the shoulder. It was supposed to be shoulder, not neck. And then they show the video of the knee on the neck. We've all seen it. And they go, look, look, the knee is not on his neck. It's on his shoulder. It's on his shoulder. And you're looking

at it going, yeah, that's right, it's on his shoulder. Wait a minute. What am I looking at with these other ones, where I know I saw it on the knee, on the neck? And you look it up. Yep, there it is. The knee is on the neck.

6.137 They're showing me a clip where—because he's moving around, and the top is moving around. They're just showing the one clip of the knee on the shoulder, but not the other. Come on, guys. That's just—come on. This is, I find, very frustrating. But maybe this is, eventually we'd now get to the truth. He's probably not going to get a retrial, but maybe somebody will say, OK, here's what we think really happened. You got this one stream view, you got this. Those are hard problems, really. That's why we need free speech, open inquiry, adversarial system. Let's hear all voices, because even if you're getting mainstream stories, they're stories. I saw it on an ABC news. OK, what if you made a big mistake? If you say no, make mistakes? OK.

Lecture 7: Miracles, Myths, and Meaning

At a glance — Miracles, myths, and meaning.

- 7.1** Good morning. We're back for some more, last two lectures in our eight-lecture series, the course Skepticism 101: How to Think Like a Scientist, about anything and everything we can. Just a quick overview. Recall that the first four to five lectures were mostly about how we determine what is true in science and what areas are probably perhaps off-limits for science, and there are different kinds of truths. The last lecture, we talked about conspiracies and conspiracy theories, to what extent we can apply rationality and scientific thinking to figure out which ones are true or not.
- 7.2** These last two lectures, I also want to cover actual topics that we can address using reason, rationality, and science to see how far we can get. These, in lecture seven here, are miracles, myths, and religion, how to think about the divine. Then in the last lecture, I will talk about the God question, the biggest of all. Let's get started. Do you believe in miracles? Well, I do, sort of, sometimes, at least in sports. I recall in the 1980 Winter Olympics when the US hockey team beat the mighty Soviet Union hockey team and Al Michaels on ABC shouted out as the clock ticked down to zero, "Do you believe in miracles?"
- 7.3** We use that term colloquially to mean something like, this is unbelievable or really unlikely or, wow, I can't believe what just happened, something like that. People often describe this as this word miracle. Like, "It's a miracle I won the lottery." I mean, pretty much everybody that's ever won the lottery goes, "It's a miracle." Or the person that survived, some of the people that survived 9/11 in the World Trade Center, it collapses, afterwards said, "It was miraculous. I went down the right stairwell instead of the left one, and those people died, and I survived. It's a miracle."
- 7.4** Whether they think it was a religious miracle or just the miracle of the forces of the universe or whatever, they often use that word. You often hear this word used by cancer survivors. That is, "We prayed and my cancer went away, or I tried this thing, it was a miracle that my cancer went away," and so forth. Are those really miracles? Well, recall in the last, two lectures ago, we talked about coincidences and probabilities and statistics, and that if you have a large enough number of things going on, the extremely unusual will inevitably happen. This is the law of large numbers and so forth, that the sum of all possible coincidences is certainty. If we have everything that could happen, then one of them is going to happen.
- 7.5** In the case of the sports miracle, there's a reason they play the games rather than just declare the winner before the tournament starts, because every once in a while, somebody beats the greatest of all time or they beat the unbeatable team and so on. In the case of 9/11, you have to wonder, the people that went down the right stairwell and declared their survival miraculous, well, what about the people that went to the

left? How come they weren't miraculously spared? Or the people that pray and their cancer goes away, well, cancers do go into remission periodically. Not often, but they do. It's not statistically impossible.

- 7.6** And so by chance, you have a thousand people with cancer, maybe 10 of them, their cancer just goes into remission no matter what they're doing. So whoever, of course, experiences that, like the person that won the lottery thinks, "What are the odds that I would do it, I would win?" The way to think about it is that, before the lottery, the odds that you would win or you would win are very low. But after you've won, the odds are one. You won. You're the one, 100%. That can't be what we mean by miracle.
- 7.7** Biblically and historically, people often spoke of miracles in terms of signs and wonders for all sorts of things, from normal births to virgin births. Recall our reviewed study of the women who were impregnated somehow miraculously without sex, and it turns out that was not the case. Or when it rains or deluges, these are miracles. Or from famines to feasts, these are signs and wonders of the providence of the divine in the world. And of course, if we lived 500 years ago, that's how you and I would have thought. We think in terms of, like with the weather, meteorology and air pressure and forces and winds and the rain and deluges and so on, have natural explanations, but they didn't then.
- 7.8** Or why did the cow die? How come there's a famine? What happened with this pandemic? And so on. We now have science to help us understand that; people at the time didn't. So they described those things as miraculously good things or miraculously bad things. The problem is, if everything's a miracle, then nothing's a miracle, and then the word becomes meaningless.
- 7.9** Let's turn to our old friend, David Hume, one of the heroes of this course, the great Enlightenment Scottish philosopher, who wrote an entire treatise on miracles in his great book, in which he said, "Nothing is esteemed a miracle if it ever happened in the common course of nature." That is to say, "it's no miracle that a man seemingly in good health should die on a sudden because such a kind of death, although more unusual than any other, has yet been frequently observed to happen." Sometimes people just keel over and die.
- 7.10** OK, so continuing with David Hume's *Of Miracles*, he says, "It's no miracle that a man seemingly in good health should die on a sudden because such a death, though more unusual than any other, has yet been frequently observed to happen. But it is a miracle," he continues, "that a dead man should come to life because that has never been observed in any age or country." One of my *Scientific American* essays, I called Hume's Maxim, based on his own quote that he kind of put as a maxim, in terms of the "what's more likely?" question. I've been asking that question a lot in this course. What's more likely? It's this or this.

- 7.11** It comes from Hume. In his, again, this is that 18th-century long-winded way of writing these paragraph-long sentences and so on. But let's see what he wrote and how we can understand this. "The plain consequence is, and it is a general maxim worthy of our attention—" Here, Hume was not a modest man. He's putting his own words in quotes and calling it a maxim. This is funny. "That no testimony is sufficient to establish a miracle unless the testimony be of such a kind that its falsehood would be more miraculous than the fact which it endeavors to establish."
- 7.12** When anyone tells me that he saw a dead man restored to life, I immediately consider with myself whether it be more probable that this person should either deceive or be deceived, or that the fact which he relates should really have happened. I weigh the one miracle against the other, and according to the superiority which I discover, I pronounce my decision and always reject the greater miracle." And close quote, "If the falsehood of his testimony be more miraculous than the event which he relates, then and not until then can he pretend to command my opinion or belief."
- 7.13** In other words, Hume is asking, what's more likely? That someone who is really and truly dead came back to life, or that someone is mistaken about what they think happened? OK, and his point is that we've never seen anybody actually come back to life from being dead. But we see people all the time make mistakes about what they think they saw, what they remember. I had this entire lecture on how thinking goes wrong, and we gave dozens of examples of how observers make mistakes about what they think happened. What's more likely? That's his point.
- 7.14** OK, what is a miracle? He defines it as a violation of a law of nature. All right, so the universe is determined by all these laws of nature. Something happens that violates the law of nature. That's his definition. But not randomly so. It's at the behest or providence of a divine being or a deity or an invisible agent of some kind, a transgression of a law of nature by a particular volition of the deity or some invisible agent. That's his definition.
- 7.15** Now, this was a long time ago that Hume wrote this, and philosophers, as you can imagine, have dissected this and offered their own opinions. One comes from my friend David Kyle Johnson, a philosopher, who clarifies a miracle as, "For any given event, if we knew that God took special care to cause it, we would and should call that event a miracle, regardless of whether it involved a violation of natural law." So he's tweaking Hume a little bit, although even there, if it could have happened by natural law and God intervenes to stir the particles to whatever, cure the cancer or something like that, what would be the difference between that happening and it just happened by randomness?
- 7.16** Now we're back to Sagan's dragon. How can you tell the difference between this and this, if there's no test we could run? Anyway, I'm just giving you a few snippets here to show you how philosophers think about things like miracles. Let's take some of the miracles of Jesus and how we might think about them. That Jesus of Nazareth existed.

This is not a particularly extraordinary claim. There were people named Yeshua at the time. He was Jewish. There was a large Jewish community in the Roman Empire.

- 7.17** That he was crucified is also not an extraordinary claim. The Romans crucified everybody for anything. I mean, this was just their criminal justice system. As one of those country Western songs put it, just hang them all and let God sort it out later. OK. Fortunately, we don't practice that kind of criminal justice, but the Romans did. My friends who are historians of religion and theologians and so on, even the ones that are atheists agree, there was probably somebody named Jesus, who really existed, although there is this little branch of philosophers who said he probably didn't exist, but I don't think they make a good argument. And that he probably was crucified, and we don't need a lot of evidence for that.
- 7.18** On the other hand, that he walked on water, that he turned water into wine, that he raised the dead, and himself was resurrected from the dead, these are extraordinary claims. These, in Hume's way of thinking, they don't happen every day. You just don't see this happening. Now, of course, I have seen somebody walk on water. It was Teller, of Penn and Teller. OK, they're magicians, and it was for a commercial. There's probably this clear plastic platform one inch below the water, and there's Teller walking across the— all right, I know those guys are magicians. Probably a magic trick. We do have to control for those kinds of factors that could explain it.
- 7.19** Or turning water into wine. Again, I've seen magicians turn water into wine. You get this little special cup from the magic store, you put this powder in there, you pour the water in there, it miraculously turns red. OK. For it to be a real miracle, we have to dismiss all those other explanations. With these miracles of Jesus, we have two possibilities. The extraordinary event actually happened. Really walked on water, really raised the dead, and so forth. Or that the witnesses and narrators of the story are just mistaken, exaggerating, hallucinating, misreported. They saw a conjuring trick, or so on.
- 7.20** There's many stories in the Bible about Jews and Christians, don't be fooled by the magicians. There were magicians at that time. I don't want to be fooled by the magicians to think what they're doing is real and not just magic. But how can I tell if I don't know how it's done? That's part of the problem. I tend to give the benefit of the doubt to believers and observers of alleged miracles that they're not lying. They're not just making it up in our time to sell books or get on TV or whatever. I think most people I've met in my 30 years of talking to people and interviewing people in this space, most people genuinely believe what they say happened to them. I've talked to a lot of alien abductees. I don't think they're making it up. I think they've really had an experience, in this case, probably a sleep paralysis-type experience.
- 7.21** Of the 100 billion people who lived before us, there's about 8 billion of us today, all of them died. Not one of them has come back from the dead. The claim that one of them actually did is truly extraordinary, and we can even put a number on it, 100 billion to one. Is the evidence for the resurrection proportional to the 100 billion to one? No.

Remember our principle of proportionality, that we should apportion our confidence in the claim being true in a Bayesian way, according to the amount of evidence. The more evidence, the more confidence we should have, the less evidence the less confidence we should have.

7.22 Again, I'm just talking about how to think about these claims from a scientific, a rational perspective. Later in this lecture, I'm going to come at it as a different way of thinking about it. The evidence for the resurrection is not only not extraordinary, it's not even ordinary. Here I'm just quoting from the philosopher Larry Shapiro, who wrote a book about this called *The Miracle Myth*. "Evidence for the resurrection"—by the way, it's a great book. He starts off in the preface saying, "I'm Jewish. I was raised Jewish. I accept the Old Testament and so on and so forth. We were taught that there will be a Messiah, but it's not Jesus," and so on and so forth.

7.23 Now, he's not particularly a believer at this point, but he gives us some background about this and why. We could take this as a good argument for why Jews don't accept the resurrection, if the evidence and arguments are so good. I'll come back to that. "Evidence of the resurrection is nowhere near as complete or convincing as the evidence on which historians rely to justify belief in other historical events, such as the destruction of Pompeii in ancient Rome. What about the eyewitnesses? We're told that people saw Jesus resurrected, or did they? They said they saw something. Post-death apparition. They think they saw him the next day or the next week.

7.24 Shapiro continues and says, "Well, maybe they were superstitious or credulous, or they saw what they wanted to see." We've already seen in this course a dozen of examples of priming subjects to see what we want the subject to see. I can make you, as I showed you in the demo yesterday, say something that, give a number that I think I want you to give, a higher number or a lower number, but just by priming you. "Or maybe they reported only feeling Jesus in spirit. Over the decades, their testimony was altered to suggest that they saw Jesus in the flesh."

7.25 I've seen this happen a lot. I was at an event recently with a fellow named Rupert Sheldrake, who believes that the soul leaves the body afterwards and that the information that represents who you are does not die with the death of the body and brain. It continues. It always exists in some other space, maybe a quantum field, something like that, he thinks. He was telling me about his next series of research projects on people who are on death's door and dying, who have Alzheimer's or dementia, and they're pretty much just gone for months or years due to the deterioration of the brain.

7.26 They have a sudden spike in sentience and clarity, and they're all of a sudden talking, and the memories all come back, and then they pass and so on. This is what he considers to be evidence for a near-death experience in a way that you continue on after the death of the brain and body. See, he's telling me all this—and that it happened to him. One of his loved ones, who was in the hospital dying, completely

gone due to dementia for a long, long time, all of a sudden, wakes up and starts communicating clearly, as if it was 20 years before, something like that.

- 7.27** I said, "Wow, it's really amazing. I mean, you were right there? You spoke to this person?" "Well, it was during Covid, so I was in another room, and there was a glass partition, and there the person was, and I witnessed this." I said, "You mean they actually just woke up and just popped up and started talking?" "Well, no, no. It was more telepathically. They were just lying there, and I could sense that they were talking to me." I'm like, OK, all right, telepathically. Right. In other words, that's just not the same. Now, you can imagine him writing all this down and leaving that part out because there's nobody like me pressing him. I'm like, "What exactly happened?" It sounds like X happened when in fact, when you push it, it's not quite that clear.
- 7.28** Or maybe the accounts of the resurrection never appeared in the original Gospels and were added in later centuries. Here it helps to have a little information about how the Bible was written. Here I recommend Bart Ehrman, my colleague and friend, and one of the great biblical scholars of our time. He's written a dozen books on these various subjects. Here I'll quote from his book, *Jesus, Interrupted*. Bart shows how, in a way, the Bible is a wiki. It's a compilation, edited over decades and centuries, depending on whether you're talking about the Old Testament or the New Testament, and that the authors are writing about various topics. They're writing of their own time. They're writing against each other. The New Testament authors are writing in response to the Old Testament authors and so on.
- 7.29** Here's how Bart thinks about this. "Our very first reference," he writes, "to Jesus' tomb being empty is in the gospel of Mark, which was written 40 years later by someone living in a different country who had heard it was empty," that the tomb was empty. How would he know? Anyway, suppose that it was empty. "How did it get empty?" he asks. "Suppose that Jesus was buried by Joseph of Arimathea in Joseph's own family tomb, and then a couple of Jesus' followers, not among the twelve, decided that night to move the body somewhere more appropriate." Now, here Bart's just speculating in a Dan Brown novelish way to make a point. He doesn't actually believe this.
- 7.30** Is this scenario likely? No, not at all. Am I proposing that this is what really happened? Absolutely not. Is it more probable that something like this happened than that a miracle happened and Jesus left the tomb to ascend to heaven? Yes, absolutely. From a purely historical point of view, a highly unlikely event is far more probable than a virtually impossible one. Here's a nice example, Bart reasoning like a Bayesian reasoner. Let's put a probability on it. Let's say our credence is high or medium or low, and look at the evidence and make that comparison.
- 7.31** There are people that write—I've mentioned this *Holy Blood, Holy Grail* as the non-fiction version of a Dan Brown novel, about how Jesus didn't actually die and was whisked off the cross and into a tomb, and then he was rescued later and moved and

married Mary Magdalene and so on and so forth. Does any of this explain away the resurrection—no, but it's a way of thinking about, well, what's more or less likely?

- 7.32** "The extraordinary resurrection miracle claim then hangs on two early testimonies from two ancient authors, neither of whom actually saw Jesus rise from the dead, and neither of whom touched him or sat down with him for a face-to-face conversation. Furthermore, we have nothing written by the Romans," or the Jews, for that matter. It's good to remember that Jesus was Jewish. People seem to forget that. It's one of the odder things in the long, dark history of anti-Semitism, that there was this meme going around that "the Jews killed our Lord." The Jews killed—well, first of all, Jesus was Jewish. In any case, it was the Romans that did it, not the Jews.
- 7.33** Finally, if you believe that Jesus had to die for your sins to be forgiven for the original sin of Adam and Eve, shouldn't you be thanking the Jews for killing Jesus? Because that had to happen. That's all part of the cosmic story that God laid down. In any case, that's just a little side bit that, like, what? People also think Matthew, Mark, Luke, and John, that was the name of these. No, these are later authors. The earliest one is, as Bart Ehrman referenced, Mark. That's 40 years after Jesus' death. Then the other Gospel authors, Matthew, Mark, and Luke, they were even decades later. It may be as late as 90 years after the fact.
- 7.34** We've already seen how unreliable memory could be. Again, the JFK assassination, right there, an hour later, people are misremembering what they just saw happen. Imagine that was days, weeks, years, or decades after the fact. In any case, you would think if something that big and miraculous happened, some Roman scribe might have exclaimed something like, "Et tu, again, Jesus? What? You're back?" Or some Jew would have snapped, "Oy vey, it's Jesus! He's back!" We have nothing like that. The accounts from Josephus, Pliny the Elder, and Tacitus that Christians cite and evidentiary support were from the second century. Now we're talking about a century later after the fact.
- 7.35** Did they reference Christ, the Messiah? Again, people say, "Jesus Christ," like it's his last name. People didn't have last names. It's just this person from, Jesus from Nazareth. Leonardo from Da Vinci. OK, from Vinci. The last names are a new thing. Christ, or Christos, is just another word for Messiah. This was prophesied in the Old Testament, and that's why Christians believe it and Jews don't.
- 7.36** Continuing, what about the 500 people, the apostle Paul says, who saw the risen Jesus? Well, note that we don't have 500 independent witnesses, 500 different written sources, or 500 different interviews or whatever, that said they saw Jesus after the crucifixion. We have one account, saying that there were 500 people. OK, that's very different. In terms of our Bayesian assessment of evidence, 500 independent accounts would be way better than one account that says there were 500 people who saw it.

- 7.37 In any case, this was an evangelist who was highly motivated to write a history in support of the newfound religion. Recall that Paul's original name was Saul, and he converted to Christianity on the road to Damascus. This is not an unusual thing to happen. Then, of course, evangelizing and proselytizing in support of a new faith, you're going to support your case by making claims.
- 7.38 It's also good to remember that the idea of history or journalism has a goal of recording an accurate, veridical account of what actually happened. That's a pretty new idea. Really, it's maybe 20th century before people said, "You know what? We should try to be as accurate as we can." Before that, everybody was—it was just clear. You just write the history or accounts in support of what you wanted to happen, what you think it is, in support of our tribe, our nation, our religion, and so on.
- 7.39 Any case, again, to make this point, if the evidence for the resurrection of Jesus is so good, why don't Jews believe it? They believe in the same God, Yahweh. They even believe in the same holy book, at least the Old Testament, which Jews call the Hebrew Bible. They even believe in the Messiah. They just don't think that the carpenter from Nazareth is him. You can watch a show I did with Ben Shapiro, who's a Jew, who is also quite politically active in conservative circles on the Daily Wire now. I did his Sunday special show. He was interviewing me about one of my books, and I said, turned it on him, and I go, "OK, tell me, why don't you accept Jesus as the Messiah? Come on. You believe in God, you believe in the Bible, you believe the holy book, you believe God is active in the world, and so on. Why don't you accept Jesus?"
- 7.40 And he just laid it out for me. That's not what the Old Testament prophesied about what the Messiah would be like. We like Christians, we're all on the same team and so on, but this is just not our thing. My point of that is that I've done a lot of debates with theists who insist that—and here's where I'm going to make the transition to thinking about these slightly differently—they insist that this has to be an empirical, objective truth. It really happened. What's the knowledge, justified true belief? What's your justification for believing that these things happened in reality, really happened?
- 7.41 To answer that, they have to marshal evidence and arguments because I've just asked them, "Tell me your arguments. What's your evidence?" Then they give me their evidence. Like I've just reviewed here, and I counter, "Well, that's not very good evidence. It was 40 years after the fact, 100 years after the fact. The eyewitnesses were proselytized or they might have been biased," and so on. But I don't really need to do any of that. All I have to say is, if your arguments and evidence are so good, how come Jews don't believe it?
- 7.42 Then sometimes I'll hear things like, "Well, if they just understood the evidence and arguments, they would become Christians." It's like, no, they do understand the arguments and evidence. You're telling me that these learned rabbis—this is all they do, is think about the Bible and God and all these big subjects, and they study for a living those topics. They don't understand your arguments? I understand your

arguments, and I'm not a biblical scholar. Come on. Maybe it's because your arguments and evidence are just not that good. OK.

7.43 Here's how Bart Ehrman documents it. "There are many stories in ancient myths about God's temporarily assuming human form to meet, speak, and interact with humans, and that these stories in many ways are similar to later Christian beliefs about Christ being a preexistent divine being who came to Earth as a human, only later to return to the heavenly realm." What Bart is talking about here—this is one of his other books about what the ancient world was really like. Again, the world we live in is so different from the way it was 2,000 years ago, not just in terms of health and technology and so on. I mean, just what people believed about things.

7.44 "To ancient Romans, it was sometimes understood that gods could and would come down to Earth in human form to make a temporary visit for purposes of their own. Sometimes it was understood that a person was born from the sexual union of a god and a mortal, thus that the person was in some sense part divine and part human. Sometimes it was understood that a human was elevated by the gods to their realm, usually after death, and at that point divinized or made into a god."

7.45 OK, his point is that, this idea of being born a virgin, being resurrected, a woman being impregnated by God and so on, that sounds really out there for us, and therefore a miracle. But in fact, what Bart is saying is that this was pretty common. Roman gods and Greek gods, they came down, they had sex, and they threw lightning bolts, and they got jealous. They were just like us, except more powerful, smarter, and so on. OK.

7.46 All right, let's look at some other miracle claims. Let's fast forward to the 20th century. This is the miracle of the apparition of Our Lady of Fatima. It's a very famous—if you're Catholic, Catholics know this story very well. In 1917, in Fatima, Portugal, three children claimed to have seen a vision of the Virgin Mary, who told them, to one of them, that she would return on October 13th to perform a miracle. And according to the kids here, which, again, here they are pictured, the account on the left of this event that happened, they said that 70,000 people gathered and saw—remember, 70,000, that sounds like a lot, but it's one account that says there were 70,000, not 70,000 different interviews by journalists that recorded this.

7.47 Here's what they said they saw. "The sun seemed to tear itself from the heavens and come crashing down upon the horrified multitude. And just when it seemed that the ball of fire would fall upon and destroy them, the miracle ceased and the sun resumed its normal place in the sky, shining for them as peacefully as ever." All right, so here we have our two possibilities again. It really happened. The sun really did move about the sky, it came crashing down toward the crowd; or 70,000 witnesses were mistaken or misrepresented; or the one account that said that 70,000 people were standing there and they all saw it was itself a misrepresentation of what happened.

- 7.48 What's more likely? So now we're back to Hume's "what's more likely" question. You can pretty much answer that on yourself. As impressive as it is that 70,000 people witnessed something this miraculous, it's an even greater miracle that the sun would stop. A little basic astronomy. The sun is rotating at a thousand miles an hour. This is why you want to launch your rockets from as close to the equator as you can. The closer to the equator, thousand miles an hour, and you want to launch your rockets eastward because the sun, Earth is turning that way, so you get a little boost like that.
- 7.49 What would happen if you just stopped it? Well, the oceans would just go sloshing right off the Earth. Wouldn't people have noticed this? 1917. We had photography and journalists and writers and observers everywhere. No one else reported anything unusual like this happening. If it did, there would be catastrophic consequences. Again, you'd see, oh, my gosh, at that particular time, at that particular day, people everywhere in the world were talking about this incredible thing that happened, and there's not a single account like that. That tells us something else is probably going on, not a miracle of this Earth stopping and so on.
- 7.50 OK, let me just give you a fun example of an apparent miracle and how to think about it. Penn and Teller's double bullet catch. How should we think about this? This is a little bit of a tongue-in-cheek analogy, but I think it makes my point. I was there the first time they did the double bullet catch at the CBS studios in Los Angeles, in which they have a studio like this divided up with a big wide line that goes right down middle. They have a studio audience, volunteer from the audience on this side, volunteer from the audience on this side.
- 7.51 "Do you have some expertise or experience with firearms?" Yes. OK, get this person, get that person. I happen to know Penn and Teller never use plants. So these truly are random people. Penn says he doesn't do it because it costs too much to pay a plant. Because that would take the fun out of it. They don't need to do it because they're doing it some other way. Then Penn stands over here on the other side of the yellow line. He stands over here. The volunteers come up. They're allowed to choose any bullet they want from this case of bullets. They can inspect the gun.
- 7.52 Each of them takes one of the bullets out of the pack, and then Penn instructs them to write on the bullet casing, some, like, happy face or their initials or anything they want to write on there and also put something on the bullet itself. You have the bullet case and the bullet. Write something on there, a little happy face or your initials or whatever. Same thing on the other side. Then the volunteers sit down, the guns are loaded. Penn and Teller stand there, and they point the guns with their little laser pointer at each other. They have two panes of glass between the two of them. Then Penn says, "Hold your ears because here it comes."
- 7.53 357 Magnums. Then bam, guns go off, the glass breaks. Both of them go like this. Then they had the bullet right there in the tooth. Volunteer from the audience, here take this, and so on. Then you come over here. Is that your bullet? Is that your bullet? Yes.

The bullets really change positions. The bullet catch really happened. OK. So, two possibilities. Penn and Teller really catch bullets in their teeth, fired from 357 Magnum handguns every night at the Rio, five nights a week in Las Vegas, or it's a magic trick. It's Penn and Teller. It's what they do.

- 7.54** The fact that we don't know how they did it doesn't make it a miracle. It just means that was a cool trick. It looks like a miracle, but it's not a miracle. Let me just comment on this because I've known magicians since I've been in the skeptic business. It's a nice connection between skepticism and magic because magicians know how to fool people. That's the whole point of what they do. They know a lot about human psychology and how easy it is to deceive people if you know the tricks. Again, these simple things, like if I take the coin and I go like that, I'm not actually moving the coin to this hand, but because I'm looking at it, you're following my hand.
- 7.55** There's a hundred things like that that magicians do that make people think one thing is happening when it's not actually happening. When Copperfield says, "OK, I'm going to move the Statue of Liberty," does anybody think he's actually moving the atoms that make up the statue? No. Everybody knows it's a trick and they enjoy it. But for some reason, close-up magic, things like spoon-bending— I remember when Uri Geller was doing his, Israeli psychic, was doing his spoon-bending bit, it's pretty impressive how he does it, but it's a magic trick. I've since then had magicians show me how to do it, and then I practiced it and I started doing it. But for some reason, a lot of people think, well, that's not a magic trick because it was right there in front of me. I saw the spoon bending or the key bending or whatever.
- 7.56** Anyway, so funny story. I was on one of these cruise ships through the Bermuda Triangle, so to speak, for a Scientific American cruise, and I was one of their guest speakers. I'm telling this story at the dinner table one night, like the stories I just told you, but without the spoon-bending part. And he goes, this guy, "OK, I'm the skeptic, too. But I saw this guy do this thing with this spoon. I can't explain. I mean, it truly was miraculous. I mean, he didn't bend the spoon like this way. He bent it like this way. I mean, I've never seen anything like—" As he's telling me this, I take the spoon on my table, I bring it down here, and I do the little move that the magicians told me.
- 7.57** You basically take the spoon thing and you go like that. It looks like the neck is twisted this way, but in fact, it's twisted this way, which is easy to do. You couldn't do it that way. As he's finishing the story, I said, "Did it look anything like this?" And he goes, "Oh, no. Crap, darn. I got fooled." And it's like, yeah, the point is that even skeptics, scientists, and so on, when they can't explain something, the temptation to think "this is paranormal, this is supernatural, this is miraculous."
- 7.58** OK, what if the miracle accounts we've been talking about are not to be treated as empirical truths, but as something else? OK, back to the first lecture. I talked about different kinds of truths. What if these stories—and that's what they are, they're accounts, they're narratives, they're stories— are treated or should be treated as

mythical accounts? That is, what if they're not meant to be taken literally, but rather mythically? Or metaphorically, or psychologically? Now the stories become, in my mind, a lot more interesting and important.

7.59 Like, take the account of Apollonius of Tyana. This is a prophet who lived in the first century AD, close to Jesus' time, slightly after, a few decades after Jesus. Apollonius' followers claimed he was the son of God. He could walk through walls, and he could heal the sick. He could cast out demons. He raised a dead girl back to life. He was accused of witchcraft. He was sent to Rome before the court, and after he died, his followers claimed he appeared to them and then ascended to heaven. It's a very similar account to the Jesus account.

7.60 Now, Christians today say, well, he ripped off the Jesus story and copied it. OK, maybe. But in fact, resurrection myths predate the Christian story by a lot. Let me introduce you to Osiris, the ancient Egyptian god, who was the Egyptian god of life, death, and fertility, appears in pyramid text 2,400 years before Jesus, before the Common Era, was said to be the giver of life in this world, the redeemer and merciful judge of the dead in the next world. Egyptian kings believe that, as Osiris rose from the dead, so they would as well, in union with him, inheriting eternal life.

7.61 Now, initially, it was only the pharaohs who got to be resurrected and live forever in eternal life. In the later kingdoms, by the New Kingdom, everyone believed that if they accepted Osiris as their God, they too would be resurrected from the dead. I think it's always good for kings and pharaohs and dictators to give the regular people some hope for the future. They work a lot harder on your pyramids. Just a little side note.

7.62 Or take virgin birth myths. Here are a few of the ancient gods who are said to have been born of a virgin. Dionysus, Perseus, Buddha, Attis, Krishna, Horus, Mercury, Romulus, and of course, Jesus. What is it about virginity that these guys are so obsessed with? You could just take that all the way up to post-9/11, and we go, "Why do they do this?" Because they get to have the 72 virgins in the next life. What is it about that? That's another story. Dionysius is the ancient Greek god said to have been born from a virgin woman. Again, let's go back to Bart Ehrman, saying, well, actually, that was common in the ancient Greek and Roman world, that gods came down to heaven and so forth.

7.63 Said to have been fathered by the king of heaven, transformed water into wine, introduced the idea of eating and drinking the flesh and blood of God, was the liberator of his people, and then returned from the dead. Flood myths are also not that uncommon in ancient times. The classic epic of Gilgamesh comes from ancient Babylon. 1800 BCE, that's 400 years before the Noachian flood story was written down in what we now call the Old Testament or the Hebrew Bible. The Babylonian god Ea warns that the gods were about to destroy all life by a flood. Utnapishtim is instructed to build an ark, take aboard one pair of each living creature, and so on. OK.

- 7.64 Why do these stories keep recurring? Let's channel our inner Joseph Campbell or maybe Jordan Peterson here, since this is his academy, and think about, what is a myth? Well, the Oxford English Dictionary defines it as a purely fictitious narrative, usually involving supernatural powers, actions, or events, and embodying some popular idea concerning natural or historical phenomenon. I like Joseph Campbell. I went through my Joseph Campbell stage in the 1980s. I read all his books, *Hero With a Thousand Faces* and *The Power of Myth*, and I really got a lot out of that.
- 7.65 Again, just autobiographically, I was not raised religious, just in a secular home. Then I became a born-again Christian when I was in high school, and I went to Pepperdine University's Church of Christ school. I was totally into it for about seven years. Then I lost my religion for various reasons. Then I started studying myths. Well, I took a comparative mythology course, a comparative world religions and anthropology, when I was in graduate school, just for fun. Discovered Joseph Campbell and went, oh, everybody's got mythic stories. What are the chances I got the right one and all these other people are just telling stories? Or maybe they're all stories. Maybe these are not empirical truths like science, these are something else. That's what I think is the case.
- 7.66 Anyway, in Campbell's great work, *The Power of Myth*, he offers four functions of myths. What is the point of these stories? Well, one is mystical. These stories serve to awaken and maintain the individual sense of awe and gratitude in relation to the mystery dimension of the universe. Not so that one lives in fear of it, but so that he recognizes that he participates in it. Maybe this hints a little bit about how I closed the last lecture, about just bad things happen in life. Entropy, it's just the way things go. We don't need some explanation in the form of gods or demons making things happen, or witches. They just happen. We have to deal with it ourselves, and that's what these mythic stories help us do. Now, come to grips, why do these things happen? What can I do about it?
- 7.67 Then there's explanatory myths, an image of the universe which will be in accord with the knowledge of the time, the sciences and the fields of action of the folk to whom the mythology is addressed. Joseph Campbell's third purpose of myths is normative. Normative, to validate, support, and imprint the norms of a given specific moral order, that namely of the society in which the individual is to live. In other words, how should I live? What should I do? I want to do the right thing. What is the right thing? Well, these myths, stories, narrative accounts help us deal with that.
- 7.68 Finally, related to that, guidance. To guide him stage by stage in health, strength, and harmony of spirit through the whole foreseeable course of a useful life. OK, so a little shoutout to Jordan here. I see he has a book coming out in 2024 called *We Who Wrestle With God*. He posted on Twitter a few snippets of this and a few other sources that nicely tied in to my point here. I put this slide in to give him a little bit of a shoutout here, because I think he is the Joseph Campbell of our time. Here's what Jordan writes about this issue of wrestling with God.

- 7.69 "The Biblical corpus, like all collections of fascinating and memorable literature, presents a series of romantic adventures—a series, alternatively, of hero narratives" — like Campbell's *Hero With a Thousand Faces*—"which follow precisely the pattern that makes them narratives or stories instead of mere listings of dry factual events. In other words, mythical truths inhabit a realm different from that of empirical truths, and as such include archetypes representing eternal truths about the human condition."
- 7.70 Here's what Jordan writes in conclusion about his point here. "There is a uniting meta-narrative. It exists as a platonic ideal behind any corpus of stories, broad enough to be truly representative of stories per se. It has a structure which was not arbitrarily constructed by an author or group of authors. Its structure is the central adaptive pattern of mankind, the representation in dramatic imagination, and then an explicit semantic formulation of the nature of man and woman as such."
- 7.71 This, I think, is insightful because I think people are reading these accounts of miraculous events and so on in the wrong way. They're taking them literally, and I think that's a big mistake. They're missing the point. In this sense, to ask if the characters and events in a story are real is to miss the point of the story. I mean, were there really Karamazov Brothers of the 19th-century Russia, about which the Russian author, Fyodor Dostoevsky, wrote in his 1880 work, *The Brothers Karamazov*? What a ridiculous question. It's a novel, right, that probes the depths of human morality, patricide, free will, and God in the context of a modernizing Russia in the late 19th century.
- 7.72 To ask if there really existed Karamazov brothers, Fyodor, Dimitri, Ivan, and Alexei, is no more relevant than wondering if there really is a Middle Earth in J.R.R. Tolkien's fantasy work, *The Lord of the Rings*. You read the *Lord of the Rings*. OK, where is this? Middle Earth. Is it in the middle of the Earth? Where should I go? Like the story of Atlantis. You've missed the point of the story. You're asking the wrong question. If you want to be really ridiculous about it, is there really a King's Cross Platform Nine and Three Quarters, visible only to non-Muggles as they begin their journey to Hogwarts, as J.K. Rowling outlined in her *Harry Potter* books? No, of course not. What an idiotic question.
- 7.73 Searching for natural explanations for apparently supernatural events, I'm claiming, is missing the point. Things like the parting of the Red Sea, or Jonah and the Great Fish, or Jesus' resurrection, to come back to that. So in the 30 years of publishing *Skeptic* magazine, every once in a while, I get articles submitted for publication consideration. "I think I figured out the natural explanation for, say, the parting of the Red Sea. You see, there was this big earthquake that happened, and it shook things, or there was a meteor strike and that caused them." And they go on and on about this.
- 7.74 They're interesting. It's like, well, maybe. But what if somebody just made up the story to make a point about the oppression of the peoples, and they're moving from one place to another, divine intervention because we're destined to be free, and so on and

so forth. Or did Jonah really—how does a person survive inside of a whale or great fish or whatever it's supposed to be? On Twitter yesterday, there was a picture of this guy that got swallowed by one of those huge whales, and then he survived for hours in there, and he managed. He is still alive. There's a picture of him on Twitter in the hospital bed. Wow. OK.

- 7.75 But could you last for three days inside the belly of a fish? Well, OK. Of course not. But you're missing the point of the story. That's not what the story is about. I'm claiming that the story about Jesus' resurrection, like that of any of the other myths, has some deeper meaning. It's a meaning I explored in my book, *How We Believe*, which I called the oppression redemption myths. There are lots of these. We've just covered some of the ancient Roman ones. This is the most famous one in modern times, the Native American ghost dance in 1890, as the western United States was closing. By closing, I mean closing off to the natives who originally lived there and the migration of Westerners, East Coast people, into the West and so on and so forth.
- 7.76 As you know from history, the treatment of the Native Americans by the US government and the military and civilians, European immigrants and so on is not good. It's a really dark part of our history. It's really bad. You only have to read a few books like *Bury My Heart at Wounded Knee*, and then just from there, just go down that pathway of literature on this. It's terrible. In 1890, this gentleman, Wovoka, got a vision during a solar eclipse. He might have been having some magic mushrooms and ayahuasca, who knows? Something else was going on there.
- 7.77 But he had a vision, like, the Messiah is coming. We're going to have a second coming. The white men are all going to disappear. Our ancestors are all coming back to life out of the ground. The buffalo will return. This is at the height of the massacre of buffalos that almost caused their extinction. If you wear this vest, it's bulletproof, that white men's bullets cannot penetrate this vest. OK, this resulted in huge gatherings of Native Americans and their tribes, in which they began to do this ghost dance. That if they dance long enough into the night and so on, day after day, week after week, these miraculous events will happen. The buffalo will return, the white men will disappear, our ancestors will rise from the ground and so on.
- 7.78 The US military became very concerned about this. They went to investigate, they sent soldiers with guns and so on and so forth, and one thing led to another. It's not clear exactly who shot first or a gun just went off. Then there was a massive massacre that left most of the Sioux Indians on the reservation dead. This is the massacre at Wounded Knee that's so famous. It resulted from this ghost dance, from this belief that we are oppressed and we're going to—something good is going to happen. The Smithsonian sent its chief anthropologist, James Mooney, out to investigate what happened, and he wrote an account about this in 1896 called *The Ghost Dance Religion and the Sioux Outbreak of 1890*. Here's how he put it rather poetically.

- 7.79 "When the race lies crushed and groaning beneath an alien yoke, how natural it is the dream of a redeemer; an Arthur, who shall return from exile or awake from some long sleep to drive out the usurper and win back for his people what they have lost. The hope becomes a faith, and the faith becomes the creed of the priests and the prophets, until the hero is a god, and the dream a religion, looking to some great miracle of nature for its culmination and accomplishment. The doctrines of the Hindu avatar, the Hebrew Messiah, the Christian millennium, and the Hesunanim of the Indian Ghost Dance are essentially the same and have their origin in a hope and longing common to all humanity."
- 7.80 A similar story in a Maori village in New Zealand in August 1934. This is not an actual picture of this. This is a more modern descendant of the Maori. A visionary member of the tribe had a dream in which an angel told him that a holy ghost would deliver his people from the whites and return their confiscated lands to them. If you're not familiar with this history, it's just as dark as the American history, that is, the almost complete eradication of the natives who first migrated to Australia and New Zealand, probably at least 60,000 years ago.
- 7.81 The whites didn't get there until really a few centuries ago. Then wiped them out, again, through guns, germs, and steel, as Jared Diamond put it. But whatever the cause of the death was, they were almost gone. By 1934, we have this vision. For days following the dream, the Maori fasted, chanted, danced, and waited for the day of deliverance. Then Maori villagers gave away their belongings, and white administrators became concerned when they got wind of the ceremonies and came to investigate. Finding starving children and deprivation-crazed adults, they declared the visionary insane and shipped him off to a mental hospital.
- 7.82 Oppression, redemption. Or think of the cargo cults that grew up in the southern Pacific following World War II. During World War II, the US military was bringing cargo to all these different islands in which most of the natives had no idea what an airplane was or how this could possibly work. All they knew is these godlike figures showed up with all this cargo. John from America, John from America. So he became John from. And these natives started to, after the Americans left, after the war, started to construct airfields and faux airplanes like this, believing that somehow, if we go through the motions and we construct these modern Western-looking things like airplanes and airfields, the cargo will come. It's the same kind of magical thinking there.
- 7.83 Or just to go even more recently, around the same time as the Maori Messiah was having visions, in India, there was the Shirdi Sai Baba, who performed miracles witnessed by and attested to by many people, including his ability to levitate, read minds, perform exorcisms, cure the dying, and even raise the dead. Shirdi Sai Baba, who's thought to be a guru, a saint, and a fakir who many millions of Hindu and Muslim followers believe is the incarnation of Lord Shiva.

- 7.84 Again, when people in my debates say, "Well, what about these claims by the followers of Jesus? They can't just be making this stuff up, can they? I mean, it must have really happened." Well, but then you put it in context. Well, what about this person and this account and this story and so on? It's very similar. People in the West don't believe that Shirdi Sai Baba was actually able to perform these miracles. But of course, his followers do. Or if we want to just take modern Muslims, and today there's about 1.5 billion Muslims today. They believe in Muhammad's miracles just as strongly as Christians believe in the miracle of Jesus.
- 7.85 For example, that Muhammad split the moon; caused blindness to these warriors assembled at his door to assassinate him; caused a horse of an enemy pursuing him to sink into the mud; quenched the thirst of thousands of his soldiers; caused it to rain during a drought in Medina. He could understand the language of animals; didn't cast a shadow. He could heal the sick and cure the blind only by touching the patient. And then, of course, most famously, at the end of his life, he flew on a winged horse from Mecca to Jerusalem on his way to heaven.
- 7.86 Again, I ask my Christian interlocutors in my debates, "Do you believe any of this?" And their answer is, "Well, no, of course, of course not." But 1.5 billion Muslims believe it. What's the difference? If you were an anthropologist from Mars and you came here to study different religions on Earth, and you saw that, well, 2 billion people believe the Jesus story, but 1.5 billion people believe the Muhammad story, which is the right one? Each side says, "Well, we're the right one." How do you know which is the right one? "Well, we really, really believe it. We have 2 billion people." Well, they have 1.5 billion. Do you have any evidence for this? "Well, we got this story. We got this account, and people said they—" Yeah, I know. Of course, the anthropologist from Mars is facing not just two stories. There's thousands of stories like this.
- 7.87 Let's reexamine the crucifixion resurrection story as a type of oppression, redemption myths. Again, I'm not trying to discount it or take away anything from those who believe it. It's just, what does it represent? I think it's a deeper, more meaningful story if you treat it mythically, rather than empirically. By the first century, the Jewish people were engulfed within the Roman Empire and feared for their existence. The regions around and including Nazareth were ruled by King Herod, who responded with violence to numerous Jewish revolts. And by CE 6, the Common Era 6, Judea was under direct Roman rule.
- 7.88 Now, when Jesus was born, it's a little bit under dispute, but we know that by, let's say, 27 to 30 CE, he was doing his preaching. A youthful Jesus must have been painfully aware of the tensions between his people and their oppressors, as well as the Biblical promise of a Messiah. It's right there in the Old Testament. It's right there in the Book of Daniel. The Messiah is going to come to rescue our oppressed Jewish communities. They would drive out the Romans and reestablish the Kingdom of God on Earth. By the time of Jesus' three-year ministry, from CE 37 to 30, more or less—we're not super sure

of the exact dates, but close to that—after which he was executed by the Romans, Jesus had codified a new theology, an eschatology, to sustain his followers.

- 7.89** OK, here's how theologian Burton Mack identified in this new theology three interconnected ideas that arose following Jesus' death. Burton wrote a great book on this. One, a perfect society conceptualized as a kingdom. The Jesus people latched onto this idea and acted as if the kingdom they imagined was a real possibility despite the Romans. They called this the Kingdom of God. Any individual, number two here, no matter of what extraction, status, or innate capacity, was fit for this kingdom and could act accordingly, if only one would. In other words, this is for everybody, not just for the true believers. That was very more universal as a religious ethic. So the Christianity expanded the moral sphere in that sense.
- 7.90** Number three, the novel notion that a mixture of people was exactly what the Kingdom of God should look like. This was a notion that many groups had used to imagine a better way to live than suffering under the Romans, Burton Mack says. OK, so what if the greatest religious truth in all of Western Christendom, that if you accept Jesus as your savior, you go to heaven where you will spend an eternity with God, was never meant to be taken literally? OK, at this point, you can imagine that Christian theologians go, "Hang on, Shermer. No, no, no, no. Let's go back to the empirical basis. Here's my argument." OK, hang on. Just let me finish.
- 7.91** What if it has a deeper meaning, if you don't take it literally? Perhaps it illuminates this passage in Matthew 16:28, in which Jesus told his disciples, "Verily I say unto you, there be some standing here, which shall not taste of death, till they see the Son of Man coming in his kingdom." A lot of atheists like to quote this and go, "Well, where is he? He said this 2,000 years ago and he's still not here." I think that's the wrong statement to make by atheists. I think, again, they're missing the point, too.
- 7.92** Maybe the kingdom to which Jesus refers to is the heaven within ourselves or the heavenly communities that we build here on Earth. This may help illuminate the passage in Luke 17:20-21, where Jesus seems to infer that heaven is a state of mind. How else would you read this? "And when he was demanded of the Pharisees, when the kingdom of God should come, he answered them and said, the kingdom of God cometh not with observation; neither shall they say, lo here or lo there! For behold, the Kingdom of God is within you."
- 7.93** And this is the conclusion I wrote in the end of my book, *Heavens on Earth: Scientific Search for the Afterlife, Immortality, and Utopia*. I think both religious people and scientific people, these are the people that want to have you chronically frozen or upload your connectome into the cloud and turn you on, you get to live forever. All these tech billionaires that are working on these, so you get to live. OK, maybe. But I think you're missing the point in trying to do this, both religiously and secularly, because "heaven is not a paradisiacal state in the next world, but a better life in this world. Heaven is not a place to go, but a way to be. Here. Now."

7.94 I mean, it bothers me to no end. Even a lot of my scientific colleagues are just obsessed with this whole thing about living forever. It's like, dude, don't forget to live now. I mean, now. You're here now. Don't miss it. Oh, I'm thinking about in centuries to come, I'm going to be alive. How about just think about tomorrow? That might be a good step, right? "Since no one, not even the devoutly religious, knows for certain what happens after we die," remember Cromwell's rule— in Bayesian reasoning, never assign a zero to one, you never know—"Jews, Christians, Muslims, and everyone else might as well work toward creating heavens on earth."

Lecture 8: The Question of God

At a glance — The question of God.

- 8.1** All right, everybody. Welcome back. This is lecture 8 of our eight-lecture series. We're at the finish line, almost. Again, this is the Skepticism 101: How to Think Like a Scientist course. I want to thank the Peterson Academy and Jordan for this opportunity. I do think this is part of the future of education, is that it should come in many forms under many platforms, in which you don't have to go to Harvard to get a good education. You could do it many different ways. I love consuming online content. I walk around with an earbud constantly when I'm driving, running errands, working out, whatever. It's a great way to be an autodidact. I hope that people watching this use it that way as well.
- 8.2** This last lecture is the truth about God. How should we think about the God question? I've written two books that address this, *How We Believe: Science, Skepticism, and the Search for God*, and *The Believing Brain*, in which I address a lot of the issues here that I'm going to discuss today, as well as in my next book on truth, which is going to have a whole chapter on the God question.
- 8.3** First of all, what are we talking about here when we talk about God? Lots of people mean different things by this. I think generally, in the West, most people think of God as omnipotent, that is, all-powerful, all-knowing, omniscient, all-good, omnibenevolent, who created out of nothing the universe and everything in it, who is uncreated and eternal, a perfect non-corporeal spirit who created, loves, and grants eternal life to humans. Now, the definitions of God vary slightly on this, but not much. I've done dozens of debates with theists on the question. Mostly, more or less, this is what they mean when they talk about God, and most Westerners mean that as well.
- 8.4** Then the question becomes, do you believe that this God exists? That is, you're a theist, it's theism. Do you believe that this God does not exist? That's strong atheism. That is, you assert that there is no such God. Or do you believe that you just simply do not believe that this God exists? This is called weak atheism. There's two kinds of atheism. Strong, I believe there is no God, versus weak atheism, I just don't believe in God.
- 8.5** Or do you believe it's not possible to know if God exists? This is what agnosticism means. This word, agnosticism, is fairly new. It was coined in 1869 by Thomas Henry Huxley, who was a colleague of Charles Darwin. He was an anthropologist and biologist and social commentator on all kinds of issues related to science and religion and morality and so on and so forth. Here's what he said about why he coined this word. He explained what the word means.
- 8.6** "One who holds that the existence of anything beyond and behind material phenomenon is unknown, and so far as can be judged unknowable, and especially that a first cause in an unseen world are subjects of which we know nothing." He said, surveying the landscape of what people believe about on the God question, you have

these people that are certain that there is a God, and there are people that are certain that there is no God. What about those of us who just don't know?

- 8.7** What he meant by agnosticism is not just, I don't know now. I'm just not sure I haven't studied it, climate change or gun control or whatever. I don't know. I'm agnostic at the moment, but I'll wait and see if more information comes in, and I'll then make my decision. He doesn't mean that. He means that it's not knowable. You can't know in any scientific sense. We'll come back to that position of agnosticism.
- 8.8** There's a few other positions you could take on the God question. There's actually dozens, but these are the big ones that people that are not strictly theists or atheists sometimes hold. Deism, the belief that the existence of a supreme being, but not necessarily a personal God. A lot of the founding fathers, like Thomas Jefferson and Benjamin Franklin and a few others, they called themselves deists. There was pretty much no such thing as atheism in the 18th century.
- 8.9** There was a famous quote from David Hume, who attended a dinner party and said, basically, "I think I'm the only atheist here," and they're like, "Well, there may be some more, but no one wants to talk about it." It's like, that's not a word that people use. That's very much a late 20th-century position. Deism is the more common one that people who were not strictly theists any time before, let's say, the 20th century, held that position.
- 8.10** Or related to that, pantheism, the belief that the universe itself is a divine unity. Now, notice, neither one of these has an anthropocentric position on this. There's no personal God. There's no agent out there that's sentient, conscious, thinking about us and so on and so forth. They mean something different than that definition I gave of God. Then my favorite, amusingly, this is really a position people take, apathyism. They just don't care. "I'm apathetic about the whole thing." OK, well, assuming you're not an apathiest and you care about whether or not God exists, what should you believe and why?
- 8.11** OK, recall we began this course in lecture one by acknowledging that I'm not God, and neither are you. So no one knows for sure if there is a God or not. OK, given that, why is it that people believe? Well, we believe a lot of things, even when we don't have enough evidence, so there's no reason why that can't apply to the God question. I've explored this considerably, again, in my books and some papers that I've published, why people believe in God. Well, it turns out, just like in any question about human existence, society, human behavior, and so on, there's never one causal variable. As we saw, it's never just X causes Y, end of the story. Never like that.
- 8.12** In the case of belief in God, we know that there are genetic factors. There's neuroscience, how the brain operates factors. A predictor of somebody's religious beliefs is that of their parents, parent beliefs. If you have parental conflict, you're likely to believe something different than what your parents believe. Education seems to

have some influence on belief in God and religiosity. Age, people change as they get older, their religious beliefs and their commitment to religion. There are intellectual and emotional reasons people offer for why they believe and why they think other people believe. Then there is, of course, history, culture, geography, and so on.

- 8.13** I'll go through each of these fairly briefly because it's an interesting subject. Genetics, is there a God gene? There's actually a book called *The God Gene*. Is God in the genes? Remember Betteridge's law of headlines? You can answer the question with the word no. There are influences, of course, from complex gene sequences and so on, but there's no God gene. In fact, there's not a gene, one gene, for anything. There's no gay gene or height gene or so on. Even height, it turns out, there's thousands of genes that have to come online that make you taller or shorter or whatever, because of all the different changes in your body that have to be made in the bones and ligaments and muscles and so on, to determine how tall you are.
- 8.14** If it's that case with something as simple as height, it's just an obvious physical thing, it's got to be way more complex for things like intelligence and temperament and personality and belief in God. Nevertheless, again, back to our behavior genetic studies with identical twins raised together, identical twins raised apart, siblings raised together, siblings raised apart, and so on, these studies indicated about 50% of the observed variance in religious attitudes and belief in God arise from genetic factors. That is, about half the differences between people and their religious attitudes, interests, and values is accounted for by their genes.
- 8.15** Again, I need to qualify this again. You can't say it enough. I'm not saying that 50% of your genes shape your religious beliefs. It's 50% of the variance between people. Let's say this group or that group is going to be accounted for by genes, the rest is environment. It's an on-average thing. Again, not for you or you, but for the whole collective population we're studying. Then there's, of course, neuroscience—how the brain is wired up to see certain things. We've encountered these already. My idea of patternicity, the tendency to see meaningful patterns in random noise, and agenticity, the tendency to infuse those patterns with meaning, intention, and agency.
- 8.16** So, pretty much in the entire history of humanity, the belief in spirits and animism or the belief that everything in the world is animated, trees and rocks and the animals and the clouds and everything, it's kind of a pantheistic-type belief, although it's more commonly seen as this animism, or maybe polytheism, where there's lots of different gods. But the idea is that nothing happens randomly. Everything happens for a reason, and the reason is being directed from outside by some source.
- 8.17** You connect the dots and the patterns, everything happens for a reason, and that reason has to do with this agent outside. Whether it's a monotheistic God as I defined, or polytheistic gods, or the entire universe is constructing things to happen a certain way, that's what most people think. I think our brains are wired up to go in that

direction. Therefore, not believing there's any pattern or not believing there's any agency behind things, I think that's unusual.

- 8.18** Let's talk about your parental influences. This is a study conducted by my friend and colleague, Frank Sulloway, at UC Berkeley. We undertook studies to see to what extent religiosity, the curve on the left, or belief in God, the curve on the right, is predicted by one's parental beliefs. That is, if you were raised religious, and religiosity goes up, the more you were raised religious, the more religious you are, the more your parents believed in God, the more you believe in God. Obviously, this is not exactly like a shocker. You would expect that because that's true for most things.
- 8.19** Unless, as Frank and I saw, thanks to his wisdom to ask the question in our survey, what was your relationship like with your parents on a one to seven scale? Would you describe your relationship with your parents, good or bad or in between or so on? We then correlated that with the subjects' religiosity scores, by which you can objectify by measuring how often do you pray, how often do you go to church, do you have a Bible or a holy book in your home, and how important is religion to your personal life? And so on and so on. You can code those and correlate them with measures like, to what extent you're in high or low conflict with your parents.
- 8.20** Here we get an interactive effect. So, if you're raised religiously, but you're in high conflict with your parents, your religion and belief in God goes down. If you're in a good relationship with your parents, then you're more likely to adhere to what they believe. That makes common sense. People go in the same direction, opposite direction, depending on how good or bad their relationship is. We found a small effect with education. The more education somebody has, the less religious they became or the less likely they are to believe in God. Not a huge amount. It just goes from a 4 to a 3. It's not a large measure on a 10-point scale, but a little bit of an effect there.
- 8.21** The age effect we found was quite interesting. Religiosity and belief in God goes down throughout the age, throughout the life span, until around age 80, and then all of a sudden it goes back up. As we jokingly said, what, is he cramming for the final, just in case? Then Frank and I also included with our survey a handwritten essay portion. That is, we asked people, "Why do you believe in God?" Just in your own words, just type it out or write it out and tell us. Then we also asked a second question. "Why do you think other people believe in God?"
- 8.22** We got a pretty interesting effect here that I'll have to deconstruct for you here. The first question, why people say they believe in God, is in the white bars. Then why people say they think other people believe in God is in the darker blue bars. The effects here are basically to summarize a fairly complex data chart, is that having good reasons or good design of the universe is a huge reason why people say they believe in God, but pretty low for why they think other people believe in God.

- 8.23** The same thing with the experience of God in everyday life. That was very powerful for people who say why they believe in God, less so for why they think other people believe in God. But look at this one. This is, brings comfort and purpose to life. Not so much the reason I believe in God, but other people, it brings comfort, or they were raised to believe. Huge effect for why people think other people believe, but not why they themselves believe. Just think about that for a minute.
- 8.24** If I ask you, "Why do you believe in God?", most people don't say, "Well, my parents believe it, so I guess I believe it, or it just makes me feel better." Most people don't say that. They say things like "the good design of the universe," and so on. I'll show you some of these reasons people wrote us. This is an attribution bias. "I believe for these kinds of reasons, but other people believe for other reasons. My reasons are more intellectually based. I recognize in others the influence of emotions and upbringing and culture and so on." It's an interesting attribution bias.
- 8.25** Here's just three examples. Here's a 30-year-old male Jewish teacher with strong religious convictions. He self-reported an 8 on a 9-point scale of religiosity. Told us he believes in God because "I believe in the Big Bang. When you believe in the Big Bang, you have to ask yourself, what came before that? A creation implies a creator." Yet, when we asked him, "Why do you think other people believe in God?", he said, "I think most people believe out of an emotional need, although there is a significant minority of rational, even skeptical believers such as myself." Again, those people believe for these reasons, but I believe for these intellectual reasons.
- 8.26** Here's another one, a 37-year-old female Catholic with strong religious convictions. She rated herself as an 8 on a 9-point scale, said she believes in God because how else could you explain our origins? "Only God could create a world and universe out of nothing. There are miracles every day that science cannot explain." But when asked, "Why do you think other people believe?", she said, "Because it gives them hope."
- 8.27** Finally, third example, a 41-year-old male Baptist, strong religious convictions. He rated himself as a 9 on a 9-point scale. He told us that he believes in God "due to the evidence of his magnificent creation and the extraordinary order of the universe." Whereas other people believe, he said, "because without God, there's no purpose for their lives or the universe." I just find this so fascinating. It's just so much what's called the attribution bias. I attribute to myself certain kinds of motives and values, but I don't attribute those the same to other people.
- 8.28** We've seen this affects like, why are you successful? Why do you think other people are successful? Most people go, "I'm successful because I'm hardworking and creative," and so on. Why do you think other people are successful? "Well, he was at the right place at the right time. He got lucky and he was influenced and he inherited money." In other words, they see in themselves internal values; they see in others external values. There's a whole body of research on attribution bias in psychology. It's just fascinating.

We just see the world differently for ourselves than we do others. Frank and I made this discovery, intellectual reasons versus emotional reasons to believe.

8.29 Then, of course, our historical, cultural, geographical factors. Just think about it. Had you and I lived 10,000 years ago, nobody would be a Jew, a Christian, or a Muslim, since no such beliefs existed anywhere in the world. Had you been born today, say, in India, China, or Japan, instead of the United States or some other Western country, chances are very high that few people would be Jewish, Christian, or Muslim. Even in the West, there are over 10,000 distinct religions of 10 general types, each of which can be further subdivided and classified.

8.30 You can just look at any of these encyclopedias of religious beliefs. There's lots of them. Here's just one map. This is a map of religions around the world. In the blue, it's mostly Roman Catholic, in Mexico and South America and parts of Africa. The red is mostly Protestant. Yellow is mostly Eastern Orthodox. Then down here, you have the Shia and Sunni Muslims. Then in the east, you have Hinduism, Judaism, sorry, Hinduism, Buddhism, Chinese religions, and so on. OK.

8.31 The fact that you can have a map of different religions, but there's nothing like this for physics or biology. There's nothing like, well, there's the Indians and how they think about physics, and then the British, they have a different kind of physics. The Canadians, well, they have their Canadian physics. There's nothing like that. There's just physics. That's the difference between discovering something that's really out there versus something else, constructing something for other reasons.

8.32 In that sense, really in that sense only, I believe in the God that exists in the minds of believers, since for them, that God is all-knowing, all-powerful, and all-good, and affects their lives in measurable ways, both good and evil. I guess people—I should just put my own cards on the table. Again, as I mentioned, I was raised in a secular home, not religious or anti-religious at all. My parents just weren't believers or anti-believers. Then in high school, I became a born-again Christian. It was the thing to do. 1971, a bunch of my friends were becoming born-again Christians, or they were raised religious, and they were constantly telling me about it.

8.33 I had two friends, George and Frank. George talked me into going to church with him and his family one Sunday morning, the Presbyterian church in Glendale, California. The whole family was going, and George, well, he had this sister. And she was really cute. I thought, OK, I'll just go and hang out. Then before I knew it, it's like, this guy, the preacher, is so convincing. He is so compelling. I think this is right. He did the call up to the front, and I went. And my friend George is like, "Oh my god, he's going." I did the whole thing. I accept, John 3:16, so "God so loved the world he gave his only begotten son, for you to have eternal life." I did the whole thing.

8.34 For the next seven years, I was a believer. Although there was a little hint the next day at school. I saw my friend Frank. I said, "Frank, I did it. I went to church and I accepted

Jesus." He goes, "Well, which church?" I said, "Well, the Presbyterian church." He goes, "No, that's the wrong one." I went, "What?" It turns out, he was a Jehovah's Witness. He's like, no, no. I'm like, the wrong one? OK. So that always sat in the back of my mind. The wrong one? All right.

- 8.35** Anyway, it's a long story I won't bore you with, about how I lost my religion. But basically, in graduate school, I started taking classes in comparative world religions mythology, as I mentioned, Joseph Campbell and so forth. I just remember thinking, what's the chances that this one I believe is the right one, and all these others got it wrong? Maybe it's more likely these are socially constructed beliefs. Maybe people believe for non-empirical reasons, and that's OK, but I just don't think I believe anymore.
- 8.36** The problem of evil was always a troubling one to me. It still is today. I don't think theists have a good argument for why bad things happen, especially to good people or to children or innocent people like that. They have arguments. I've read them. They're good arguments, but they don't really convince me. I've told this story before publicly, so I'll just say it again. I wrote about it in *The Believing Brain*. My sweetheart at the time, my college sweetheart, and we were now out of college, she was in a bad car accident and broke her back and was paralyzed for life.
- 8.37** I remember I wasn't really deeply religious still at that point, but still a Christian. I remember just endless days in the emergency room and then in the hospital where she's just laid out hanging upside down with oxygen tanks and all this stuff to try to get her back to come back. Never did. I prayed for her, and nothing happened. Now, it wasn't like a big test. It was just like, why should something this bad happen to this sweet, wonderful woman? If there really is a God, why would he not heal her?
- 8.38** Again, it wasn't a big test. I just felt so bad for her, like, somebody do something, and nothing happened. To me, it was just like, you know what? I just think bad things happen, and that there's nobody out there calling the shots. That's just the way it goes. It's sad, but that's true. Her story is not the only one. She was paralyzed for life. I still talk to her periodically, and it's just, things happen.
- 8.39** All right. In my public debates with theists, I just did one a few weeks ago at an evangelical church in Arkansas, in which I was debating a theist who had all his ducks lined up. Here's the six reasons I believe in the resurrection. Here's the 10 reasons I believe there really is a God, and that God is Yahweh, and so on and so forth. A very sharp guy. We're in this gym with 3,000 people, 2,999 of them are born-again evangelical Christians and just me. I said, all right, just hypothetically, what if it turns out you thought my arguments were better than his? Would you then stop being a Christian? Would you go, you know what? The Shermer guy, he's right on. I give up. I'm not going to believe anymore. Of course, nobody raised their hand. It's like, well, no.

- 8.40** The reason, I think, is because that's not why people believe. They believe for other reasons. They don't come to it and go, well, here's all the arguments on this side, here's all the arguments on that side. I guess that's the better one, something like that. Most people don't do that. After you believe, then you back into it with reasons. Again, the attribution bias. Nobody says, I believe because, I don't know, makes me feel better. Most people go, I believe because the Big Bang, the fine-tuned universe, the origins of morality, consciousness, whatever.
- 8.41** And not just religion. If you ask somebody, why are you a Republican or why are you a Democrat? They'll give you reasons that have to do with intellectual arguments and so on. But that's not really why. Most people gravitate toward one political end of the spectrum or the other for other reasons, emotional, social, personal, family, emotions, whatever. Then after the fact, you go, OK, here's the reasons I'm going to come up with to justify this.
- 8.42** I studied gun control as an issue. It looks like, from all the research, most people gravitate to one position or the other. For some reason, it just vibes with who they are and what they stand for. "I'm a freedom guy, and I think people should have guns." But then you go, well, why? Then they'll come up with arguments. Well, because if people have guns, then the criminals are less likely to come to your house and so on and so on. But those are after the fact. They believe for other reasons. I think religion is along those lines.
- 8.43** But nevertheless, there are arguments for and against God, since most people attribute their own belief in God and religiosity to what they perceive to be cogent arguments and evidence. And of course, like this on the search for truth, where we're looking at science and reason, in evidence, let's look at some of the arguments that are presented. The most famous are St. Thomas Aquinas' five proofs of God. This is in his great Summa Theologica, 14th-century work. And almost every theist today, if they are going to articulate some set of reasons for why they believe in God, it's some variation on Aquinas' arguments.
- 8.44** They are the prime mover argument, the first cause argument, the possibility and necessity argument. I'll go through each of these individually. The perfection ontological argument and the design teleological argument. Again, pretty much every argument I've ever heard from a theist is some variation of these. Let's look at that first one, the prime mover argument. The prime mover and first cause are pretty much the same argument. That is, everything in the universe has a cause. It is in motion. Therefore, there must be a first mover and a first cause that is not moved by or caused by any other mover or causal agent, and this is what we call God. That's the prime mover, first cause. You just regress the causal chain all the way back. There had to be a first mover that needs no other cause behind it.
- 8.45** OK, so here's—first of all, I should premise this by saying there are entire books written about each of these arguments and their rebuttals, so I'm necessarily being very brief

here. Here's my brief rebuttal to this. Well, the universe is everything that is, ever was, or ever shall be. That's what the word means. Thus, God must be within the universe or is the universe. In either case, God would himself need to be moved and caused, and thus the regress to a prime mover and a first cause just begs the question of what moved and caused God.

- 8.46** Now, I can tell you what the theist rebuttal to that is, is that God is that which does not need to be caused or moved. Yeah, OK, to which I say, if God does not need to be moved and caused, then apparently not everything in the universe needs to be moved and caused. Perhaps the universe itself does not need a prime mover, a prime causal agent.
- 8.47** Philosophers also point out that when people talk about the movement of objects in the world or the cause of objects, what caused that? These are everyday things that we have in life. When you're talking about the universe, this is a different category. Those kinds of arguments don't apply to the one and only thing there is. It's not like, as one wit said, "There's lots of blueberries. What's the cause of the blueberries? There's only one universe." It's not the same as, there's not, like, a blueberries of universe.
- 8.48** Third argument, possibility and necessity. Here's this argument. In nature, it is possible for things to be or not to be. But not everything could be in the realm of possible, for then there could be nothing. Here's Aquinas. "Therefore, if everything is possible not to be, then at one time there could have been nothing in existence." Now here, Aquinas goes on and on for pages in his book. It touches on that problem I talked about in a previous lecture. How can you conceive of nothing? I mean, the word itself, no thing, implies there's a thing that doesn't exist. But then there's a thing. Where did that idea come from?
- 8.49** What Aquinas is after here is, at some point, we just hit an epistemological wall. I don't even know what these words mean, if there's not even the meaning of meaning, and the words can't exist. There's no Platonic concept. There's no logic or mathematics. Where would God even be? Because God is a concept, is an archetype. It's an idea. If you've taken everything away to get to nothing, then there can't even be that. At some point, again, as I've said, I don't even know what I'm talking about when I use these words. I think that's what bothers Aquinas here.
- 8.50** My response to that is, why is it not equally plausible that God, like the universe, is possible but not necessary, and as such could just as well not exist? In any case, Aquinas' argument is presuming a monotheistic deity that must be supernatural or somehow part of our universe and outside of the universe, but how can that be? If God is outside of space and time and part of the universe, which is in space and time, how can that work? If the outside entity reaches in to our world to stir the particles, to whatever, cure somebody's cancer or something, how does this deity do that? What laws of nature uses, used, electromagnetism to move the gravity? What's going on here? There are no answers to that.

- 8.51** I like the way Carl Sagan expressed it in *The Varieties of Scientific Experience*, a personal view of his search for God. These were initially a series of lectures that he gave, and that became a posthumously published book. Carl died in 1996, so this book came out several years later, in which his widow, Ann Druyan, took the transcripts of the lectures and converted them into chapters. It's a great book. Here's how Carl thinks about it.
- 8.52** "Now, think again of all the possibilities. Worlds without gods, gods without worlds, gods that are made by preexisting gods, gods that were always here, gods that never die, gods that do die, gods that die more than once. Different degrees of divine intervention in human affairs. Zero, one, or many prophets. Zero, one, or many saviors. Zero, one, or many resurrections. Zero, one, or many gods." See, again, I think we just end up at some epistemological wall. I just don't know how else to think about this. There's just so many possibilities. The more you open it up, it's like, oh, I see. That particular religious belief is just this one little thing carved out here. But if you widen your horizon, there's a lot of possibilities that could be.
- 8.53** OK, number four, the perfection ontological argument. Aquinas argued that there are gradations, from less to more good, true, and noble. Here's Aquinas. "There is then something which is truest, something best, something noblest. Therefore, there must be also something which is to all beings the cause of their being, goodness and every other perfection. And this we call God." OK. Some of these are hard to wrap your mind around. I think what he's arguing is that I can conceive of something, gradations of perfection, and then perfection. So whatever God, it must be beyond that.
- 8.54** Well, OK. Why couldn't the opposite be true in that case? Here's how I rewrote this. "There is then something which is falsest, something worst, something ignoblest. Therefore, there must also be something which is to all beings the cause of falsity, badness, and ignobility. And this we call God." Nobody thinks God is these things. But if you're going to use the argument to get to the one, why not flip the argument the other direction?
- 8.55** Finally, the design teleological argument. This is, in my mind, the best of the theist arguments. Things look designed in nature. Eyes are designed to see. Wings are designed to fly. Here's Aquinas. "Therefore, some intelligent being exists by whom all natural things are ordered to their end, and this being we call God." Here's my rebuttal to that. "Such design arguments from nature are untenable because they're based on designed human artifacts whose design could be traced back to the natural designer, such as a watchmaker."
- 8.56** OK, here we go. Let's say I'm walking along a path. Here I'm going to give you William Paley's argument, called the watchmaker argument. Let's say I'm walking along a path and I come upon a stone, and I pick up the stone and go, "Well, it's probably always been here. It looks like it's just part of nature. It's pretty simple." But if I come across a watch sitting there on the path, I wouldn't think, "Well, that looks like it's part of

nature. It looks like it's designed." It has all these intricate parts that had to come together in just the right way for it to work, and the little thing in the back has to swirl around to keep it going and so on and so forth.

- 8.57** Sure enough, when you check, you can go and find out who made it. You just look on there and go, "Oh, let's see. It's a TAG Heuer. OK. I know where those watches are made," and I can go online and watch them make them. There's photographs of it. I could travel there myself and see the watchmakers making the watches. The problem is, how do I do that for the world, the universe? Where are universes made? What factory can I go to, to see the artifices being made?
- 8.58** In other words, it's what's called a category mistake in philosophy, that you've shifted over here, made things look designed, therefore there must be a designer, and you can check. Sure enough, there is. But now you're taking something that's in a different category, universes or worlds. How are those designed? We have no evidence whatsoever of a supernatural designer beyond the circular argument that natural objects appear designed, and thus a designer is needed to explain them.
- 8.59** In other words, why is it not reasonable to ask, "Well, who designed the designer?" In other words, if you have a designer, if the world is complex and looks integrally designed, and therefore, the best inference is that there must be an intelligent designer, should we not then infer that an intelligent designer must itself have been designed? We have to go searching for a super intelligent designer. And guess what you have to then look for? A super-duper intelligent designer that made the super-intelligent designer, that made the intelligent designer, that made us, and so on.
- 8.60** Now, you can do this ad infinitum. The theist will say, "No, no, no, come on, come on. You got to stop the causal chain somewhere. I'm stopping it right there at the first one." To which you say, "Well, yeah, that's an arbitrary place to draw the line. Why not just take it back one more step? That's reasonable. You're the one who's making the argument that causal chains go back. I'm just taking it back one more step." At some point, you're just left going, "I don't really know." How far back? Is the universe infinite? Can we go back infinite steps? And so on. At some point, we just hit a wall and we can't go back any further because that's all we can think of.
- 8.61** All right, let me rifle through a few of the other arguments for God's existence. Those are the five. Here's five more that I often hear, or you could pick up any book on why people should believe in God, and you'll see these. The miracles argument. Well, we already dismissed that one in the last lecture. There's no evidence for these beyond statistical improbabilities, most of which can be explained. Or the answer is, I don't know how that happened. Because we can't explain everything. There's always a residue of unexplained anomalies in any field, and it's OK to say I don't know, but that's not proof of something beyond that.

- 8.62** There's something called Pascal's wager. You're probably familiar with it. This is the famous philosopher Pascal, who made the argument that it's almost a signal detection problem, two-by-two grid. OK, so, if there is a God and it's the Christian God, and you wager in favor of that, and you believe you get eternal life, and then in the other cell, there really is the Christian God, and there is a possibility of eternal life, and you reject it, bad decision. You're going to hell forever. Or that one is not true, and you elect to believe in that one anyway, you've wasted a lot of time because you don't get eternal life. Or that one's not true and you reject it. That's a correct rejection in our two-by-two matrix for the signal detection problem.
- 8.63** So Pascal's argument is that the benefits of the payoff, the payoff is so huge, that you might as well just pick the one cell just in case. What have you got to lose? I mean, come on. Well, you don't want to spend eternity in hell, and the benefits are pretty great. Now, he didn't exactly mean, just pretend to believe, because an omniscient being is presumably going to know you're pretending. Come on. But his argument was that, if you go through the motions, you will come to believe, really truly believe, and God will know you truly believe, and then you're in.
- 8.64** To which I say, that's all fine and good. But really, that's a human way of thinking about the divine, which I think, if there is a God who's omniscient, omni-benevolent and so on, really, he's going to punish me for picking the wrong two-by-two cell, the wrong cell in the two-by-two matrix? Really? I mean, come on. You gave me this brain to think, if that's how it came about. You just didn't provide enough evidence for me to make the switch over from doubt to belief in the matrix there.
- 8.65** I'm not really worried about it. I really am not. I just think I've tried my best. I tried this, I tried that. This is where I'm at now. If I die and I wake up, and turns out there is a God, it's like, OK, all right. So where am I going? I mean, come on. I'd argue with him. Come on, man. I just tried to reason the best I could, and so I picked the wrong one. So what are you going to do? An all-loving being cannot possibly punish most people for choosing a religion that they had no choice over. Really, they were just born into this. They were born in India; they were born 10,000 years ago, and so on and so forth.
- 8.66** Mystical experiences of God. We talked about that when we were talking about mystical truths and why we can't ever validate those. So again, for you personally, if you go, "I believe in God because I had this experience," this is very common. Francis Collins, the head of the National Institutes of Health till recently, and somebody I know, wrote this book about his life in science. He was one of the pioneers in the human genome project. This is a smart guy, very successful guy, and a super nice guy, good guy. He and I have had debates and discussions about this.
- 8.67** Basically, the last chapter of his book is—his book is called *The Language of God*—the first set of chapters, all the chapters except the last one, could have been written by any atheist scientist. It's like, here is why the theory of evolution is true. Here's what we know about the genome. Here's how the world works and so on. He gets to the last

chapter and says, like my friend Ken Miller I talked about, "And by the way, I believe in God. I accept Jesus as my savior because here's what happened. I was on this hike. I was having some personal problems, and I saw this waterfall, this frozen waterfall, and I realized Jesus is the way, I accepted him and, wow, life is so much better."

- 8.68** OK. I'm not even going to ask, did you have a video camera, the frozen waterfall, and there was an image of—no, it's nothing like that. It's just a personal feeling he had, and it makes his life better. OK, fine. I have no objections to that. He's not trying to convert anybody. He's just going, "This is what happened with me." That's it, full stop.
- 8.69** More recently, I had this exchange with Ayaan Hirsi Ali, the great champion of women's rights. She was born and raised in Islam in Somalia. She was subjected to a forced marriage. She underwent female genital mutilation. The whole horror story there. She's been campaigning against this her whole life. She's a lifelong atheist, after she left the Islamic religion. She was one of the new atheists, along with Christopher Hitchens and Dawkins and Dennett, and Harris and others, myself. We would all meet and talk about whatever we talk about, atheism, whatever.
- 8.70** It's sort of a weird thing. There used to be this, the Richard Dawkins Foundation would give this award, the Atheist of the Year Award. It's like, who gets that award? The person who really did not believe in God the most all year. I never believed once all year. It's like, OK. This is the problem. Atheism is not a worldview. It's not a thing. It's nothing. It's just, I just don't believe in God, full stop. OK, what do you believe? Well, universal human rights, civil rights, women's rights, gay rights, animal rights, whatever I believe in, equal treatment under the law, universal principles that apply to everybody and so on. That's humanism or secular humanism or Enlightenment humanism. That's not atheism. This is a mistake people make.
- 8.71** Anyway, a couple of weeks ago, Ayaan wrote this essay saying, Why I am now a Christian. It was like, what? The Internet Twittersphere blew up like, "Oh my god, one of the world's most famous atheists is now a Christian." Wow, OK. So I read it. Then basically, she said, it was a political argument. The West is under attack from various illiberal forces. Atheism is doing nothing about this. In fact, a lot of these far left atheists, they're making it even worse. Christians, Christianity, the Judeo-Christian worldview, at least believes in right and wrong and true values that are universal and so on. So they are the ones standing up to this assault, the liberal assault on the West. Therefore, I go along with that. That's her argument.
- 8.72** Something like, I don't know what you think about Christianity, but at least they know what a woman is. That was one of her points. It was like, what is a woman? I can't answer the question. At least Christians can answer the question. That's her argument. Anyway, I had a private exchange with her and invited her on my podcast. Well, I don't really—she doesn't want to debate because she's not making an argument that I switched because of all these really good reasons. It was just like, politically, this gels with me.

- 8.73** And by the way, then she gave a podcast interview with the platform that published her essay. When she said, "Well, I've also had a lot of personal problems in my life, with depression and alcohol, and so on and so forth. This is what got me out of that." OK. To which I go, whatever gets you through the night. Life is hard, entropy, bad things happen. People need what they need. I'm never going to try to take that away from anybody.
- 8.74** I remember being at a Thanksgiving dinner at my friend Michael Cole's house in Atlanta, Georgia, and he was friends with Isaac Hayes, the great singer, Theme from Shaft, Grammy award-winner, and he was the voice of Chef on South Park, and I'd never met him, but there he is, sitting right across from me at the Thanksgiving table because my buddy was close to him. He's famous at the time, Scientologist. This is one of the things I debunk in Skeptic, and I'm like, "Oh my god, here's Mr. Scientology himself." But I didn't want to be a jerk about it. I just, "So Isaac, why are you a Scientologist?"
- 8.75** He gave me a really thoughtful answer. It was that, "Well, Michael, I had all this fame and fortune. I was top of the game here, Grammy award-winning, and on and on, and I lost it all. Just lost all my money, family, alcoholism, drugs, the whole thing. Scientology pulled me out of that, and they gave me my life back." In other words, these are not empirical truth claims. These are something else. I don't want to take that away. That's related to this, this consoling, this consoling argument, credo quia consolans, and it's, I believe because it's consoling to me. In a way, that's what that is. It's a personal truth, a subjective internal truth, to which all you can say is OK. There's really not much more to go on that.
- 8.76** The last one I often hear is how to explain the origins of morality. Where does right and wrong come from? We'll save that for another. That's a whole lecture. I've written quite a bit about that. Then one final position on this before we wrap this up. I introduced you in an earlier lecture to Martin Gardner. There he is. I knew Martin, and he is really one of the giants of the modern skeptical movement. Again, as I mentioned, he wrote the Scientific American column before I got it for 25 years.
- 8.77** He's highly regarded among scientists and secularists, but he also believed in God, to which most atheists are like, "What? How can you believe? Martin, you debunk all the supernatural, the paranormal, but why this one?" He called himself a fideist. In those atheists, theist, atheist, deism, pantheism, and so on, fideism is an actual position. This has to do with believing for pragmatic reasons. It's a pragmatic truth. Here's Martin.
- 8.78** "As a fideist, I don't think there are any arguments that prove the existence of God or the immortality of the soul. Even more than that, I agree with Unamuno," who's a philosopher that influenced him, "that atheists have the better argument. So it is a case of quixotic emotional belief that is really against the evidence and against the odds." Then Martin attributes this to William James, the great founder of modern psychology,

whose book is probably the most famous on this, that's launched the whole study of the psychology of religion. It's called *The Will to Believe*.

- 8.79** James' argument, in essence, is that if you have strong emotional reasons for a metaphysical belief, and it's not strongly contradicted by science or logical reasons, then you have a right to make a leap of faith if it provides sufficient satisfaction. Then he explains a little bit more about how that works. "It has to be a leap of faith about something that has overwhelming importance to an individual. Second, it has to be something for which there isn't any strong empirical evidence or logical arguments against it." There is something radically different about belief in a mind behind the universe and the whole cluster of beliefs that the New Age movement presents.
- 8.80** Martin's reference to this is all the psychics, the spoon-benders, the astrologers, and people that talk to the dead, and the tarot card readers, and all that stuff that the skeptics have been debunking for decades, of which Martin was one of the pioneers to do that. But he's saying, this is something different. This is in a different category, and there aren't many things you could put in that. Really, it's just the God question and maybe free will and determinism. You can't prove that I don't make choices. So even if we live in a determined universe, it's OK to believe that you are free, even if you can't prove it. That would be something like that.
- 8.81** And in conclusion, he says, "If you ask me for details about the nature of God, I would have to answer, I don't know. The God I believe in is so completely transcendent and so wholly other that you really can't say anything about God's nature. To ask, for example, whether God is inside or outside of time, I have no idea what this means or how to reply to it." There, I think he's echoing Paul Tillich, the great theologian, who made this argument that, I've been reading this for a long time going, what is he talking about? And now I think I figured it out.
- 8.82** He says, "God does not exist. He is being itself beyond essence and existence. Therefore, to argue that God exists is to deny him. The question of the existence of God can neither be asked nor answered. If asked, it is a question about that which, by its very nature, is above existence. Therefore, the answer, whether negative or affirmative, implicitly denies the nature of God. It is as atheistic to affirm the existence of God as it is to deny it. God is being itself, not a being."
- 8.83** Now, I'm not sure what category this would go in. Maybe pantheism, the entire universe is God. But I think it's even more than that. It's like, whatever God is, it can't just be more of this stuff, like us, but smarter. Like all the Greek and Roman gods, they get jealous, they get angry, they have sex with people. They're just like us, but a little different. I think what Tillich and Martin Gardner are talking about is something completely different. It's like, you can't even really conceive of it. Therefore, there's, it's possibly one of those unknowables. Back to Huxley's definition of agnosticism, we'll just never know. What do you mean by these words?

- 8.84** All right, I'm going to finish up this lecture in the course with what I call my God Gambit, which I originally call Shermer's Last Law in one of my Scientific American essays, which goes like this. "Any sufficiently advanced extraterrestrial intelligence or far future human is indistinguishable from God." I originally just called it Shermer's Last Law, but you shouldn't name anything after yourself. So, I've just called it my God Gambit. It's based on this.
- 8.85** This is Arthur C. Clarke's famous three laws. Number one, when a distinguished but elderly scientist states that something is possible, he is almost certainly right. When he states that something is impossible, he is very probably wrong. His second law, the only way of discovering the limits of the possible is to venture a little way past them into the impossible. Of course, that's what scientists do. Then three, the important one here for me, is any sufficiently advanced technology is indistinguishable from magic.
- 8.86** All right, so let's think of a current example of this. You can plant chips in people's brains now. You've heard of cochlear implants. They have the thing on the outside here, so you can change the batteries and so on, but it's connected inside to the brain to help deaf people hear again. It's quite miraculous. Or there's people that have Parkinson's, and they put little chips on their motor cortex to stop the trembling caused by the disease.
- 8.87** There's even some of these examples of people that are paralyzed for life, not just paraplegic, but quadriplegics, where they put a chip in the motor cortex, so the guy can control his hands or control the computer-aided assistance moving his hands, just by thinking. He just thinks the thought. The neurons fire, the motor cortex neurons fire, send the signal to the computer that moves the object in the hand together, and they pick up the cup or whatever.
- 8.88** OK, so here's the reason I'm bringing this up. If you didn't know he had a chip in the brain and you know he's paralyzed and all of a sudden he's moving, that looks paranormal. That looks like magic. When you don't know the technology, it just looks like magic. OK, so Clarke's idea is that, the more—I mean, he wrote this back in the '50s. If you can only imagine how far AI and computers have come now, that it looks even more magical. Just picture this. You have a chip in your brain that just thinks, like, you walk into the house and I think, I want to hear Mozart.
- 8.89** You can already do this with Alexa. Alexa, play Mozart, and boom, on it comes. Imagine the next step, you just think it, and it comes on. Now, if you don't know what the technology is, even now, if you didn't know there's a little Alexa and you're just talking to the walls and somebody, a time traveler from a century ago, just sees you talking to the wall and all of a sudden the music comes on, that looks like magic, because you don't know behind the wall is the little Alexa thing. Anyway, that's the point of that.
- 8.90** OK, Here's my gambit. Biological evolution is relatively slow. It's been three and a half billion years since the evolution of bacteria, 500 million years since complex life arose

on Earth, 35 million years since the primates first arose. Seven million years since the first hominids branched off from the primates that led to our lineage. Three million years since bipedalism arose with Lucy and Australopithecine afarensis in Africa. One and a half million years since the use of tools. 100,000 years of our species. That's a long, slow development.

8.91 By contrast, cultural evolution is relatively fast. It's been only 10,000 years since the agricultural revolution, 5,000 years since writing, 500 years since the printing press, 200 years since the industrial revolution. We went from the Wright brothers to the landing on the moon in only 66 years. You just take this principle, Moore's Law. Gordon Moore, one the cofounders of Intel, in the '60s, identified a trend he had already spotted. That is, the amount of computing power of a computer for the same price doubles about every 12 to 18 months, and it's down to about a year now, 12 months.

8.92 In other words, for, I don't know, a thousand bucks, you can buy this amount of computing power. Every year to a year and a half, you can buy double that amount of computing power for the same thousand bucks, and so on. You can see this in the history of computing. Now, at some point, the curve has to—it's an asymptotic curve. It's going to run out of space, atomic limitations of space on the chips. But that's the idea.

8.93 You can take that principle and apply it to all kinds of technologies, like genome sequencing. I mentioned the Human Genome Project. Well, this was billions of dollars to do. You can now get your cheek swab with 23andMe for, I don't know, it's like 100 bucks or something. It's nothing. That's what happens to almost all science and technology, technologies based on science, is that the knowledge itself gets applied, and then you have these asymptotic curves where you get more and more of the technology for cheaper and cheaper, and this applies across the human landscape. Therefore, how far can we go?

8.94 Here's my next step. If you do encounter an extraterrestrial intelligence, the odds are close to zero that it will be anything like us. I've made jokes about this, like the aliens landed in Roswell in 1947, and they had technology that was just five years ahead of us, and they gave us transistors or back-engineer. I mean, this is not at all what would happen. The universe is so vast, the distances between the stars are so great, that the chances of some other species evolving exactly on the same parallel path as us, bacteria to multicellular organisms, to mammals, to primates, to humans, to tool use, and so on, in just the right, exact same—and somehow they get here and find us, and they're just close to us in culture and technology. No, it won't be anything like that.

8.95 In fact, how much more advanced is it going to be? Well, SETI scientists estimate something like 10,000 to 100,000 years or more. It could be millions of years ahead of us. Now, they could be behind us, but of course, we're not going to encounter them unless we go to their planet. If they come here, or we encounter them out there somewhere when we're a space-faring civilization, they're going to be way, way more

advanced than us. Therefore, just extrapolate Moore's law out 10,000, 100,000 years. Just think of ChatGPT and the introduction of these AI programs in just the last year. Just astonishingly advanced over what we just had two years ago.

- 8.96** Now, just take that out 10,000 years or 100,000 years or a million years. One final piece of this is Nikolai Kardashev's civilization types. He was a Russian astronomer who thought big about the future of human civilizations or extraterrestrial civilizations around the galaxy or cosmos, based on energy consumption. Again, second law of thermodynamics. Entropy, the first law of life is to consume energy somewhere to get calories and push things around in the world, to carve out a little niche of order in the chaos of the universe.
- 8.97** Our civilization has done this largely through using dead plants and animals, fossil fuels. That's it. We are a type zero civilization so far. Well, Sagan calculated that we're about a 0.7 civilization, because we're now starting to get some renewables, nuclear power, and so on. If we can get that, we can get off fossil fuels. We're not going to get off fossil fuels until we have a really solid, cheap alternative to them because they're so good for getting people out of abject poverty and into the industrial world.
- 8.98** I do find it, just as a parenthetical note, unfair for Westerners, particularly Western environmentalists, to tell third-world people, "You can't use fossil fuels. You can keep living in poverty because we don't want to burn anymore and cause greenhouse gases." That's not very fair. The alternative to that is, let's get them as rich as possible, as quickly as possible, through fossil fuels and then help them make the transition to renewables and nuclear power. And so, oh, no, you can't use nuclear power. OK. I don't want to go down. I don't want to get in that debate.
- 8.99** But Carl's idea is that, at some point, if you want to be an advanced civilization, you got to make the transition to using the energy of an entire planet. That is, its geo forces, weather forces, renewables, sun, wind, waves, just anything that moves, water, and so forth. And so, scientists who think about these things have calculated we will become a type one civilization, probably in about a century or so. I think that's possible. I think that's very doable.
- 8.100** What's the next step after that? It's a stellar civilization. That is, you control the energy of an entire star, which is about a thousand years into the future. What are they talking about here? They're talking about something, it's called a Dyson sphere. It's featured on Star Trek, for example. Freeman Dyson was one of the great physicists the 20th century, who came up with this idea that if you really want to capture energy for your civilization, you should just build a gigantic solar panel that's the size of the entire orbit of the Earth. Or a planetary size, or solar panels that are the size of continents, and you put them in space to collect all that energy.
- 8.101** Because basically the sun is a big battery up there, just burning energy, just free energy, just pouring out. Just get as much of it as you can. Sort of like if you go, if you fly, like

I did on this trip, you go near Vegas and you can see the Tonopah solar panel array. It is huge. I mean, it's just like tens of square miles of nothing but solar panels. You can see it from hundreds of miles away. It's like this bright light coming up. Imagine if you had the entire state of Nevada is a giant solar panel. OK, well, we want to use some of that land because you can't do that in all parts of the country. So, we put it in space and look for that.

8.102 Now, astronomers, who are interested in knowing if we're alone in the cosmos, are tuning their telescopes, the Hubble telescope, and now the James Webb telescope, to look for what they call technosignatures. It is something that looks a little different than anything else we've seen. It doesn't look like a comet, doesn't look like another planet or a meteor, like a broken-up planet or anything like that. It looks something artificial because it gives off this radiation in the area of the electromagnetic spectrum that looks like solar panels or looks like something that could be human-made.

8.103 Now, we haven't found anything like this yet, but there's good reason to think that we're not the first. The Copernican principle, we're not the special. Therefore, look how far we've come, and we're now becoming a space-faring civilization. We have several dozen spacecraft we've sent all over the solar system. We've been to every planet. The Voyager, both Voyager spacecraft, launched in 1976, are now far beyond our solar system. They've reached the heliopause, that is, the place where the sun's radiation pushing out is no more stronger than the radiation from other stars pushing this way.

8.104 That's pretty good. I just read yesterday that the signal is going dead. It's because it was never supposed to last more than a few decades. Here we are, 50 years later, and it's still a little nuclear power station on the Voyager still ticking away, sending this little signal out and so on. That's pretty cool. But imagine if we had 10,000 years of this. We were sending—just think of what Elon has been able to accomplish in just a few decades. He's launched more rockets with SpaceX than all space agencies everywhere in the world combined, in just the last five years. This is quite doable.

8.105 You just need somebody, a good leader that says, "We're going to just do this. We're going to Mars." What? And we are. It's going to happen. Imagine that. Again, carry this out a thousand years, 10,000 years, 100,000 years. And assuming we're not alone and we're not first, others are out there already doing this. Why don't we look and see if there's anything weird out there that gives off radiation that's not normal? Therefore, that might be a Dyson sphere or a giant solar panel. I just find this so interesting.

8.106 There was an object found, I don't know, maybe about five years ago. It was labeled Tabby Star because this female astronomer named Tabby had found it. It has to do with how they look for extra-solar system planets, which is to say, between us and some other star, a planet passes between us and their star. The light is dimmed ever so little bit because of the occultation of the light by their planet. Sort of like a solar eclipse here with our moon, but tiny. The instruments on these telescopes are so sensitive.

They can see just the tiny, tiny, tiny bit of light that dips because of the planet passing between us and the star. It's astonishing they can do these calculations.

8.107 Just like what's called the occultation of Venus. Venus passes between us and the sun every once in a while, whatever the calculation is there. Astronomers now can detect it. You can see the dip in the light. Anyway, or you can actually see with a home telescope if you used a filter so you don't burn your eyes looking at the sun, and you can see the little disk of Venus moving right across the face. It's really quite incredible.

8.108 Anyway, so Tabby found this dip in the light curve for this particular star that was 1,500 light years away. The light from that left 1,500 years ago. It just got here. It didn't look like comets or meteors or small planets or broken-up planet. It didn't look like anything anyone had ever seen in the astronomical literature. This became like front page news. Tabby Star. Oh, my God, it might be the aliens. It might be a technosignature. It might be a Dyson sphere. It might be solar panels and so on.

8.109 I remember doing a radio show, KBC Talk Radio, DriveTime LA with Mark and Gillian. They said, "Dr. Shermer, what's going on here?" It's so funny to see how quickly people go to the alien answer. I'm trying to explain this like I am to you. Gillian goes, "So what are they like?" I'm like, "What is who like?" "The aliens." "No, we don't know there's aliens. All we have is a dip in the light curve. That's all we have." In any case, it's 1,500 light years away. If we've discovered it really is aliens and you go, "Hey, we're out here," it takes 1,500 years for our signal to get there, and then 1,500 years for them to answer. That's not exactly a snappy conversation. That takes 3,000 years for a hello. Anyway, it's an amusing thing, but that's possible.

8.110 By the way, Kardashev also proposed that there could be galactic civilizations. These would be hundreds of thousands or maybe millions of years in the future, in which they control the energy of the entire galaxy. Imagine what you could do with that energy. Now we're on the page of pure science fiction, but there's no laws of nature violated here. We could do all of this with enough knowledge because that's basically what all of science is about. It's just knowledge. If we know the right solution to the problem, we can solve it.

8.111 Others have taken this and run with it. There's a Type 4 civilization that controls the energy of the entire universe, and maybe there's multiple universes, and so the Type 5 would be, you get to control. This is like the Borg in Star Trek or I don't know what. Now, I don't even know what we're talking about. It's so far out there. But that's not my point. I love all this stuff. It's super interesting.

8.112 My point is that, what if we did that, or what if somebody else did that? OK, one little sidebar before I finish here because we're just about done. If what I said is true, and that we're not the first, surely we're in the middle of the bell curve, half are behind us, half are in front of us, where is everybody? They should have already— many, many civilizations should have already become type 2, let's say, and sent their probes out and

managed to explore the planets of every star in the galaxy. This has been calculated. You could do this in a million years or so. It would take about a million years to send out these probes to every star in the galaxy.

- 8.113** The galaxy is about 100,000 light years across, about 50,000 light years wide, and maybe 10,000 light years thick. They should be able to do this, even if it was several million years. Again, on an evolutionary time scale of billions of years, it's nothing. This is called Fermi's Paradox, because the first guy to think about this was the Italian-American physicist Enrico Fermi, who worked on the Manhattan Project. After the war, when he and his buddies like Oppenheimer were having lunch one day, they started doing this thought experiment, like I just did for you.
- 8.114** They go, "Yeah, if that's true, they should be here. Where is everybody?" That led to this whole Fermi paradox. There's about 50 different answers to this question, one of which is, "Well, we're alone. We're first. Somebody has to be first. We're the ones." Not likely, right? Or it's just big and empty, and they just missed us. Or they do know we're here, and it's like a zoo. They're up in the mountains cloaked and they're watching us carefully. Or they have probes in space monitoring us and so on. And this goes on and on.
- 8.115** I think it's that they probably are out there. I think they just haven't found us. Really, it's mostly empty space. I gave you that analogy yesterday about the Earth the size of an orange in LA, the closest one would be in Chicago. You could do this various ways. The Voyager spacecraft I mentioned, it is really moving. It is fast. It's moving, I think it's something like 50,000 miles an hour. If it was going to the closest star, the Alpha Centauri system, Proxima Centauri system, that's the closest star. It would take something like 70,000 years to get there. All right, that's a long time to sit there with your tray table down and your seat back. It's mostly empty space.
- 8.116** OK, so my gambit. What would you call an entity or being capable of creating life, planet, stars, and universes? There are speculations about how this could be done. Really, we're already on the verge of creating life forms. Just in a test, there's already been alternative manmade artifactual genomes. Well, if we could do that, you could create cells and so on. All it is, is a technological problem. With enough knowledge, you could do it. You could create planets and stars by pushing stuff around.
- 8.117** A lot of these science fiction writers, not just science fiction writers, people like Elon Musk and others, Bob Zubrin, who has a whole book out about how to colonize Mars and how to get out to the moons of Jupiter and colonize them, how we can move on to the next solar system and so on—this is all doable with enough knowledge and energy to do it. One possibility, one theory for the origin of the universe is a collapsed black hole, because at the center of the black hole, there's this thing called the singularity. This is Stephen Hawking's idea, that maybe the entire universe was once a singularity, and that a collapsing black hole in some universe gave rise to our universe, something like that.

- 8.118** Anyway, my point about that is, what would you call an entity, an agency, a smart civilization, that could do it? If you don't know the technology, you call it God. If we do know the technology, we call it an extraterrestrial intelligence, or maybe in humanity's far future, we will become, in essence, omniscient, omnipotent, and hopefully omnibenevolent. Either way, let's hope we rise to the challenge of encountering or becoming divine. And so I end by offering us to, with difficulty, go to the stars. That's what we could do with the right knowledge. Thank you.
- 8.119** Either pertaining to this topic or any topic that we've discussed, can you give me two book recommendations, one that you have published and one that someone else has published?
- 8.120** On this general subject? Yeah, or any subject. Yeah, OK. I hate to plug my own books. I've already done enough of that. People often ask me, what's your favorite book that you've written? The answer is always the last one I wrote because I like to think I'm getting better. I don't know. I guess my own, maybe *The Moral Arc*, about moral progress, which is a whole other course I think I'll maybe do here. *The Believing Brain* deals with a lot of these topics. The next book is really, this course outlines much of the next book, which is just called *Truth*. But forget that.
- 8.121** Stephen Pinker's book *Rationality*, and his previous book, *Enlightenment Now*, I think really outlines to what extent knowledge applied, scientific knowledge applied, technologically speaking, really has made the world a better place, just measurable on every level. It's just astonishing. He starts off with David Deutsch's book. He cites David Deutsch's book, which is called *The Beginning of Infinity*. It's David Deutsch. He's a British quantum physicist. He makes the argument with that, if it doesn't violate the laws of nature, we could do anything with the right knowledge. Therefore, error detection is the most important thing we could do, because most of the knowledge we have, we get wrong. The solutions come from getting it right. We have to have reliable knowledge and so forth.
- 8.122** Steve quotes Deutsche saying that, and then says, "Now, let's see what we have done with good knowledge for the last 300 years." The average life span used to be 30-something, now it's more than double. How did that happen? Solving problems. Why are there pandemics? This goes back to Snow's discovery in the 18th century that people using that well right there, and you can map the spread of the disease from that well. OK, so maybe we have a germ theory of disease applied to help. It gets dispersed in water, understanding how the public health systems like sewers should be structured and so on. All this has led to all these public health measures and so on. You could just go across the board on almost any technology.
- 8.123** Anyway, that's Pinker's *Enlightenment Now*, and then his last book, *Rationality*, about much of the subjects we've been talking about here. Maybe who else? Let's see. I have so many favorite books. It's often whoever I had on my podcast. But I think Dan Dennett's books. I've cited Dan. He's a good writer. Of course, you can't go wrong with

Christopher Hitchens. He's one of the most compelling writers I've ever read. And fortunately, he reads his own books on Audible.com, so you can listen to the great Hitch in his British accent, writing about all these, God and politics and so on.

- 8.124** Lastly, I guess, Richard Dawkins. Well, *The God Delusion*, if you're into this subject. But pretty much anything on evolutionary biology. *The Selfish Gene* was his book that put him on the map. And that's one of these—it's one of these books that changed all of how biologists think about life and evolution. I mean, it was a total game changer, 1976. But it's also written in a way that anybody could read it. You could just pick it up at an airport book stall, and read it on the plane. I think that's just so admirable to be able to do that, to write for your own colleagues in a very professional way that changes the way they think, and anybody could read it. Those are my recommendations.
- 8.125** I was just wondering, other than critical thinking and scientific reasoning, what would be the next foundational skill that one should seek to master to contribute to progress?
- 8.126** Well, I guess it would just depend on the field and how they operate. But how can you improve, I don't know, the criminal justice system or journalism? Journalism is undergoing something of a revolution now because of the dispersal across all platforms, of anybody could be a journalist now. Well, where's the fact-checking and the editing, something like that? Medicine, a lot of problems in the healthcare system. So figuring out how to solve health care problems to me is huge.
- 8.127** I just had Angus Deaton on my podcast. He's a Nobel Prize-winning economist who wrote an entire book about health care reform. It's astonishing how bad and inefficient our health care system is, despite all the medical advances we made. I think each field has its own set of problems to solve. Just taking the principle of, well, what's the metric we're looking for? Well, we want, in this case, affordable health care for everybody. That should be the goal. Why? Because of human flourishing is our ultimate goal. But that's often not the case.
- 8.128** There's a perverse incentive to make as much money as possible in individual units or doctors themselves. So you end up with these ridiculously priced anything and everything. The Band-Aid is \$30 or something in the hospital. It's just crazy. Just my own personal experiences of this, they just try to ding you every way they can because the incentive is for them to do that. It would depend on the field.
- 8.129** I remember I went into a hand physician in Santa Barbara, a little carpal tunnel just from cycling. Here, he's like, "Oh, yeah, well, we could do this and that surgery." I'm like, "I don't really want to cut up my hands. Let's just see if it goes away." He said, "Well, I'll give you these splints that you can wear, these little grips, things that hold your thumb out like this, and it comes up to your wrist. It's mostly neoprene, like a wetsuit, with just a little bit of carbon here just to make it stiff." I don't know. He gave me a pair, and it looked like it'd probably cost five bucks to make these things in China

or something like that. I go, "Well, I ride a lot, so these will get sweaty. Maybe you have another pair you can give me when I'm not riding my bike?" "Oh, yeah, sure. Here's a pair."

- 8.130** Then a month later, I get the bill through my health care provider, and they charged me \$800 for the pair of these things. I'm like, OK, I still have the box. I'm going online to see what it would cost me to buy these personally. They were like 20 bucks. I called my healthcare provider Blue Shield, and I said, "Do you know what? They're dinging you. They are screwing you." I finally got somebody who got the doctor's office on the phone. I mean, this was one of these great moments.
- 8.131** It's like, "OK, Dr. Shermer, hold on. I'm going to get the other guy on the line." It gets not the doctor, but the office manager or whatever. The three of us are going, and this woman on the other end, she's like, "There was a typo or something. We made a mistake. We'll correct it." OK. I happened to stumble into this one thing. How much does this go on that nobody who doesn't bother to follow up? To me, those are huge problems, just the efficiency of it.
- 8.132** My question is regarding how you mentioned, with your Sunday special with Ben Shapiro, how Biblical scholars will disagree, and that could be reflective of lacking evidence on either side of the argument. I wanted to know, in your professional opinion, how this applies to science, because we see scientific disagreements all the time. So would that be reflective of inadequate evidence?
- 8.133** I think so, yes. I think the difference is, in science, at least in principle, there has to be a way to test it. You claim you have data, I claim I have data. OK, you run the experiment, see if you can figure out which of us is wrong and so on. There's some way to get at it. My opinion, anyway, these religious differences are not resolvable in some experimental way or some, here's a set of arguments we should all agree. It never happens. I think the reason is because it's a fundamentally different area of knowledge than science.
- 8.134** I just think, again, the anthropologist from Mars goes, "Well, which is the right one?" Well, we are. No, we are. Well, is there something I can do to see which is the right one? And the answer is no. Whereas if it's climate change or vaccines or whatever the empirical question is, there has to be, at least in principle, some way we can figure it out and go, "That's the right answer." It's like in the '50s and '60s, there was a big debate between the Big Bang theorists and the steady-state theorists. It wasn't clear which theory was right for the origin of the universe, where all this matter comes from, and so on and so forth.
- 8.135** Well, within about 20 years or so, it was clear the Big Bang theorists were right because of all this new evidence that was coming in, and the steady-state theorists just accumulated no more evidence in support of their theory, and that was pretty much the end of that. There are a few Big Bang skeptics around, but not many. Any field,

there's always going to be a few people on the outside. No, I still don't believe it. Theory of evolution, there's still people that doubt it, that are not creationists, that they're just inside. But most, almost everybody accepts it.

- 8.136** That tells us there's a convergence of evidence toward a position that eventually gets settled. That almost usually happens in most areas of science. When it doesn't, it tells me that they really don't know. There's something wrong with the whole paradigm. You're asking the wrong question. There's something off here while you're not getting at it. Thank you so much.